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Prevalence of Sarcopenia among the Elderly in China: A Systematic Review and Meta-analysis

Journal:	*BMJ Open*
Manuscript ID	bmjopen-2020-041879
Article Type:	Original research
Date Submitted by the Author:	22-Jun-2020
Complete List of Authors:	Xin, Caihong; The Fourth People's Hospital of Shenyang Sun, Xin; First Affiliated Hospital of Soochow University Lu, Li; The People's Hospital of China Medical University Shan, Ling; The People's Hospital of China Medical University,
Keywords:	GERIATRIC MEDICINE, EPIDEMIOLOGY, General endocrinology < DIABETES & ENDOCRINOLOGY

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Prevalence of Sarcopenia among the Elderly in China: A Systematic Review and Meta-analysis

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Running title: Prevalence of sarcopenia among

Key words: Prevalence; sarcopenia; China

Word count of the manuscript: 1624

Figures number: 7

Tables number: 2

Disclosure Statement: The authors have nothing to disclose.

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Abstract

Objectives: Sarcopenia is characterized by lean muscle mass wastage and muscle function decline that occurs with aging, with a high rate of adverse outcomes. Several studies have investigated the prevalence of sarcopenia among elderly Chinese people, and results were inconsistent. This meta-analysis was conducted to estimate the overall prevalence of sarcopenia among the elderly in China.

Design: Systematic review and meta-analysis.

Setting: A literature research was conducted using PubMed, Web of Science, Embase, CENTRAL, CNKI, VIP, CBM, and Wanfang literature databases. The population-based studies that reported the prevalence of sarcopenia among elderly Chinese people were selected.

Primary and secondary outcome measures: Random effect model was used to estimate the prevalence of sarcopenia. The sex-specific prevalence of sarcopenia at 95% confidence interval (CI) was also calculated.

Results: In total, 28 articles were included in this meta-analysis, involving 26,626 participants. On the basis of the Asian Working Group for Sarcopenia criterion, the overall prevalence of sarcopenia among the elderly in China was 18% (95% CI, 15%–20%). The prevalence of sarcopenia was higher in Chinese females than in males (18% vs. 15%).

Conclusions: This systematic review is the first estimation of the pooled prevalence of sarcopenia among the elderly in China. The results suggest that China may have the largest sarcopenia patient count. Therefore, more attention should be given to the prevention and control of sarcopenia.

Strengths and limitations of this study:

- This systematic review is the first estimation of the pooled prevalence of sarcopenia among the elderly in China, suggesting that China may have the largest sarcopenia patient count.
- The main data are from urban populations and studies on the prevalence of sarcopenia in Chinese rural areas are rare, impacting the calculation of the total prevalence of sarcopenia.
- Not all studies presented data according to age group, so we cannot accurately estimate the prevalence in terms of age subgroup.

Introduction

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Sarcopenia is characterized by the deterioration of muscle mass and function that occurs with aging. Several studies have shown that with age increase, skeletal muscle mass and strength will be lost at a rate of 1%–2% annually, possibly leading to the decline of physical function, increasing the disability, and mortality rates of elderly people and bringing serious economic and social burdens[1–4].

A practical clinical definition was reported by the European Working Group on Sarcopenia in Older People (EWGSOP) in 2010 [5]. Later in 2011, the International Working Group on Sarcopenia (IWGS) published another consensus [6]. However, Asia is a vast region with different races, countries, and cultures. Developing screening and diagnostic strategies for the Asian population is imperative. The Asian Working Group for Sarcopenia (AWGS) was established in 2013, and the consensus of AWGS was released based on 16 epidemiologic reports on sarcopenia in six countries or regions, leading to the development of a diagnosis strategy for sarcopenia in Asia[7]. The diagnostic criteria recommended by AWGS and EWGSOP are similar, including muscle mass, muscle strength, and activity measurements. The difference lies in the specific cutting value.

According to statistics, the elderly population (≥60 years old) has reached 16.1% of China's total population, which is close to 250 million. In the face of a severe aging situation, people are paying more attention to sarcopenia, but the prevalence of sarcopenia was inconsistent in different studies [8–12]. At present, there is no large-scale epidemiologic survey of sarcopenia in China. The purpose of this study is to systematically reveal the prevalence of sarcopenia in elderly Chinese and to provide a scientific basis for the work of clinical departments and concerned government agencies.

Methods

Search strategy

We searched the following electronic databases: PubMed, Web of Science, Embase, Cochrane Central Register of Controlled Trials (CENTRAL), China National Knowledge Infrastructure (CNKI), VIP, China Biology Medicine (CBM), and Wanfang. The search terms in the title or abstract were identified as follows: "sarcopenia," "muscle mass," and "muscle strength" in combination with the terms "prevalence," "epidemiology," "frequency," "incidence," and "China." All studies published from January 2010 to January 2020 were included. The references of the retrieved articles were examined to

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identify additional eligible studies, but unpublished studies were not included. The search languages were limited to English and Chinese. The completed Preferred Reporting Items for Systematic Reviews and Meta-Analyses checklist is presented in the Supplementary data.

Inclusion and exclusion criteria

Inclusion of the selected studies was based on the following criteria: (1) a cross-sectional study or retrospective study, (2) a sample size of ≥100 elderly Chinese participants; (3) sufficient information on the sample size and crude prevalence rate of sarcopenia at the corresponding 95% confidence interval (CI) and *P* value, and (4) the definition of sarcopenia is in line with the AWGS, EWGSOP, or IWGS standards.

Exclusion of the selected studies was based on the following criteria: (1) the same data from different studies, (2) studies must define sarcopenia only by reduction of muscle mass or strength, and (3) the subjects were a specific population, such as diabetes patients or patients with disability. Reviews, case reports, letters, and animal researches were excluded.

Data extraction and risk of bias

Information was extracted by two independent researchers from the included studies, including first author's name, publication year, study period, sample, diagnostic criteria, participants, details, and prevalence. Two independent reviewers who were not blinded to the authors or journals assessed the risk of bias in the included studies using the Newcastle–Ottawa scale (NOS) [13]. The NOS is recommended by the Cochrane Handbook for Systematic Reviews of Interventions [14]. Each included study was judged in terms of three domains using the "star system," namely, representativeness of the study group selection, group comparability, and exposure or outcome ascertainment. NOS scores range from zero to nine stars. The two researchers independently assessed the studies by discussion, comparison of findings, and resolution of differences through consensus formation. If no consensus can be reached, a third researcher was commissioned to resolve the difference.

Statistical analysis

We used a systematic analysis approach to calculate the pooled prevalence of sarcopenia for all eligible studies, and heterogeneity among these studies was assessed using Cochran's Q test and the I^2

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statistical index. Random effect model was used in combining the prevalence of sarcopenia. We additionally performed sensitivity analysis to evaluate the influence of any given study on the pooled estimate. Publication bias was evaluated using Begg's test. Subgroup analyses were also performed. The significance level was set at $P < 0.05$. All statistical analyses were performed using Stata software version 12.0 (College Station, Texas, USA).

Results

Study characteristics

In total, 1536 studies were retrieved from the aforementioned databases. After screening, a total of 28 articles were included based on the criteria shown in Figure 1 [8–12, 15–37]. Altogether, the 28 articles included 26,626 participants in China between 2010 and 2020. The selected studies' characteristics are summarized in Table 1. On the basis of the NOS scale, the selected studies' quality was acceptable; therefore, all articles were included.

Results of the meta-analysis

According to the AWGS standard, the total prevalence of sarcopenia among Chinese elderly people was 18% (95% CI, 15%–20%, Figure 2), and the total prevalence rates of sarcopenia among the elderly in China based on the IWGS and EWGSOP standards were 20% (95% CI, 15%–25%, Figure 3) and 11% (95% CI, 8%–15%, Figure 4), respectively. The prevalence rates of severe sarcopenia and obesity-related sarcopenia among the elderly population in China were 18% (95% CI, 15%–25%) and 18% (95% CI, 15%–25%), respectively.

Subgroup analyses were performed to determine the influence of sex on the prevalence of sarcopenia. The prevalence of sarcopenia among elderly females was higher than that in males in all three standards (Figures 5–7). Table 2 shows the subgroup analyses results based on different diagnostic methods. Except the AWGS standard, the prevalence of sarcopenia estimated using the Bio-electrical Impedance Analysis (BIA) method was higher than that estimated using the Dual Energy X-Ray Absorptiometry (DEXA) method in the other two standards.

Sensitivity analysis and publication bias

Sensitivity analysis was performed to examine the influence of each study. We found no significant difference between the results of the sensitivity analysis and our previous estimates, indicating that our statistical results were relatively credible (Figures S1–S3). The articles obtained from the databases

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were carefully and comprehensively searched. Begg's test was also conducted to determine whether potential publication bias existed in the reviewed literature. The results ($P > 0.05$) suggested that there were no publication bias.

Discussion

The present study is the first meta-analysis of the prevalence of sarcopenia in elderly Chinese people. Sarcopenia is a type of age-related disease characterized by progressive deterioration in skeletal muscle fiber quality, strength, and function. Currently, sarcopenia has been internationally recognized as a new type of elderly syndrome, attracting worldwide attention. In 2017, Shafeee et al. conducted a meta-analysis on the prevalence of sarcopenia. Their results showed that the prevalence of sarcopenia among the elderly in communities worldwide was about 10%, which was higher among non-Asian individuals than in Asians in both sexes [38]. A similar study was performed in Japan, and it was found that the prevalence of sarcopenia among the elderly Japanese was 9.9% (95% CI, 6.2%–15.4%) according to the AWGS standard [39]. The results of our study showed that the prevalence of sarcopenia was 18% (95% CI, 15%–20%) among the elderly Chinese, which was much higher than the results of the aforementioned studies [40, 41]. China may have the largest sarcopenia case number all over the world because it is the most populated country in the world. Therefore, more attention should be given to the prevention and control of sarcopenia.

Here, we found that the prevalence of sarcopenia among females was higher than that among males, which was consistent with other meta-analyses [38–40]. This may be due to men's eating habits and more exercise. Most of the studies included in our meta-analysis used the BIA method because of its affordability and portable equipment. BIA is a widely used non-invasive method for measuring human body composition. The principle behind this method is to record the different electrical resistances of different tissues using electrodes placed on the body surface and then to measure muscle mass using an image reconstruction method. DEXA is also a commonly used method to evaluate muscle mass, which can accurately distinguish whole-body and local muscles, as well as fats and bones. The limited number of studies using DEXA included in this meta-analysis may influence our results. Furthermore, we noticed that aging increases the prevalence of sarcopenia. However, owing to the different age classifications in the included studies, we cannot obtain any conclusion.

The limitation of the present study is that the main data are from urban populations and

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studies on the prevalence of sarcopenia in Chinese rural areas are rare, impacting the calculation of the total prevalence of sarcopenia. Second, not all studies presented data according to age group, so we cannot accurately estimate the prevalence in terms of age subgroup. Third, most of the studies were performed in large cities or economically developed areas. All of these factors affected our results, and as such, they should be interpreted cautiously as further research is needed.

Conclusions

To the best of our knowledge, the present systematic review is the first to estimate the pooled prevalence of sarcopenia among the elderly population in China. The results of the present systematic review reveal that the total prevalence of sarcopenia among the elderly in China is 14.8% according to the AWGS standard, indicating that China may have the largest number of sarcopenia cases. Therefore, the prevention and control of sarcopenia should be comprehensively addressed.

Author contributions statement

SL designed the study. SX and LL searched databases and collected the data. XCH and SL assessed the quality of the study. SL performed the analysis. SX and XCH wrote the manuscript. All authors contributed to this systematic review and meta-analysis.

Supporting Information

Table S1. PRISMA Checklist.

Figure S1. The sensitivity analysis results according to the Asian Working Group for Sarcopenia (AWGS) standard.

Figure S2. The sensitivity analysis results according to the the International Working Group on Sarcopenia (IWGS) standard.

Figure S3. The sensitivity analysis results according to the European Working Group on Sarcopenia in Older People (EWGSOP) standard.

Funds

None.

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Acknowledgments

We thank Enago for English language editing.

Conflict of interests

None declared.

Patient and Public Involvement:

No patient involved.

Reference

[1] Cederholm, T.; Morley, J.E. Sarcopenia: the new definitions. *Current Opinion in Clinical Nutrition and Metabolic Care.* **2015**, *18*, 1–4.

[2] Frontera, W.R.; Hughes, V.A.; Fielding, R.A.; Fiatarone, M.A.; Evans, W.J.; Roubenoff, R. Aging of skeletal muscle: a 12-yr longitudinal study. *J Appl Physiol (1985)*. **2000**, *88*, 1321-1326.

[3] Delmonico, M.J.; Harris, T.B.; Lee, J.S.; Visser, M.; Nevitt, M.; Kritchevsky, S.B.; Tylavsky, F.A.; Newman, A.B. Alternative definitions of sarcopenia, lower extremity performance, and functional impairment with aging in older men and women. *J. Am. Geriatr. Soc.* **2007**, *55*, 769-774..

[4] Woo, J. Sarcopenia. *Clin. Geriatr. Med*. **2017**, *33*, 305–314.

[5] Cruz-Jentoft, A.J.; Baeyens, J.P.; Bauer, J.M.; Boirie, Y.; Cederholm, T.; Landi, F.; Martin, F.C.; Michel, J.P.; Rolland, Y.; Schneider, S.M.; Topinkova, E.; Vandewoude, M.; Zamboni, M. Sarcopenia: European consensus on definition and diagnosis: Report of the European Working Group on Sarcopenia in Older People. *Age Ageing*. **2010**, *39*, 412-423.

[6] Fielding, R.A.; Vellas, B.; Evans, W.J.; Bhasin, S.; Morley, J.E.; Newman, A.B.; Abellan, V.K.G.; Andrieu, S.; Bauer, J.; Breuille, D.; Cederholm, T.; Chandler, J.; De Meynard, C.; Donini, L.; Harris, T.; Kannt, A.; Keime, G.F.; Onder, G.; Papanicolaou, D.; Rolland, Y.; Rooks, D.; Sieber, C.; Souhami, E.; Verlaan, S.; Zamboni, M. Sarcopenia: an undiagnosed condition in older adults. Current consensus definition: prevalence, etiology, and consequences. International working group on sarcopenia. *J. Am. Med. Dir. Assoc.* **2011**, *12*, 249-256.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

[7] Chen, L.K.; Liu, L.K.; Woo, J.; Assantachai, P.; Auyeung, T.W.; Bahyah, K.S.; Chou, M.Y.; Chen, L.Y.; Hsu, P.S.; Krairit, O.; Lee, J.S.; Lee, W.J.; Lee, Y.; Liang, C.K.; Limpawattana, P.; Lin, C.S.; Peng, L.N.; Satake, S.; Suzuki, T.; Won, C.W.; Wu, C.H.; Wu, S.N.; Zhang, T.; Zeng, P.; Akishita, M.; Arai, H. Sarcopenia in Asia: consensus report of the Asian Working Group for Sarcopenia. *J. Am. Med. Dir. Assoc.* **2014**, *15*, 95-101.

[8] Lee, W.; Liu, L.; Peng, L.; Lin, M.; Chen, L. Comparisons of Sarcopenia Defined by IWGS and EWGSOP Criteria Among Older People: Results From the I-Lan Longitudinal Aging Study. *J. Am. Med. Dir. Assoc.* **2013**, *14*, 521-528.

[9] Wu, I.; Lin, C.; Hsiung, C.A.; Wang, C.; Wu, C.; Chan, D.D.; Li, T.; Lin, W.; Huang, K.; Chen, C.; Hsu, C. Epidemiology of sarcopenia among community-dwelling older adults in Taiwan: A pooled analysis for a broader adoption of sarcopenia assessments. *Geriatr. Gerontol. Int.* **2014**, *14*, 52-60.

[10] Wu, C.; Chen, K.; Hou, M.; Chang, Y.; Chang, C.; Liu, P.; Wu, S.; Chiu, C.; Jou, I.; Chen, C. Prevalence and associated factors of sarcopenia and severe sarcopenia in older Taiwanese living in rural community: The Tianliao Old People study 04. *Geriatr. Gerontol. Int.* **2014**, *14*, 69-75.

[11] Li, M.; Hu, Y.; Dong, H.Y.; Zhang, Y.; Fan, L.; Zhang, M.X.; Sun, J.; Han, X.Q.; Liu, Y.X.; Ma, X.N. Compare different measurement for prevalence of sarcopenia in a cohort of healthy community dwellingely in Beijing area. *Chin. J. Health Care Med.* **2014**, *16*, 426-429(Chinese article).

[12] Meng, N.; Li, C.; Liu, C.; Lin, C.; Lin, W.; Chang, C.; Li, T.; Lin, C. Comparison of height- and weight-adjusted sarcopenia in a Taiwanese metropolitan older population. *Geriatr. Gerontol. Int.* **2015**, *15*, 45-53.

[13] Wells, G.A.; Shea, B.; O'Connell, D.; Peterson, J.; Welch, V.; Losos, M.; Tugwell, P. The Newcastle-Ottawa Scale (NOS) for assessing the quality of nonrandomised studies in meta-analyses. Available: http://www.ohri.ca/programs/clinical_epidemiology/oxford.asp. Accessed 2014 Aug 5

[14] Higgins, J.P.T.; Green, S. Cochrane Handbook for Systematic Reviews of Interventions Version 5.1.0.Available: http://www.cochrane-handbook.org. Accessed 2014 Aug

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

[15] Wen, X.; An, P.; Chen, W.C.; Lv, Y.; Fu, Q. Comparisons of sarcopenia prevalence based on different diagnostic criteria in Chinese older adults. *The journal of nutrition, health & aging*. **2015**, *19*, 342-347.

[16] Zeng, P.; Wu, S.; Han, Y.; Liu, J.; Zhang, Y.; Zhang, E.; Zhang, Y.; Gong, H.; Pang, J.; Tang, Z.; Liu, H.; Zheng, X.; Zhang, T. Differences in body composition and physical functions associated with sarcopenia in Chinese elderly: Reference values and prevalence. *Arch. Gerontol. Geriat.* **2015**, *60*, 118-123.

[17] Gao, L.; Jiang, J.; Yang, M.; Hao, Q.; Luo, L.; Dong, B. Prevalence of Sarcopenia and Associated Factors in Chinese Community-Dwelling Elderly: Comparison Between Rural and Urban Areas. *J. Am. Med. Dir. Assoc.* **2015**, *16*, 1001-1003.

[18] Wang, Y.; Wang, Y.; Zhan, J.; Tang, Z.; He, J.; Tan, P.; Deng, H.; Huang, W.; Liu, Y. Sarco-Osteoporosis: Prevalence and Association with Frailty in Chinese Community-Dwelling Older Adults. *Int. J. Endocrinol.* **2015**, *2015*, 1-8.

[19] Woo, J.; Leung, J.; Morley, J.E. Defining Sarcopenia in Terms of Incident Adverse Outcomes. *J. Am. Med. Dir. Assoc.* **2015**, *16*, 247-252.

[20] Wang, H.; Hai, S.; Cao, L.; Zhou, J.; Liu, P.; Dong, B. Estimation of prevalence of sarcopenia by using a new bioelectrical impedance analysis in Chinese community-dwelling elderly people. *BMC Geriatr.* **2016**, *16*.

[21] Han, P.; Kang, L.; Guo, Q.; Wang, J.; Zhang, W.; Shen, S.; Wang, X.; Dong, R.; Ma, Y.; Shi, Y.; Shi, Z.; Li, H.; Li, C.; Ma, Y.; Wang, L.; Niu, K. Prevalence and Factors Associated With Sarcopenia in Suburb-dwelling Older Chinese Using the Asian Working Group for Sarcopenia Definition. *The Journals of Gerontology Series A: Biological Sciences and Medical Sciences*. **2016**, *71*, 529-535.

[22] Han, D.; Chang, K.; Li, C.; Lin, Y.; Kao, T.; Tsai, K.; Wang, T.; Yang, W. Skeletal muscle mass adjusted by height correlated better with muscular functions than that adjusted by body weight in defining sarcopenia. *Sci. Rep.-UK*. **2016**, *6*.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

[23] Xia, Z.W.; Meng, L.P.; Man, Q.Q.; Li, L.X.; Song, P.K.; Li, Y.Q.; Gao, Y.X.; Jia, S.S.; Zhang, J. Analysis of the dietary factors on sarcopenia in elderly in Beijing. *J Hygiene Res.* **2016**, *45*, 388-393 (Chinese article).

[24] Chen, Z.; Dai, R.; Gu, X.H. Prevalence of sarcopenia and quality of life in retired cadres of a military unit. *Chin. J. Osteopo. Rosis. & Bone Miner. Res.* **2016**, *9*, 264-268 (Chinese article).

[25] Bai, H.; Sun, J.; Chen, M.; Xu, D.; Xie, H.; Yu, Z.; Bao, Z.; Chen, J.; Pan, Y.; Lu, D.; Cheng, S. Age-related decline in skeletal muscle mass and function among elderly men and women in Shanghai, China: A cross sectional study. *Asia Pac. J. Clin. Nutr.* **2016**, *25*, 326-332..

[26] Hai, S.; Wang, H.; Cao, L.; Liu, P.; Zhou, J.; Yang, Y.; Dong, B. Association between sarcopenia with lifestyle and family function among community-dwelling Chinese aged 60 years and older. *BMC Geriatr.* **2017**, *17*.

[27] Ma, Y.; Bian, A.L.; Wang, X.F.; Wang, J.X.; Zhou, X.Z. Difference of the prevalence rate of sarcopenia in elderly people with different gender and age in Tianjin. *Chin. General Practice* **2017**, *20*, 3267-3271 (Chinese article).

[28] Wu, Z.N.; Wang, Y.B.; Shu, F.Q.; Song, B.; Li, G.; Zhu, H.M. Epidemiological investigation of sarcopenia in the nursing homes in Fengxian District and analysis of their risk factors. *Shanghai Med.* **2017**, *38*, 37-40 (Chinese article).

[29] Huang, L.J.; Liu, Y.C.; Wen, Y.F.; Liu, K. State and associated factors of sarcopenia in Jinzhou community elderly people. *Modern Preve. Med.* **2017**, *44*, 4458-4460 (Chinese article).

[30] Yang, M.; Hu, X.; Xie, L.; Zhang, L.; Zhou, J.; Lin, J.; Wang, Y.; Li, Y.; Han, Z.; Zhang, D.; Zuo, Y.; Li, Y.; Wu, L. Screening Sarcopenia in Community-Dwelling Older Adults: SARC-F vs SARC-F Combined With Calf Circumference (SARC-CalF). *J. Am. Med. Dir. Assoc.* **2018**, *19*, 271-277.

[31] Zeng, Y.; Hu, X.; Xie, L.; Han, Z.; Zuo, Y.; Yang, M. The Prevalence of Sarcopenia in Chinese Elderly Nursing Home Residents: A Comparison of 4 Diagnostic Criteria. *J. Am. Med. Dir. Assoc.* **2018**, *19*, 690-695.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

[32] Wang, Y.N.; Xu, J.L.; Song, H.L. Influencing factors of sarcopenia among elderly patients. *Modern Clinical Nursing* **2018**, *17*, 45-50 (Chinese article).

[33] Yang, L.; Wu, G.; Yang, Y.; Wu, Y.; Zhang, L.; Wang, M.; Mo, L.; Xue, G.; Wang, C.; Weng, X. Nutrition, Physical Exercise, and the Prevalence of Sarcopenia in Elderly Residents in Nursing Homes in China. *Med. Sci. Monitor*. **2019**, *25*, 4390-4399.

[34] Du, Y.; Wang, X.; Xie, H.; Zheng, S.; Wu, X.; Zhu, X.; Zhang, X.; Xue, S.; Li, H.; Hong, W.; Tang, W.; Chen, M.; Cheng, Q.; Sun, J. Sex differences in the prevalence and adverse outcomes of sarcopenia and sarcopenic obesity in community dwelling elderly in East China using the AWGS criteria. *BMC Endocr. Disord.* **2019**, *19*.

[35] Wang, H.; Hai, S.; Liu, Y.; Liu, Y.X.; Zhou, J.H.; Yang, Y.; Dong, B.R.; Yue, J.R. Prevalence of sarcopenia and associated factors in community-dwelling elderly populations in Chengdu China. *J. Sichuan Univ.* **2019**, *50*, 224-228 (Chinese article).

[36] Zhang, Y.; Chen, X.; Hou, L.; Lin, X.; Qin, D.; Wang, H.; Hai, S.; Cao, L.; Dong, B. Prevalence and Risk Factors Governing the Loss of Muscle Function in Elderly Sarcopenia Patients: A longitudinal Study in China with 4 Years of Follow-Up. *The journal of nutrition, health & aging*. **2020**, *24*, 518-524.

[37] Liu, X.; Hao, Q.; Yue, J.; Hou, L.; Xia, X.; Zhao, W.; Zhang, Y.; Ge, M.; Ge, N.; Dong, B. Sarcopenia, Obesity and Sarcopenia Obesity in Comparison: Prevalence, Metabolic Profile, and Key Differences: Results from WCHAT Study. *The journal of nutrition, health & aging*. **2020**, *24*, 429-437.

[38] Shafiee, G.; Keshtkar, A.; Soltani, A.; Ahadi, Z.; Larijani, B.; Heshmat, R. Prevalence of sarcopenia in the world: a systematic review and meta- analysis of general population studies. *Journal of Diabetes & Metabolic Disorders*. **2017**, *16*.

[39] Makizako, H.; Nakai, Y.; Tomioka, K.; Taniguchi, Y. Prevalence of sarcopenia defined using the Asia Working Group for Sarcopenia criteria in Japanese community-dwelling older adults: A systematic review and meta-analysis. *Physical therapy research*. **2019**, *22*, 53-57.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

[40] Diz, J.B.M.; Leopoldino, A.A.O.; Moreira, B.D.S.; Henschke, N.; Dias, R.C.; Pereira, L.S.M.; Oliveira, V.C. Prevalence of sarcopenia in older Brazilians: A systematic review and meta-analysis. *Geriatr. Gerontol. Int.* **2017**, *17*, 5-16.

[41] Papadopoulou, S.K.; Tsintavis, P.; Potsaki, G.; Papandreou, D. Differences in the Prevalence of Sarcopenia in Community-Dwelling, Nursing Home and Hospitalized Individuals. A Systematic Review and Meta-Analysis. *The journal of nutrition, health & aging*. **2020**, *24*, 83-90.

Legend

Figure 1. Flowchart of the detailed procedure for the inclusion or exclusion of selected studies.

Figure 2. Forest plots for the total prevalence of sarcopenia among the elderly Chinese population according to the Asian Working Group for Sarcopenia (AWGS) standard. Diamond represents the pooled odds ratio at 95% CI.

Figure 3. Forest plots for the total prevalence of sarcopenia among the elderly Chinese population according to the the International Working Group on Sarcopenia (IWGS) standard. Diamond represents the pooled odds ratio at 95% CI.

Figure 4. Forest plots for the total prevalence of sarcopenia among the elderly Chinese population according to the European Working Group on Sarcopenia in Older People (EWGSOP) standard. Diamond represents the pooled odds ratio at 95% CI.

Figure 5. Forest plots for the prevalence of sarcopenia among male and female elderly people according to the Asian Working Group for Sarcopenia (AWGS) standard. Diamond represents the pooled odds ratio at 95% CI.

Figure 6. Forest plots for the prevalence of sarcopenia among male and female elderly people according to the the International Working Group on Sarcopenia (IWGS) standard. Diamond represents the pooled odds ratio at 95% CI.

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Figure 7. Forest plots for the prevalence of sarcopenia among male and female elderly people according to the European Working Group on Sarcopenia in Older People (EWGSOP) standard. Diamond represents the pooled odds ratio at 95% CI.

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Table 1 Study characteristics of the studies included in the meta-analysis

Study	Publication Year	Participate(n)			Region	Setting	Method	Diagnose Criteria	Prevalence(%)		
		Total	Male	Female					Total	Male	Female
Lee WJ	2013	386	223	163	Taiwan	Urban Community	DEXA	IWGS	11.1	10.8	11.7
								EWGSOP	16.6	14.9	19.0
Wu IC	2014	2155	1055	1100	Taiwan	Urban Community	BIA	EWGSOP	3.9	5.4	2.5
Wu CH	2014	549	285	264	Taiwan	Rural Community	BIA	EWGSOP	12.7	8.1	17.8
Li M	2014	169	169	0	Beijing	Urban Community	DEXA	EWGSOP	33.3	33.3	-
Meng NH	2015	771	359	712	Taiwan	Urban Community	DEXA	EWGSOP	9.7	0.7	12.5
Gao LL	2015	612	254	358	Sichuan	Rural and urban Community	Calf circumference	AWGS	9.8	6.7	12.0
Wen X	2015	286	136	150	Hangzhou	Urban Community	DEXA	IWGS	6.0	7.4	4.7
								EWGSOP	0.4	0.8	0.01

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							AWGS	3.1	5.9	0.7
Woo J	2015	4000	2000	Hongkong	Urban Community	DEXA	IWGS	20.2	22.1	18.3
							EWGSOP	9.0	9.5	8.6
							AWGS	7.3	5.3	9.3
Zeng P	2015	395	199	196	Beijing	Rural and urban Community	BIA	AWGS	24.3	18.1	30.6
Wang YJ	2015	316	164	152	Changsha	Urban Community	DEXA	AWGS	29.7	26.2	33.6
Han PP	2016	1069	467	602	Tianjin	Urban Community	BIA	AWGS	9.3	6.4	17.5
Wang H	2016	636	316	320	Chengdu	Urban Community	BIA	AWGS	10.4	8.2	12.5
Bai HJ	2016	415	212	203	Shanghai	Urban Community	BIA	AWGS	28.7	33.5	23.6
Xia ZW	2016	683	239	444	Beijing	Urban Community	BIA	AWGS	20.1	17.2	21.6
Chen Z	2016	281	203	78	Chongqing	Urban Community	BIA	IWGS	27.8	-	-
Han DS	2016	878	402	476	Taiwan	Urban Community	BIA	EWGSOP	9.8	4.5	14.3
Hai S	2017	834	415	419	Chengdu	Urban Community	BIA	AWGS	10.6	11.3	9.8
Wu ZN	2017	786	320	466	Shanghai	Urban Nursing home	BIA	AWGS	25.3	20.0	29.0

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Huang LJ	2017	193	65	128	Jinzhou	Urban Community	BIA	AWGS	11.4	7.7	13.3
Ma Y	2017	3276	2325	951	Tianjin	Urban Out-patient	BIA	IWGS	15.7	15.5	15.8
								AWGS	12.8	11.1	17.0
Yang M	2018	384	160	224	Chengdu	Urban Community	BIA	AWGS	15.9	11.9	18.8
								IWGS	25.0	24.4	25.4
								EWGSOP	11.7	10.6	12.5
Zeng YL	2018	277	83	194	Chengdu	Urban Nursing home	BIA	AWGS	34.3	15.7	42.3
								IWGS	38.3	28.9	42.3
								EWGSOP	32.5	25.3	35.6
Wang YN	2018	407	197	210	Nantong	Urban Out-patient	BIA	AWGS	22.6	19.8	25.2
Yang LJ	2019	316	112	204	Suzhou	Urban Nursing home	BIA	AWGS	28.8	30.4	27.9
Du YP	2019	631	213	418	Shanghai	Urban Community	BIA	AWGS	12.2	19.2	8.6
Wang H	2019	947	465	482	Chengdu	Urban Community	BIA	AWGS	7.7	6.5	8.9
Zhang Y	2020	474	238	236	Chengdu	Urban Community	BIA	AWGS	15.4	-	-

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Liu XL	2020	4500	1627	2873	Yunnan, Guizhou, Sichuan, and Xinjiang	Urban Community	BIA	AWGS	19.3	22.1	17.8

DEXA, dual-energy x-ray absorptiometry; BIA, bioelectrical impedance analysis; IWGS, International Working Group on Sarcopenia; AWGS, Asia Working Group for Sarcopenia; EWGSOP, European Working Group on Sarcopenia in Older People.

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Table 2 The prevalence of sarcopenia diagnosed by different criteria using DEXA and BIA methods(%).

	DEXA	BIA
AWSG	23(4-41)	17(14-20)
IWSG	12(3-22)	26(16-36)
EWGSOP	9(3-15)	14(7-20)

DEXA, dual-energy x-ray absorptiometry; BIA, bioelectrical impedance analysis; IWGS, International Working Group on Sarcopenia; AWGS, Asia Working Group for Sarcopenia; EWGSOP, European Working Group on Sarcopenia in Older People.

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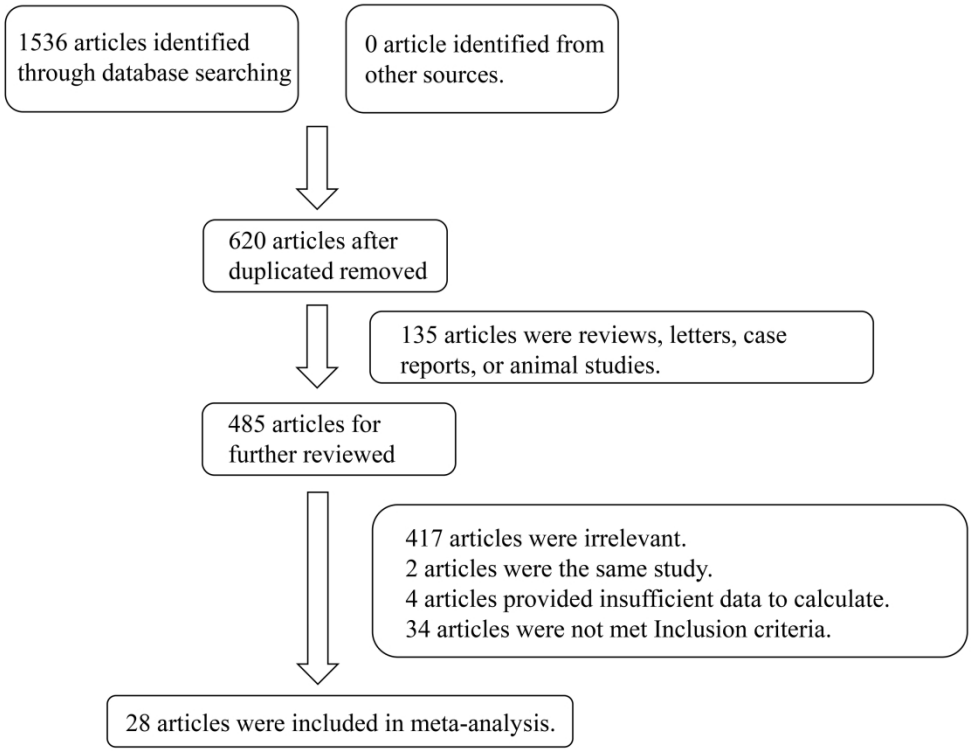


Figure 1. Flowchart of the detailed procedure for the inclusion or exclusion of selected studies.

252x190mm (300 x 300 DPI)

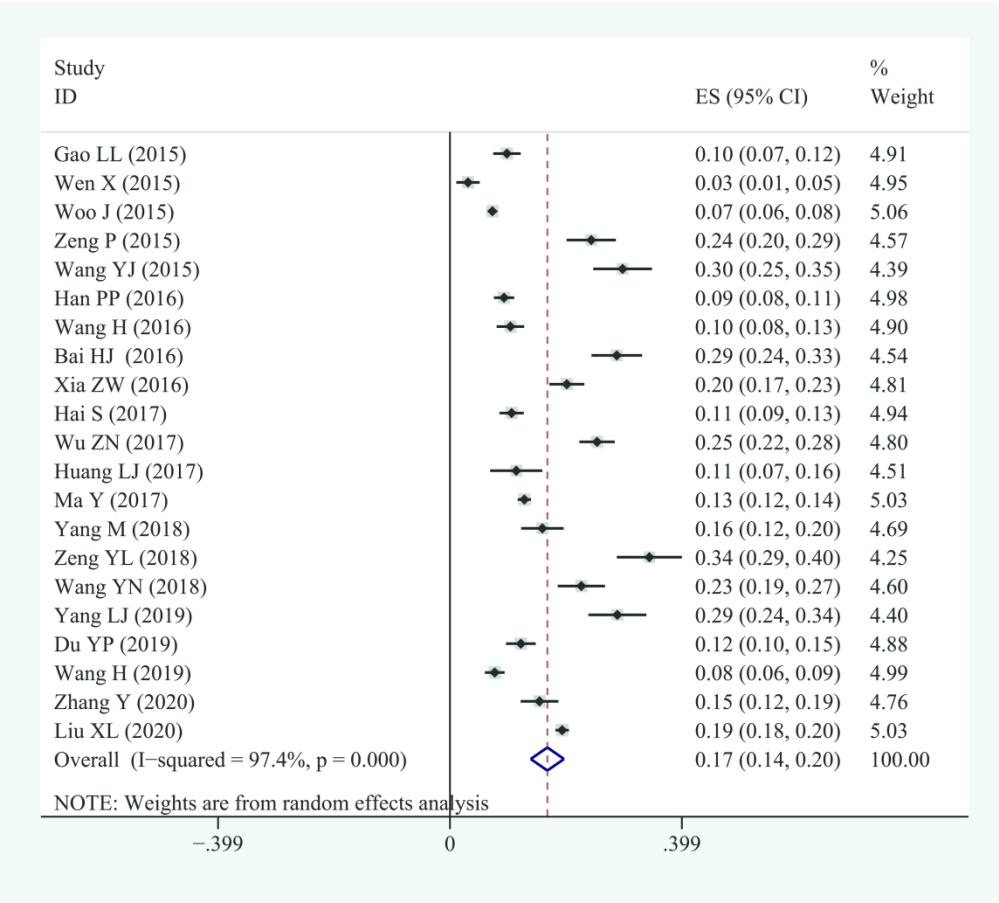


Figure 2. Forest plots for the total prevalence of sarcopenia among the elderly Chinese population according to the Asian Working Group for Sarcopenia (AWGS) standard. Diamond represents the pooled odds ratio at 95% CI.

283x255mm (300 x 300 DPI)

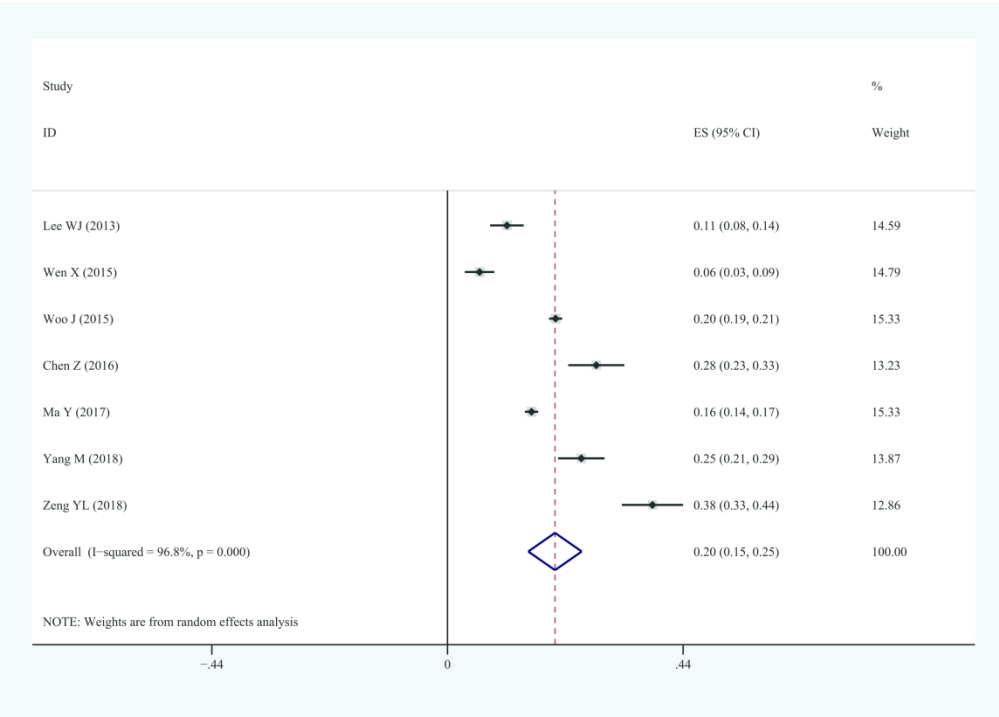


Figure 3. Forest plots for the total prevalence of sarcopenia among the elderly Chinese population according to the the International Working Group on Sarcopenia (IWGS) standard. Diamond represents the pooled odds ratio at 95% CI.

273x195mm (300 x 300 DPI)

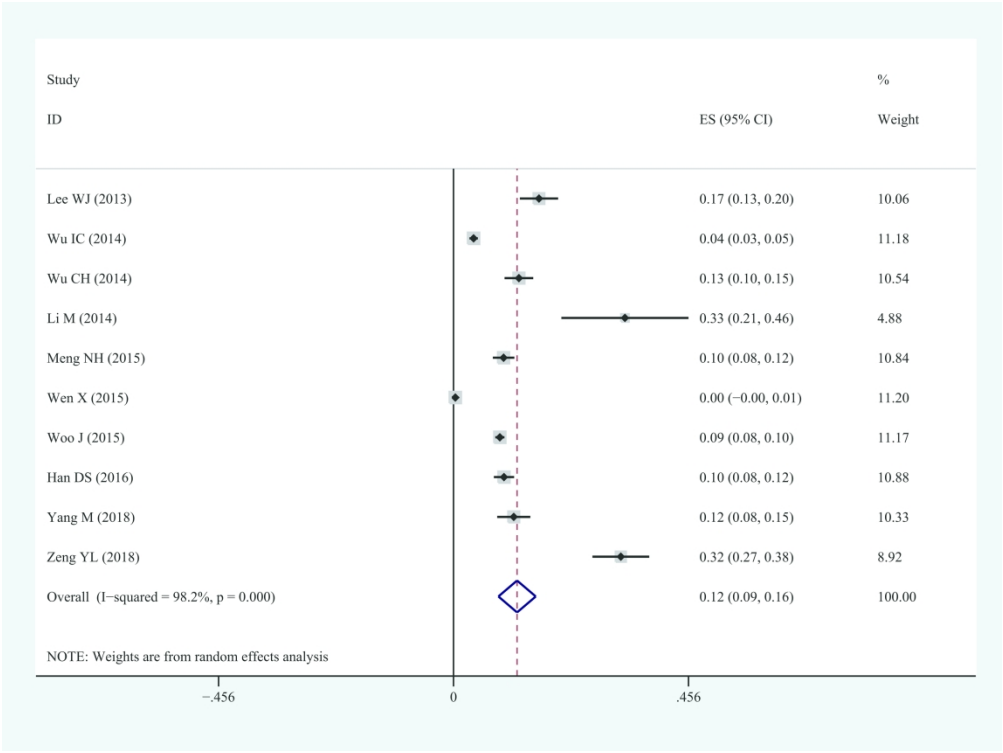


Figure 4. Forest plots for the total prevalence of sarcopenia among the elderly Chinese population according to the European Working Group on Sarcopenia in Older People (EWGSOP) standard. Diamond represents the pooled odds ratio at 95% CI.

275x206mm (300 x 300 DPI)

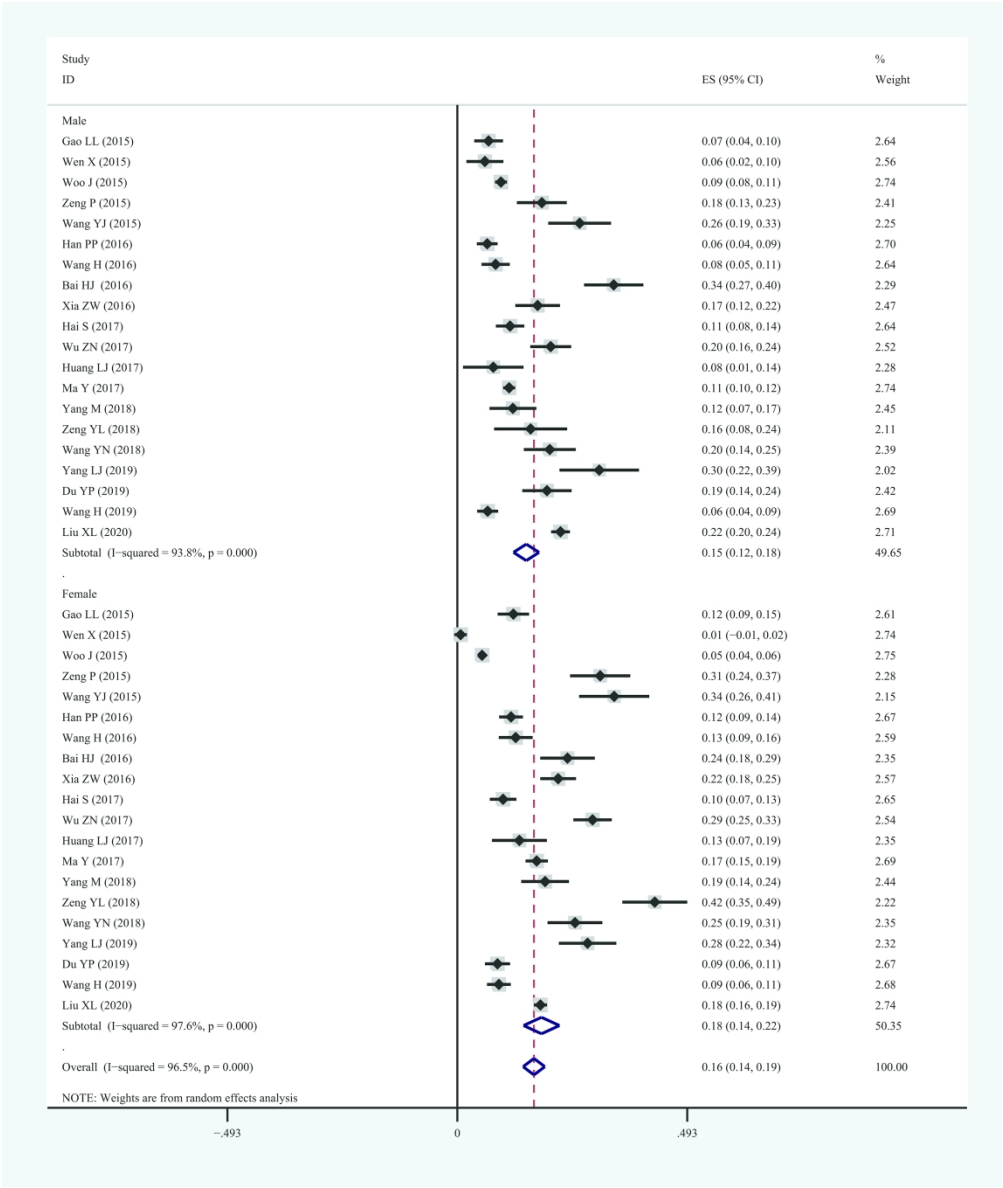


Figure 5. Forest plots for the prevalence of sarcopenia among male and female elderly people according to the Asian Working Group for Sarcopenia (AWGS) standard. Diamond represents the pooled odds ratio at 95% CI.

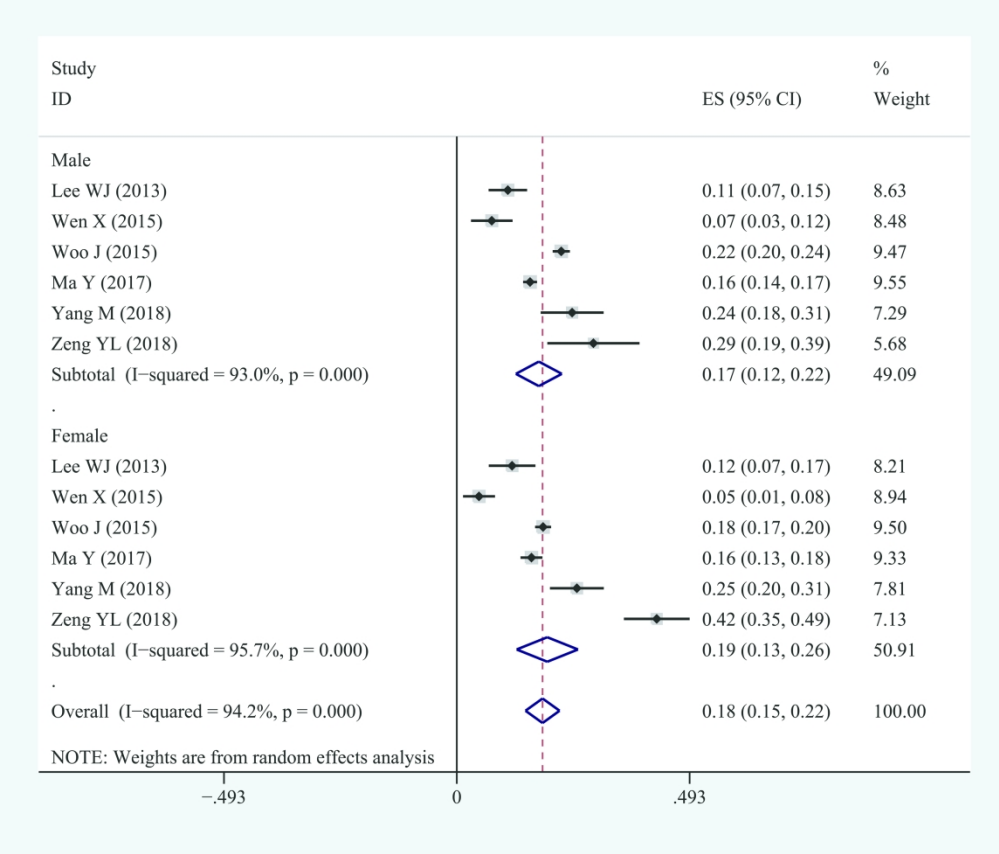


Figure 6. Forest plots for the prevalence of sarcopenia among male and female elderly people according to the the International Working Group on Sarcopenia (IWGS) standard. Diamond represents the pooled odds ratio at 95% CI.

281x240mm (300 x 300 DPI)

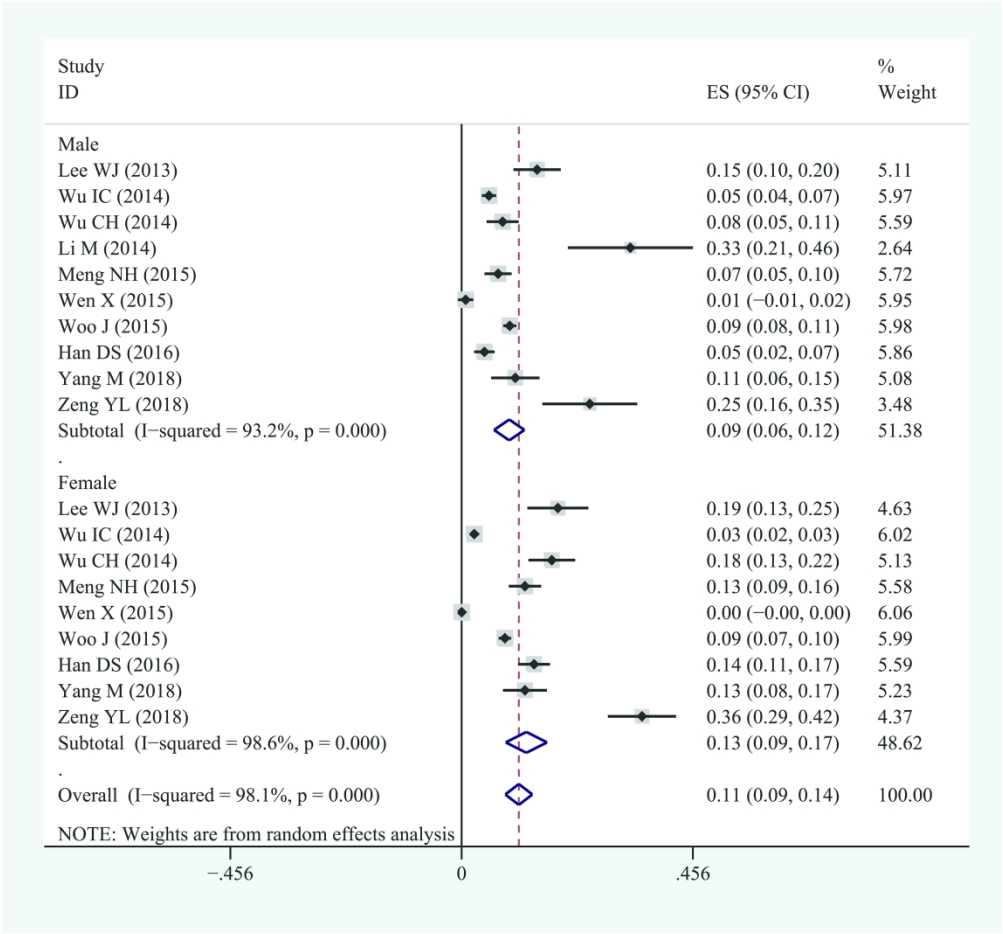


Figure 7. Forest plots for the prevalence of sarcopenia among male and female elderly people according to the European Working Group on Sarcopenia in Older People (EWGSOP) standard. Diamond represents the pooled odds ratio at 95% CI.

Table S1. PRISMA Checklist.

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	3
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	3,4
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	4
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	4
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	4
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	4
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	4
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	5
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	5
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	5
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	5

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Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	5

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	5
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	5
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	6
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	6
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	6
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	6
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	6,7
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	7
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	7
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	7,8
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	8
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	8
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	9

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Figure S1. The sensitivity analysis results according to the Asian Working Group for Sarcopenia (AWGS) standard.

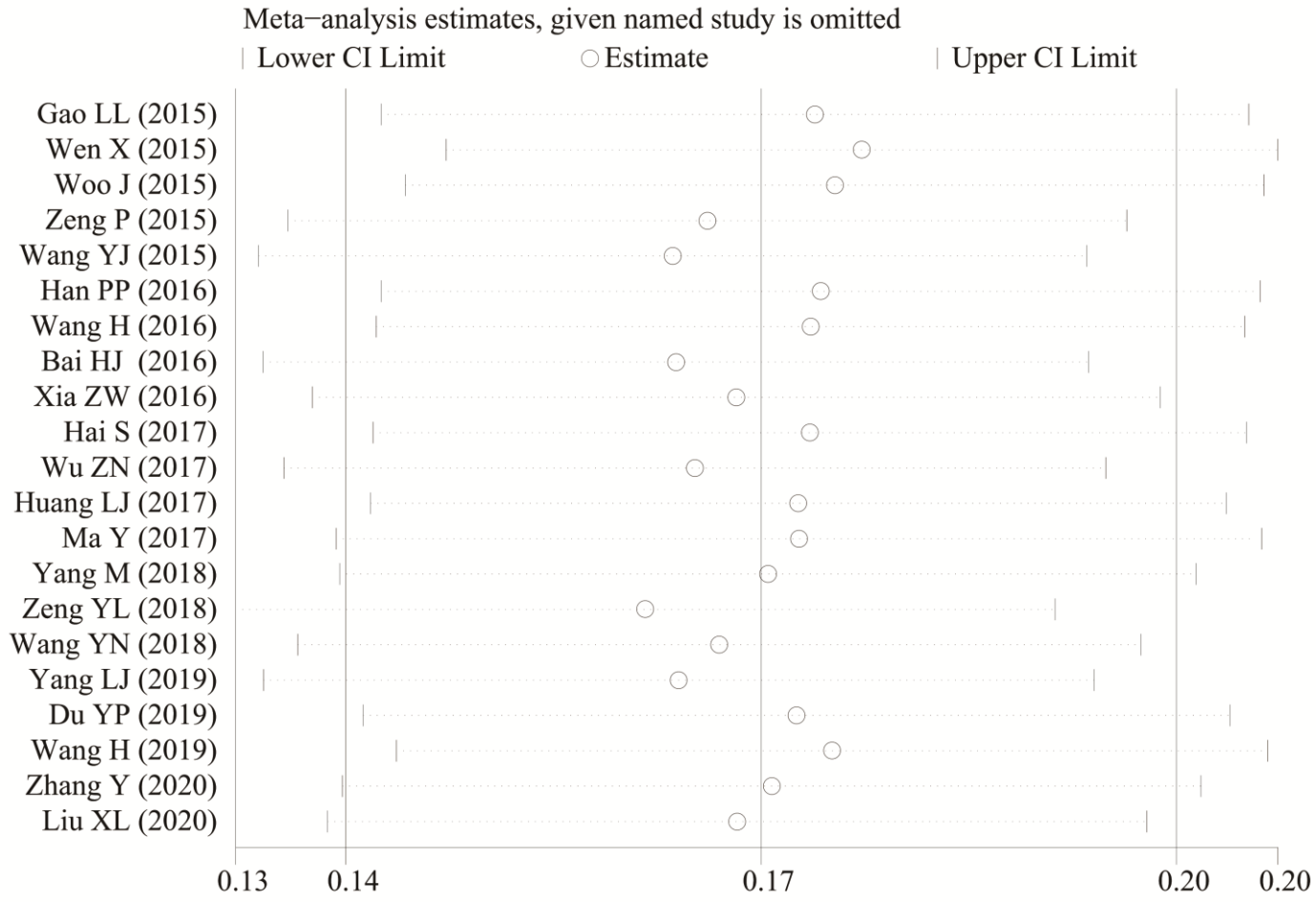


Figure S2. The sensitivity analysis results according to the International Working Group on Sarcopenia (IWGS) IWGS standard.

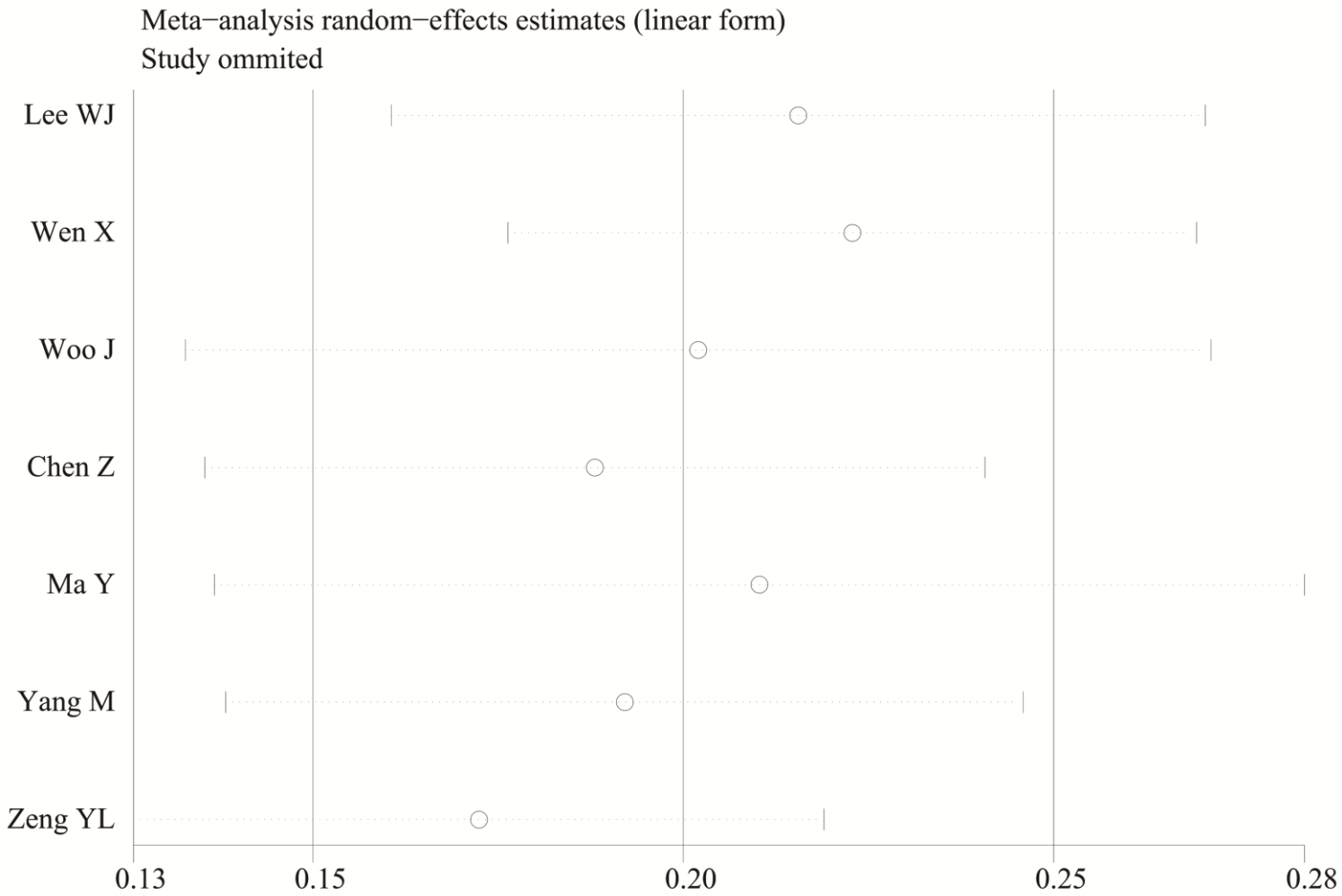
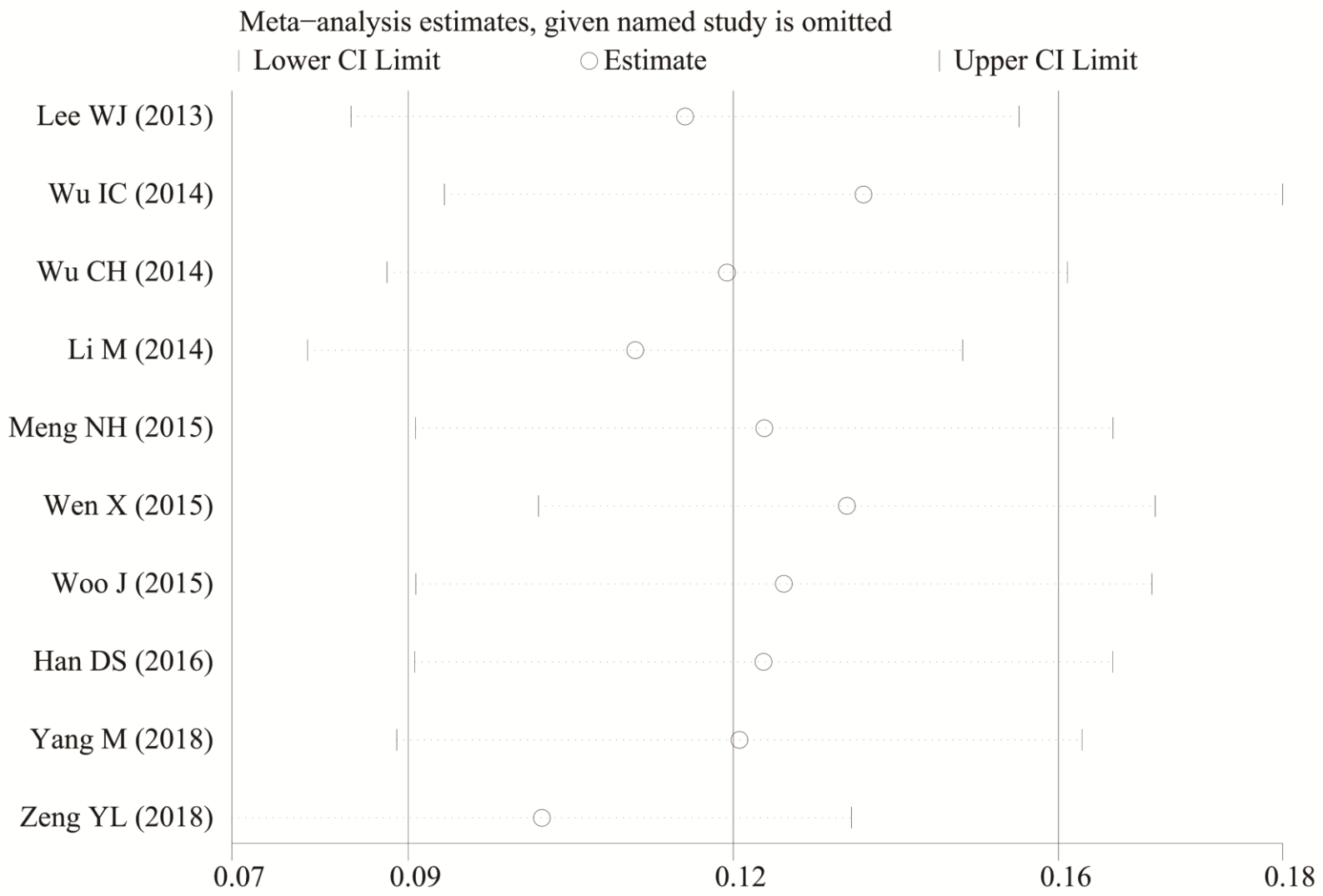


Figure S3. The sensitivity analysis results according to the European Working Group on Sarcopenia in Older People (EWGSOP) standard.



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Prevalence of Sarcopenia in Older Chinese Adults: A Systematic Review and Meta-analysis

Journal:	*BMJ Open*
Manuscript ID	bmjopen-2020-041879.R1
Article Type:	Original research
Date Submitted by the Author:	10-Dec-2020
Complete List of Authors:	Xin, Caihong; Fourth People's Hospital in Shenyang Sun, Xin; First Affiliated Hospital of Soochow University Lu, Li; People's Hospital of Liaoning Province Shan, Ling; People's Hospital of Liaoning Province
Primary Subject Heading:	Geriatric medicine
Secondary Subject Heading:	Epidemiology, Global health, Public health
Keywords:	GERIATRIC MEDICINE, EPIDEMIOLOGY, General endocrinology < DIABETES & ENDOCRINOLOGY

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Prevalence of Sarcopenia in Older Chinese Adults: A Systematic Review and Meta-analysis

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Running title: Prevalence of sarcopenia in

Key words: Prevalence; sarcopenia; China

Word count of the manuscript: 1606

Figures number: 7

Tables number: 2

Disclosure Statement: The authors have nothing to disclose.

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Abstract

Background: Sarcopenia is characterized by lean muscle mass wastage and muscle function decline that occurs with aging, with a high rate of adverse outcomes. Several studies have investigated the prevalence of sarcopenia in Chinese older people, and results were inconsistent.

Objective: This meta-analysis was conducted to estimate the overall prevalence of sarcopenia in older Chinese adults.

Methods: A literature research was conducted using PubMed, Web of Science, Medline, CNKI, CQVIP, and Wanfang databases. The search terms in abstract were identified as follows: "sarcopenia," in combination with the terms "prevalence," "epidemiology," and "China." Random effect model was used to estimate the prevalence of sarcopenia. The sex-specific prevalence of sarcopenia at 95% confidence interval (CI) was also calculated by different criteria for the definition of sarcopenia.

Results: In total, 23 articles were included in this meta-analysis, involving 21,564 participants. On the basis of the Asian Working Group for Sarcopenia (AWGS) criterion, the overall prevalence of sarcopenia among the elderly in China was 14% (95% CI, 11%–18%). The prevalence of sarcopenia was higher in Chinese females than in males (15% vs. 14%) according to AWGS criterion.

Conclusions: This systematic review is the first estimation of the pooled prevalence of sarcopenia in older Chinese adults. The results suggest that China may have the largest sarcopenia patient count. Therefore, more attention should be given to the prevention and control of sarcopenia.

Strengths and limitations of this study:

This systematic review is the first estimation of the pooled prevalence of sarcopenia in Chinese older people, suggesting that China may have the largest sarcopenia patient count.

The main data are from urban populations and studies on the prevalence of sarcopenia in Chinese rural areas are rare, impacting the calculation of the total prevalence of sarcopenia.

Not all studies presented data according to age group, so we cannot accurately estimate the prevalence in terms of age subgroup.

Introduction

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Sarcopenia is characterized by the deterioration of muscle mass and function that occurs with aging. Several studies have shown that with age increase, skeletal muscle mass and strength will be lost at a rate of 1%–2% annually, possibly leading to the decline of physical function, increasing the disability, and mortality rates of elderly people and bringing serious economic and social burdens [1–4].

A practical clinical definition was reported by the European Working Group on Sarcopenia in Older People (EWGSOP) in 2010 [5]. Later in 2011, the International Working Group on Sarcopenia (IWGS) published another consensus [6]. However, Asia is a vast region with different races, countries, and cultures. Developing screening and diagnostic strategies for the Asian population is imperative. The Asian Working Group for Sarcopenia (AWGS) was established in 2013, and the consensus of AWGS was released based on 16 epidemiologic reports on sarcopenia in six countries or regions, leading to the development of a diagnosis strategy for sarcopenia in Asia [7]. The diagnostic criteria recommended by AWGS and EWGSOP are similar, including muscle mass, muscle strength, and activity measurements. The difference lies in the specific cut-off value.

According to statistics, the elderly population (≥60 years old) has reached 16.1% of China's total population, which is close to 250 million. In the face of a severe aging situation, people are paying more attention to sarcopenia, but the prevalence of sarcopenia was inconsistent in different studies [8–12]. At present, there is no large-scale epidemiologic survey of sarcopenia in China. The purpose of this study is to systematically reveal the prevalence of sarcopenia in older Chinese adults.

Methods

Search strategy

We searched the following electronic databases: PubMed, Web of Science, Medline, China National Knowledge Infrastructure (CNKI), CQVIP database, and Wanfang databases. The search terms in the title or abstract were identified as follows: "sarcopenia," "muscle mass," and "muscle strength" in combination with the terms "prevalence," "epidemiology," and "China". The full search strategy was included in supplementary file (File S1). All studies published from January 2010 to January 2020 were included. The references of the retrieved articles were examined to identify additional eligible studies, but unpublished studies were not included. The search languages were limited to English and

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Chinese. The completed Preferred Reporting Items for Systematic Reviews and Meta-Analyses checklist is presented in the Supplementary data (Table S1).

Patient and Public Involvement:

No patient involved.

Inclusion and exclusion criteria

Inclusion of the selected studies was based on the following criteria: (1) a cross-sectional study, retrospective study or prospective study, (2) a sample size of ≥100 elderly Chinese participants (age ⩾ 60 years); (3) sufficient information of sarcopenia at the corresponding 95% confidence interval (CI) and *P* value, and (4) the definition of sarcopenia is in line with the AWGS, EWGSOP, or IWGS standards.

Exclusion of the selected studies was based on the following criteria: (1) the same data from different studies, (2) studies must define sarcopenia only by reduction of muscle mass or strength, and (3) the subjects were a specific population, such as patients living in nursing home or patients with disability. Reviews, case reports, letters, and animal researches were excluded.

Data extraction and Methodological quality assessment

Information was extracted by two independent researchers from the included studies, including first author's name, publication year, sample, diagnostic criteria, participants, setting, and prevalence. Two independent reviewers who were not blinded to the authors or journals assessed the risk of bias in the included studies using the Newcastle–Ottawa scale (NOS) [13]. The NOS is recommended by the Cochrane Handbook for Systematic Reviews of Interventions [14]. Each included study was judged in terms of three domains using the "star system," namely, representativeness of the study group selection, group comparability, and exposure or outcome ascertainment. NOS scores range from zero to nine stars. The two researchers independently assessed the studies by discussion, comparison of findings, and resolution of differences through consensus formation. If no consensus can be reached, a third researcher was commissioned to resolve the difference.

Statistical analysis

We used a systematic analysis approach to calculate the pooled prevalence of sarcopenia for all eligible

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studies, and heterogeneity among these studies was assessed using Cochran's Q test and the I^2 statistical index. Random effect model was used in combining the prevalence of sarcopenia. We additionally performed sensitivity analysis to evaluate the influence of any given study on the pooled estimate. Publication bias was evaluated using Begg's test. Subgroup analyses were also performed. The significance level was set at $P < 0.05$. All statistical analyses were performed using Stata software version 12.0 (College Station, Texas, USA).

Results

Study characteristics

In total, 1536 studies were retrieved from the aforementioned databases. After screening, a total of 23 articles were included based on the criteria shown in Figure 1 [8–12, 15–37]. Altogether, the 23 articles included 21,564 participants in China between 2010 and 2020. The selected studies' characteristics are summarized in Table 1. On the basis of the NOS scale, the score of selected studies were all above 5 stars; therefore, all articles were included.

Results of the meta-analysis

According to the AWGS standard, the total prevalence of sarcopenia among Chinese elderly people was 14% (95% CI, 11%–18%, Figure 2), and the total prevalence rates of sarcopenia among the elderly in China based on the IWGS and EWGSOP standards were 18% (95% CI, 11%–25%, Figure 3) and 10% (95% CI, 7%–14%, Figure 4), respectively.

Subgroup analyses were performed to determine the influence of sex on the prevalence of sarcopenia. The prevalence of sarcopenia among elderly females was higher than that in males in AWGS and EWGSOP standards (Figures 5–7). Table 2 shows the subgroup analyses results based on different methods for diagnostic criteria. The prevalence of sarcopenia estimated using the Bio-electrical Impedance Analysis (BIA) method was higher than that estimated using the Dual Energy X-Ray Absorptiometry (DEXA) method in AWGS and IWGS standards.

Sensitivity analysis and publication bias

Sensitivity analysis was performed to examine the influence of each study. We found no significant difference between the results of the sensitivity analysis and our previous estimates, indicating that our statistical results were relatively credible (Figures S1–S3). The articles obtained from the databases were carefully and comprehensively searched. Begg's test was also conducted to determine whether

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potential publication bias existed in the reviewed literature. The results ($P > 0.05$) suggested that there were no publication bias.

Discussion

The present study is the first meta-analysis of the prevalence of sarcopenia in Chinese older adults. Sarcopenia is a type of age-related disease characterized by progressive deterioration in skeletal muscle fiber quality, strength, and function. Currently, sarcopenia has been internationally recognized as a new type of geriatric syndrome, attracting worldwide attention. In 2017, Shafeee et al. conducted a meta-analysis on the prevalence of sarcopenia. Their results showed that the prevalence of sarcopenia among the elderly in communities worldwide was about 10%, which was higher among non-Asian individuals than in Asians in both sexes [38]. A similar study was performed in Japan, and it was found that the prevalence of sarcopenia among the elderly Japanese was 9.9% (95% CI, 6.2%–15.4%) according to the AWGS standard [39]. The results of our study showed that the prevalence of sarcopenia was 14% (95% CI, 11%–18%) in Chinese older adults, which was much higher than the results of the aforementioned studies [40, 41]. China may have the largest sarcopenia case number all over the world because it is the most populated country in the world. High prevalence of sarcopenia in China may be related to the Chinese diet structure. Protein is the raw material for muscle synthesis. Chinese people have higher carbohydrate content and less protein intake, especially for the older people. Therefore, more attention should be given to the prevention and control of sarcopenia.

Here, we found that the prevalence of sarcopenia among females was higher than that in males, which was consistent with other meta-analysis [38–40]. This may be due to men's eating habits and more exercise. Most of the studies included in our meta-analysis used the BIA method because of its affordability and portable equipment. BIA is a widely used non-invasive method for measuring human body composition. The principle behind this method is to record the different electrical resistances of different tissues using electrodes placed on the body surface and then to measure muscle mass using an image reconstruction method. DEXA is also a commonly used method to evaluate muscle mass, which can accurately distinguish whole-body and local muscles, as well as fats and bones. The limited number of studies using DEXA included in this meta-analysis may influence our results. Furthermore, we noticed that aging increases the prevalence of sarcopenia. However, owing to the different age classifications in the included studies, we cannot obtain any conclusion.

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The limitation of the present study is that the main data are from urban populations and studies on the prevalence of sarcopenia in Chinese rural areas are rare, impacting the calculation of the total prevalence of sarcopenia. Second, the prevalence of sarcopenia in 60–70 and 70–80 years old people might be different. Not all studies presented data according to age group, so we cannot accurately estimate the prevalence in terms of age subgroup. Third, most of the studies were performed in large cities or economically developed areas. All of these factors affected our results, and as such, they should be interpreted cautiously as further research is needed.

Conclusions

To the best of our knowledge, the present systematic review is the first to estimate the pooled prevalence of sarcopenia in Chinese older adults. The results of the present systematic review reveal that China may have the largest number of sarcopenia cases. Therefore, the prevention and management of sarcopenia should be comprehensively addressed.

Author contributions statement

SL designed the study. SX and LL searched databases and collected the data. XCH and SL assessed the quality of the study. SL performed the analysis. SX and XCH wrote the manuscript. SX modified the manuscript in the revision. All authors contributed to this systematic review and meta-analysis.

Supporting Information

File S1 Search strategy in databases.

Table S1. PRISMA Checklist.

Figure S1. The sensitivity analysis results according to the Asian Working Group for Sarcopenia (AWGS) standard.

Figure S2. The sensitivity analysis results according to the the International Working Group on Sarcopenia (IWGS) standard.

Figure S3. The sensitivity analysis results according to the European Working Group on Sarcopenia in Older People (EWGSOP) standard.

Funds

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None.

Acknowledgments

We thank Enago for English language editing.

Conflict of interests

None declared.

Data availability statement

All data relevant to the study are included in the article or uploaded as supplementary information

Reference

[1] Cederholm, T.; Morley, J.E. Sarcopenia: the new definitions. *Current Opinion in Clinical Nutrition and Metabolic Care.* **2015**, *18*, 1–4.

[2] Frontera, W.R.; Hughes, V.A.; Fielding, R.A.; Fiatarone, M.A.; Evans, W.J.; Roubenoff, R. Aging of skeletal muscle: a 12-yr longitudinal study. *J Appl Physiol (1985)*. **2000**, *88*, 1321-1326.

[3] Delmonico, M.J.; Harris, T.B.; Lee, J.S.; Visser, M.; Nevitt, M.; Kritchevsky, S.B.; Tylavsky, F.A.; Newman, A.B. Alternative definitions of sarcopenia, lower extremity performance, and functional impairment with aging in older men and women. *J. Am. Geriatr. Soc.* **2007**, *55*, 769-774..

[4] Woo, J. Sarcopenia. *Clin. Geriatr. Med*. **2017**, *33*, 305–314.

[5] Cruz-Jentoft, A.J.; Baeyens, J.P.; Bauer, J.M.; Boirie, Y.; Cederholm, T.; Landi, F.; Martin, F.C.; Michel, J.P.; Rolland, Y.; Schneider, S.M.; Topinkova, E.; Vandewoude, M.; Zamboni, M. Sarcopenia: European consensus on definition and diagnosis: Report of the European Working Group on Sarcopenia in Older People. *Age Ageing*. **2010**, *39*, 412-423.

[6] Fielding, R.A.; Vellas, B.; Evans, W.J.; Bhasin, S.; Morley, J.E.; Newman, A.B.; Abellan, V.K.G.; Andrieu, S.; Bauer, J.; Breuille, D.; Cederholm, T.; Chandler, J.; De Meynard, C.; Donini, L.; Harris, T.; Kannt, A.; Keime, G.F.; Onder, G.; Papanicolaou, D.; Rolland, Y.; Rooks, D.; Sieber,

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

C.; Souhami, E.; Verlaan, S.; Zamboni, M. Sarcopenia: an undiagnosed condition in older adults. Current consensus definition: prevalence, etiology, and consequences. International working group on sarcopenia. *J. Am. Med. Dir. Assoc.* **2011**, *12*, 249-256.

[7] Chen, L.K.; Liu, L.K.; Woo, J.; Assantachai, P.; Auyeung, T.W.; Bahyah, K.S.; Chou, M.Y.; Chen, L.Y.; Hsu, P.S.; Krairit, O.; Lee, J.S.; Lee, W.J.; Lee, Y.; Liang, C.K.; Limpawattana, P.; Lin, C.S.; Peng, L.N.; Satake, S.; Suzuki, T.; Won, C.W.; Wu, C.H.; Wu, S.N.; Zhang, T.; Zeng, P.; Akishita, M.; Arai, H. Sarcopenia in Asia: consensus report of the Asian Working Group for Sarcopenia. *J. Am. Med. Dir. Assoc.* **2014**, *15*, 95-101.

[8] Lee, W.; Liu, L.; Peng, L.; Lin, M.; Chen, L. Comparisons of Sarcopenia Defined by IWGS and EWGSOP Criteria Among Older People: Results From the I-Lan Longitudinal Aging Study. *J. Am. Med. Dir. Assoc.* **2013**, *14*, 521-528.

[9] Wu, I.; Lin, C.; Hsiung, C.A.; Wang, C.; Wu, C.; Chan, D.D.; Li, T.; Lin, W.; Huang, K.; Chen, C.; Hsu, C. Epidemiology of sarcopenia among community-dwelling older adults in Taiwan: A pooled analysis for a broader adoption of sarcopenia assessments. *Geriatr. Gerontol. Int.* **2014**, *14*, 52-60.

[10] Wu, C.; Chen, K.; Hou, M.; Chang, Y.; Chang, C.; Liu, P.; Wu, S.; Chiu, C.; Jou, I.; Chen, C. Prevalence and associated factors of sarcopenia and severe sarcopenia in older Taiwanese living in rural community: The Tianliao Old People study 04. *Geriatr. Gerontol. Int.* **2014**, *14*, 69-75.

[11] Li, M.; Hu, Y.; Dong, H.Y.; Zhang, Y.; Fan, L.; Zhang, M.X.; Sun, J.; Han, X.Q.; Liu, Y.X.; Ma, X.N. Compare different measurement for prevalence of sarcopenia in a cohort of healthy community dwellingely in Beijing area. *Chin. J. Health Care Med.* **2014**, *16*, 426-429(Chinese article).

[12] Meng, N.; Li, C.; Liu, C.; Lin, C.; Lin, W.; Chang, C.; Li, T.; Lin, C. Comparison of height- and weight-adjusted sarcopenia in a Taiwanese metropolitan older population. *Geriatr. Gerontol. Int.* **2015**, *15*, 45-53.

[13] Wells, G.A.; Shea, B.; O'Connell, D.; Peterson, J.; Welch, V.; Losos, M.; Tugwell, P. The Newcastle-Ottawa Scale (NOS) for assessing the quality of nonrandomised studies in meta-analyses. Available: http://www.ohri.ca/programs/clinical_epidemiology/oxford.asp. Accessed 2014 Aug 5

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

[14] Higgins, J.P.T.; Green, S. Cochrane Handbook for Systematic Reviews of Interventions Version 5.1.0.Available: http://www.cochrane-handbook.org. Accessed 2014 Aug

[15] Wen, X.; An, P.; Chen, W.C.; Lv, Y.; Fu, Q. Comparisons of sarcopenia prevalence based on different diagnostic criteria in Chinese older adults. *The journal of nutrition, health & aging*. **2015**, *19*, 342-347.

[16] Zeng, P.; Wu, S.; Han, Y.; Liu, J.; Zhang, Y.; Zhang, E.; Zhang, Y.; Gong, H.; Pang, J.; Tang, Z.; Liu, H.; Zheng, X.; Zhang, T. Differences in body composition and physical functions associated with sarcopenia in Chinese elderly: Reference values and prevalence. *Arch. Gerontol. Geriat.* **2015**, *60*, 118-123.

[17] Gao, L.; Jiang, J.; Yang, M.; Hao, Q.; Luo, L.; Dong, B. Prevalence of Sarcopenia and Associated Factors in Chinese Community-Dwelling Elderly: Comparison Between Rural and Urban Areas. *J. Am. Med. Dir. Assoc.* **2015**, *16*, 1001-1003.

[18] Wang, Y.; Wang, Y.; Zhan, J.; Tang, Z.; He, J.; Tan, P.; Deng, H.; Huang, W.; Liu, Y. Sarco-Osteoporosis: Prevalence and Association with Frailty in Chinese Community-Dwelling Older Adults. *Int. J. Endocrinol.* **2015**, *2015*, 1-8.

[19] Woo, J.; Leung, J.; Morley, J.E. Defining Sarcopenia in Terms of Incident Adverse Outcomes. *J. Am. Med. Dir. Assoc.* **2015**, *16*, 247-252.

[20] Wang, H.; Hai, S.; Cao, L.; Zhou, J.; Liu, P.; Dong, B. Estimation of prevalence of sarcopenia by using a new bioelectrical impedance analysis in Chinese community-dwelling elderly people. *BMC Geriatr.* **2016**, *16*.

[21] Han, P.; Kang, L.; Guo, Q.; Wang, J.; Zhang, W.; Shen, S.; Wang, X.; Dong, R.; Ma, Y.; Shi, Y.; Shi, Z.; Li, H.; Li, C.; Ma, Y.; Wang, L.; Niu, K. Prevalence and Factors Associated With Sarcopenia in Suburb-dwelling Older Chinese Using the Asian Working Group for Sarcopenia Definition. *The Journals of Gerontology Series A: Biological Sciences and Medical Sciences*. **2016**, *71*, 529-535.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

[22] Han, D.; Chang, K.; Li, C.; Lin, Y.; Kao, T.; Tsai, K.; Wang, T.; Yang, W. Skeletal muscle mass adjusted by height correlated better with muscular functions than that adjusted by body weight in defining sarcopenia. *Sci. Rep.-UK*. **2016**, *6*.

[23] Xia, Z.W.; Meng, L.P.; Man, Q.Q.; Li, L.X.; Song, P.K.; Li, Y.Q.; Gao, Y.X.; Jia, S.S.; Zhang, J. Analysis of the dietary factors on sarcopenia in elderly in Beijing. *J Hygiene Res.* **2016**, *45*, 388-393 (Chinese article).

[24] Chen, Z.; Dai, R.; Gu, X.H. Prevalence of sarcopenia and quality of life in retired cadres of a military unit. *Chin. J. Osteopo. Rosis. & Bone Miner. Res.* **2016**, *9*, 264-268 (Chinese article).

[25] Bai, H.; Sun, J.; Chen, M.; Xu, D.; Xie, H.; Yu, Z.; Bao, Z.; Chen, J.; Pan, Y.; Lu, D.; Cheng, S. Age-related decline in skeletal muscle mass and function among elderly men and women in Shanghai, China: A cross sectional study. *Asia Pac. J. Clin. Nutr.* **2016**, *25*, 326-332..

[26] Hai, S.; Wang, H.; Cao, L.; Liu, P.; Zhou, J.; Yang, Y.; Dong, B. Association between sarcopenia with lifestyle and family function among community-dwelling Chinese aged 60 years and older. *BMC Geriatr.* **2017**, *17*.

[27] Ma, Y.; Bian, A.L.; Wang, X.F.; Wang, J.X.; Zhou, X.Z. Difference of the prevalence rate of sarcopenia in elderly people with different gender and age in Tianjin. *Chin. General Practice* **2017**, *20*, 3267-3271 (Chinese article).

[28] Wu, Z.N.; Wang, Y.B.; Shu, F.Q.; Song, B.; Li, G.; Zhu, H.M. Epidemiological investigation of sarcopenia in the nursing homes in Fengxian District and analysis of their risk factors. *Shanghai Med.* **2017**, *38*, 37-40 (Chinese article).

[29] Huang, L.J.; Liu, Y.C.; Wen, Y.F.; Liu, K. State and associated factors of sarcopenia in Jinzhou community elderly people. *Modern Preve. Med.* **2017**, *44*, 4458-4460 (Chinese article).

[30] Yang, M.; Hu, X.; Xie, L.; Zhang, L.; Zhou, J.; Lin, J.; Wang, Y.; Li, Y.; Han, Z.; Zhang, D.; Zuo, Y.; Li, Y.; Wu, L. Screening Sarcopenia in Community-Dwelling Older Adults: SARC-F vs SARC-F Combined With Calf Circumference (SARC-CalF). *J. Am. Med. Dir. Assoc.* **2018**, *19*, 271-277.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

[31] Zeng, Y.; Hu, X.; Xie, L.; Han, Z.; Zuo, Y.; Yang, M. The Prevalence of Sarcopenia in Chinese Elderly Nursing Home Residents: A Comparison of 4 Diagnostic Criteria. *J. Am. Med. Dir. Assoc.* **2018**, *19*, 690-695.

[32] Wang, Y.N.; Xu, J.L.; Song, H.L. Influencing factors of sarcopenia among elderly patients. *Modern Clinical Nursing* **2018**, *17*, 45-50 (Chinese article).

[33] Yang, L.; Wu, G.; Yang, Y.; Wu, Y.; Zhang, L.; Wang, M.; Mo, L.; Xue, G.; Wang, C.; Weng, X. Nutrition, Physical Exercise, and the Prevalence of Sarcopenia in Elderly Residents in Nursing Homes in China. *Med. Sci. Monitor*. **2019**, *25*, 4390-4399.

[34] Du, Y.; Wang, X.; Xie, H.; Zheng, S.; Wu, X.; Zhu, X.; Zhang, X.; Xue, S.; Li, H.; Hong, W.; Tang, W.; Chen, M.; Cheng, Q.; Sun, J. Sex differences in the prevalence and adverse outcomes of sarcopenia and sarcopenic obesity in community dwelling elderly in East China using the AWGS criteria. *BMC Endocr. Disord.* **2019**, *19*.

[35] Wang, H.; Hai, S.; Liu, Y.; Liu, Y.X.; Zhou, J.H.; Yang, Y.; Dong, B.R.; Yue, J.R. Prevalence of sarcopenia and associated factors in community-dwelling elderly populations in Chengdu China. *J. Sichuan Univ.* **2019**, *50*, 224-228 (Chinese article).

[36] Zhang, Y.; Chen, X.; Hou, L.; Lin, X.; Qin, D.; Wang, H.; Hai, S.; Cao, L.; Dong, B. Prevalence and Risk Factors Governing the Loss of Muscle Function in Elderly Sarcopenia Patients: A longitudinal Study in China with 4 Years of Follow-Up. *The journal of nutrition, health & aging*. **2020**, *24*, 518-524.

[37] Liu, X.; Hao, Q.; Yue, J.; Hou, L.; Xia, X.; Zhao, W.; Zhang, Y.; Ge, M.; Ge, N.; Dong, B. Sarcopenia, Obesity and Sarcopenia Obesity in Comparison: Prevalence, Metabolic Profile, and Key Differences: Results from WCHAT Study. *The journal of nutrition, health & aging*. **2020**, *24*, 429-437.

[38] Shafiee, G.; Keshtkar, A.; Soltani, A.; Ahadi, Z.; Larijani, B.; Heshmat, R. Prevalence of sarcopenia in the world: a systematic review and meta-analysis of general population studies. *Journal of Diabetes & Metabolic Disorders*. **2017**, *16*.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

[39] Makizako, H.; Nakai, Y.; Tomioka, K.; Taniguchi, Y. Prevalence of sarcopenia defined using the Asia Working Group for Sarcopenia criteria in Japanese community-dwelling older adults: A systematic review and meta-analysis. *Physical therapy research*. **2019**, *22*, 53-57.

[40] Diz, J.B.M.; Leopoldino, A.A.O.; Moreira, B.D.S.; Henschke, N.; Dias, R.C.; Pereira, L.S.M.; Oliveira, V.C. Prevalence of sarcopenia in older Brazilians: A systematic review and meta-analysis. *Geriatr. Gerontol. Int.* **2017**, *17*, 5-16.

[41] Papadopoulou, S.K.; Tsintavis, P.; Potsaki, G.; Papandreou, D. Differences in the Prevalence of Sarcopenia in Community-Dwelling, Nursing Home and Hospitalized Individuals. A Systematic Review and Meta-Analysis. *The journal of nutrition, health & aging*. **2020**, *24*, 83-90.

Legend

Figure 1. Flowchart of the detailed procedure for the inclusion or exclusion of selected studies.

Figure 2. Forest plots for the total prevalence of sarcopenia among the elderly Chinese population according to the Asian Working Group for Sarcopenia (AWGS) standard. Diamond represents the pooled odds ratio at 95% CI.

Figure 3. Forest plots for the total prevalence of sarcopenia among the elderly Chinese population according to the International Working Group on Sarcopenia (IWGS) standard. Diamond represents the pooled odds ratio at 95% CI.

Figure 4. Forest plots for the total prevalence of sarcopenia among the elderly Chinese population according to the European Working Group on Sarcopenia in Older People (EWGSOP) standard. Diamond represents the pooled odds ratio at 95% CI.

Figure 5. Forest plots for the prevalence of sarcopenia among male and female elderly people according to the Asian Working Group for Sarcopenia (AWGS) standard. Diamond represents the pooled odds ratio at 95% CI.

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Figure 6. Forest plots for the prevalence of sarcopenia among male and female elderly people according to the International Working Group on Sarcopenia (IWGS) standard. Diamond represents the pooled odds ratio at 95% CI.

Figure 7. Forest plots for the prevalence of sarcopenia among male and female elderly people according to the European Working Group on Sarcopenia in Older People (EWGSOP) standard. Diamond represents the pooled odds ratio at 95% CI.

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Table 1 Study characteristics of the studies included in the meta-analysis

Author	Publication Year	Participants(n)			Region	Setting	Measurement	Diagnostic Criteria	Prevalence (%)		
		Total	Male	Female					Total	Male	Female
Lee WJ	2013	386	223	163	Taiwan	Urban Community	DEXA	IWGS	11.1	10.8	11.7
								EWGSOP	16.6	14.9	19.0
Wu IC	2014	2155	1055	1100	Taiwan	Urban Community	BIA	EWGSOP	3.9	5.4	2.5
Wu CH	2014	549	285	264	Taiwan	Rural Community	BIA	EWGSOP	12.7	8.1	17.8
Li M	2014	169	169	0	Beijing	Urban Community	DEXA	EWGSOP	33.3	33.3	-
Meng NH	2015	771	359	712	Taiwan	Urban Community	DEXA	EWGSOP	9.7	0.7	12.5
Gao LL	2015	612	254	358	Sichuan	Rural and urban Community	Calf circumference	AWGS	9.8	6.7	12.0
Wen X	2015	286	136	150	Hangzhou	Urban Community	DEXA	IWGS	6.0	7.4	4.7
								EWGSOP	0.4	0.8	0.01

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							AWGS	3.1	5.9	0.7
Woo J	2015	4000	2000	Hongkong	Urban Community	DEXA	IWGS	20.2	22.1	18.3
							EWGSOP	9.0	9.5	8.6
							AWGS	7.3	5.3	9.3
Zeng P	2015	395	199	196	Beijing	Rural and urban Community	BIA	AWGS	24.3	18.1	30.6
Wang YJ	2015	316	164	152	Changsha	Urban Community	DEXA	AWGS	29.7	26.2	33.6
Han PP	2016	1069	467	602	Tianjin	Urban Community	BIA	AWGS	9.3	6.4	17.5
Wang H	2016	636	316	320	Chengdu	Urban Community	BIA	AWGS	10.4	8.2	12.5
Bai HJ	2016	415	212	203	Shanghai	Urban Community	BIA	AWGS	28.7	33.5	23.6
Xia ZW	2016	683	239	444	Beijing	Urban Community	BIA	AWGS	20.1	17.2	21.6
Chen Z	2016	281	203	78	Chongqing	Urban Community	BIA	IWGS	27.8	-	-
Han DS	2016	878	402	476	Taiwan	Urban Community	BIA	EWGSOP	9.8	4.5	14.3
Hai S	2017	834	415	419	Chengdu	Urban Community	BIA	AWGS	10.6	11.3	9.8

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Huang LJ	2017	193	65	128	Jinzhou	Urban Community	BIA	AWGS	11.4	7.7	13.3
Yang M	2018	384	160	224	Chengdu	Urban Community	BIA	AWGS	15.9	11.9	18.8
								IWGS	25.0	24.4	25.4
								EWGSOP	11.7	10.6	12.5
Du YP	2019	631	213	418	Shanghai	Urban Community	BIA	AWGS	12.2	19.2	8.6
Wang H	2019	947	465	482	Chengdu	Urban Community	BIA	AWGS	7.7	6.5	8.9
Zhang Y	2020	474	238	236	Chengdu	Urban Community	BIA	AWGS	15.4	-	-
Liu XL	2020	4500	1627	2873	Yunnan, Guizhou, Sichuan, and Xinjiang	Urban Community	BIA	AWGS	19.3	22.1	17.8

DEXA, dual-energy x-ray absorptiometry; BIA, bioelectrical impedance analysis; IWGS, International Working Group on Sarcopenia; AWGS, Asia Working Group for Sarcopenia; EWGSOP, European Working Group on Sarcopenia in Older People.

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Table 2 The prevalence of sarcopenia diagnosed by different criteria using DEXA and BIA measurement (%).

	DEXA	BIA
AWGS	13(5−21)	15(12−19)
IWGS	12(3−22)	26(23−29)
EWGSOP	12(6−18)	9(5−14)

DEXA, dual-energy x-ray absorptiometry; BIA, bioelectrical impedance analysis; IWGS, International Working Group on Sarcopenia; AWGS, Asia Working Group for Sarcopenia; EWGSOP, European Working Group on Sarcopenia in Older People.

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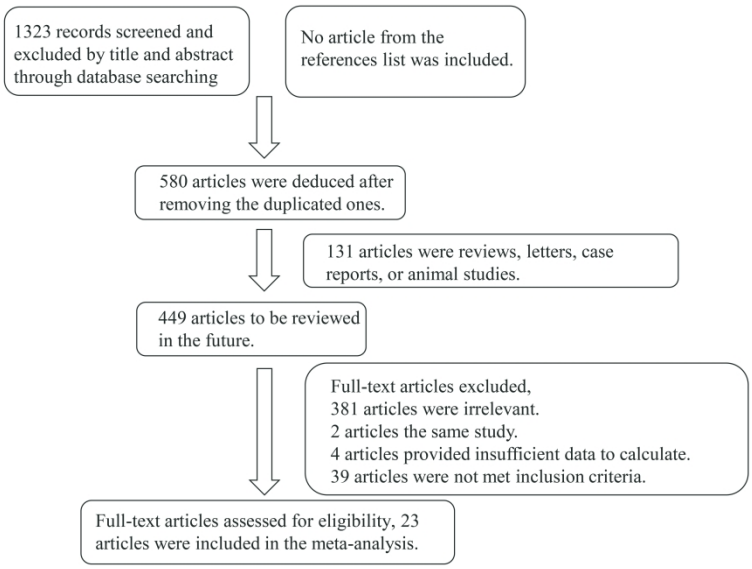


Figure 1. Flowchart of the detailed procedure for the inclusion or exclusion of selected studies.

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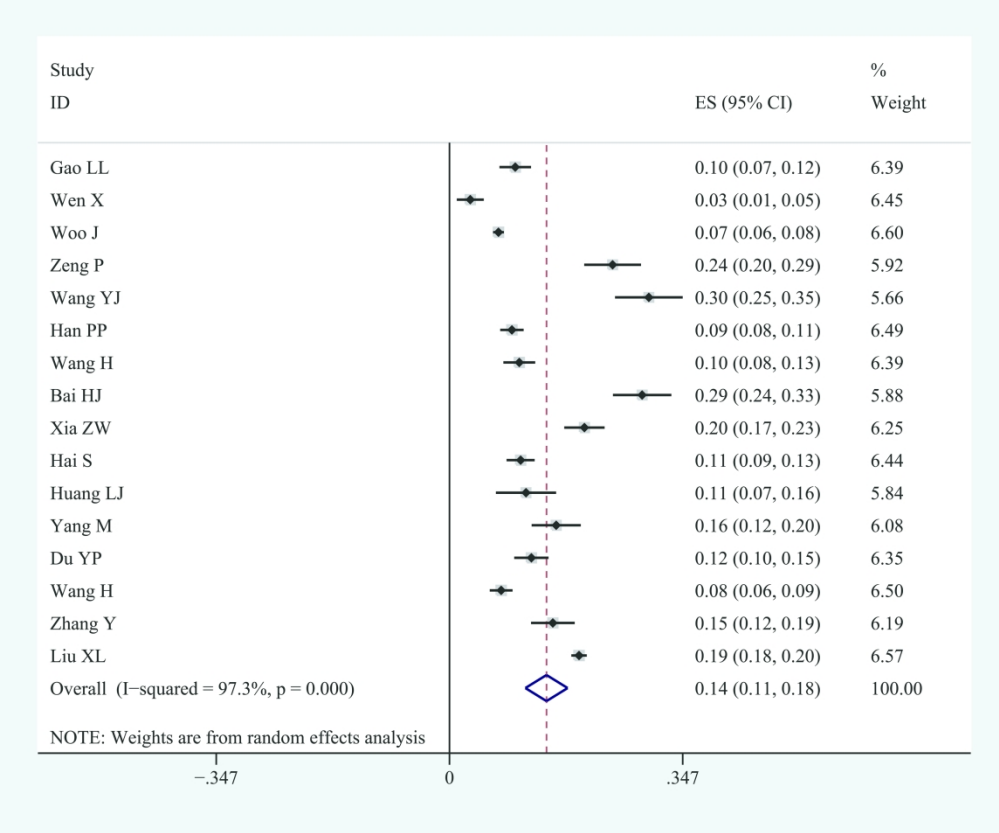


Figure 2. Forest plots for the total prevalence of sarcopenia among the elderly Chinese population according to the Asian Working Group for Sarcopenia (AWGS) standard. Diamond represents the pooled odds ratio at 95% CI.

280x233mm (300 x 300 DPI)

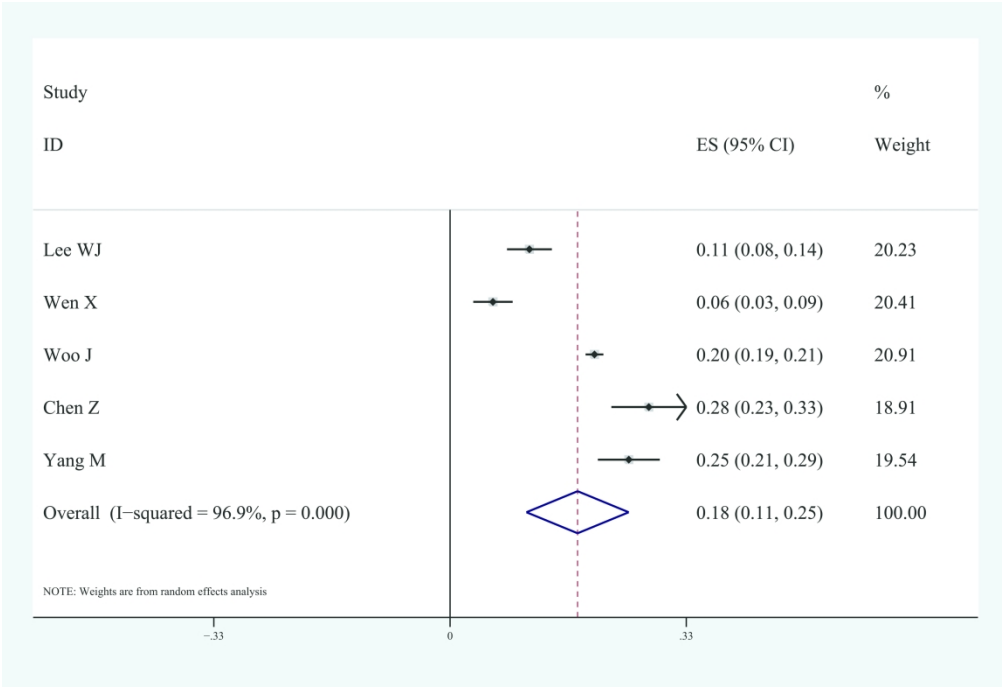


Figure 3. Forest plots for the total prevalence of sarcopenia among the elderly Chinese population according to the International Working Group on Sarcopenia (IWGS) standard. Diamond represents the pooled odds ratio at 95% CI.

272x186mm (300 x 300 DPI)

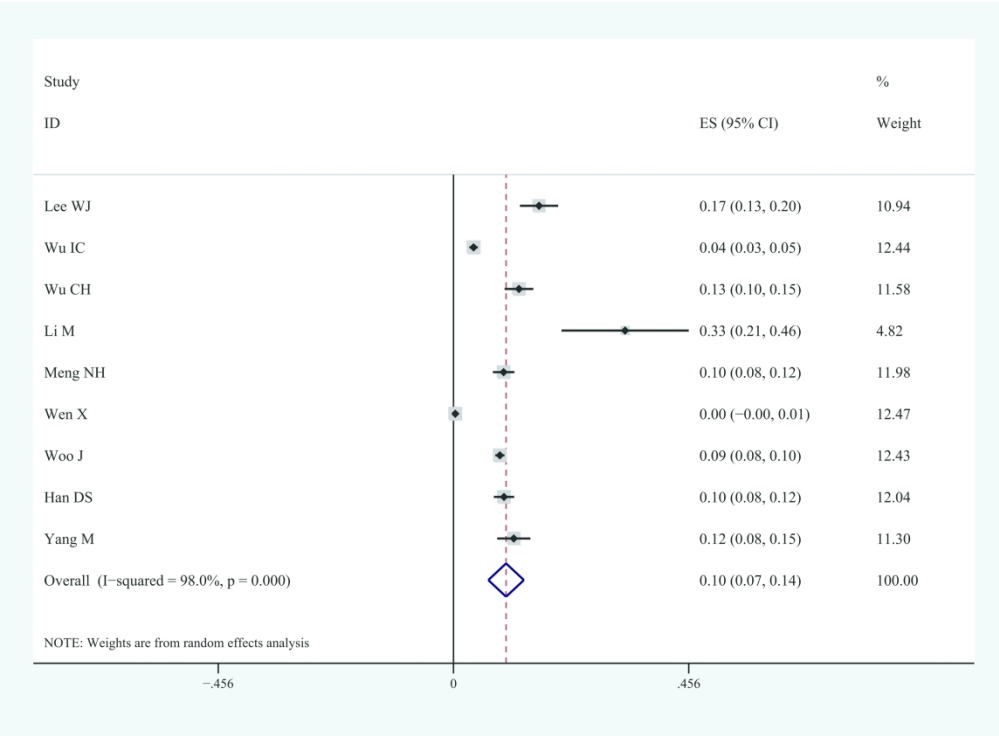


Figure 4. Forest plots for the total prevalence of sarcopenia among the elderly Chinese population according to the European Working Group on Sarcopenia in Older People (EWGSOP) standard. Diamond represents the pooled odds ratio at 95% CI.

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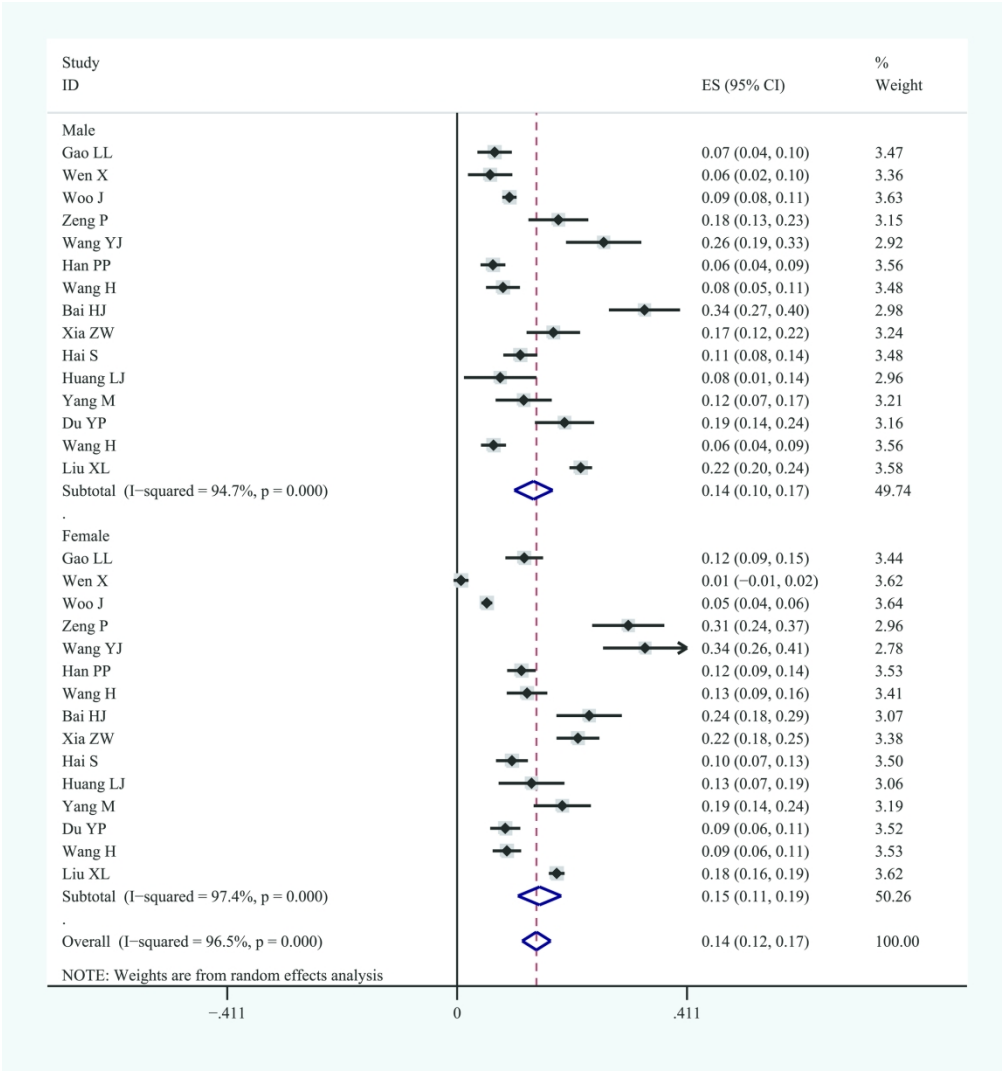


Figure 5. Forest plots for the prevalence of sarcopenia among male and female elderly people according to the Asian Working Group for Sarcopenia (AWGS) standard. Diamond represents the pooled odds ratio at 95% CI.

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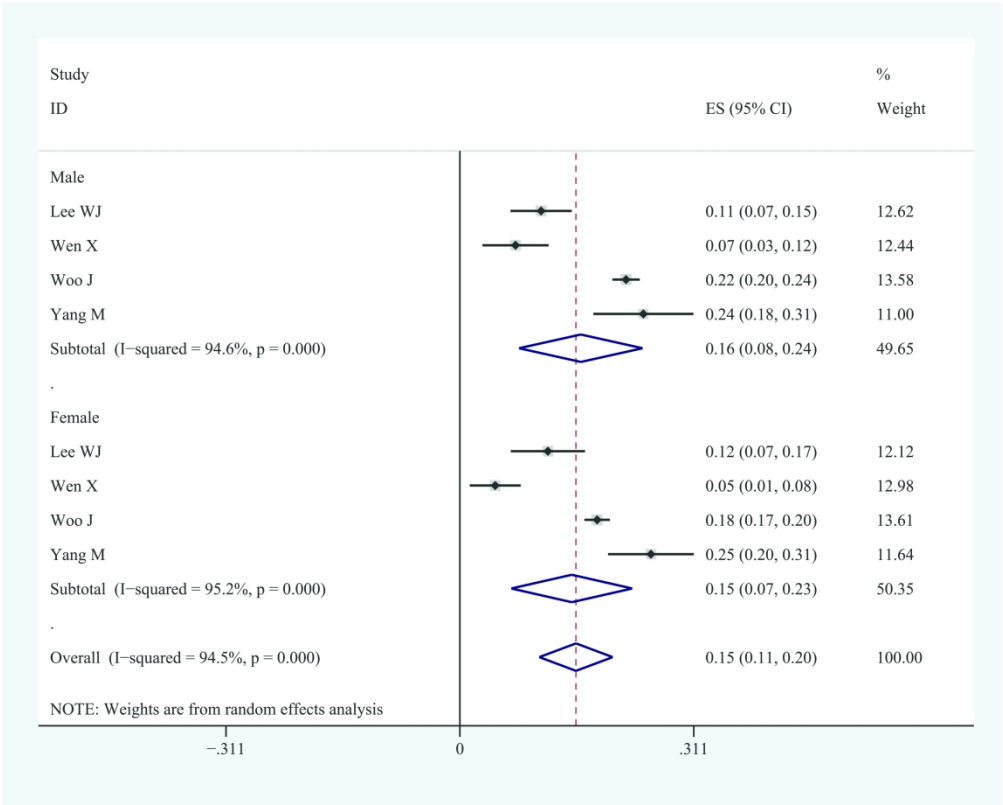


Figure 6. Forest plots for the prevalence of sarcopenia among male and female elderly people according to the International Working Group on Sarcopenia (IWGS) standard. Diamond represents the pooled odds ratio at 95% CI.

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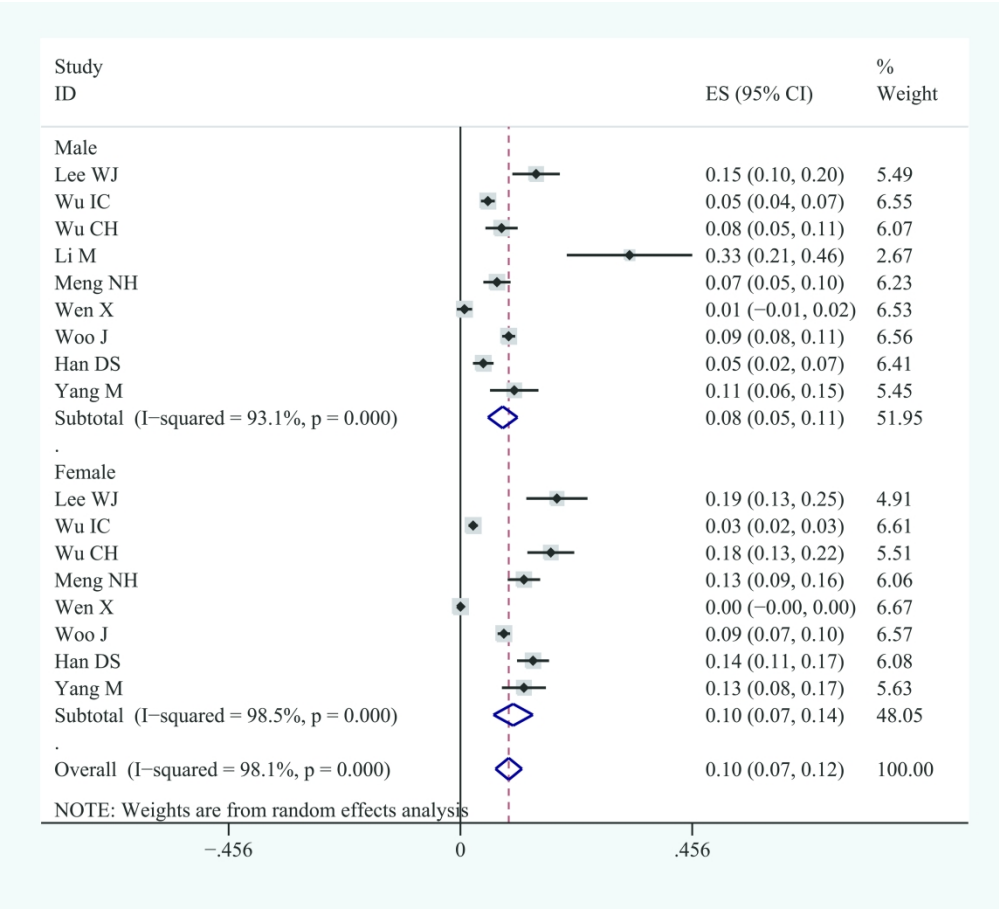


Figure 7. Forest plots for the prevalence of sarcopenia among male and female elderly people according to the European Working Group on Sarcopenia in Older People (EWGSOP) standard. Diamond represents the pooled odds ratio at 95% CI.

284x257mm (300 x 300 DPI)

File S1 Search strategy in databases

PubMed: (((sarcopenia[Title/Abstract]) OR “muscle mass [Title/Abstract]) OR muscle strength[Title/Abstract]) AND ((prevalence[Title/Abstract]) OR epidemi*[Title/Abstract]) AND ((Chinese[Title/Abstract]) OR China[Title/Abstract])

Web of Science: (((TS=sarcopenia) OR TS=muscle mass) OR TS=muscle strength) AND ((TS=prevalence) OR TS=epidemi*) AND ((TS=Chinese) OR TS=China)

MEDLINE: (((TS=sarcopenia) OR TS=muscle mass) OR TS=muscle strength) AND ((TS=prevalence) OR TS=epidemi*) AND ((TS=Chinese) OR TS=China)

CNKI: (AB =sarcopenia OR AB =muscle mass OR AB =muscle strength) AND (AB =prevalence OR AB =epidemiology)

CQVIP: (R=sarcopenia +R=muscle mass+ R=muscle strength) *(R=prevalence +R=epidemiology)

Wangfang: Abstract: (sarcopenia + muscle mass + muscle strength) * (prevalence + epidemiology)

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Table S1. PRISMA Checklist.

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	3
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	3,4
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	4
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	4
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	4
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	4
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	4
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	5
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	5
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	5
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	5

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Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	5

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	5
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	5
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	6
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	6
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	6
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	6
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	6,7
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	7
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	7
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	7,8
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	8
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	8
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	9

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Figure S1. The sensitivity analysis results according to the Asian Working Group for Sarcopenia (AWGS) standard.

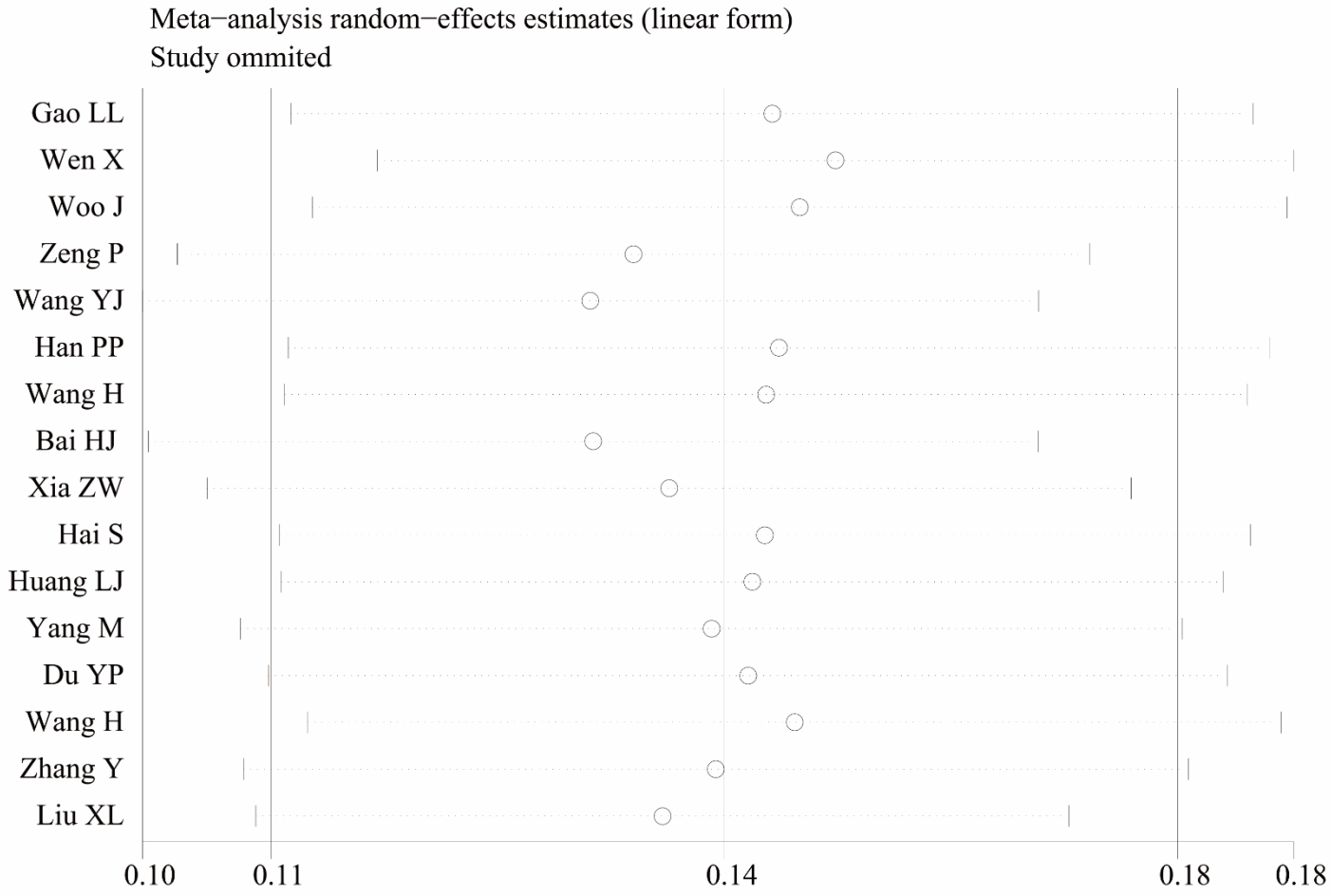


Figure S2. The sensitivity analysis results according to the International Working Group on Sarcopenia (IWGS) IWGS standard.

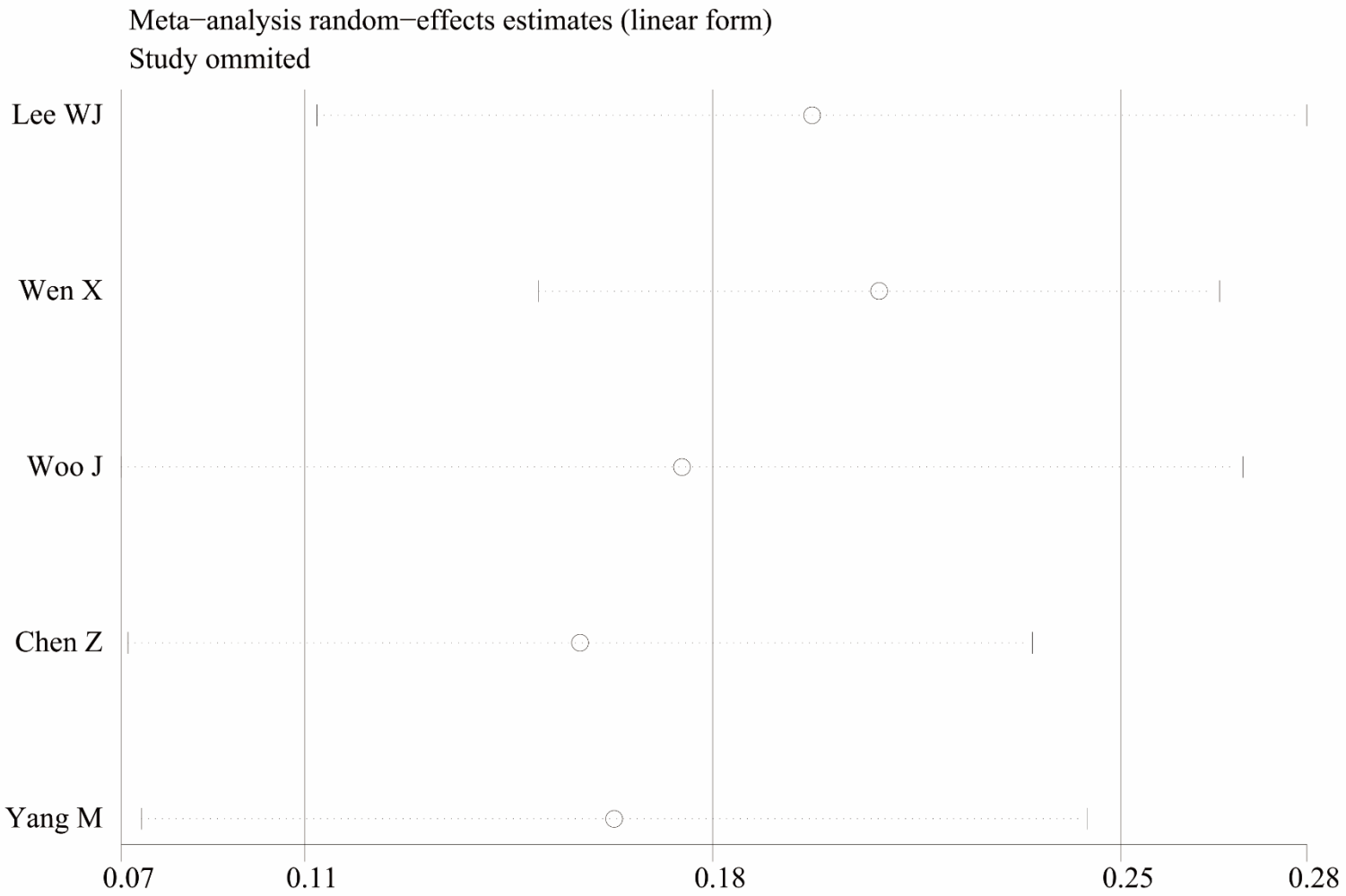
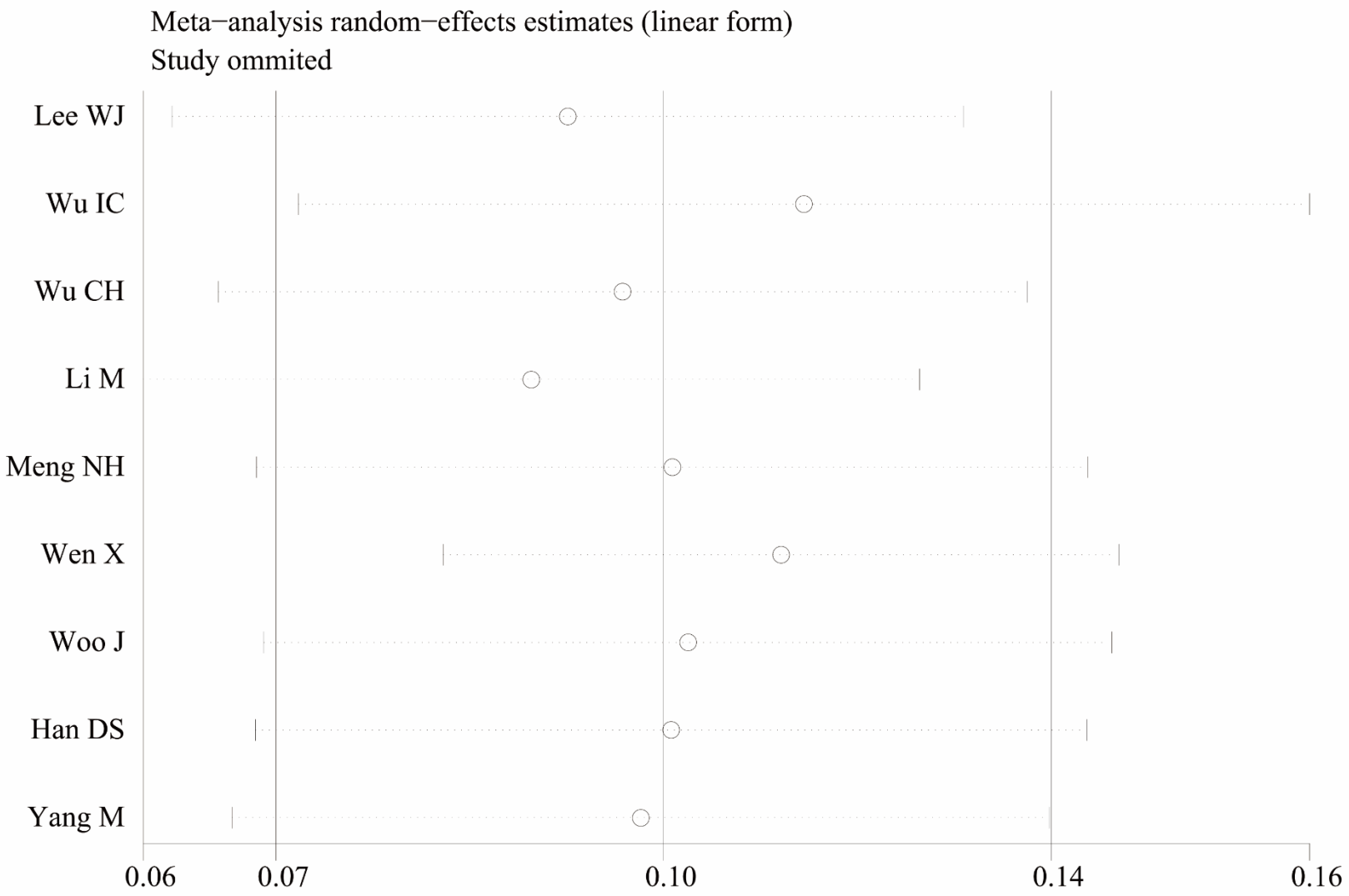


Figure S3. The sensitivity analysis results according to the European Working Group on Sarcopenia in Older People (EWGSOP) standard.



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Prevalence of Sarcopenia in Older Chinese Adults: A Systematic Review and Meta-analysis

Journal:	*BMJ Open*
Manuscript ID	bmjopen-2020-041879.R2
Article Type:	Original research
Date Submitted by the Author:	08-Apr-2021
Complete List of Authors:	Xin, Caihong; Fourth People's Hospital in Shenyang Sun, Xin; First Affiliated Hospital of Soochow University; People's Hospital of Liaoning Province Lu, Li; People's Hospital of Liaoning Province Shan, Ling; People's Hospital of Liaoning Province
Primary Subject Heading:	Geriatric medicine
Secondary Subject Heading:	Epidemiology, Global health, Public health
Keywords:	GERIATRIC MEDICINE, EPIDEMIOLOGY, General endocrinology < DIABETES & ENDOCRINOLOGY

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Prevalence of Sarcopenia in Older Chinese Adults: A Systematic Review and Meta-analysis

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Running title: Prevalence of sarcopenia in

Key words: Prevalence; sarcopenia; China

Word count of the manuscript: 1703

Figures number: 7

Tables number: 2

Disclosure Statement: The authors have nothing to disclose.

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Abstract

Objective: This meta-analysis was conducted to estimate the overall prevalence of sarcopenia in older Chinese adults.

Design: Systematic review and meta-analysis.

Participants: A literature research was conducted using PubMed, Web of Science, CNKI, CQVIP, and Wanfang databases. The search terms in abstract were identified as follows: "sarcopenia," in combination with the terms "prevalence," "epidemiology," and "China." All studies published from January 2010 to January 2020 were included. Random effect model was used to estimate the prevalence of sarcopenia. The sex-specific prevalence of sarcopenia at 95% confidence interval (CI) was also calculated using different criteria for defining sarcopenia.

Primary outcome measures: The overall prevalence of sarcopenia in older Chinese adults.

Results: In total, 23 articles were included in this meta-analysis, involving 21,564 participants. On the basis of the Asian Working Group for Sarcopenia (AWGS) criterion, the overall prevalence of sarcopenia among the elderly in China was 14% (95% CI, 11%–18%). According to AWGS criterion, the prevalence of sarcopenia was higher in Chinese females than in males (15% vs. 14%).

Conclusions: This systematic review is the first estimation of the pooled prevalence of sarcopenia in older Chinese adults. The results suggest that China may have the largest sarcopenia patient count. Therefore, more attention should be given to the prevention and control of sarcopenia.

Trial registration: PROSPERO CRD42020223405

Strengths and limitations of this study:

To our knowledge, this systematic review is the first estimation of the pooled prevalence of sarcopenia in Chinese older people.

The search languages were limited to English and Chinese.

A limited number of high-quality studies.

A high level of heterogeneity among included studies.

Introduction

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Sarcopenia is characterized by the deterioration of muscle mass and function that occurs with aging. Several studies have shown that with age increase, skeletal muscle mass and strength will be lost at a rate of 1%–2% annually, possibly leading to the decline of physical function, increasing the disability, and mortality rates of elderly people and bringing serious economic and social burdens [1–4].

A practical clinical definition was reported by the European Working Group on Sarcopenia in Older People (EWGSOP) in 2010 [5]. Based on the EWGSOP criteria, slow walking speed (0.8 m/sec) or low grip strength (<30 kg for men and <20 kg for women) are indications for muscle mass measurement. Later in 2011, another consensus was published by the International Working Group on Sarcopenia (IWGS) [6]. However, Asia is a vast region with different races, countries, and cultures. Developing screening and diagnostic strategies for the Asian population is imperative. The Asian Working Group for Sarcopenia (AWGS) was established in 2013, and the consensus of AWGS was released based on 16 epidemiologic reports on sarcopenia in six countries or regions, leading to the development of a diagnosis strategy for sarcopenia in Asia [7]. The diagnostic criteria recommended by AWGS and EWGSOP are similar, including muscle mass, muscle strength, and activity measurements. The difference lies in the specific cut-off value. The cut-off values for muscle strength (<26 kg for men and <18 kg for women), walking speed (0.8 m/sec), and muscle mass (7.0 kg/m^2 for men and 5.4 kg/m^2 for women measured using dual X-ray absorptiometry and 7.0 kg/m^2 for men and 5.7 kg/m^2 for women measured using bioimpedance analysis) have been revised according to data from regional cohort studies among Asian populations.

According to statistics, the elderly population (≥60 years old) has reached 16.1% of China's total population, which is close to 250 million. In the face of a severe aging situation, people are paying more attention to sarcopenia, but the prevalence of sarcopenia was inconsistent in different studies [8–12]. At present, there is no large-scale epidemiologic survey of sarcopenia in China. The purpose of this study is to systematically reveal the prevalence of sarcopenia in older Chinese adults.

Methods

Search strategy

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We searched the following electronic databases: PubMed, Web of Science, China National Knowledge Infrastructure (CNKI), CQVIP database, and Wanfang databases. The search terms in the title or abstract were identified as follows: "sarcopenia," "muscle mass," and "muscle strength" in combination with the terms "prevalence," "epidemiology," and "China." The complete search strategy is included in supplementary file (File S1). All studies published from January 2010 to January 2020 were included. The references of the retrieved articles were examined to identify additional eligible studies, but unpublished studies were not included. The search languages were limited to English and Chinese. This systematic review and meta-analysis was registered in PROSPERO (registration number: CRD42020223405). The completed Preferred Reporting Items for Systematic Reviews and Meta-Analyses checklist is presented in the supplementary data. This meta-analysis was conducted to estimate the overall prevalence of sarcopenia in older Chinese adults.

Patient and public involvement:

No patients were involved in this study.

Inclusion and exclusion criteria

Inclusion of the selected studies was based on the following criteria: (1) a cross-sectional study or retrospective study, (2) a sample size of ≥100 elderly Chinese participants (age ⩾ 60 years); (3) sufficient information of sarcopenia at the corresponding 95% confidence interval (CI) and *P* value, and (4) the definition of sarcopenia is in line with the AWGS, EWGSOP, or IWGS standards.

Exclusion of the selected studies was based on the following criteria: (1) same data from different studies, (2) studies must define sarcopenia only by reduction of muscle mass or strength, and (3) the subjects were a specific population, such as patients living in a nursing home or those with disability. Reviews, case reports, letters, and animal researches were excluded.

Data extraction and methodological quality assessment

Information was extracted by two independent researchers from the included studies, including first author's name, publication year, sample, diagnostic criteria, participants, setting, and prevalence. Two independent reviewers who were not blinded to the authors or journals assessed the risk of bias in the included studies using the Newcastle–Ottawa scale (NOS) [13]. The NOS is recommended by the

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Cochrane Handbook for Systematic Reviews of Interventions [14]. Each included study was judged in terms of three domains using the "star system," namely, representativeness of the study group selection, group comparability, and exposure or outcome ascertainment. NOS scores range from zero to nine stars. The two researchers independently assessed the studies by discussion, comparison of findings, and resolution of differences through consensus formation. If no consensus can be reached, a third researcher was commissioned to resolve the difference.

Statistical analysis

We used a systematic analysis approach to calculate the pooled prevalence of sarcopenia for all eligible studies, and heterogeneity among these studies was assessed using Cochran's Q test and the I^2 statistical index. Random effect model was used in combining the prevalence of sarcopenia. We additionally performed sensitivity analysis to evaluate the influence of any given study on the pooled estimate. Publication bias was evaluated using Egger's test. Subgroup analyses were also performed. The significance level was set at $P < 0.05$. All statistical analyses were performed using Stata software version 12.0 (College Station, Texas, USA).

Results

Study characteristics

In total, 1213 studies were retrieved from the aforementioned databases. After screening, a total of 23 articles were included based on the criteria shown in Figure 1 [8–12, 15–37]. Altogether, the 23 articles included 21,564 participants in China between 2010 and 2020. The selected studies' characteristics are summarized in Table 1. On the basis of the NOS scale, the score of selected studies were all above five stars; therefore, all articles were included.

Results of the meta-analysis

According to the AWGS standard, the total prevalence of sarcopenia among Chinese elderly people was 14% (95% CI, 11%–18%, Figure 2), and the total prevalence rates of sarcopenia among the elderly in China based on the IWGS and EWGSOP standards were 18% (95% CI, 11%–25%, Figure 3) and 10% (95% CI, 7%–14%, Figure 4), respectively.

Subgroup analyses were performed to determine the influence of sex on the prevalence of sarcopenia.

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The prevalence of sarcopenia among elderly females was higher than that in males in AWGS and EWGSOP standards (Figures 5–7). Table 2 shows the subgroup analyses results based on different methods for diagnostic criteria. The prevalence of sarcopenia estimated using the Bio-electrical Impedance Analysis (BIA) method was higher than that estimated using the Dual Energy X-Ray Absorptiometry (DEXA) method in AWGS and IWGS standards.

Sensitivity analysis and publication bias

Sensitivity analysis was performed to examine the influence of each study. We found no significant difference between the results of the sensitivity analysis and our previous estimates, indicating that our statistical results were relatively credible (Figures S1–S3). The articles obtained from the databases were carefully and comprehensively searched. Egger's test was also conducted to determine whether potential publication bias existed in the reviewed literature. The results ($P > 0.05$) suggested that there were no publication bias (Figures S4–S6).

Discussion

The present study is the first meta-analysis of the prevalence of sarcopenia in older Chinese adults. Sarcopenia is a type of age-related disease characterized by progressive deterioration in skeletal muscle fiber quality, strength, and function. Currently, sarcopenia has been internationally recognized as a new type of geriatric syndrome, attracting worldwide attention. In 2017, Shafeee et al. conducted a meta-analysis on the prevalence of sarcopenia. Their results showed that the prevalence of sarcopenia among the elderly in communities worldwide was about 10%, which was higher among non-Asian individuals than in Asians in both sexes [38]. A similar study was performed in Japan, and it was found that the prevalence of sarcopenia among the elderly Japanese was 9.9% (95% CI, 6.2%–15.4%) according to the AWGS standard [39]. The results of our study showed that the prevalence of sarcopenia was 14% (95% CI, 11%–18%) in older Chinese adults, which was much higher than the results of the aforementioned studies [40, 41]. China may have the largest sarcopenia case number all over the world because it is the most populated country in the world. The high prevalence of sarcopenia in China may be related to the Chinese diet structure. Although protein is the raw material for muscle synthesis, Chinese people, especially the older population, have a higher carbohydrate intake and a less protein intake [42]. Therefore, more attention should be given to the prevention and control of sarcopenia in

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older Chinese adults.

Here, we found that the prevalence of sarcopenia among females was higher than that in males, which was consistent with other meta-analyses [38–40]. This is mainly because men eat and exercise more than women, and it helps muscle synthesis. Most of the studies included in our meta-analysis used the BIA method because of its affordability and portable equipment. BIA is a widely used non-invasive method for measuring human body composition. The principle behind this method is to record the different electrical resistances of different tissues using electrodes placed on the body surface and then to measure muscle mass using an image reconstruction method. DEXA is also a commonly used method to evaluate muscle mass, which can accurately distinguish whole-body and local muscles, as well as fats and bones. The limited number of studies using DEXA included in this meta-analysis may influence our results. As we know, aging increases the prevalence of sarcopenia, however, owing to the different age classifications in the included studies, we cannot obtain any conclusion.

The limitation of the present study is that the main data are from urban populations and studies on the prevalence of sarcopenia in Chinese rural areas are rare, impacting the calculation of the total prevalence of sarcopenia. Second, the prevalence of sarcopenia in people aged 60–70 years and in those aged 70–80 years might be different. Not all studies presented data according to age group, so we cannot accurately estimate the prevalence in terms of age subgroup. Third, most of the studies were performed in large cities or economically developed areas. All of these factors affected our results, and as such, they should be interpreted cautiously as further research is needed.

Conclusions

To the best of our knowledge, the present systematic review is the first to estimate the pooled prevalence of sarcopenia in older Chinese adults. The results of the present systematic review reveal that China may have the largest number of sarcopenia cases. Therefore, the prevention and management of sarcopenia should be comprehensively addressed.

Author contributions statement

SL designed the study. SX and LL searched databases and collected the data. XCH and SL assessed the

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quality of the study. SL performed the analysis. SX and XCH wrote the manuscript. SX modified the manuscript in the revision. All authors contributed to this systematic review and meta-analysis.

Supporting Information

File S1 Search strategy in databases.

Table S1. PRISMA Checklist.

Figure S1. The sensitivity analysis results according to the Asian Working Group for Sarcopenia (AWGS) standard.

Figure S2. The sensitivity analysis results according to the the International Working Group on Sarcopenia (IWGS) standard.

Figure S3. The sensitivity analysis results according to the European Working Group on Sarcopenia in Older People (EWGSOP) standard.

Figure S4. The publication bias results according to the Asian Working Group for Sarcopenia (AWGS) standard.

Figure S5. The publication bias results according to the the International Working Group on Sarcopenia (IWGS) standard.

Figure S6. The publication bias results according to the European Working Group on Sarcopenia in Older People (EWGSOP) standard.

Ethics approval statement

Not applicable.

Funds

This research was financially supported by the National Natural Science Foundation of China (grant No. 81800697).

Acknowledgments

We thank Enago for English language editing.

Conflict of interests

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None declared.

Data availability statement

All data relevant to the study are included in the article or uploaded as supplementary information

Reference

[1] Cederholm, T.; Morley, J.E. Sarcopenia: the new definitions. *Current Opinion in Clinical Nutrition and Metabolic Care*. **2015**, *18*, 1–4.

[2] Frontera, W.R.; Hughes, V.A.; Fielding, R.A.; Fiatarone, M.A.; Evans, W.J.; Roubenoff, R. Aging of skeletal muscle: a 12-yr longitudinal study. *J Appl Physiol (1985)*. **2000**, *88*, 1321-1326.

[3] Delmonico, M.J.; Harris, T.B.; Lee, J.S.; Visser, M.; Nevitt, M.; Kritchevsky, S.B.; Tylavsky, F.A.; Newman, A.B. Alternative definitions of sarcopenia, lower extremity performance, and functional impairment with aging in older men and women. *J. Am. Geriatr. Soc.* **2007**, *55*, 769-774..

[4] Woo, J. Sarcopenia. *Clin. Geriatr. Med*. **2017**, *33*, 305–314.

[5] Cruz-Jentoft, A.J.; Baeyens, J.P.; Bauer, J.M.; Boirie, Y.; Cederholm, T.; Landi, F.; Martin, F.C.; Michel, J.P.; Rolland, Y.; Schneider, S.M.; Topinkova, E.; Vandewoude, M.; Zamboni, M. Sarcopenia: European consensus on definition and diagnosis: Report of the European Working Group on Sarcopenia in Older People. *Age Ageing*. **2010**, *39*, 412-423.

[6] Fielding, R.A.; Vellas, B.; Evans, W.J.; Bhasin, S.; Morley, J.E.; Newman, A.B.; Abellan, V.K.G.; Andrieu, S.; Bauer, J.; Breuille, D.; Cederholm, T.; Chandler, J.; De Meynard, C.; Donini, L.; Harris, T.; Kannt, A.; Keime, G.F.; Onder, G.; Papanicolaou, D.; Rolland, Y.; Rooks, D.; Sieber, C.; Souhami, E.; Verlaan, S.; Zamboni, M. Sarcopenia: an undiagnosed condition in older adults. Current consensus definition: prevalence, etiology, and consequences. International working group on sarcopenia. *J. Am. Med. Dir. Assoc.* **2011**, *12*, 249-256.

[7] Chen, L.K.; Liu, L.K.; Woo, J.; Assantachai, P.; Auyeung, T.W.; Bahyah, K.S.; Chou, M.Y.; Chen, L.Y.; Hsu, P.S.; Krairit, O.; Lee, J.S.; Lee, W.J.; Lee, Y.; Liang, C.K.; Limpawattana, P.; Lin, C.S.; Peng, L.N.; Satake, S.; Suzuki, T.; Won, C.W.; Wu, C.H.; Wu, S.N.; Zhang, T.; Zeng, P.; Akishita,

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M.; Arai, H. Sarcopenia in Asia: consensus report of the Asian Working Group for Sarcopenia. *J. Am. Med. Dir. Assoc.* **2014**, *15*, 95-101.

[8] Lee, W.; Liu, L.; Peng, L.; Lin, M.; Chen, L. Comparisons of Sarcopenia Defined by IWGS and EWGSOP Criteria Among Older People: Results From the I-Lan Longitudinal Aging Study. *J. Am. Med. Dir. Assoc.* **2013**, *14*, 521-528.

[9] Wu, I.; Lin, C.; Hsiung, C.A.; Wang, C.; Wu, C.; Chan, D.D.; Li, T.; Lin, W.; Huang, K.; Chen, C.; Hsu, C. Epidemiology of sarcopenia among community-dwelling older adults in Taiwan: A pooled analysis for a broader adoption of sarcopenia assessments. *Geriatr. Gerontol. Int.* **2014**, *14*, 52-60.

[10] Wu, C.; Chen, K.; Hou, M.; Chang, Y.; Chang, C.; Liu, P.; Wu, S.; Chiu, C.; Jou, I.; Chen, C. Prevalence and associated factors of sarcopenia and severe sarcopenia in older Taiwanese living in rural community: The Tianliao Old People study 04. *Geriatr. Gerontol. Int.* **2014**, *14*, 69-75.

[11] Li, M.; Hu, Y.; Dong, H.Y.; Zhang, Y.; Fan, L.; Zhang, M.X.; Sun, J.; Han, X.Q.; Liu, Y.X.; Ma, X.N. Compare different measurement for prevalence of sarcopenia in a cohort of healthy community dwellingely in Beijing area. *Chin. J. Health Care Med.* **2014**, *16*, 426-429(Chinese article).

[12] Meng, N.; Li, C.; Liu, C.; Lin, C.; Lin, W.; Chang, C.; Li, T.; Lin, C. Comparison of height- and weight-adjusted sarcopenia in a Taiwanese metropolitan older population. *Geriatr. Gerontol. Int.* **2015**, *15*, 45-53.

[13] Wells, G.A.; Shea, B.; O'Connell, D.; Peterson, J.; Welch, V.; Losos, M.; Tugwell, P. The Newcastle-Ottawa Scale (NOS) for assessing the quality of nonrandomised studies in meta-analyses. Available: http://www.ohri.ca/programs/clinical_epidemiology/oxford.asp. Accessed 2014 Aug 5

[14] Higgins, J.P.T.; Green, S. Cochrane Handbook for Systematic Reviews of Interventions Version 5.1.0. Available: http://www.cochrane-handbook.org. Accessed 2014 Aug

[15] Wen, X.; An, P.; Chen, W.C.; Lv, Y.; Fu, Q. Comparisons of sarcopenia prevalence based on different diagnostic criteria in Chinese older adults. *The journal of nutrition, health & aging*. **2015**, *19*, 342-347.

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[16] Zeng, P.; Wu, S.; Han, Y.; Liu, J.; Zhang, Y.; Zhang, E.; Zhang, Y.; Gong, H.; Pang, J.; Tang, Z.; Liu, H.; Zheng, X.; Zhang, T. Differences in body composition and physical functions associated with sarcopenia in Chinese elderly: Reference values and prevalence. *Arch. Gerontol. Geriat.* **2015**, *60*, 118-123.

[17] Gao, L.; Jiang, J.; Yang, M.; Hao, Q.; Luo, L.; Dong, B. Prevalence of Sarcopenia and Associated Factors in Chinese Community-Dwelling Elderly: Comparison Between Rural and Urban Areas. *J. Am. Med. Dir. Assoc.* **2015**, *16*, 1001-1003.

[18] Wang, Y.; Wang, Y.; Zhan, J.; Tang, Z.; He, J.; Tan, P.; Deng, H.; Huang, W.; Liu, Y. Sarco-Osteoporosis: Prevalence and Association with Frailty in Chinese Community-Dwelling Older Adults. *Int. J. Endocrinol.* **2015**, *2015*, 1-8.

[19] Woo, J.; Leung, J.; Morley, J.E. Defining Sarcopenia in Terms of Incident Adverse Outcomes. *J. Am. Med. Dir. Assoc.* **2015**, *16*, 247-252.

[20] Wang, H.; Hai, S.; Cao, L.; Zhou, J.; Liu, P.; Dong, B. Estimation of prevalence of sarcopenia by using a new bioelectrical impedance analysis in Chinese community-dwelling elderly people. *BMC Geriatr.* **2016**, *16*.

[21] Han, P.; Kang, L.; Guo, Q.; Wang, J.; Zhang, W.; Shen, S.; Wang, X.; Dong, R.; Ma, Y.; Shi, Y.; Shi, Z.; Li, H.; Li, C.; Ma, Y.; Wang, L.; Niu, K. Prevalence and Factors Associated With Sarcopenia in Suburb-dwelling Older Chinese Using the Asian Working Group for Sarcopenia Definition. *The Journals of Gerontology Series A: Biological Sciences and Medical Sciences*. **2016**, *71*, 529-535.

[22] Han, D.; Chang, K.; Li, C.; Lin, Y.; Kao, T.; Tsai, K.; Wang, T.; Yang, W. Skeletal muscle mass adjusted by height correlated better with muscular functions than that adjusted by body weight in defining sarcopenia. *Sci. Rep.-UK*. **2016**, *6*.

[23] Xia, Z.W.; Meng, L.P.; Man, Q.Q.; Li, L.X.; Song, P.K.; Li, Y.Q.; Gao, Y.X.; Jia, S.S.; Zhang, J. Analysis of the dietary factors on sarcopenia in elderly in Beijing. *J Hygiene Res.* **2016**, *45*, 388-393 (Chinese article).

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

[24] Chen, Z.; Dai, R.; Gu, X.H. Prevalence of sarcopenia and quality of life in retired cadres of a military unit. *Chin. J. Osteopo. Rosis. & Bone Miner. Res.* **2016**, *9*, 264-268 (Chinese article).

[25] Bai, H.; Sun, J.; Chen, M.; Xu, D.; Xie, H.; Yu, Z.; Bao, Z.; Chen, J.; Pan, Y.; Lu, D.; Cheng, S. Age-related decline in skeletal muscle mass and function among elderly men and women in Shanghai, China: A cross sectional study. *Asia Pac. J. Clin. Nutr.* **2016**, *25*, 326-332..

[26] Hai, S.; Wang, H.; Cao, L.; Liu, P.; Zhou, J.; Yang, Y.; Dong, B. Association between sarcopenia with lifestyle and family function among community-dwelling Chinese aged 60 years and older. *BMC Geriatr.* **2017**, *17*.

[27] Ma, Y.; Bian, A.L.; Wang, X.F.; Wang, J.X.; Zhou, X.Z. Difference of the prevalence rate of sarcopenia in elderly people with different gender and age in Tianjin. *Chin. General Practice* **2017**, *20*, 3267-3271 (Chinese article).

[28] Wu, Z.N.; Wang, Y.B.; Shu, F.Q.; Song, B.; Li, G.; Zhu, H.M. Epidemiological investigation of sarcopenia in the nursing homes in Fengxian District and analysis of their risk factors. *Shanghai Med.* **2017**, *38*, 37-40 (Chinese article).

[29] Huang, L.J.; Liu, Y.C.; Wen, Y.F.; Liu, K. State and associated factors of sarcopenia in Jinzhou community elderly people. *Modern Preve. Med.* **2017**, *44*, 4458-4460 (Chinese article).

[30] Yang, M.; Hu, X.; Xie, L.; Zhang, L.; Zhou, J.; Lin, J.; Wang, Y.; Li, Y.; Han, Z.; Zhang, D.; Zuo, Y.; Li, Y.; Wu, L. Screening Sarcopenia in Community-Dwelling Older Adults: SARC-F vs SARC-F Combined With Calf Circumference (SARC-CalF). *J. Am. Med. Dir. Assoc.* **2018**, *19*, 271-277.

[31] Zeng, Y.; Hu, X.; Xie, L.; Han, Z.; Zuo, Y.; Yang, M. The Prevalence of Sarcopenia in Chinese Elderly Nursing Home Residents: A Comparison of 4 Diagnostic Criteria. *J. Am. Med. Dir. Assoc.* **2018**, *19*, 690-695.

[32] Wang, Y.N.; Xu, J.L.; Song, H.L. Influencing factors of sarcopenia among elderly patients. *Modern Clinical Nursing* **2018**, *17*, 45-50 (Chinese article).

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

[33] Yang, L.; Wu, G.; Yang, Y.; Wu, Y.; Zhang, L.; Wang, M.; Mo, L.; Xue, G.; Wang, C.; Weng, X. Nutrition, Physical Exercise, and the Prevalence of Sarcopenia in Elderly Residents in Nursing Homes in China. *Med. Sci. Monitor*. **2019**, *25*, 4390-4399.

[34] Du, Y.; Wang, X.; Xie, H.; Zheng, S.; Wu, X.; Zhu, X.; Zhang, X.; Xue, S.; Li, H.; Hong, W.; Tang, W.; Chen, M.; Cheng, Q.; Sun, J. Sex differences in the prevalence and adverse outcomes of sarcopenia and sarcopenic obesity in community dwelling elderly in East China using the AWGS criteria. *BMC Endocr. Disord.* **2019**, *19*.

[35] Wang, H.; Hai, S.; Liu, Y.; Liu, Y.X.; Zhou, J.H.; Yang, Y.; Dong, B.R.; Yue, J.R. Prevalence of sarcopenia and associated factors in community-dwelling elderly populations in Chengdu China. *J. Sichuan Univ.* **2019**, *50*, 224-228 (Chinese article).

[36] Zhang, Y.; Chen, X.; Hou, L.; Lin, X.; Qin, D.; Wang, H.; Hai, S.; Cao, L.; Dong, B. Prevalence and Risk Factors Governing the Loss of Muscle Function in Elderly Sarcopenia Patients: A longitudinal Study in China with 4 Years of Follow-Up. *The journal of nutrition, health & aging*. **2020**, *24*, 518-524.

[37] Liu, X.; Hao, Q.; Yue, J.; Hou, L.; Xia, X.; Zhao, W.; Zhang, Y.; Ge, M.; Ge, N.; Dong, B. Sarcopenia, Obesity and Sarcopenia Obesity in Comparison: Prevalence, Metabolic Profile, and Key Differences: Results from WCHAT Study. *The journal of nutrition, health & aging*. **2020**, *24*, 429-437.

[38] Shafiee, G.; Keshtkar, A.; Soltani, A.; Ahadi, Z.; Larijani, B.; Heshmat, R. Prevalence of sarcopenia in the world: a systematic review and meta-analysis of general population studies. *Journal of Diabetes & Metabolic Disorders*. **2017**, *16*.

[39] Makizako, H.; Nakai, Y.; Tomioka, K.; Taniguchi, Y. Prevalence of sarcopenia defined using the Asia Working Group for Sarcopenia criteria in Japanese community-dwelling older adults: A systematic review and meta-analysis. *Physical therapy research*. **2019**, *22*, 53-57.

[40] Diz, J.B.M.; Leopoldino, A.A.O.; Moreira, B.D.S.; Henschke, N.; Dias, R.C.; Pereira, L.S.M.; Oliveira, V.C. Prevalence of sarcopenia in older Brazilians: A systematic review and meta-analysis. *Geriatr. Gerontol. Int.* **2017**, *17*, 5-16.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

[41] Papadopoulou, S.K.; Tsintavis, P.; Potsaki, G.; Papandreou, D. Differences in the Prevalence of Sarcopenia in Community-Dwelling, Nursing Home and Hospitalized Individuals. A Systematic Review and Meta-Analysis. *The journal of nutrition, health & aging*. **2020**, *24*, 83-90.

[42] Han, A.; Sun, T.; Ming, J.; Chai, L.; Liao, X. Are the Chinese moving toward a healthy diet? Evidence from macro data from 1961 to 2017. *Int J Environ Res Public Health,* **2020**, *17*, 5294

Legend

Figure 1. Flowchart of the detailed procedure for the inclusion or exclusion of selected studies.

Figure 2. Forest plots for the total prevalence of sarcopenia among the elderly Chinese population according to the Asian Working Group for Sarcopenia (AWGS) standard. Diamond represents the pooled odds ratio at 95% CI.

Figure 3. Forest plots for the total prevalence of sarcopenia among the elderly Chinese population according to the International Working Group on Sarcopenia (IWGS) standard. Diamond represents the pooled odds ratio at 95% CI.

Figure 4. Forest plots for the total prevalence of sarcopenia among the elderly Chinese population according to the European Working Group on Sarcopenia in Older People (EWGSOP) standard. Diamond represents the pooled odds ratio at 95% CI.

Figure 5. Forest plots for the prevalence of sarcopenia among male and female elderly people according to the Asian Working Group for Sarcopenia (AWGS) standard. Diamond represents the pooled odds ratio at 95% CI.

Figure 6. Forest plots for the prevalence of sarcopenia among male and female elderly people according to the International Working Group on Sarcopenia (IWGS) standard. Diamond represents the pooled odds ratio at 95% CI.

Figure 7. Forest plots for the prevalence of sarcopenia among male and female elderly people according to the European Working Group on Sarcopenia in Older People (EWGSOP) standard. Diamond represents the pooled odds ratio at 95% CI.

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Table 1 Study characteristics of the studies included in the meta-analysis

Author	Publication Year	Participants(n)			Region	Setting	Measurement	Diagnostic Criteria	Prevalence (%)			NOS
		Total	Male	Female					Total	Male	Female	
Lee WJ	2013	386	223	163	Taiwan	Urban Community	DEXA	IWGS	11.1	10.8	11.7	6
								EWGSOP	16.6	14.9	19.0	
Wu IC	2014	2155	1055	1100	Taiwan	Urban Community	BIA	EWGSOP	3.9	5.4	2.5	6
Wu CH	2014	549	285	264	Taiwan	Rural Community	BIA	EWGSOP	12.7	8.1	17.8	6
Li M	2014	169	169	0	Beijing	Urban Community	DEXA	EWGSOP	33.3	33.3	-	5
Meng NH	2015	771	359	712	Taiwan	Urban Community	DEXA	EWGSOP	9.7	0.7	12.5	6
Gao LL	2015	612	254	358	Sichuan	Rural and urban Community	Calf circumference	AWGS	9.8	6.7	12.0	6
Wen X	2015	286	136	150	Hangzhou	Urban Community	DEXA	IWGS	6.0	7.4	4.7	5
								EWGSOP	0.4	0.8	0.01	
								AWGS	3.1	5.9	0.7	

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Woo J	2015	4000	2000	2000	Hongkong	Urban Community	DEXA	IWGS	20.2	22.1	18.3	6
								EWGSOP	9.0	9.5	8.6	
								AWGS	7.3	5.3	9.3	
Zeng P	2015	395	199	196	Beijing	Rural and urban Community	BIA	AWGS	24.3	18.1	30.6	5
Wang YJ	2015	316	164	152	Changsha	Urban Community	DEXA	AWGS	29.7	26.2	33.6	6
Han PP	2016	1069	467	602	Tianjin	Urban Community	BIA	AWGS	9.3	6.4	17.5	5
Wang H	2016	636	316	320	Chengdu	Urban Community	BIA	AWGS	10.4	8.2	12.5	5
Bai HJ	2016	415	212	203	Shanghai	Urban Community	BIA	AWGS	28.7	33.5	23.6	6
Xia ZW	2016	683	239	444	Beijing	Urban Community	BIA	AWGS	20.1	17.2	21.6	5
Chen Z	2016	281	203	78	Chongqing	Urban Community	BIA	IWGS	27.8	-	-	6
Han DS	2016	878	402	476	Taiwan	Urban Community	BIA	EWGSOP	9.8	4.5	14.3	6
Hai S	2017	834	415	419	Chengdu	Urban Community	BIA	AWGS	10.6	11.3	9.8	6
Huang	2017	193	65	128	Jinzhou	Urban Community	BIA	AWGS	11.4	7.7	13.3	6

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LJ												
Yang M	2018	384	160	224	Chengdu	Urban Community	BIA	AWGS	15.9	11.9	18.8	6
								IWGS	25.0	24.4	25.4	
								EWGSOP	11.7	10.6	12.5	
Du YP	2019	631	213	418	Shanghai	Urban Community	BIA	AWGS	12.2	19.2	8.6	5
Wang H	2019	947	465	482	Chengdu	Urban Community	BIA	AWGS	7.7	6.5	8.9	5
Zhang Y	2020	474	238	236	Chengdu	Urban Community	BIA	AWGS	15.4	-	-	5
Liu XL	2020	4500	1627	2873	Yunnan, Guizhou, Sichuan, and Xinjiang	Urban Community	BIA	AWGS	19.3	22.1	17.8	6

DEXA, dual-energy x-ray absorptiometry; BIA, bioelectrical impedance analysis; IWGS, International Working Group on Sarcopenia; AWGS, Asia Working Group for Sarcopenia; EWGSOP, European Working Group on Sarcopenia in Older People; NOS, Newcastle–Ottawa scale.

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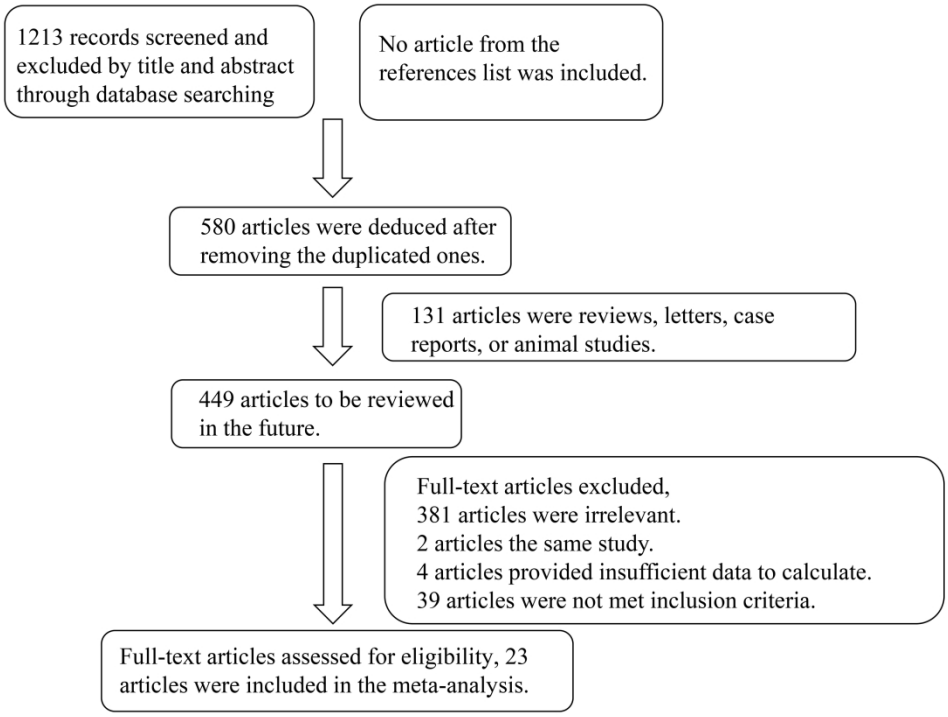
Table 2 The prevalence of sarcopenia diagnosed by different criteria using DEXA and BIA measurement (%).

	DEXA	BIA
AWGS	13(5−21)	15(12−19)
IWGS	12(3−22)	26(23−29)
EWGSOP	12(6−18)	9(5−14)

DEXA, dual-energy x-ray absorptiometry; BIA, bioelectrical impedance analysis; IWGS, International Working Group on Sarcopenia; AWGS, Asia Working Group for Sarcopenia; EWGSOP, European Working Group on Sarcopenia in Older People.

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Flowchart of the detailed procedure for the inclusion or exclusion of selected studies.

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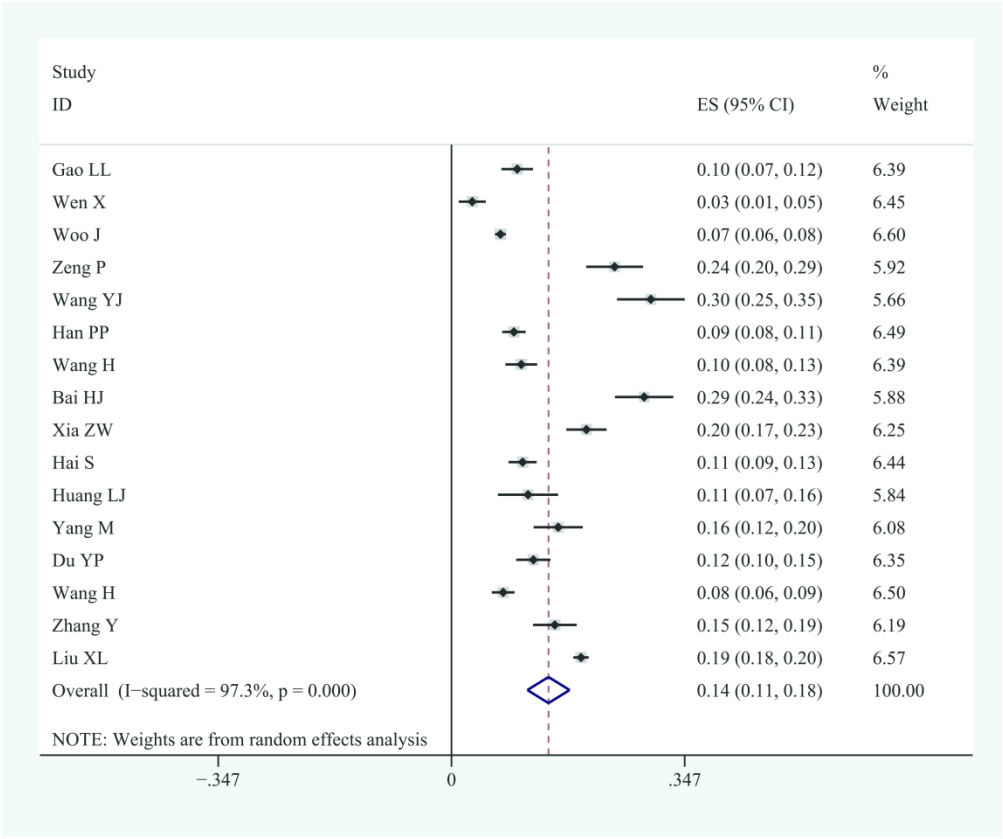


Figure 2. Forest plots for the total prevalence of sarcopenia among the elderly Chinese population according to the Asian Working Group for Sarcopenia (AWGS) standard. Diamond represents the pooled odds ratio at 95% CI.

280x233mm (600 x 600 DPI)

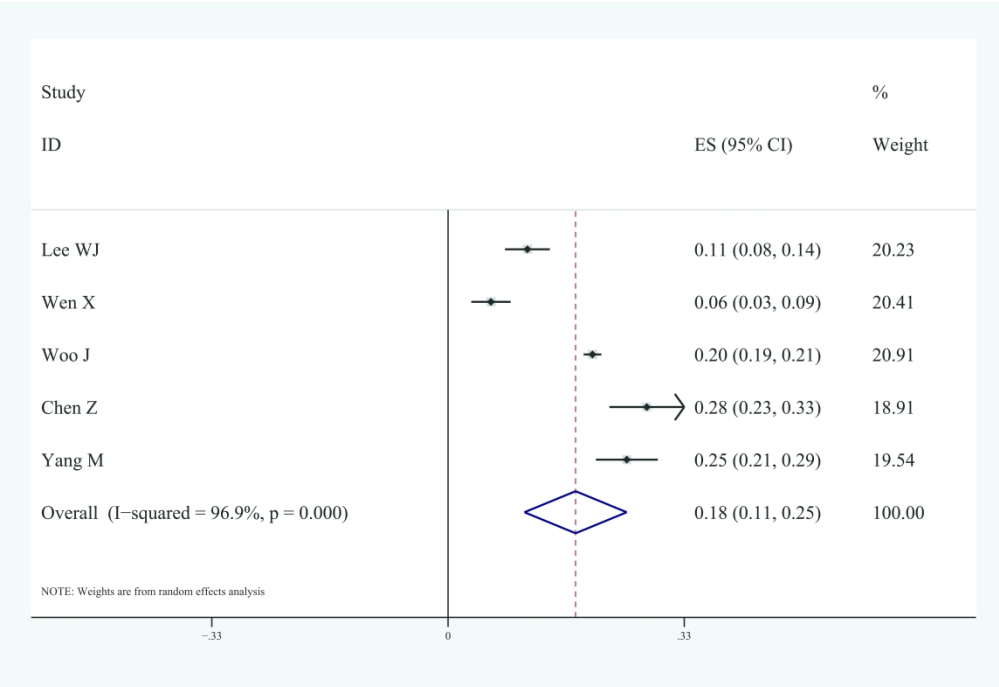


Figure 3. Forest plots for the total prevalence of sarcopenia among the elderly Chinese population according to the International Working Group on Sarcopenia (IWGS) standard. Diamond represents the pooled odds ratio at 95% CI.

272x186mm (600 x 600 DPI)

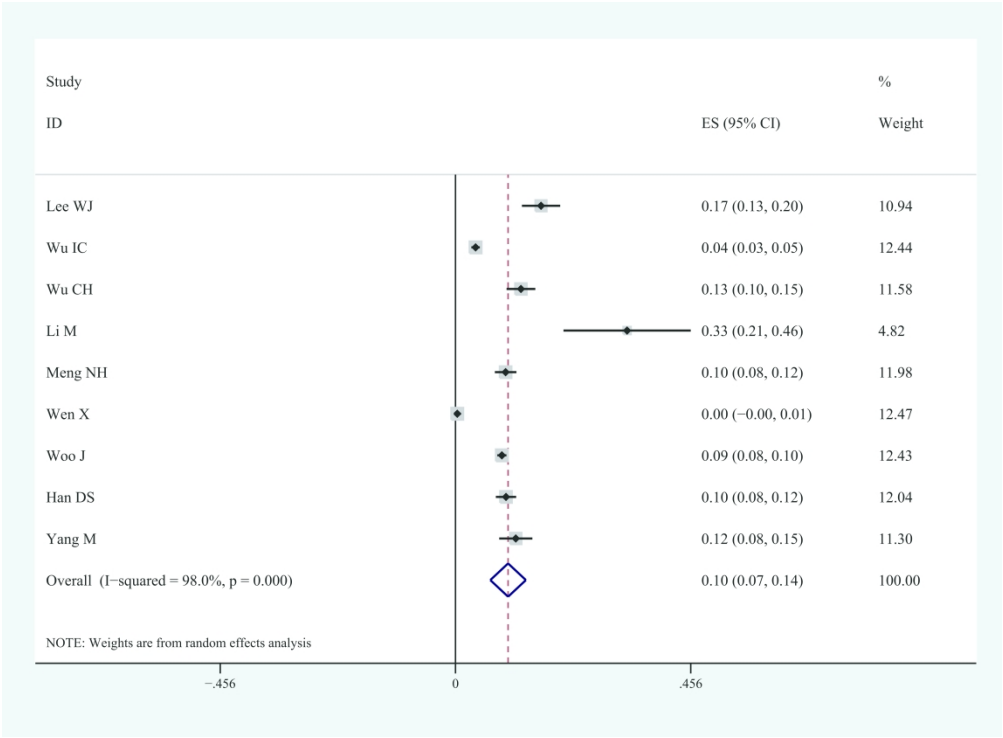


Figure 4. Forest plots for the total prevalence of sarcopenia among the elderly Chinese population according to the European Working Group on Sarcopenia in Older People (EWGSOP) standard. Diamond represents the pooled odds ratio at 95% CI.

274x201mm (600 x 600 DPI)

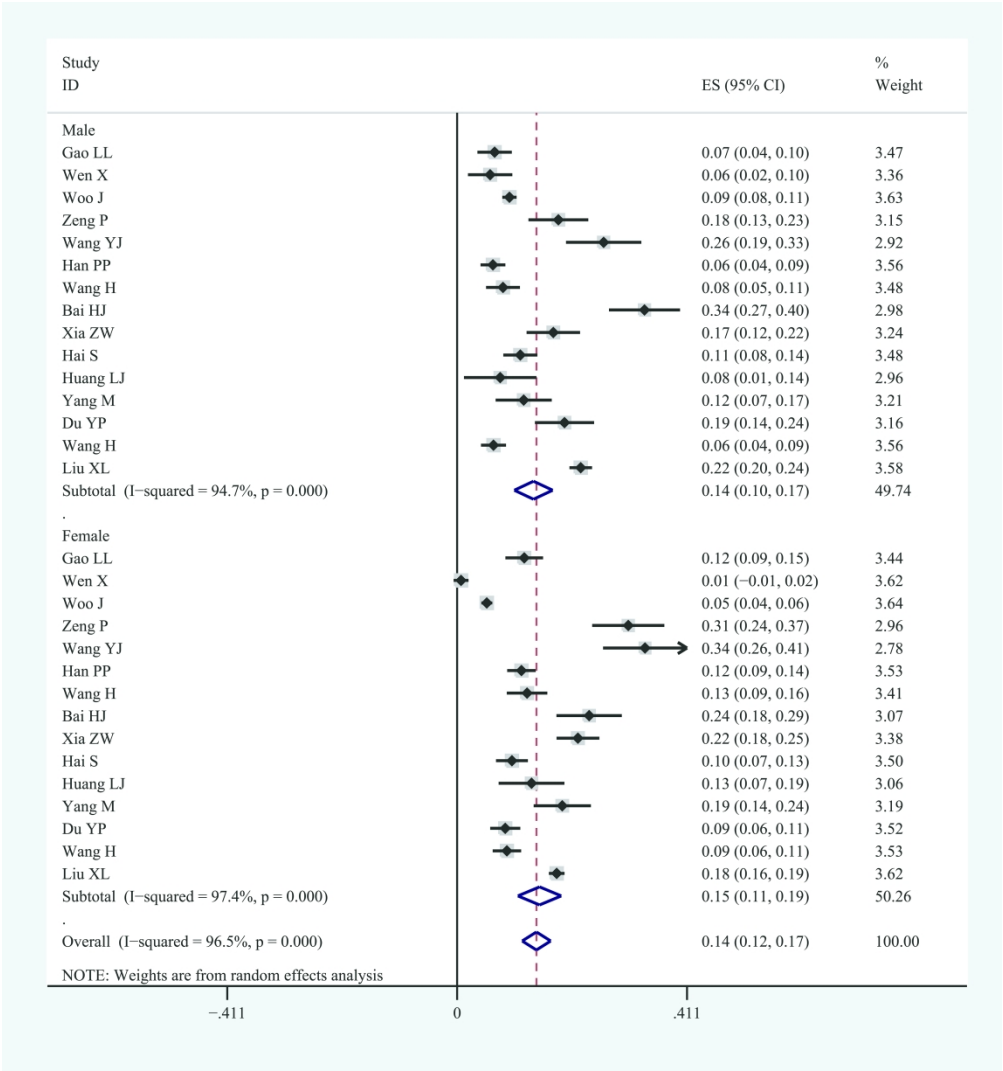


Figure 5. Forest plots for the prevalence of sarcopenia among male and female elderly people according to the Asian Working Group for Sarcopenia (AWGS) standard. Diamond represents the pooled odds ratio at 95% CI.

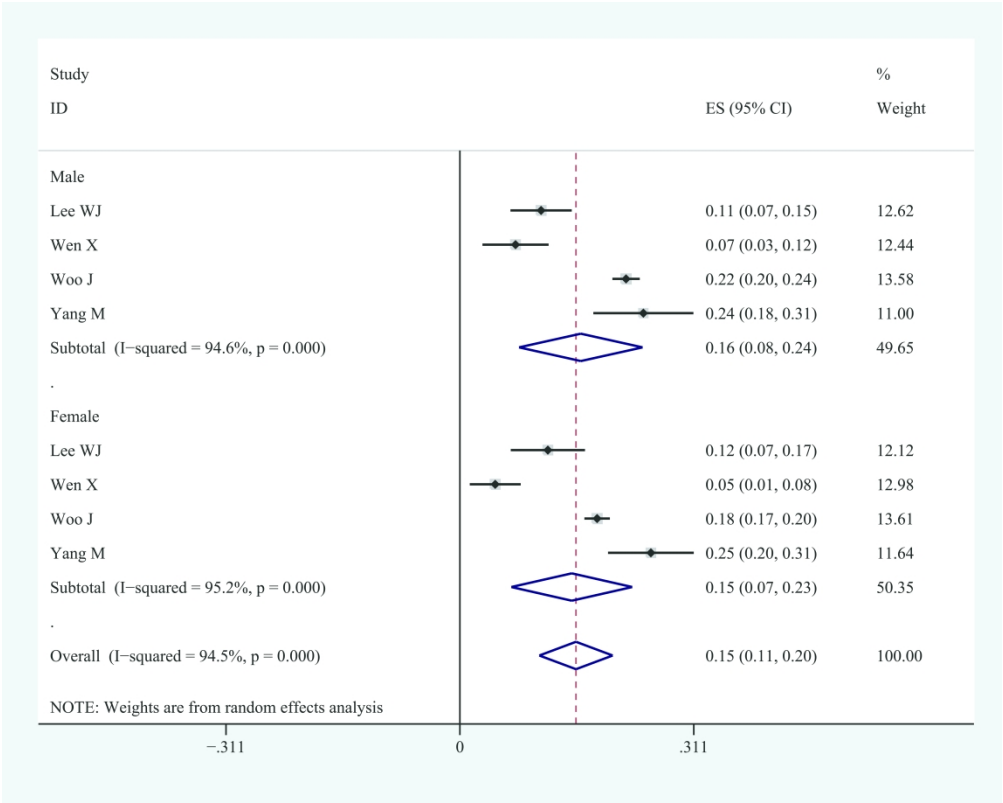


Figure 6. Forest plots for the prevalence of sarcopenia among male and female elderly people according to the International Working Group on Sarcopenia (IWGS) standard. Diamond represents the pooled odds ratio at 95% CI.

278x223mm (600 x 600 DPI)

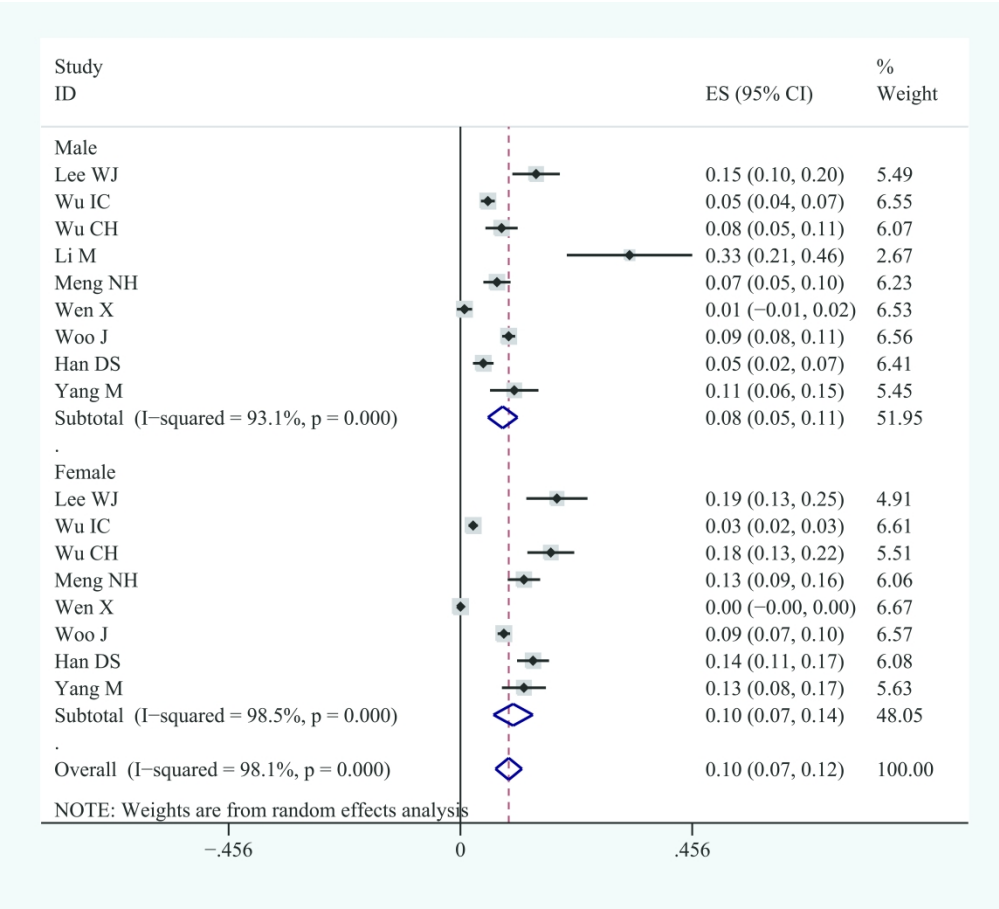


Figure 7. Forest plots for the prevalence of sarcopenia among male and female elderly people according to the European Working Group on Sarcopenia in Older People (EWGSOP) standard. Diamond represents the pooled odds ratio at 95% CI.

284x257mm (600 x 600 DPI)

File S1 Search strategy in databases

PubMed: (((sarcopenia[Title/Abstract]) OR muscle mass [Title/Abstract]) OR muscle strength[Title/Abstract]) AND ((prevalence[Title/Abstract]) OR epidemi*[Title/Abstract]) AND ((Chinese[Title/Abstract]) OR China[Title/Abstract])

Web of Science: (((TS=sarcopenia) OR TS=muscle mass) OR TS=muscle strength) AND ((TS=prevalence) OR TS=epidemi*) AND ((TS=Chinese) OR TS=China)

CNKI: (AB =sarcopenia OR AB =muscle mass OR AB =muscle strength) AND (AB =prevalence OR AB =epidemiology)

CQVIP: (R=sarcopenia +R=muscle mass+ R=muscle strength) *(R=prevalence +R=epidemiology)

Wangfang: Abstract: (sarcopenia + muscle mass + muscle strength) * (prevalence + epidemiology)

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Figure S1. The sensitivity analysis results according to the Asian Working Group for Sarcopenia (AWGS) standard.

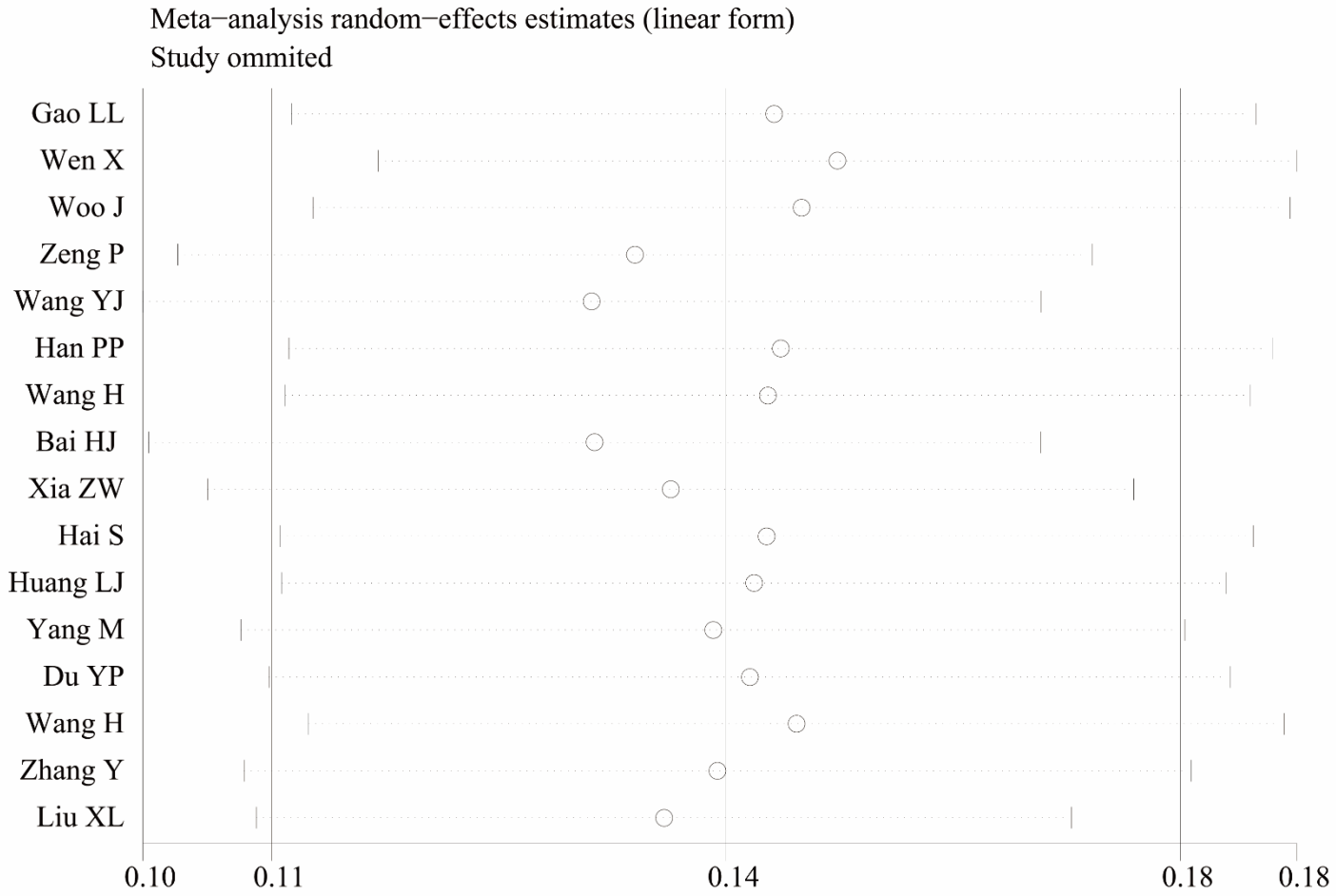


Figure S2. The sensitivity analysis results according to the International Working Group on Sarcopenia (IWGS) IWGS standard.

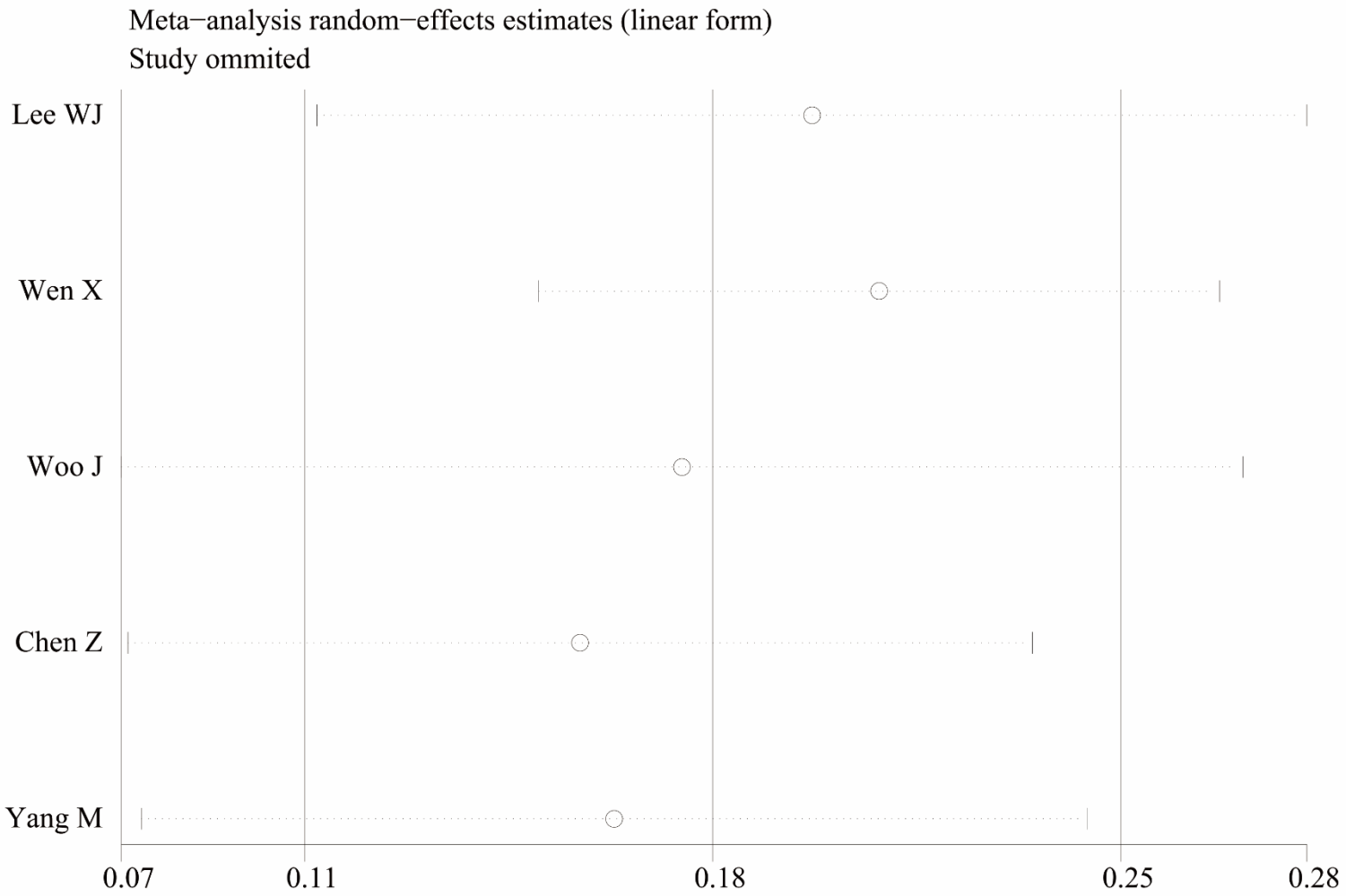


Figure S3. The sensitivity analysis results according to the European Working Group on Sarcopenia in Older People (EWGSOP) standard.

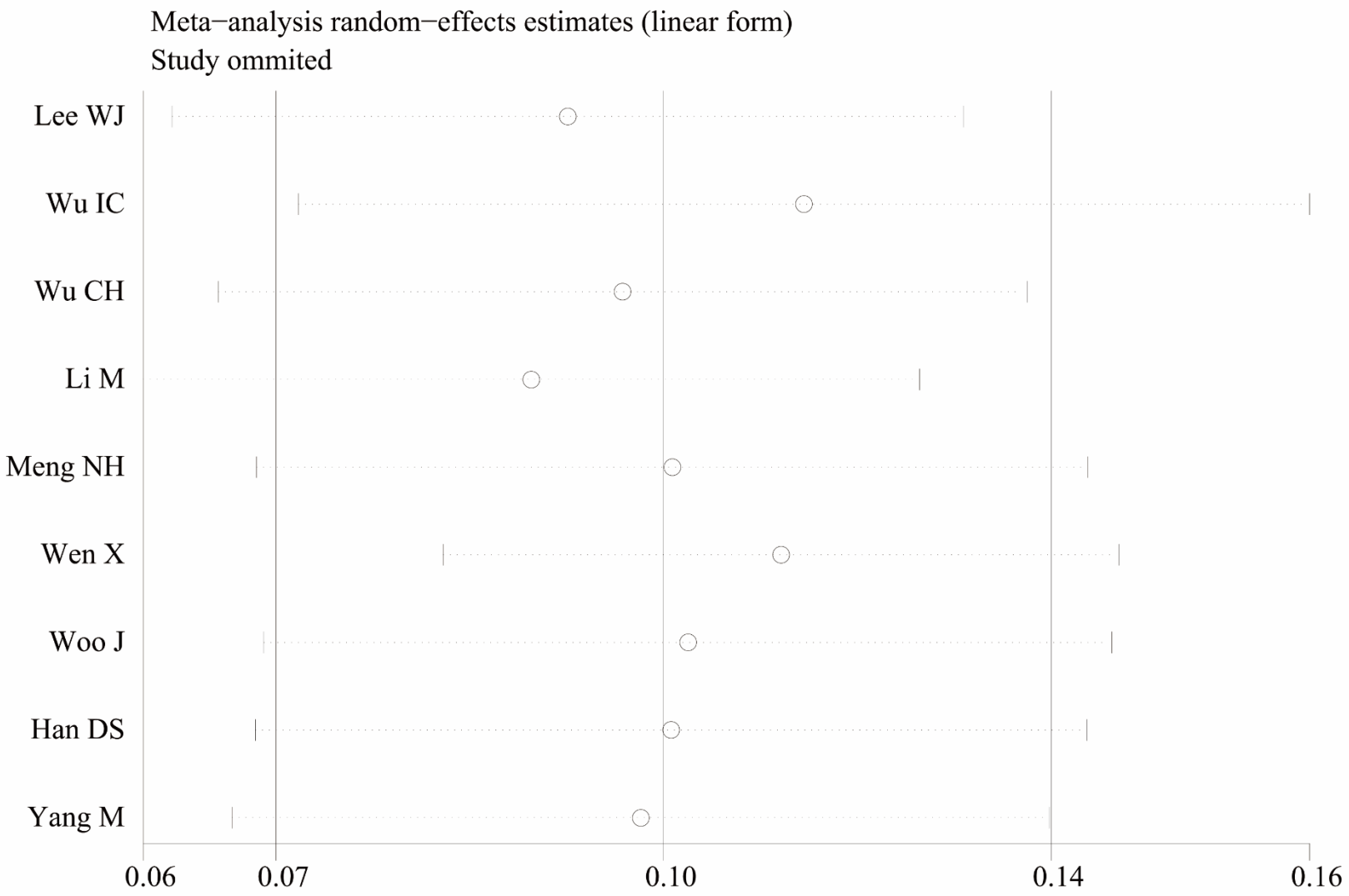


Figure S4. The publication bias results according to the Asian Working Group for Sarcopenia (AWGS) standard.

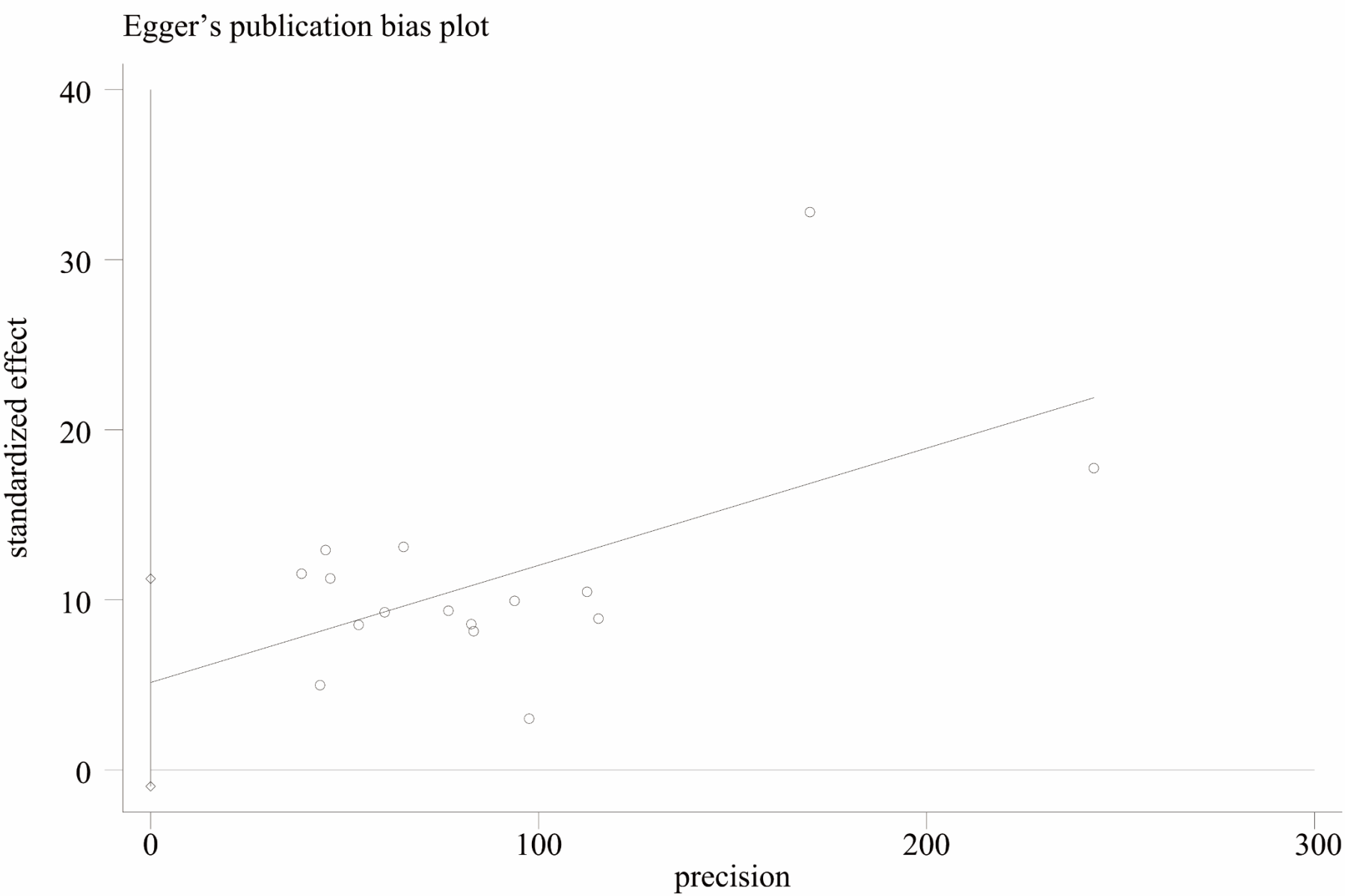


Figure S5. The publication bias results according to the the International Working Group on Sarcopenia (IWGS) standard.

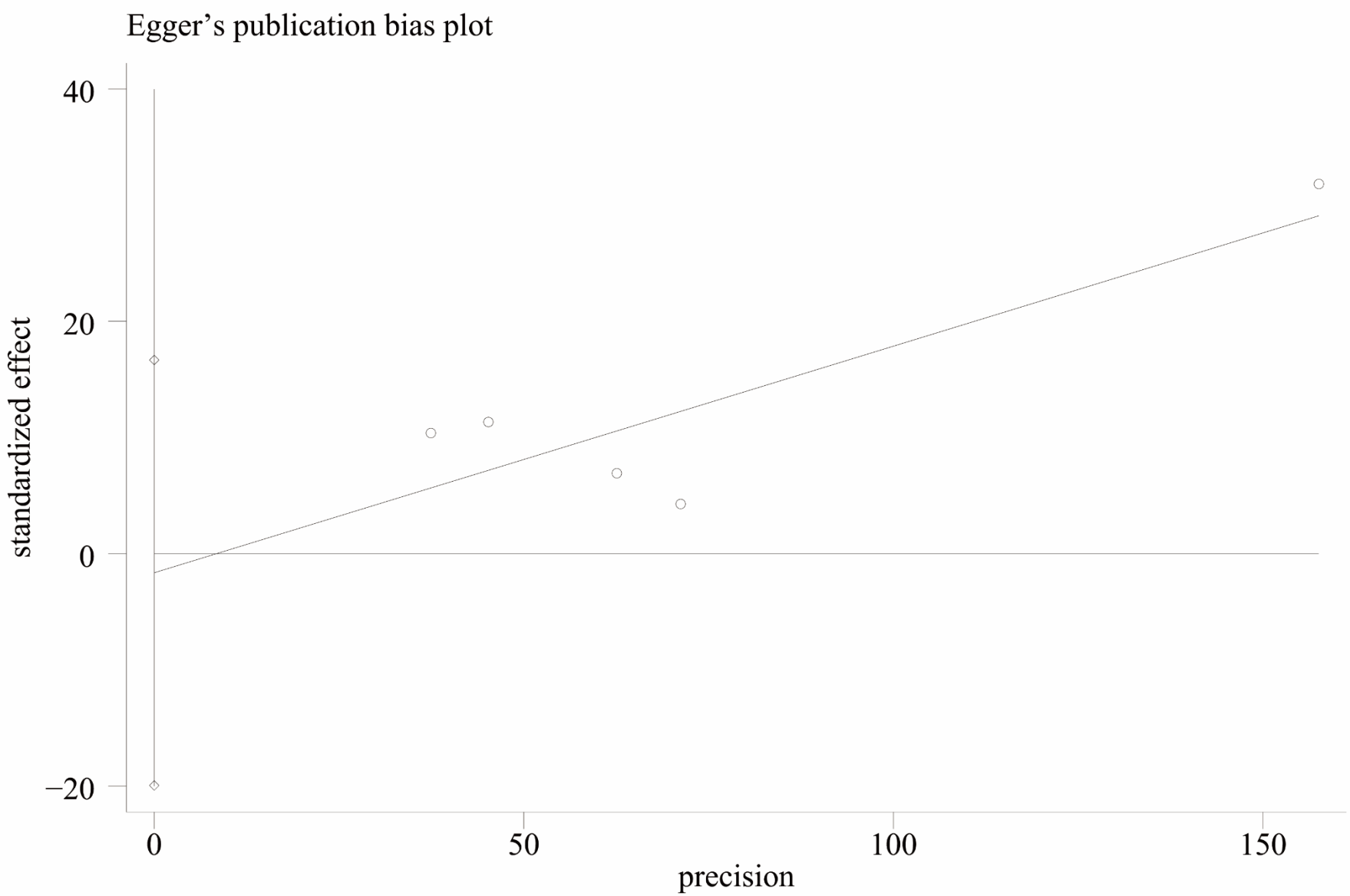
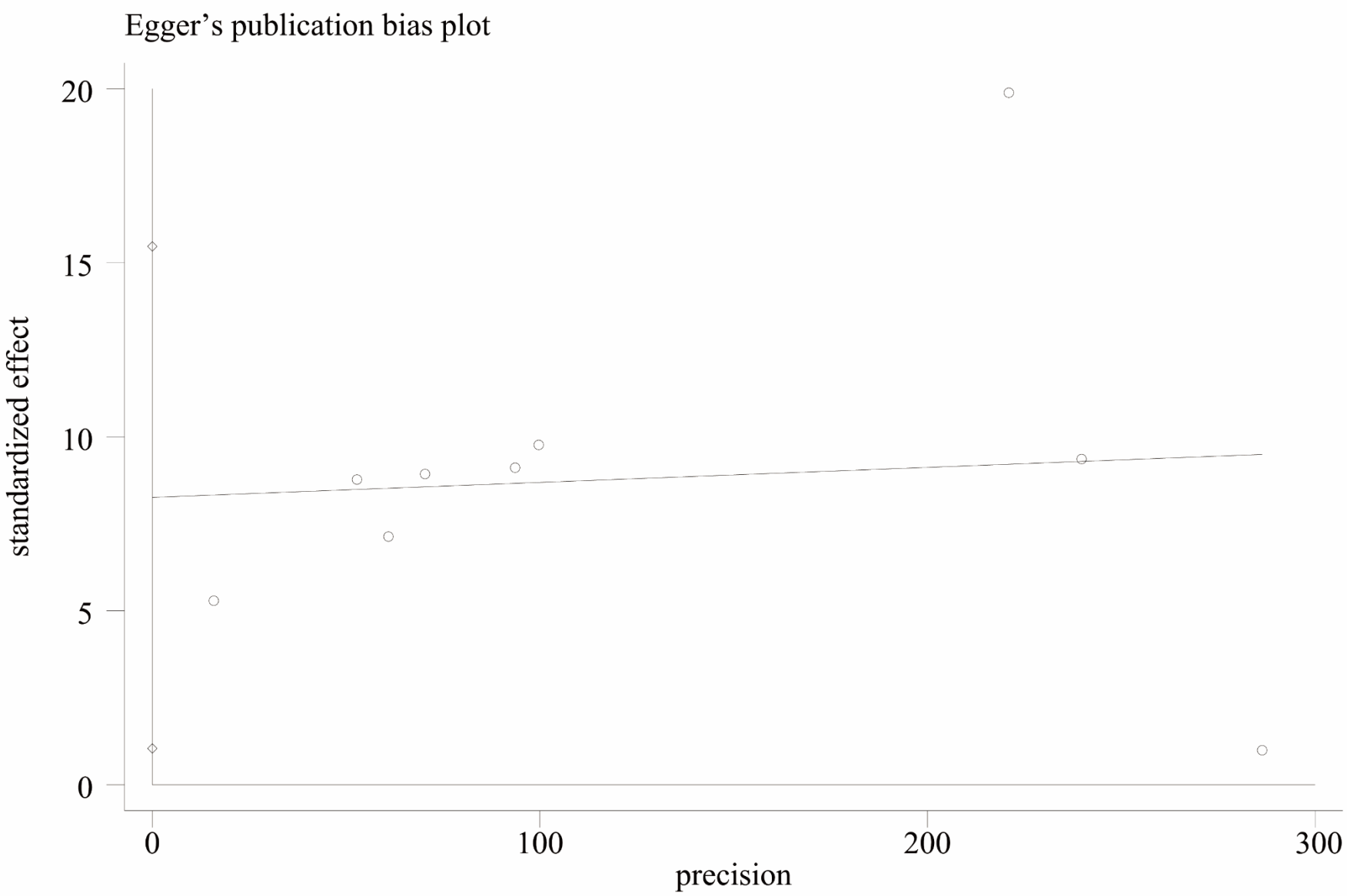


Figure S6. The publication bias results according to the European Working Group on Sarcopenia in Older People (EWGSOP) standard.



PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	3
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	3,4
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	4
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	4
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	4
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	4
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	4
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	5
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	5
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	5
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	5
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	5

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	5
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	5
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	6
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	6
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	6
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	6
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	6,7
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	7
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	7
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	7,8
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	8
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	8
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	9

From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(7): e1000097. doi:10.1371/journal.pmed1000097

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BMJ Open

Prevalence of Sarcopenia in Older Chinese Adults: A Systematic Review and Meta-analysis

Journal:	*BMJ Open*
Manuscript ID	bmjopen-2020-041879.R3
Article Type:	Original research
Date Submitted by the Author:	10-May-2021
Complete List of Authors:	Xin, Caihong; Fourth People's Hospital in Shenyang, Department of Endocrinology and Metabolism Sun, Xin; First Affiliated Hospital of Soochow University, Department of Endocrinology and Metabolism; People's Hospital of Liaoning Province, Department of Endocrinology and Metabolism Lu, Li; People's Hospital of Liaoning Province, Department of Endocrinology and Metabolism Shan, Ling; People's Hospital of Liaoning Province, Department of Endocrinology and Metabolism
Primary Subject Heading:	Geriatric medicine
Secondary Subject Heading:	Epidemiology, Global health, Public health
Keywords:	GERIATRIC MEDICINE, EPIDEMIOLOGY, General endocrinology < DIABETES & ENDOCRINOLOGY

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Prevalence of Sarcopenia in Older Chinese Adults: A Systematic Review and Meta-analysis

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Running title: Prevalence of sarcopenia in

Key words: Prevalence; sarcopenia; China

Word count of the manuscript: 1864

Figures number: 7

Tables number: 2

Disclosure Statement: The authors have nothing to disclose.

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Abstract

Objective: This meta-analysis was conducted to estimate the overall prevalence of sarcopenia in older Chinese adults.

Design: Systematic review and meta-analysis

Participants: A literature research was conducted using the PubMed, Web of Science, CNKI, CQVIP, and Wanfang databases. The following search terms in the abstract were used: "sarcopenia" in combination with the terms "prevalence," "epidemiology," and "China." All studies published from January 2010 to November 2020 were included. The random effect model was used to estimate the prevalence of sarcopenia. The sex-specific prevalence of sarcopenia at a 95% confidence interval (CI) was also calculated using different criteria for defining sarcopenia.

Primary outcome measures: The overall prevalence of sarcopenia in older Chinese adults.

Results: In total, 23 articles were included in this meta-analysis involving 21,564 participants. On the basis of the Asian Working Group for Sarcopenia (AWGS) criterion, the overall prevalence of sarcopenia among the elderly in China was 14% (95% CI, 11%–18%), the prevalence was higher in Chinese females than in males (15% vs. 14%).

Conclusions: This systematic review is the first estimation of the pooled prevalence of sarcopenia in older Chinese adults. Our results suggest that China has a large number of sarcopenia patients. These findings would be useful for sarcopenia prevention in China.

Trial registration: PROSPERO CRD42020223405

Strengths and limitations of this study:

- To our knowledge, this systematic review is the first estimation of the pooled prevalence of sarcopenia in Chinese older people.
- The search languages were limited to English and Chinese.
- We included limited number of high-quality studies.
- A high level of heterogeneity among included studies was observed.

Introduction

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Sarcopenia is characterized by the deterioration of muscle mass and function that occurs with aging. Several studies have shown that as age increases, skeletal muscle mass and strength are lost at a rate of 1%–2% annually, possibly leading to the decline of physical function and increasing the disability and mortality rates of elderly people as well as bringing serious economic and social burdens [1–4].

A practical clinical definition was reported by the European Working Group on Sarcopenia in Older People (EWGSOP) in 2010 [5]. Based on this EWGSOP criteria, slow walking speed (0.8 m/sec) or low grip strength (<30 kg for men and <20 kg for women) are indications of muscle mass decline. Later in 2011, another consensus was published by the International Working Group on Sarcopenia (IWGS) [6]. However, Asia is a vast region comprising different races, countries, and cultures, and developing screening and diagnostic strategies for the Asian population is imperative. The Asian Working Group for Sarcopenia (AWGS) was established in 2013, and the consensus of AWGS was released based on 16 epidemiologic reports on sarcopenia in six countries or regions, leading to the development of a diagnosis strategy for sarcopenia in Asia [7]. The diagnostic criteria recommended by AWGS and EWGSOP are similar, and both include muscle mass, muscle strength, and activity measurements. The only difference lies in the specific cut-off value. The cut-off values for muscle strength (<26 kg for men and <18 kg for women), walking speed (0.8 m/sec), and muscle mass (7.0 kg/m^2 for men and 5.4 kg/m^2 for women measured using dual X-ray absorptiometry and 7.0 kg/m^2 for men and 5.7 kg/m^2 for women measured using bioimpedance analysis) have been revised according to data from regional cohort studies among Asian populations.

According to statistics, the elderly population (≥60 years old) includes 16.1% of China's total population, which is close to 250 million. In the face of a severe aging situation, people are paying more attention to sarcopenia, but its prevalence is inconsistent across different studies [8–12]. At present, there is no large-scale epidemiologic survey of sarcopenia in China. The purpose of this study is to systematically determine the prevalence of sarcopenia in older Chinese adults.

Methods

Search strategy

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We searched the PubMed, Web of Science, China National Knowledge Infrastructure (CNKI), CQVIP, and Wanfang electronic databases. The search terms in the title or abstract included the following: "sarcopenia," "muscle mass," and "muscle strength" in combination with the terms "prevalence," "epidemiology," and "China." The complete search strategy are below: PubMed: (((sarcopenia [Title/Abstract]) OR muscle mass[Title/Abstract]) OR muscle strength[Title/Abstract]) AND ((prevalence[Title/Abstract]) OR epidemi*[Title/Abstract]) AND ((Chinese[Title/Abstract]) OR China[Title/Abstract]); Web of Science: (((TS=sarcopenia) OR TS=muscle mass) OR TS=muscle strength) AND ((TS=prevalence) OR TS=epidemi*) AND ((TS=Chinese) OR TS=China); CNKI: (AB =sarcopenia OR AB =muscle mass OR AB =muscle strength) AND (AB =prevalence OR AB =epidemiology); CQVIP: (R=sarcopenia +R=muscle mass+ R=muscle strength) *(R=prevalence +R=epidemiology); Wangfang: Abstract: (sarcopenia + muscle mass + muscle strength) * (prevalence + epidemiology). All studies published from January 2010 to November 2020 were included. The references of the retrieved articles were examined to identify additional eligible studies, and unpublished studies were excluded. The search languages were limited to English and Chinese. This systematic review and meta-analysis was registered in PROSPERO (registration number: CRD42020223405). This meta-analysis was conducted to estimate the overall prevalence of sarcopenia in older Chinese adults.

Patient and public involvement

No patients were involved in this study.

Inclusion and exclusion criteria

Inclusion of the selected studies was based on the following criteria: (1) a cross-sectional study or retrospective study, (2) a sample size of ≥100 elderly Chinese participants (age ≥60 years); (3) sufficient information on sarcopenia at the corresponding 95% confidence interval (CI) and *P* value, and (4) the definition of sarcopenia in line with the AWGS, EWGSOP, or IWGS standards.

Exclusion of the selected studies was based on the following criteria: (1) same data from different studies, (2) studies defining sarcopenia only as reduction of muscle mass or strength, and (3) the subjects include a specific population, such as patients living in a nursing home or those with disability. Reviews, case reports, letters, and animal researches were also excluded.

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Data extraction and methodological quality assessment

Information from the selected studies was extracted by two independent researchers, including the first author's name, publication year, sample, diagnostic criteria, participants, setting, and prevalence. Two independent reviewers who were not blinded to the authors or journals assessed the risk of bias in the included studies using the Newcastle–Ottawa scale (NOS) [13]. The NOS is recommended by the Cochrane Handbook for Systematic Reviews of Interventions [14]. Each included study was analyzed in terms of three domains, namely, representativeness of the study group selection, group comparability, and exposure or outcome ascertainment, using the "star system." NOS scores ranged from zero to nine stars. The two researchers independently assessed the studies by discussion, comparison of findings, and resolution of differences through consensus formation. If no consensus was reached, a third researcher was commissioned to resolve the difference.

Statistical analysis

We used a systematic analysis approach to calculate the pooled prevalence of sarcopenia for all eligible studies, and heterogeneity among these studies was assessed using the Cochran's Q test and I^2 statistical index. The random effect model was used to combine the prevalence of sarcopenia. We additionally performed sensitivity analysis to evaluate the influence of any given study on the pooled estimate. Publication bias was evaluated using Egger's test. Subgroup analyses were also performed. The significance level was set at $P < 0.05$. All statistical analyses were performed using Stata software version 12.0 (College Station, Texas, USA).

Results

Study characteristics

In total, 1213 studies were retrieved from the databases and 580 articles were remained after the exclusion of duplicates. After screening for eligibility based on the full-text, a total of 23 articles were included as per the criteria shown in Figure 1 [8–12, 15–37]. Altogether, the 23 articles included 21,564 participants in China between 2010 and 2020. The selected studies' characteristics are summarized in Table 1. On the basis of the NOS scale, the score of selected studies were all above five stars; therefore, all articles were included.

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Results of the meta-analysis

According to the AWGS standard, the total prevalence of sarcopenia among Chinese elderly people was 14% (95% CI, 11%–18%, Figure 2), and total prevalence rates of sarcopenia among the elderly in China based on the IWGS and EWGSOP standards were 18% (95% CI, 11%–25%, Figure 3) and 10% (95% CI, 7%–14%, Figure 4), respectively.

Subgroup analyses were performed to determine the influence of sex on the prevalence of sarcopenia. Sarcopenia prevalence among elderly females was higher than that in males as per the AWGS and EWGSOP standards (Figures 5–7). Table 2 shows the subgroup analyses results based on the different diagnostic criteria. The prevalence of sarcopenia estimated using the bio-electrical impedance analysis (BIA) method was higher than that estimated using the dual energy X-ray absorptiometry (DEXA) method as per the AWGS and IWGS standards.

Sensitivity analysis and publication bias

Sensitivity analysis was performed to examine the influence of each study. We found no significant difference between the results of the sensitivity analysis and our previous estimates, indicating that our statistical results were relatively credible (Figures S1–S3). The articles obtained from the databases were carefully and comprehensively searched. Additionally, Egger's test was conducted to determine whether potential publication bias existed in the reviewed literature. The results ($P > 0.05$) suggested that there were no publication bias (Figures S4–S6).

Discussion

To the best of our knowledge, the present study is the first meta-analysis of the prevalence of sarcopenia in older Chinese adults. Sarcopenia is an age-related disease characterized by progressive deterioration in skeletal muscle fiber quality, strength, and function. Currently, sarcopenia has been internationally recognized as a new type of geriatric syndrome and has attracted worldwide attention. In 2017, Shafeee et al. conducted a meta-analysis on the prevalence of sarcopenia. Their results showed that sarcopenia prevalence among the elderly in communities worldwide was approximately 10%; this was higher among non-Asian individuals than Asians in both sexes [38]. A similar study was performed in Japan, where sarcopenia prevalence among the elderly Japanese was 9.9% (95% CI,

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6.2%–15.4%) according to the AWGS standard [39]. The results of our study showed that the prevalence of sarcopenia was 14% (95% CI, 11%–18%) in older Chinese adults, which was much higher than the results of the aforementioned studies [40, 41]. This high prevalence may be related to the Chinese diet structure. Although protein is the raw material for muscle synthesis, Chinese people, particularly the older population, have higher carbohydrate intake and less protein intake [42]. Therefore, more attention should be given to the prevention and control of sarcopenia in older Chinese adults.

Here, we found that the prevalence of sarcopenia among females was higher than that in males, which was consistent with other meta-analyses [38–40]. This is mainly because men eat and exercise more than women; this supports muscle synthesis. Most of the studies included in our meta-analysis used the BIA method because of its affordability and portable equipment. BIA is a widely used non-invasive method for measuring human body composition. The principle behind this method is to record the different electrical resistances of different tissues using electrodes placed on the body surface and then to measure muscle mass using an image reconstruction method. DEXA is also commonly used to evaluate muscle mass. DEXA can accurately distinguish whole-body and local muscles as well as fats and bones. However, the limited number of studies using DEXA included in this meta-analysis may influence our results. As is known, aging increases the prevalence of sarcopenia; however, due to the different age classifications in the included studies, we cannot obtain any conclusion.

The results of this meta-analysis indicated high heterogeneity, which may be related to various factors, such as differences in age, race, and living environment. In addition, differences in the prevalence rates of sarcopenia have been observed in rural and urban areas. China is a country with a vast territory and multi-ethnic people. The findings of this meta-analysis should be interpreted with caution.

The limitation of the present study is that the main data are from urban populations and that studies on the prevalence of sarcopenia in Chinese rural areas are rare; this impacts the calculation of the total sarcopenia prevalence. Second, the prevalence of sarcopenia in people aged 60–70 years and 70–80 years might be different. Not all studies presented their data according to age group, so we cannot accurately estimate the prevalence in terms of age subgroup. Third, most of the studies were performed in large cities or economically developed areas. All of these factors affected our results, and as such, they should be interpreted cautiously as further research is needed.

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Conclusions

To the best of our knowledge, the present systematic review is the first to estimate the pooled prevalence of sarcopenia in older Chinese adults. The results of the present systematic review reveal that China has a large number of sarcopenia cases and sarcopenia is an emerging public health issue in China. These findings would be useful for sarcopenia prevention in China.

Author contributions statement

SL designed the study. SX and LL searched databases and collected the data. XCH and SL assessed the quality of the study. SL performed the analysis. SX and XCH wrote the manuscript. SX modified the manuscript in the revision. All authors contributed to this systematic review and meta-analysis.

Supporting Information

Figure S1. The sensitivity analysis results according to the Asian Working Group for Sarcopenia (AWGS) standard.

Figure S2. The sensitivity analysis results according to the the International Working Group on Sarcopenia (IWGS) standard.

Figure S3. The sensitivity analysis results according to the European Working Group on Sarcopenia in Older People (EWGSOP) standard.

Figure S4. The publication bias results according to the Asian Working Group for Sarcopenia (AWGS) standard.

Figure S5. The publication bias results according to the the International Working Group on Sarcopenia (IWGS) standard.

Figure S6. The publication bias results according to the European Working Group on Sarcopenia in Older People (EWGSOP) standard.

Ethics approval statement

Not applicable. The manuscript does not contain any personal or medical information about an identifiable individual.

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Funds

This research was financially supported by the National Natural Science Foundation of China (grant No. 81800697).

Acknowledgments

We thank Enago for English language editing.

Conflict of interests

None declared.

Data availability statement

All data relevant to the study are included in the article or uploaded as supplementary information

Reference

[1] Cederholm, T.; Morley, J.E. Sarcopenia: the new definitions. *Current Opinion in Clinical Nutrition and Metabolic Care.* **2015**, *18*, 1–4.

[2] Frontera, W.R.; Hughes, V.A.; Fielding, R.A.; Fiatarone, M.A.; Evans, W.J.; Roubenoff, R. Aging of skeletal muscle: a 12-yr longitudinal study. *J Appl Physiol (1985)*. **2000**, *88*, 1321-1326.

[3] Delmonico, M.J.; Harris, T.B.; Lee, J.S.; Visser, M.; Nevitt, M.; Kritchevsky, S.B.; Tylavsky, F.A.; Newman, A.B. Alternative definitions of sarcopenia, lower extremity performance, and functional impairment with aging in older men and women. *J. Am. Geriatr. Soc.* **2007**, *55*, 769-774..

[4] Woo, J. Sarcopenia. *Clin. Geriatr. Med*. **2017**, *33*, 305–314.

[5] Cruz-Jentoft, A.J.; Baeyens, J.P.; Bauer, J.M.; Boirie, Y.; Cederholm, T.; Landi, F.; Martin, F.C.; Michel, J.P.; Rolland, Y.; Schneider, S.M.; Topinkova, E.; Vandewoude, M.; Zamboni, M. Sarcopenia: European consensus on definition and diagnosis: Report of the European Working Group on Sarcopenia in Older People. *Age Ageing*. **2010**, *39*, 412-423.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

[6] Fielding, R.A.; Vellas, B.; Evans, W.J.; Bhasin, S.; Morley, J.E.; Newman, A.B.; Abellan, V.K.G.; Andrieu, S.; Bauer, J.; Breuille, D.; Cederholm, T.; Chandler, J.; De Meynard, C.; Donini, L.; Harris, T.; Kannt, A.; Keime, G.F.; Onder, G.; Papanicolaou, D.; Rolland, Y.; Rooks, D.; Sieber, C.; Souhami, E.; Verlaan, S.; Zamboni, M. Sarcopenia: an undiagnosed condition in older adults. Current consensus definition: prevalence, etiology, and consequences. International working group on sarcopenia. *J. Am. Med. Dir. Assoc.* **2011**, *12*, 249-256.

[7] Chen, L.K.; Liu, L.K.; Woo, J.; Assantachai, P.; Auyeung, T.W.; Bahyah, K.S.; Chou, M.Y.; Chen, L.Y.; Hsu, P.S.; Krairit, O.; Lee, J.S.; Lee, W.J.; Lee, Y.; Liang, C.K.; Limpawattana, P.; Lin, C.S.; Peng, L.N.; Satake, S.; Suzuki, T.; Won, C.W.; Wu, C.H.; Wu, S.N.; Zhang, T.; Zeng, P.; Akishita, M.; Arai, H. Sarcopenia in Asia: consensus report of the Asian Working Group for Sarcopenia. *J. Am. Med. Dir. Assoc.* **2014**, *15*, 95-101.

[8] Lee, W.; Liu, L.; Peng, L.; Lin, M.; Chen, L. Comparisons of Sarcopenia Defined by IWGS and EWGSOP Criteria Among Older People: Results From the I-Lan Longitudinal Aging Study. *J. Am. Med. Dir. Assoc.* **2013**, *14*, 521-528.

[9] Wu, I.; Lin, C.; Hsiung, C.A.; Wang, C.; Wu, C.; Chan, D.D.; Li, T.; Lin, W.; Huang, K.; Chen, C.; Hsu, C. Epidemiology of sarcopenia among community-dwelling older adults in Taiwan: A pooled analysis for a broader adoption of sarcopenia assessments. *Geriatr. Gerontol. Int.* **2014**, *14*, 52-60.

[10] Wu, C.; Chen, K.; Hou, M.; Chang, Y.; Chang, C.; Liu, P.; Wu, S.; Chiu, C.; Jou, I.; Chen, C. Prevalence and associated factors of sarcopenia and severe sarcopenia in older Taiwanese living in rural community: The Tianliao Old People study 04. *Geriatr. Gerontol. Int.* **2014**, *14*, 69-75.

[11] Li, M.; Hu, Y.; Dong, H.Y.; Zhang, Y.; Fan, L.; Zhang, M.X.; Sun, J.; Han, X.Q.; Liu, Y.X.; Ma, X.N. Compare different measurement for prevalence of sarcopenia in a cohort of healthy community dwellingely in Beijing area. *Chin. J. Health Care Med.* **2014**, *16*, 426-429(Chinese article).

[12] Meng, N.; Li, C.; Liu, C.; Lin, C.; Lin, W.; Chang, C.; Li, T.; Lin, C. Comparison of height- and weight-adjusted sarcopenia in a Taiwanese metropolitan older population. *Geriatr. Gerontol. Int.* **2015**, *15*, 45-53.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

[13] Wells, G.A.; Shea, B.; O'Connell, D.; Peterson, J.; Welch, V.; Losos, M.; Tugwell, P. The Newcastle-Ottawa Scale (NOS) for assessing the quality of nonrandomised studies in meta-analyses. Available: http://www.ohri.ca/programs/clinical_epidemiology/oxford.asp. Accessed 2014 Aug 5

[14] Higgins, J.P.T.; Green, S. Cochrane Handbook for Systematic Reviews of Interventions Version 5.1.0. Available: http://www.cochrane-handbook.org. Accessed 2014 Aug

[15] Wen, X.; An, P.; Chen, W.C.; Lv, Y.; Fu, Q. Comparisons of sarcopenia prevalence based on different diagnostic criteria in Chinese older adults. *The journal of nutrition, health & aging*. **2015**, *19*, 342-347.

[16] Zeng, P.; Wu, S.; Han, Y.; Liu, J.; Zhang, Y.; Zhang, E.; Zhang, Y.; Gong, H.; Pang, J.; Tang, Z.; Liu, H.; Zheng, X.; Zhang, T. Differences in body composition and physical functions associated with sarcopenia in Chinese elderly: Reference values and prevalence. *Arch. Gerontol. Geriat.* **2015**, *60*, 118-123.

[17] Gao, L.; Jiang, J.; Yang, M.; Hao, Q.; Luo, L.; Dong, B. Prevalence of Sarcopenia and Associated Factors in Chinese Community-Dwelling Elderly: Comparison Between Rural and Urban Areas. *J. Am. Med. Dir. Assoc.* **2015**, *16*, 1001-1003.

[18] Wang, Y.; Wang, Y.; Zhan, J.; Tang, Z.; He, J.; Tan, P.; Deng, H.; Huang, W.; Liu, Y. Sarco-Osteoporosis: Prevalence and Association with Frailty in Chinese Community-Dwelling Older Adults. *Int. J. Endocrinol.* **2015**, *2015*, 1-8.

[19] Woo, J.; Leung, J.; Morley, J.E. Defining Sarcopenia in Terms of Incident Adverse Outcomes. *J. Am. Med. Dir. Assoc.* **2015**, *16*, 247-252.

[20] Wang, H.; Hai, S.; Cao, L.; Zhou, J.; Liu, P.; Dong, B. Estimation of prevalence of sarcopenia by using a new bioelectrical impedance analysis in Chinese community-dwelling elderly people. *BMC Geriatr.* **2016**, *16*.

[21] Han, P.; Kang, L.; Guo, Q.; Wang, J.; Zhang, W.; Shen, S.; Wang, X.; Dong, R.; Ma, Y.; Shi, Y.; Shi, Z.; Li, H.; Li, C.; Ma, Y.; Wang, L.; Niu, K. Prevalence and Factors Associated With Sarcopenia in Suburb-dwelling Older Chinese Using the Asian Working Group for Sarcopenia

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Definition. *The Journals of Gerontology Series A: Biological Sciences and Medical Sciences*. **2016**, *71*, 529-535.

[22] Han, D.; Chang, K.; Li, C.; Lin, Y.; Kao, T.; Tsai, K.; Wang, T.; Yang, W. Skeletal muscle mass adjusted by height correlated better with muscular functions than that adjusted by body weight in defining sarcopenia. *Sci. Rep.-UK*. **2016**, *6*.

[23] Xia, Z.W.; Meng, L.P.; Man, Q.Q.; Li, L.X.; Song, P.K.; Li, Y.Q.; Gao, Y.X.; Jia, S.S.; Zhang, J. Analysis of the dietary factors on sarcopenia in elderly in Beijing. *J Hygiene Res.* **2016**, *45*, 388-393 (Chinese article).

[24] Chen, Z.; Dai, R.; Gu, X.H. Prevalence of sarcopenia and quality of life in retired cadres of a military unit. *Chin. J. Osteopo. Rosis. & Bone Miner. Res.* **2016**, *9*, 264-268 (Chinese article).

[25] Bai, H.; Sun, J.; Chen, M.; Xu, D.; Xie, H.; Yu, Z.; Bao, Z.; Chen, J.; Pan, Y.; Lu, D.; Cheng, S. Age-related decline in skeletal muscle mass and function among elderly men and women in Shanghai, China: A cross sectional study. *Asia Pac. J. Clin. Nutr.* **2016**, *25*, 326-332..

[26] Hai, S.; Wang, H.; Cao, L.; Liu, P.; Zhou, J.; Yang, Y.; Dong, B. Association between sarcopenia with lifestyle and family function among community-dwelling Chinese aged 60 years and older. *BMC Geriatr.* **2017**, *17*.

[27] Ma, Y.; Bian, A.L.; Wang, X.F.; Wang, J.X.; Zhou, X.Z. Difference of the prevalence rate of sarcopenia in elderly people with different gender and age in Tianjin. *Chin. General Practice* **2017**, *20*, 3267-3271 (Chinese article).

[28] Wu, Z.N.; Wang, Y.B.; Shu, F.Q.; Song, B.; Li, G.; Zhu, H.M. Epidemiological investigation of sarcopenia in the nursing homes in Fengxian District and analysis of their risk factors. *Shanghai Med.* **2017**, *38*, 37-40 (Chinese article).

[29] Huang, L.J.; Liu, Y.C.; Wen, Y.F.; Liu, K. State and associated factors of sarcopenia in Jinzhou community elderly people. *Modern Preve. Med.* **2017**, *44*, 4458-4460 (Chinese article).

[30] Yang, M.; Hu, X.; Xie, L.; Zhang, L.; Zhou, J.; Lin, J.; Wang, Y.; Li, Y.; Han, Z.; Zhang, D.; Zuo, Y.; Li, Y.; Wu, L. Screening Sarcopenia in Community-Dwelling Older Adults: SARC-F vs SARC-F Combined With Calf Circumference (SARC-CalF). *J. Am. Med. Dir. Assoc.* **2018**, *19*,

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

271-277.

[31] Zeng, Y.; Hu, X.; Xie, L.; Han, Z.; Zuo, Y.; Yang, M. The Prevalence of Sarcopenia in Chinese Elderly Nursing Home Residents: A Comparison of 4 Diagnostic Criteria. *J. Am. Med. Dir. Assoc.* **2018**, *19*, 690-695.

[32] Wang, Y.N.; Xu, J.L.; Song, H.L. Influencing factors of sarcopenia among elderly patients. *Modern Clinical Nursing* **2018**, *17*, 45-50 (Chinese article).

[33] Yang, L.; Wu, G.; Yang, Y.; Wu, Y.; Zhang, L.; Wang, M.; Mo, L.; Xue, G.; Wang, C.; Weng, X. Nutrition, Physical Exercise, and the Prevalence of Sarcopenia in Elderly Residents in Nursing Homes in China. *Med. Sci. Monitor*. **2019**, *25*, 4390-4399.

[34] Du, Y.; Wang, X.; Xie, H.; Zheng, S.; Wu, X.; Zhu, X.; Zhang, X.; Xue, S.; Li, H.; Hong, W.; Tang, W.; Chen, M.; Cheng, Q.; Sun, J. Sex differences in the prevalence and adverse outcomes of sarcopenia and sarcopenic obesity in community dwelling elderly in East China using the AWGS criteria. *BMC Endocr. Disord.* **2019**, *19*.

[35] Wang, H.; Hai, S.; Liu, Y.; Liu, Y.X.; Zhou, J.H.; Yang, Y.; Dong, B.R.; Yue, J.R. Prevalence of sarcopenia and associated factors in community-dwelling elderly populations in Chengdu China. *J. Sichuan Univ.* **2019**, *50*, 224-228 (Chinese article).

[36] Zhang, Y.; Chen, X.; Hou, L.; Lin, X.; Qin, D.; Wang, H.; Hai, S.; Cao, L.; Dong, B. Prevalence and Risk Factors Governing the Loss of Muscle Function in Elderly Sarcopenia Patients: A longitudinal Study in China with 4 Years of Follow-Up. *The journal of nutrition, health & aging*. **2020**, *24*, 518-524.

[37] Liu, X.; Hao, Q.; Yue, J.; Hou, L.; Xia, X.; Zhao, W.; Zhang, Y.; Ge, M.; Ge, N.; Dong, B. Sarcopenia, Obesity and Sarcopenia Obesity in Comparison: Prevalence, Metabolic Profile, and Key Differences: Results from WCHAT Study. *The journal of nutrition, health & aging*. **2020**, *24*, 429-437.

[38] Shafiee, G.; Keshtkar, A.; Soltani, A.; Ahadi, Z.; Larijani, B.; Heshmat, R. Prevalence of sarcopenia in the world: a systematic review and meta-analysis of general population studies. *Journal of Diabetes & Metabolic Disorders*. **2017**, *16*.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

[39] Makizako, H.; Nakai, Y.; Tomioka, K.; Taniguchi, Y. Prevalence of sarcopenia defined using the Asia Working Group for Sarcopenia criteria in Japanese community-dwelling older adults: A systematic review and meta-analysis. *Physical therapy research*. **2019**, *22*, 53-57.

[40] Diz, J.B.M.; Leopoldino, A.A.O.; Moreira, B.D.S.; Henschke, N.; Dias, R.C.; Pereira, L.S.M.; Oliveira, V.C. Prevalence of sarcopenia in older Brazilians: A systematic review and meta-analysis. *Geriatr. Gerontol. Int.* **2017**, *17*, 5-16.

[41] Papadopoulou, S.K.; Tsintavis, P.; Potsaki, G.; Papandreou, D. Differences in the Prevalence of Sarcopenia in Community-Dwelling, Nursing Home and Hospitalized Individuals. A Systematic Review and Meta-Analysis. *The journal of nutrition, health & aging*. **2020**, *24*, 83-90.

[42] Han, A.; Sun, T.; Ming, J.; Chai, L.; Liao, X. Are the Chinese moving toward a healthy diet? Evidence from macro data from 1961 to 2017. *Int J Environ Res Public Health,* **2020**, *17*, 5294

Legend

Figure 1. Flowchart of the detailed procedure for the inclusion or exclusion of selected studies.

Figure 2. Forest plots for the total prevalence of sarcopenia among the elderly Chinese population according to the Asian Working Group for Sarcopenia (AWGS) standard. Diamond represents the pooled odds ratio at 95% CI.

Figure 3. Forest plots for the total prevalence of sarcopenia among the elderly Chinese population according to the International Working Group on Sarcopenia (IWGS) standard. Diamond represents the pooled odds ratio at 95% CI.

Figure 4. Forest plots for the total prevalence of sarcopenia among the elderly Chinese population according to the European Working Group on Sarcopenia in Older People (EWGSOP) standard. Diamond represents the pooled odds ratio at 95% CI.

Figure 5. Forest plots for the prevalence of sarcopenia among male and female elderly people according to the Asian Working Group for Sarcopenia (AWGS) standard. Diamond represents the pooled odds ratio at 95% CI.

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Figure 6. Forest plots for the prevalence of sarcopenia among male and female elderly people according to the International Working Group on Sarcopenia (IWGS) standard. Diamond represents the pooled odds ratio at 95% CI.

Figure 7. Forest plots for the prevalence of sarcopenia among male and female elderly people according to the European Working Group on Sarcopenia in Older People (EWGSOP) standard. Diamond represents the pooled odds ratio at 95% CI.

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Table 1 Study characteristics of the studies included in the meta-analysis

Author	Publication Year	Participants(n)			Region	Setting	Measurement	Diagnostic Criteria	Prevalence (%)			NOS
		Total	Male	Female					Total	Male	Female	
Lee WJ	2013	386	223	163	Taiwan	Urban Community	DEXA	IWGS	11.1	10.8	11.7	6
								EWGSOP	16.6	14.9	19.0	
Wu IC	2014	2155	1055	1100	Taiwan	Urban Community	BIA	EWGSOP	3.9	5.4	2.5	6
Wu CH	2014	549	285	264	Taiwan	Rural Community	BIA	EWGSOP	12.7	8.1	17.8	6
Li M	2014	169	169	0	Beijing	Urban Community	DEXA	EWGSOP	33.3	33.3	-	5
Meng NH	2015	771	359	712	Taiwan	Urban Community	DEXA	EWGSOP	9.7	0.7	12.5	6
Gao LL	2015	612	254	358	Sichuan	Rural and urban Community	Calf circumference	AWGS	9.8	6.7	12.0	6
Wen X	2015	286	136	150	Hangzhou	Urban Community	DEXA	IWGS	6.0	7.4	4.7	5
								EWGSOP	0.4	0.8	0.01	
								AWGS	3.1	5.9	0.7	

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Woo J	2015	4000	2000	2000	Hongkong	Urban Community	DEXA	IWGS	20.2	22.1	18.3	6
								EWGSOP	9.0	9.5	8.6	
								AWGS	7.3	5.3	9.3	
Zeng P	2015	395	199	196	Beijing	Rural and urban Community	BIA	AWGS	24.3	18.1	30.6	5
Wang YJ	2015	316	164	152	Changsha	Urban Community	DEXA	AWGS	29.7	26.2	33.6	6
Han PP	2016	1069	467	602	Tianjin	Urban Community	BIA	AWGS	9.3	6.4	17.5	5
Wang H	2016	636	316	320	Chengdu	Urban Community	BIA	AWGS	10.4	8.2	12.5	5
Bai HJ	2016	415	212	203	Shanghai	Urban Community	BIA	AWGS	28.7	33.5	23.6	6
Xia ZW	2016	683	239	444	Beijing	Urban Community	BIA	AWGS	20.1	17.2	21.6	5
Chen Z	2016	281	203	78	Chongqing	Urban Community	BIA	IWGS	27.8	-	-	6
Han DS	2016	878	402	476	Taiwan	Urban Community	BIA	EWGSOP	9.8	4.5	14.3	6
Hai S	2017	834	415	419	Chengdu	Urban Community	BIA	AWGS	10.6	11.3	9.8	6
Huang	2017	193	65	128	Jinzhou	Urban Community	BIA	AWGS	11.4	7.7	13.3	6

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LJ												
Yang M	2018	384	160	224	Chengdu	Urban Community	BIA	AWGS	15.9	11.9	18.8	6
								IWGS	25.0	24.4	25.4	
								EWGSOP	11.7	10.6	12.5	
Du YP	2019	631	213	418	Shanghai	Urban Community	BIA	AWGS	12.2	19.2	8.6	5
Wang H	2019	947	465	482	Chengdu	Urban Community	BIA	AWGS	7.7	6.5	8.9	5
Zhang Y	2020	474	238	236	Chengdu	Urban Community	BIA	AWGS	15.4	-	-	5
Liu XL	2020	4500	1627	2873	Yunnan, Guizhou, Sichuan, and Xinjiang	Urban Community	BIA	AWGS	19.3	22.1	17.8	6

DEXA, dual-energy x-ray absorptiometry; BIA, bioelectrical impedance analysis; IWGS, International Working Group on Sarcopenia; AWGS, Asia Working Group for Sarcopenia; EWGSOP, European Working Group on Sarcopenia in Older People; NOS, Newcastle–Ottawa scale.

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Table 2 The prevalence of sarcopenia diagnosed by different criteria using DEXA and BIA measurement (%).

	DEXA	BIA
AWGS	13(5−21)	15(12−19)
IWGS	12(3−22)	26(23−29)
EWGSOP	12(6−18)	9(5−14)

DEXA, dual-energy x-ray absorptiometry; BIA, bioelectrical impedance analysis; IWGS, International Working Group on Sarcopenia; AWGS, Asia Working Group for Sarcopenia; EWGSOP, European Working Group on Sarcopenia in Older People.

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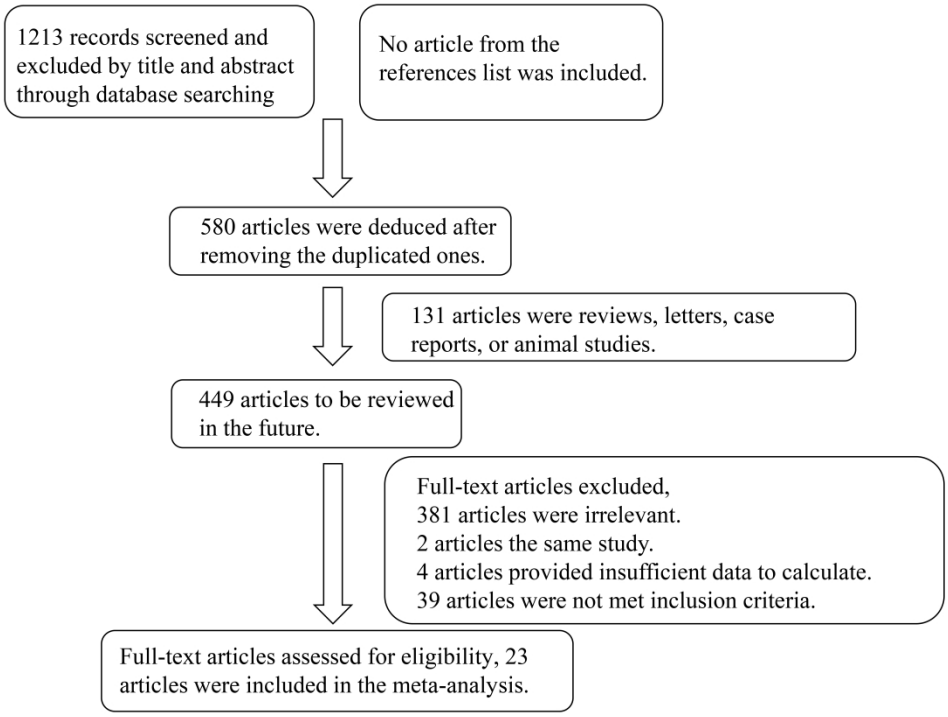


Figure 1. Flowchart of the detailed procedure for the inclusion or exclusion of selected studies.

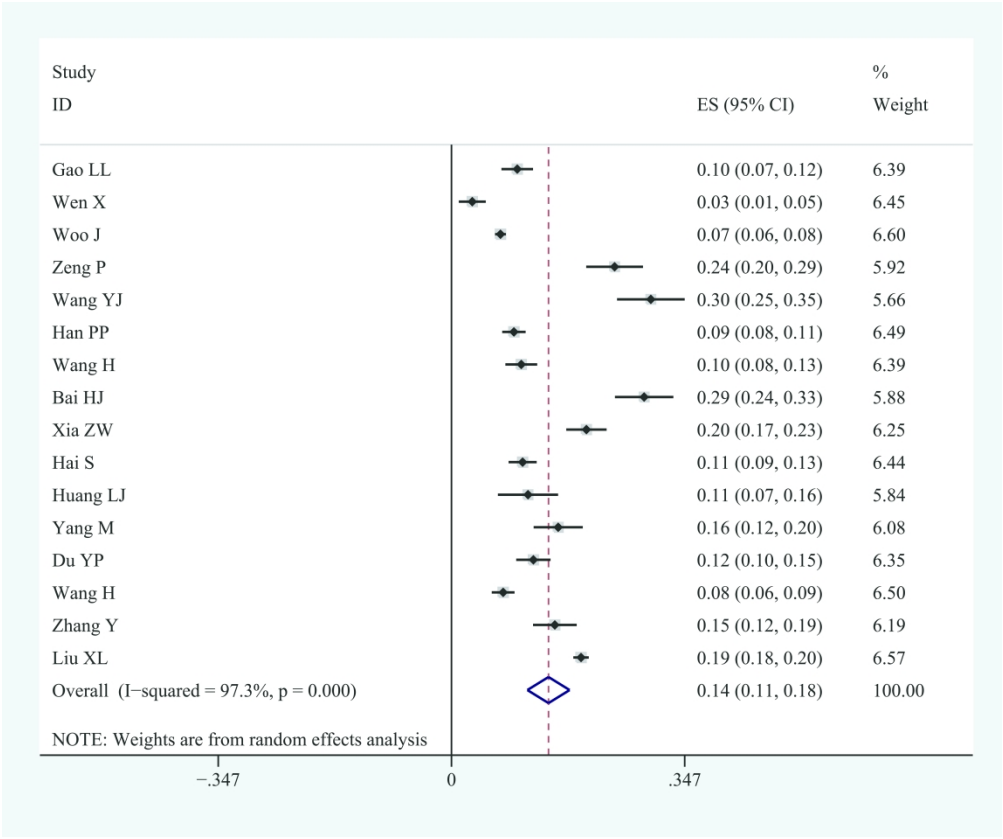


Figure 2. Forest plots for the total prevalence of sarcopenia among the elderly Chinese population according to the Asian Working Group for Sarcopenia (AWGS) standard. Diamond represents the pooled odds ratio at 95% CI.

280x233mm (600 x 600 DPI)

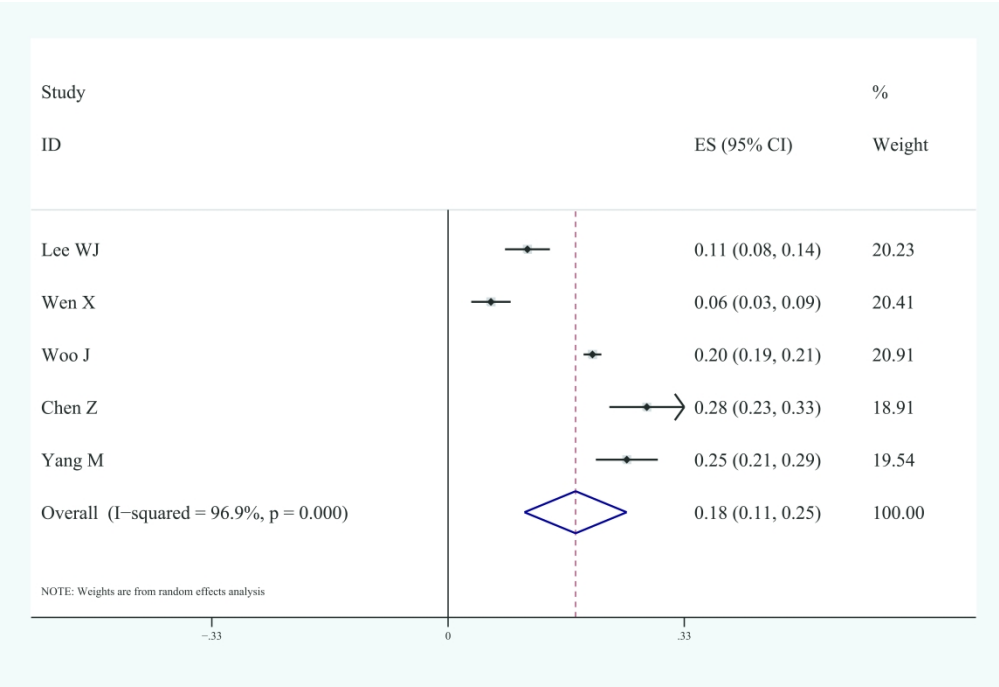


Figure 3. Forest plots for the total prevalence of sarcopenia among the elderly Chinese population according to the International Working Group on Sarcopenia (IWGS) standard. Diamond represents the pooled odds ratio at 95% CI.

272x186mm (600 x 600 DPI)

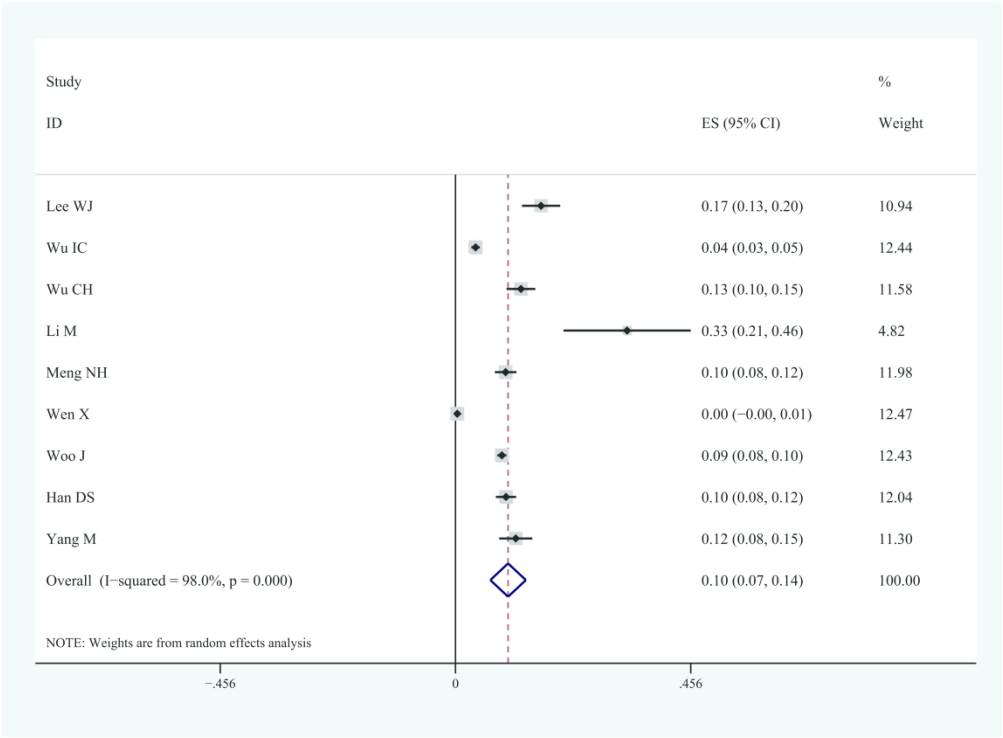


Figure 4. Forest plots for the total prevalence of sarcopenia among the elderly Chinese population according to the European Working Group on Sarcopenia in Older People (EWGSOP) standard. Diamond represents the pooled odds ratio at 95% CI.

274x201mm (600 x 600 DPI)

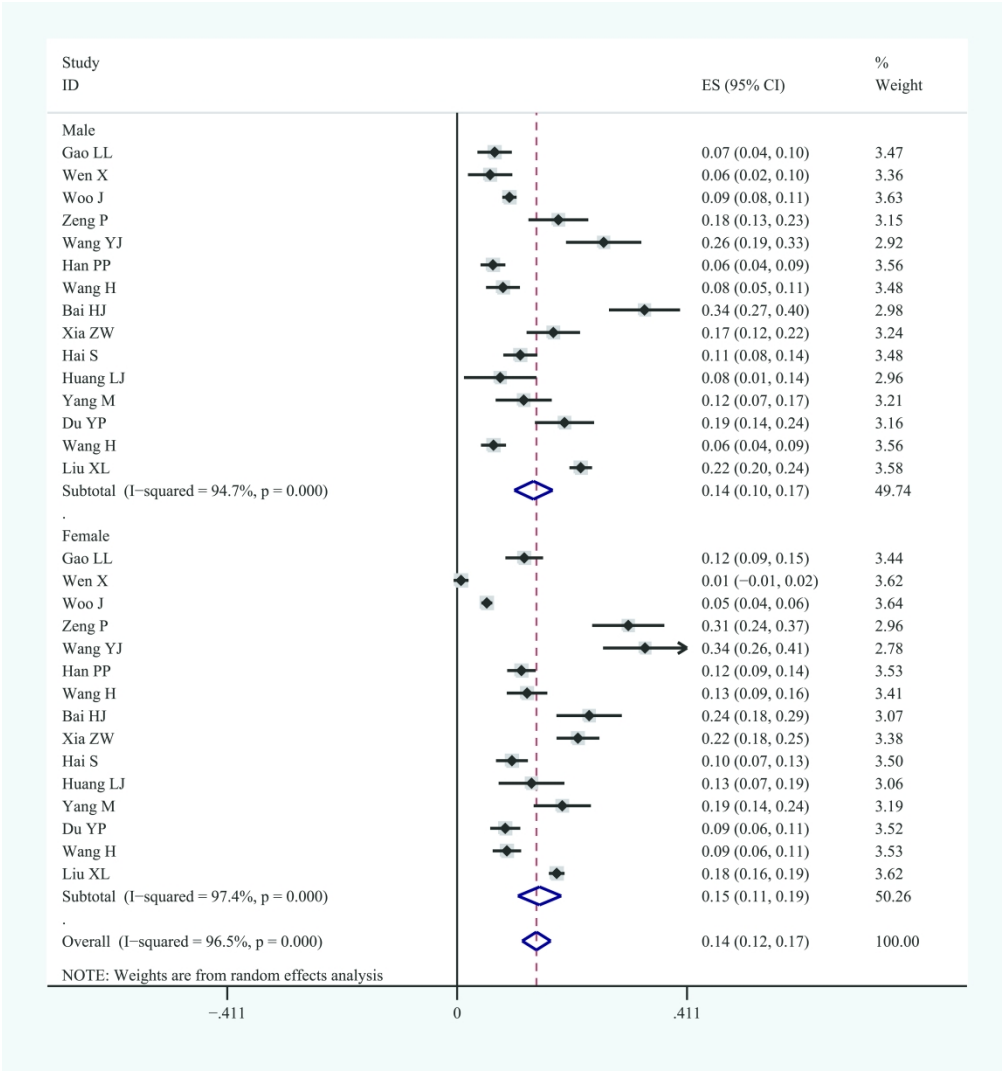


Figure 5. Forest plots for the prevalence of sarcopenia among male and female elderly people according to the Asian Working Group for Sarcopenia (AWGS) standard. Diamond represents the pooled odds ratio at 95% CI.

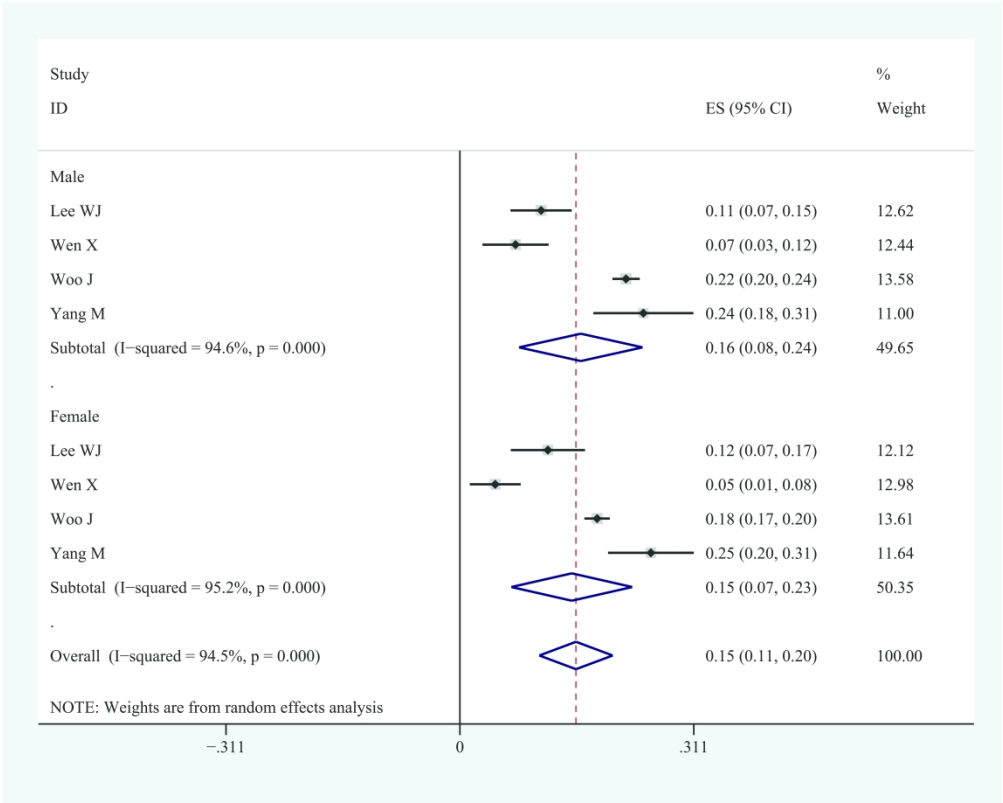


Figure 6. Forest plots for the prevalence of sarcopenia among male and female elderly people according to the International Working Group on Sarcopenia (IWGS) standard. Diamond represents the pooled odds ratio at 95% CI.

278x223mm (600 x 600 DPI)

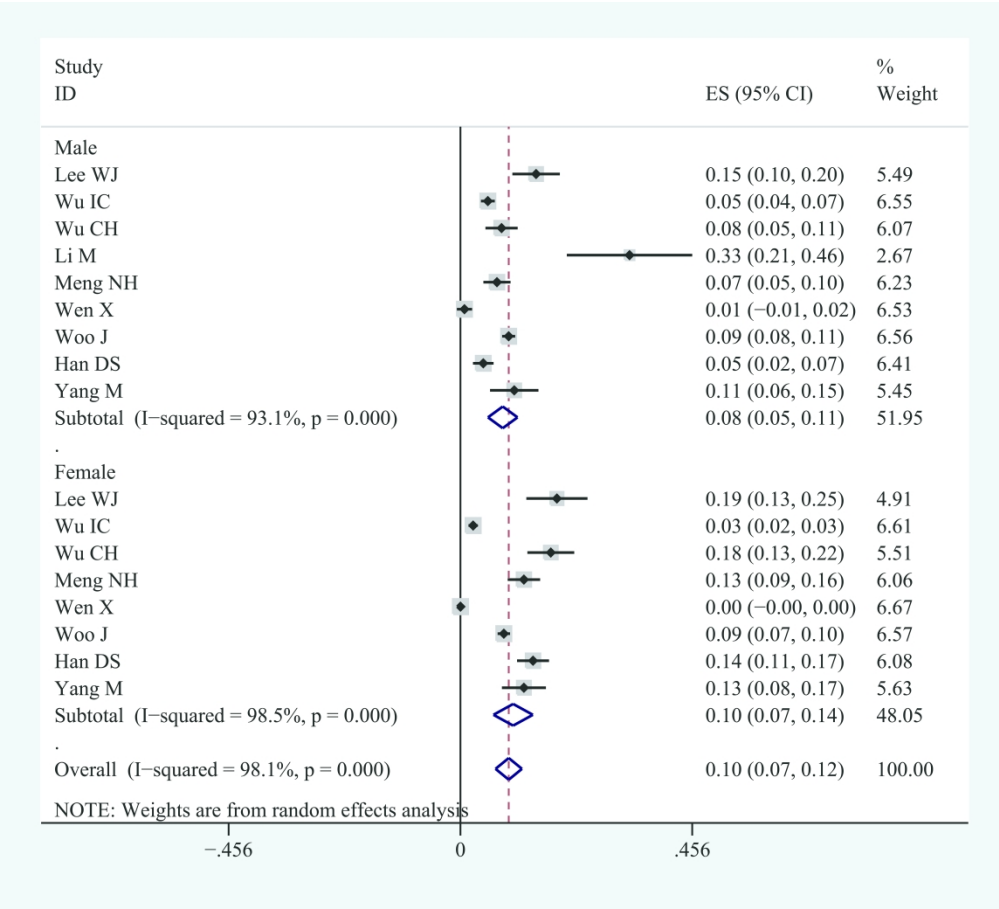


Figure 7. Forest plots for the prevalence of sarcopenia among male and female elderly people according to the European Working Group on Sarcopenia in Older People (EWGSOP) standard. Diamond represents the pooled odds ratio at 95% CI.

284x257mm (600 x 600 DPI)

Figure S1. The sensitivity analysis results according to the Asian Working Group for Sarcopenia (AWGS) standard.

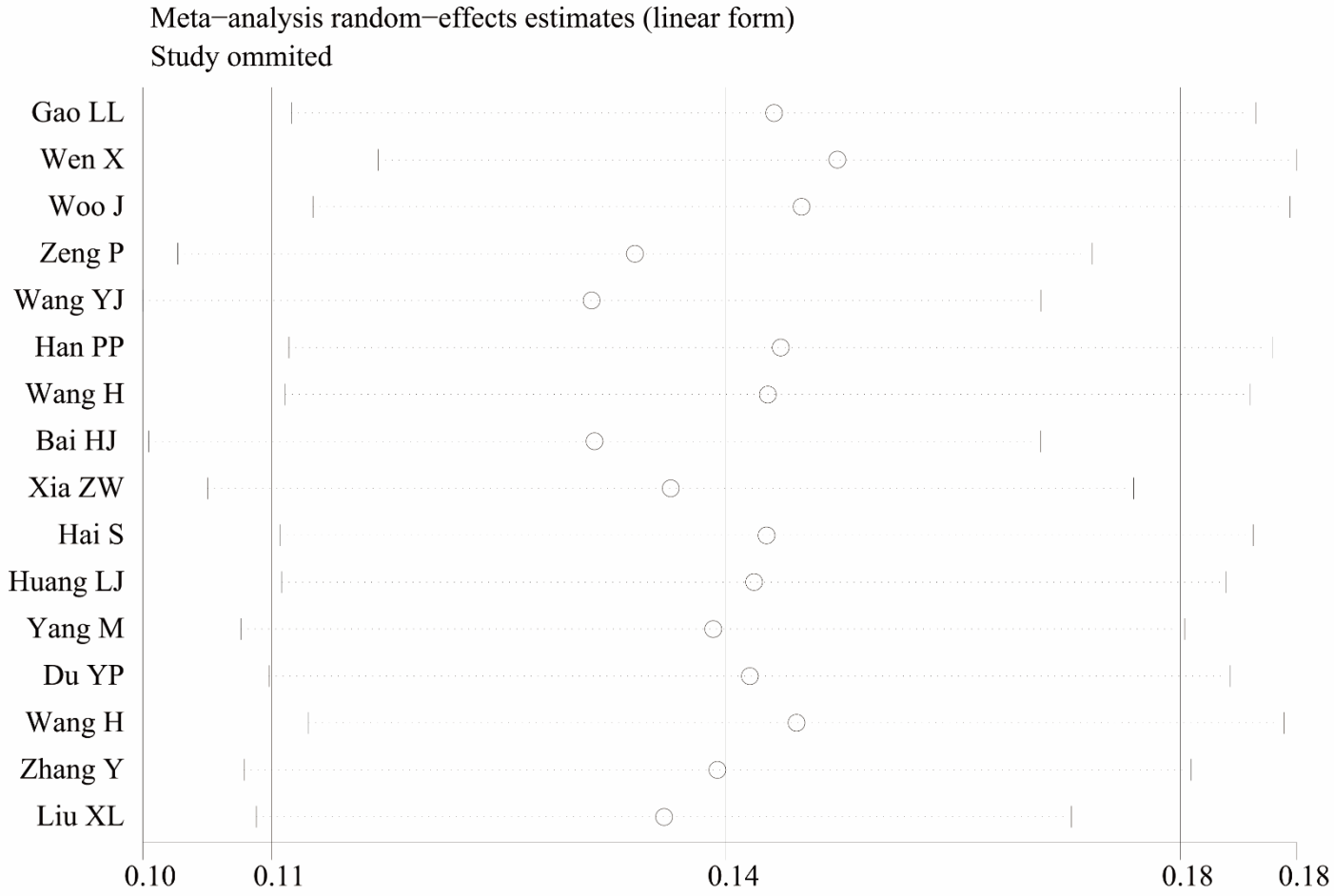


Figure S2. The sensitivity analysis results according to the International Working Group on Sarcopenia (IWGS) IWGS standard.

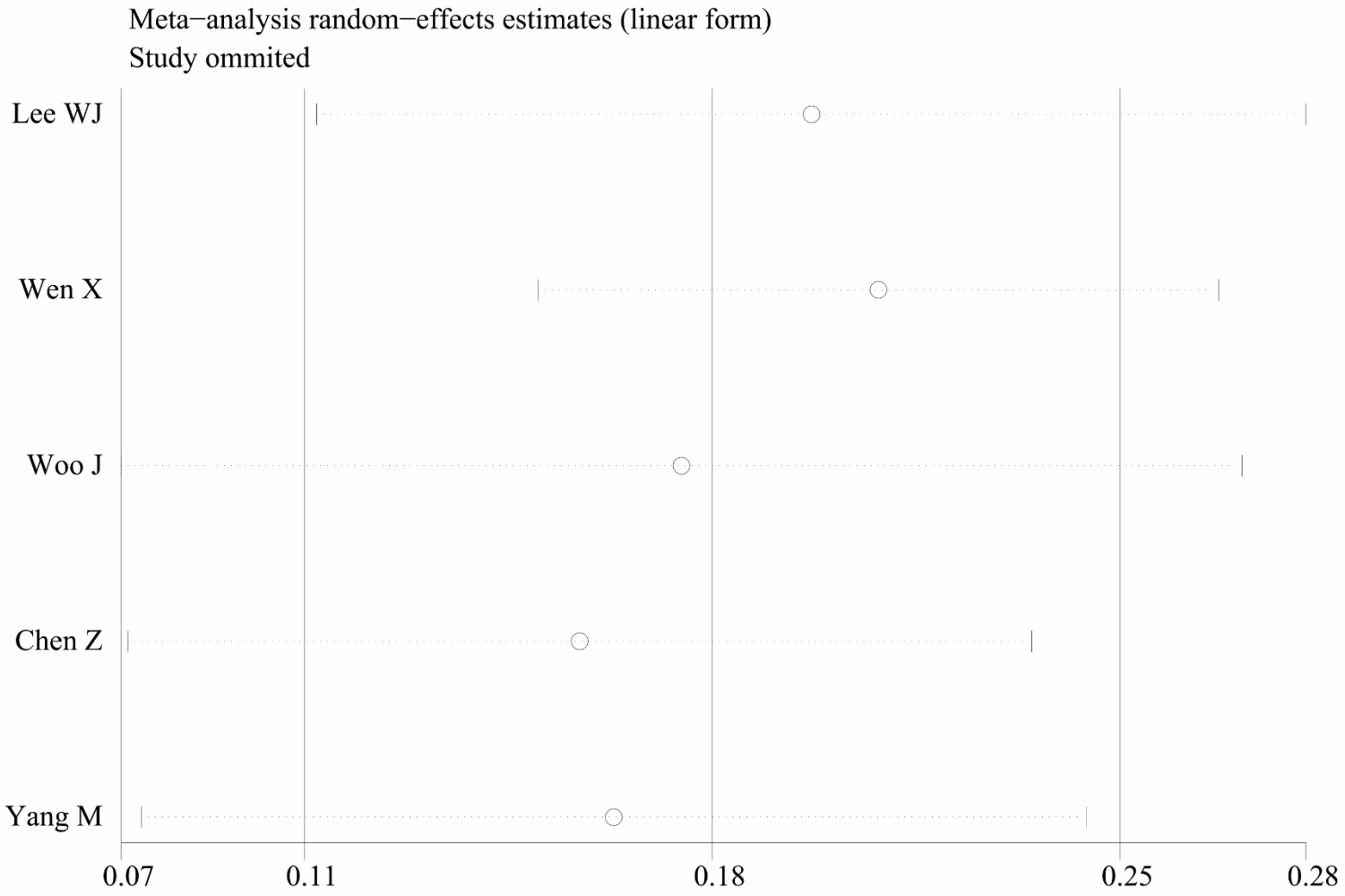


Figure S3. The sensitivity analysis results according to the European Working Group on Sarcopenia in Older People (EWGSOP) standard.

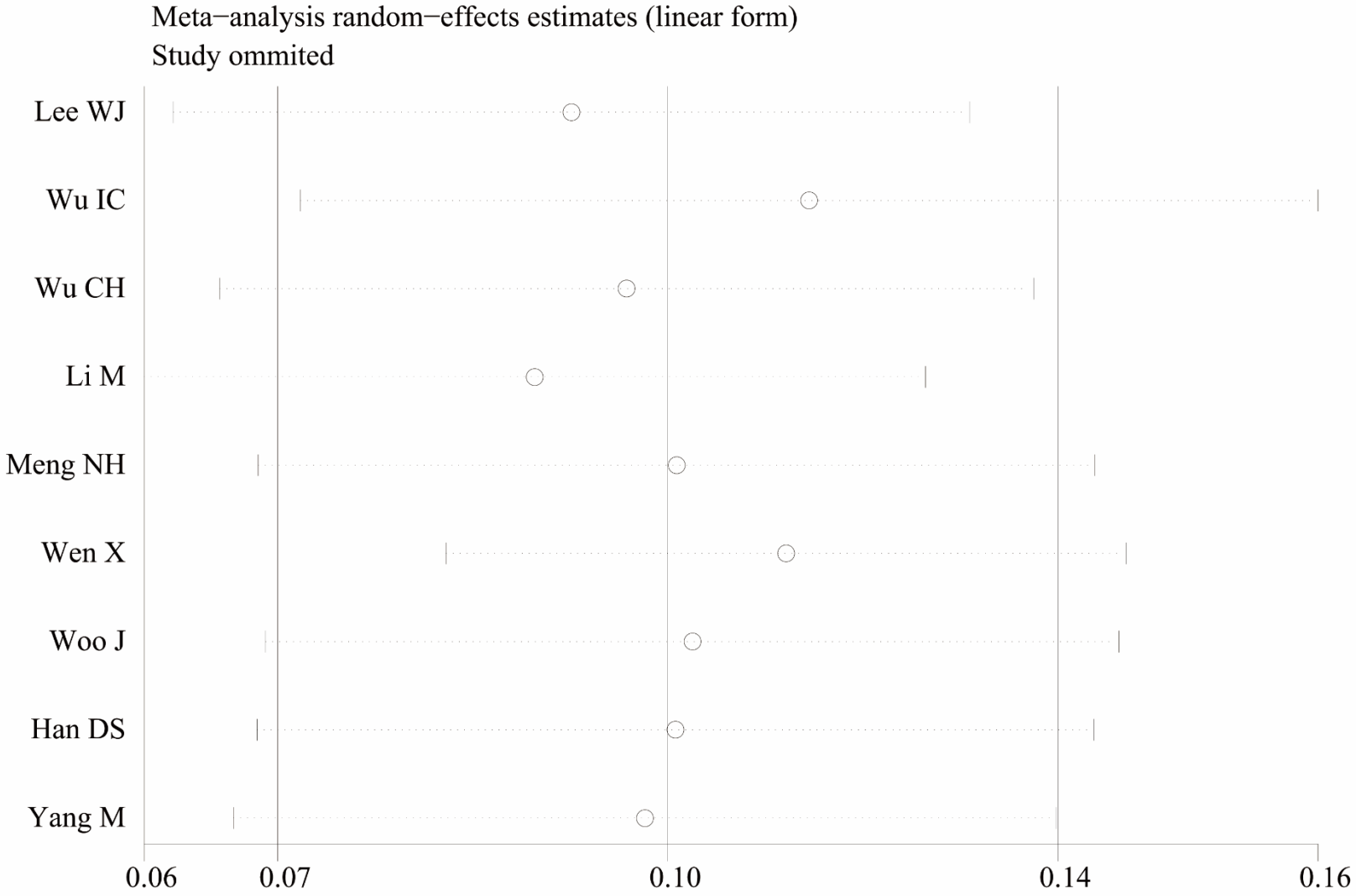


Figure S4. The publication bias results according to the Asian Working Group for Sarcopenia (AWGS) standard.

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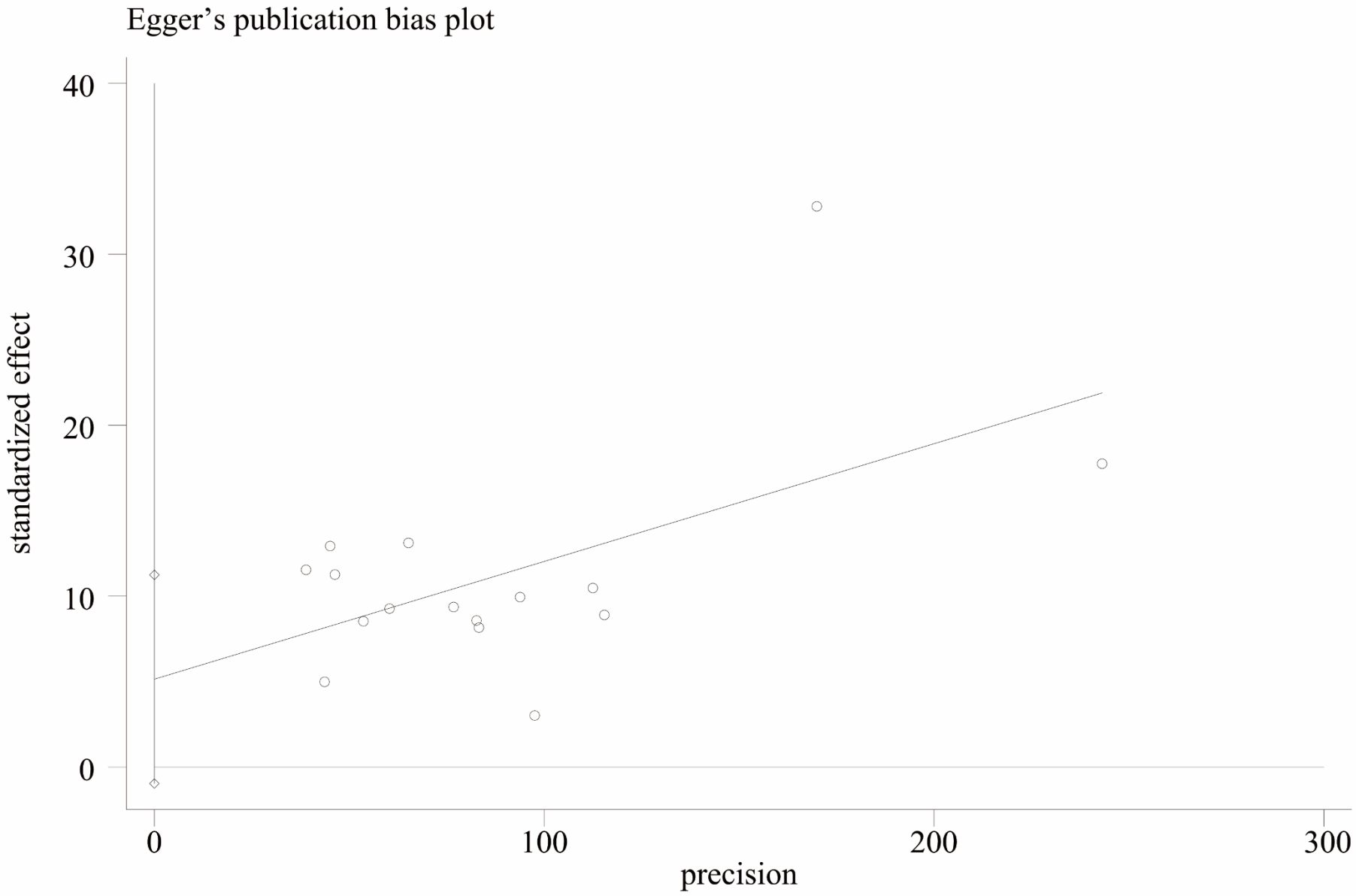


Figure S5. The publication bias results according to the the International Working Group on Sarcopenia (IWGS) standard.

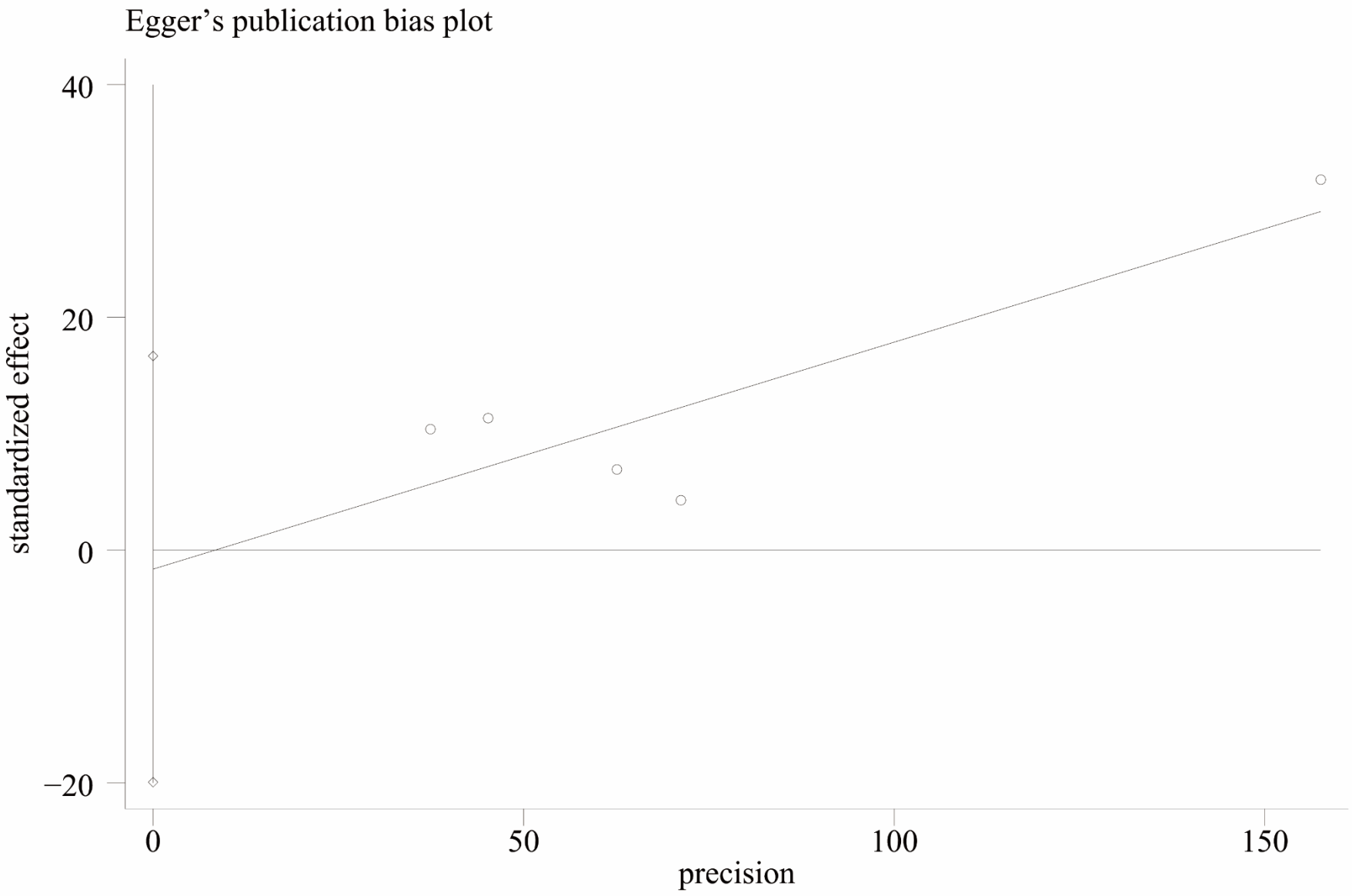
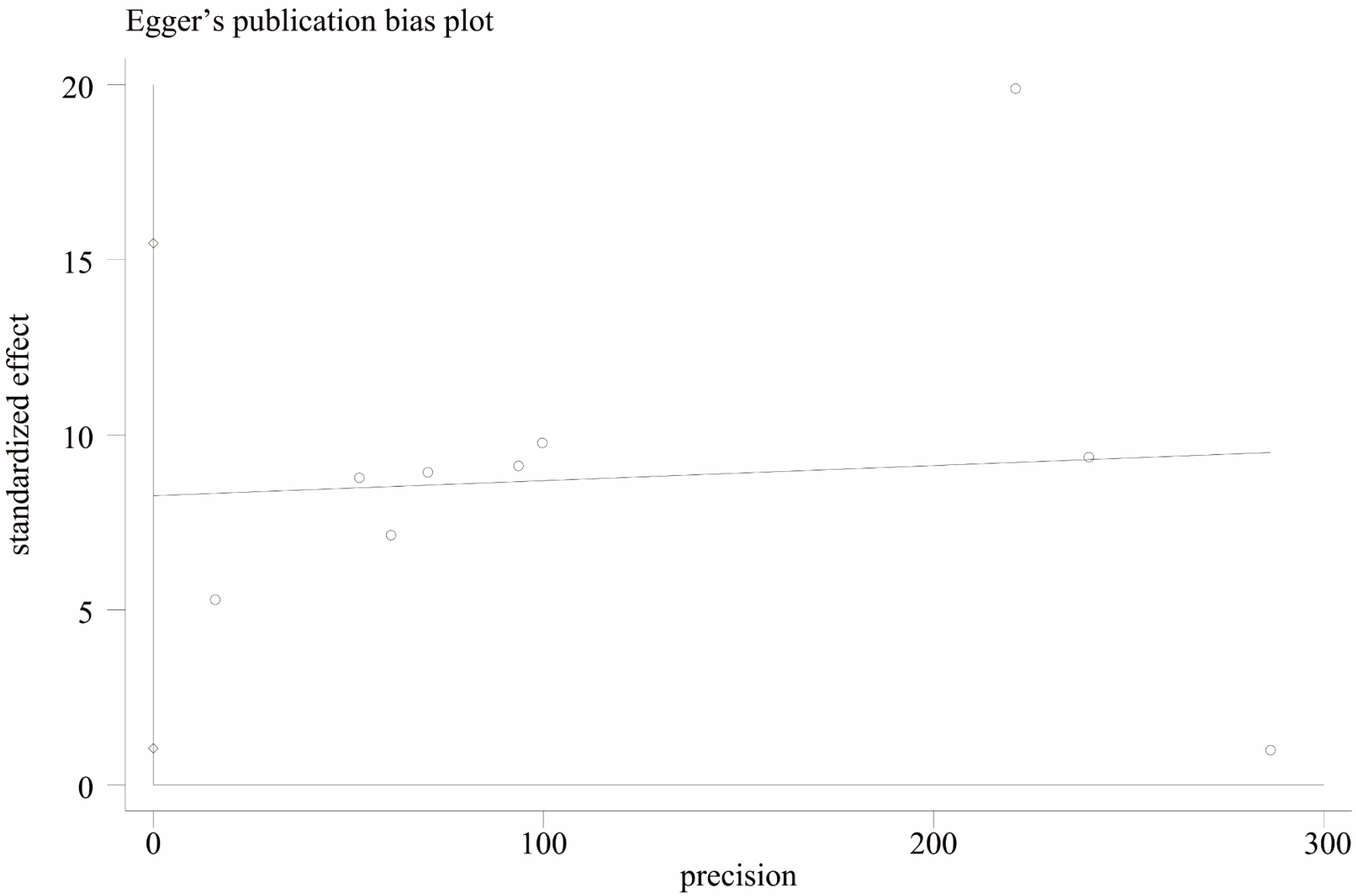


Figure S6. The publication bias results according to the European Working Group on Sarcopenia in Older People (EWGSOP) standard.


Egger's publication bias plot
standardized effect
precision
20
15
10
5
0
0
100
200
300

PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	3
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	3,4
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	4
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	4
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	4
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	4
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	4
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	5
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	5
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	5
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	5
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	5

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	5
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	5
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	6
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	6
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	6
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	6
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	6,7
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	7
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	7
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	7,8
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	8
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	8
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	9

From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(7): e1000097. doi:10.1371/journal.pmed1000097

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Prevalence of Sarcopenia in Older Chinese Adults: A Systematic Review and Meta-analysis

Journal:	*BMJ Open*
Manuscript ID	bmjopen-2020-041879.R4
Article Type:	Original research
Date Submitted by the Author:	17-May-2021
Complete List of Authors:	Xin, Caihong; Fourth People's Hospital in Shenyang, Department of Endocrinology and Metabolism Sun, Xin; First Affiliated Hospital of Soochow University, Department of Endocrinology and Metabolism; People's Hospital of Liaoning Province, Department of Endocrinology and Metabolism Lu, Li; People's Hospital of Liaoning Province, Department of Endocrinology and Metabolism Shan, Ling; People's Hospital of Liaoning Province, Department of Endocrinology and Metabolism
Primary Subject Heading:	Geriatric medicine
Secondary Subject Heading:	Epidemiology, Global health, Public health
Keywords:	GERIATRIC MEDICINE, EPIDEMIOLOGY, General endocrinology < DIABETES & ENDOCRINOLOGY

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Prevalence of Sarcopenia in Older Chinese Adults: A Systematic Review and Meta-analysis

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Running title: Prevalence of sarcopenia in

Key words: Prevalence; sarcopenia; China

Word count of the manuscript: 1864

Figures number: 7

Tables number: 2

Disclosure Statement: The authors have nothing to disclose.

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Abstract

Objective: This meta-analysis was conducted to estimate the overall prevalence of sarcopenia in older Chinese adults.

Design: Systematic review and meta-analysis

Participants: A literature research was conducted using the PubMed, Web of Science, CNKI, CQVIP, and Wanfang databases. The following search terms in the abstract were used: "sarcopenia" in combination with the terms "prevalence," "epidemiology," and "China." All studies published from January 2010 to November 2020 were included. The random effect model was used to estimate the prevalence of sarcopenia. The sex-specific prevalence of sarcopenia at a 95% confidence interval (CI) was also calculated using different criteria for defining sarcopenia.

Primary outcome measures: The overall prevalence of sarcopenia in older Chinese adults.

Results: In total, 23 articles were included in this meta-analysis involving 21,564 participants. On the basis of the Asian Working Group for Sarcopenia (AWGS) criterion, the overall prevalence of sarcopenia among the elderly in China was 14% (95% CI, 11%–18%), the prevalence was higher in Chinese females than in males (15% vs. 14%).

Conclusions: This systematic review is the first estimation of the pooled prevalence of sarcopenia in older Chinese adults. Our results suggest that China has a large number of sarcopenia patients. These findings would be useful for sarcopenia prevention in China. There is a high degree of heterogeneity and although there are a large number of cases and could be an emerging public health issue, more research is required to make these claims.

Trial registration: PROSPERO CRD42020223405

Strengths and limitations of this study:

- To our knowledge, this systematic review is the first estimation of the pooled prevalence of sarcopenia in Chinese older people.
- The search languages were limited to English and Chinese.
- We included limited number of high-quality studies.
- A high level of heterogeneity among included studies was observed.

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Introduction

Sarcopenia is characterized by the deterioration of muscle mass and function that occurs with aging. Several studies have shown that as age increases, skeletal muscle mass and strength are lost at a rate of 1%–2% annually, possibly leading to the decline of physical function and increasing the disability and mortality rates of elderly people as well as bringing serious economic and social burdens [1–4].

A practical clinical definition was reported by the European Working Group on Sarcopenia in Older People (EWGSOP) in 2010 [5]. Based on this EWGSOP criteria, slow walking speed (0.8 m/sec) or low grip strength (<30 kg for men and <20 kg for women) are indications of muscle mass decline. Later in 2011, another consensus was published by the International Working Group on Sarcopenia (IWGS) [6]. However, Asia is a vast region comprising different races, countries, and cultures, and developing screening and diagnostic strategies for the Asian population is imperative. The Asian Working Group for Sarcopenia (AWGS) was established in 2013, and the consensus of AWGS was released based on 16 epidemiologic reports on sarcopenia in six countries or regions, leading to the development of a diagnosis strategy for sarcopenia in Asia [7]. The diagnostic criteria recommended by AWGS and EWGSOP are similar, and both include muscle mass, muscle strength, and activity measurements. The only difference lies in the specific cut-off value. The cut-off values for muscle strength (<26 kg for men and <18 kg for women), walking speed (0.8 m/sec), and muscle mass (7.0 kg/m^2 for men and 5.4 kg/m^2 for women measured using dual X-ray absorptiometry and 7.0 kg/m^2 for men and 5.7 kg/m^2 for women measured using bioimpedance analysis) have been revised according to data from regional cohort studies among Asian populations.

According to statistics, the elderly population (≥60 years old) includes 16.1% of China's total population, which is close to 250 million. In the face of a severe aging situation, people are paying more attention to sarcopenia, but its prevalence is inconsistent across different studies [8–12]. At present, there is no large-scale epidemiologic survey of sarcopenia in China. The purpose of this study is to systematically determine the prevalence of sarcopenia in older Chinese adults.

Methods

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Search strategy

We searched the PubMed, Web of Science, China National Knowledge Infrastructure (CNKI), CQVIP, and Wanfang electronic databases. The search terms in the title or abstract included the following: "sarcopenia," "muscle mass," and "muscle strength" in combination with the terms "prevalence," "epidemiology," and "China." The complete search strategy are below: PubMed: (((sarcopenia [Title/Abstract]) OR muscle mass[Title/Abstract]) OR muscle strength[Title/Abstract]) AND ((prevalence[Title/Abstract]) OR epidemi*[Title/Abstract]) AND ((Chinese[Title/Abstract]) OR China[Title/Abstract]); Web of Science: (((TS=sarcopenia) OR TS=muscle mass) OR TS=muscle strength) AND ((TS=prevalence) OR TS=epidemi*) AND ((TS=Chinese) OR TS=China); CNKI: (AB =sarcopenia OR AB =muscle mass OR AB =muscle strength) AND (AB =prevalence OR AB =epidemiology); CQVIP: (R=sarcopenia +R=muscle mass+ R=muscle strength) *(R=prevalence +R=epidemiology); Wangfang: Abstract: (sarcopenia + muscle mass + muscle strength) * (prevalence + epidemiology). All studies published from January 2010 to November 2020 were included. The references of the retrieved articles were examined to identify additional eligible studies, and unpublished studies were excluded. The search languages were limited to English and Chinese. This systematic review and meta-analysis was registered in PROSPERO (registration number: CRD42020223405). This meta-analysis was conducted to estimate the overall prevalence of sarcopenia in older Chinese adults.

Patient and public involvement

No patients were involved in this study.

Inclusion and exclusion criteria

Inclusion of the selected studies was based on the following criteria: (1) a cross-sectional study or retrospective study, (2) a sample size of ≥100 elderly Chinese participants (age ≥60 years); (3) sufficient information on sarcopenia at the corresponding 95% confidence interval (CI) and *P* value, and (4) the definition of sarcopenia in line with the AWGS, EWGSOP, or IWGS standards.

Exclusion of the selected studies was based on the following criteria: (1) same data from different studies, (2) studies defining sarcopenia only as reduction of muscle mass or strength, and (3) the

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subjects include a specific population, such as patients living in a nursing home or those with disability. Reviews, case reports, letters, and animal researches were also excluded.

Data extraction and methodological quality assessment

Information from the selected studies was extracted by two independent researchers, including the first author's name, publication year, sample, diagnostic criteria, participants, setting, and prevalence. Two independent reviewers who were not blinded to the authors or journals assessed the risk of bias in the included studies using the Newcastle–Ottawa scale (NOS) [13]. The NOS is recommended by the Cochrane Handbook for Systematic Reviews of Interventions [14]. Each included study was analyzed in terms of three domains, namely, representativeness of the study group selection, group comparability, and exposure or outcome ascertainment, using the "star system." NOS scores ranged from zero to nine stars. The two researchers independently assessed the studies by discussion, comparison of findings, and resolution of differences through consensus formation. If no consensus was reached, a third researcher was commissioned to resolve the difference.

Statistical analysis

We used a systematic analysis approach to calculate the pooled prevalence of sarcopenia for all eligible studies, and heterogeneity among these studies was assessed using the Cochran's Q test and I^2 statistical index. The random effect model was used to combine the prevalence of sarcopenia. We additionally performed sensitivity analysis to evaluate the influence of any given study on the pooled estimate. Publication bias was evaluated using Egger's test. Subgroup analyses were also performed. The significance level was set at $P < 0.05$. All statistical analyses were performed using Stata software version 12.0 (College Station, Texas, USA).

Results

Study characteristics

In total, 1213 studies were retrieved from the databases and 580 articles were remained after the exclusion of duplicates. After screening for eligibility based on the full-text, a total of 23 articles were included as per the criteria shown in Figure 1 [8–12, 15–37]. Altogether, the 23 articles included 21,564 participants in China between 2010 and 2020. The selected studies' characteristics are

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summarized in Table 1. On the basis of the NOS scale, the score of selected studies were all above five stars; therefore, all articles were included.

Results of the meta-analysis

According to the AWGS standard, the total prevalence of sarcopenia among Chinese elderly people was 14% (95% CI, 11%–18%, Figure 2), and total prevalence rates of sarcopenia among the elderly in China based on the IWGS and EWGSOP standards were 18% (95% CI, 11%–25%, Figure 3) and 10% (95% CI, 7%–14%, Figure 4), respectively.

Subgroup analyses were performed to determine the influence of sex on the prevalence of sarcopenia. Sarcopenia prevalence among elderly females was higher than that in males as per the AWGS and EWGSOP standards (Figures 5–7). Table 2 shows the subgroup analyses results based on the different diagnostic criteria. The prevalence of sarcopenia estimated using the bio-electrical impedance analysis (BIA) method was higher than that estimated using the dual energy X-ray absorptiometry (DEXA) method as per the AWGS and IWGS standards.

Sensitivity analysis and publication bias

Sensitivity analysis was performed to examine the influence of each study. We found no significant difference between the results of the sensitivity analysis and our previous estimates, indicating that our statistical results were relatively credible (Figures S1–S3). The articles obtained from the databases were carefully and comprehensively searched. Additionally, Egger's test was conducted to determine whether potential publication bias existed in the reviewed literature. The results ($P > 0.05$) suggested that there were no publication bias (Figures S4–S6).

Discussion

To the best of our knowledge, this study is the first meta-analysis of the prevalence of sarcopenia in older Chinese adults. Sarcopenia is an age-related disease characterized by progressive deterioration in skeletal muscle fiber quality, strength, and function. Currently, sarcopenia has been internationally recognized as a new type of geriatric syndrome and has attracted worldwide attention. In 2017, Shafeee et al. conducted a meta-analysis on the prevalence of sarcopenia. Their results showed that sarcopenia prevalence among the elderly in communities worldwide was approximately 10%; this was higher

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among non-Asian individuals than Asians in both sexes [38]. A similar study was performed in Japan, where sarcopenia prevalence among the elderly Japanese was 9.9% (95% CI, 6.2%–15.4%) according to the AWGS standard [39]. The results of our study showed that the prevalence of sarcopenia was 14% (95% CI, 11%–18%) in older Chinese adults, which was much higher than the results of the aforementioned studies [40, 41]. This high prevalence may be related to the Chinese diet structure. Although protein is the raw material for muscle synthesis, Chinese people, particularly the older population, have higher carbohydrate intake and less protein intake [42]. Therefore, more attention should be given to the prevention and control of sarcopenia in older Chinese adults.

Here, we found that the prevalence of sarcopenia among females was higher than that in males, which was consistent with other meta-analyses [38–40]. This is mainly because men eat and exercise more than women; this supports muscle synthesis. Most of the studies included in our meta-analysis used the BIA method because of its affordability and portable equipment. BIA is a widely used non-invasive method for measuring human body composition. The principle behind this method is to record the different electrical resistances of different tissues using electrodes placed on the body surface and then to measure muscle mass using an image reconstruction method. DEXA is also commonly used to evaluate muscle mass. DEXA can accurately distinguish whole-body and local muscles as well as fats and bones. However, the limited number of studies using DEXA included in this meta-analysis may influence our results. As is known, aging increases the prevalence of sarcopenia; however, due to the different age classifications in the included studies, we cannot obtain any conclusion.

The results of this meta-analysis indicated high heterogeneity, which may be related to various factors, such as differences in age, race, and living environment. In addition, differences in the prevalence rates of sarcopenia have been observed in rural and urban areas. China is a country with a vast territory and multi-ethnic people. The findings of this meta-analysis should be interpreted with caution.

The limitation of the present study is that the main data are from urban populations and that studies on the prevalence of sarcopenia in Chinese rural areas are rare; this impacts the calculation of the total sarcopenia prevalence. Second, the prevalence of sarcopenia in people aged 60–70 years and 70–80 years might be different. Not all studies presented their data according to age group, so we cannot accurately estimate the prevalence in terms of age subgroup. Third, most of the studies were performed in large cities or economically developed areas. All of these factors affected our results, and as such,

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they should be interpreted cautiously as further research is needed.

Conclusions

To the best of our knowledge, the present systematic review is the first to estimate the pooled prevalence of sarcopenia in older Chinese adults. The results of the present systematic review reveal that China has a large number of sarcopenia cases and sarcopenia is an emerging public health issue in China. There is a high degree of heterogeneity and although there are a large number of cases and could be an emerging public health issue, more research is required to make these claims.

Author contributions statement

SL designed the study. SX and LL searched databases and collected the data. XCH and SL assessed the quality of the study. SL performed the analysis. SX and XCH wrote the manuscript. SX modified the manuscript in the revision. All authors contributed to this systematic review and meta-analysis.

Supporting Information

Figure S1. The sensitivity analysis results according to the Asian Working Group for Sarcopenia (AWGS) standard.

Figure S2. The sensitivity analysis results according to the the International Working Group on Sarcopenia (IWGS) standard.

Figure S3. The sensitivity analysis results according to the European Working Group on Sarcopenia in Older People (EWGSOP) standard.

Figure S4. The publication bias results according to the Asian Working Group for Sarcopenia (AWGS) standard.

Figure S5. The publication bias results according to the the International Working Group on Sarcopenia (IWGS) standard.

Figure S6. The publication bias results according to the European Working Group on Sarcopenia in Older People (EWGSOP) standard.

Patient consent

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Not required.

Ethics approval

This study did not require ethical approval as the data used have been published previously, and hence are already in the public domain.

Funds

This research was financially supported by the National Natural Science Foundation of China (grant No. 81800697).

Acknowledgments

We thank Enago for English language editing.

Conflict of interests

None declared.

Data availability statement

All data relevant to the study are included in the article or uploaded as supplementary information

Reference

[1] Cederholm, T.; Morley, J.E. Sarcopenia: the new definitions. *Current Opinion in Clinical Nutrition and Metabolic Care.* **2015**, *18*, 1–4.

[2] Frontera, W.R.; Hughes, V.A.; Fielding, R.A.; Fiatarone, M.A.; Evans, W.J.; Roubenoff, R. Aging of skeletal muscle: a 12-yr longitudinal study. *J Appl Physiol (1985)*. **2000**, *88*, 1321-1326.

[3] Delmonico, M.J.; Harris, T.B.; Lee, J.S.; Visser, M.; Nevitt, M.; Kritchevsky, S.B.; Tylavsky, F.A.; Newman, A.B. Alternative definitions of sarcopenia, lower extremity performance, and functional impairment with aging in older men and women. *J. Am. Geriatr. Soc.* **2007**, *55*, 769-774..

[4] Woo, J. Sarcopenia. *Clin. Geriatr. Med*. **2017**, *33*, 305–314.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

[5] Cruz-Jentoft, A.J.; Baeyens, J.P.; Bauer, J.M.; Boirie, Y.; Cederholm, T.; Landi, F.; Martin, F.C.; Michel, J.P.; Rolland, Y.; Schneider, S.M.; Topinkova, E.; Vandewoude, M.; Zamboni, M. Sarcopenia: European consensus on definition and diagnosis: Report of the European Working Group on Sarcopenia in Older People. *Age Ageing*. **2010**, *39*, 412-423.

[6] Fielding, R.A.; Vellas, B.; Evans, W.J.; Bhasin, S.; Morley, J.E.; Newman, A.B.; Abellan, V.K.G.; Andrieu, S.; Bauer, J.; Breuille, D.; Cederholm, T.; Chandler, J.; De Meynard, C.; Donini, L.; Harris, T.; Kannt, A.; Keime, G.F.; Onder, G.; Papanicolaou, D.; Rolland, Y.; Rooks, D.; Sieber, C.; Souhami, E.; Verlaan, S.; Zamboni, M. Sarcopenia: an undiagnosed condition in older adults. Current consensus definition: prevalence, etiology, and consequences. International working group on sarcopenia. *J. Am. Med. Dir. Assoc.* **2011**, *12*, 249-256.

[7] Chen, L.K.; Liu, L.K.; Woo, J.; Assantachai, P.; Auyeung, T.W.; Bahyah, K.S.; Chou, M.Y.; Chen, L.Y.; Hsu, P.S.; Krairit, O.; Lee, J.S.; Lee, W.J.; Lee, Y.; Liang, C.K.; Limpawattana, P.; Lin, C.S.; Peng, L.N.; Satake, S.; Suzuki, T.; Won, C.W.; Wu, C.H.; Wu, S.N.; Zhang, T.; Zeng, P.; Akishita, M.; Arai, H. Sarcopenia in Asia: consensus report of the Asian Working Group for Sarcopenia. *J. Am. Med. Dir. Assoc.* **2014**, *15*, 95-101.

[8] Lee, W.; Liu, L.; Peng, L.; Lin, M.; Chen, L. Comparisons of Sarcopenia Defined by IWGS and EWGSOP Criteria Among Older People: Results From the I-Lan Longitudinal Aging Study. *J. Am. Med. Dir. Assoc.* **2013**, *14*, 521-528.

[9] Wu, I.; Lin, C.; Hsiung, C.A.; Wang, C.; Wu, C.; Chan, D.D.; Li, T.; Lin, W.; Huang, K.; Chen, C.; Hsu, C. Epidemiology of sarcopenia among community-dwelling older adults in Taiwan: A pooled analysis for a broader adoption of sarcopenia assessments. *Geriatr. Gerontol. Int.* **2014**, *14*, 52-60.

[10] Wu, C.; Chen, K.; Hou, M.; Chang, Y.; Chang, C.; Liu, P.; Wu, S.; Chiu, C.; Jou, I.; Chen, C. Prevalence and associated factors of sarcopenia and severe sarcopenia in older Taiwanese living in rural community: The Tianliao Old People study 04. *Geriatr. Gerontol. Int.* **2014**, *14*, 69-75.

[11] Li, M.; Hu, Y.; Dong, H.Y.; Zhang, Y.; Fan, L.; Zhang, M.X.; Sun, J.; Han, X.Q.; Liu, Y.X.; Ma, X.N. Compare different measurement for prevalence of sarcopenia in a cohort of healthy community dwellingely in Beijing area. *Chin. J. Health Care Med.* **2014**, *16*, 426-429(Chinese article).

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

[12] Meng, N.; Li, C.; Liu, C.; Lin, C.; Lin, W.; Chang, C.; Li, T.; Lin, C. Comparison of height- and weight-adjusted sarcopenia in a Taiwanese metropolitan older population. *Geriatr. Gerontol. Int.* **2015**, *15*, 45-53.

[13] Wells, G.A.; Shea, B.; O'Connell, D.; Peterson, J.; Welch, V.; Losos, M.; Tugwell, P. The Newcastle-Ottawa Scale (NOS) for assessing the quality of nonrandomised studies in meta-analyses. Available: http://www.ohri.ca/programs/clinical_epidemiology/oxford.asp. Accessed 2014 Aug 5

[14] Higgins, J.P.T.; Green, S. Cochrane Handbook for Systematic Reviews of Interventions Version 5.1.0. Available: http://www.cochrane-handbook.org. Accessed 2014 Aug

[15] Wen, X.; An, P.; Chen, W.C.; Lv, Y.; Fu, Q. Comparisons of sarcopenia prevalence based on different diagnostic criteria in Chinese older adults. *The journal of nutrition, health & aging*. **2015**, *19*, 342-347.

[16] Zeng, P.; Wu, S.; Han, Y.; Liu, J.; Zhang, Y.; Zhang, E.; Zhang, Y.; Gong, H.; Pang, J.; Tang, Z.; Liu, H.; Zheng, X.; Zhang, T. Differences in body composition and physical functions associated with sarcopenia in Chinese elderly: Reference values and prevalence. *Arch. Gerontol. Geriat.* **2015**, *60*, 118-123.

[17] Gao, L.; Jiang, J.; Yang, M.; Hao, Q.; Luo, L.; Dong, B. Prevalence of Sarcopenia and Associated Factors in Chinese Community-Dwelling Elderly: Comparison Between Rural and Urban Areas. *J. Am. Med. Dir. Assoc.* **2015**, *16*, 1001-1003.

[18] Wang, Y.; Wang, Y.; Zhan, J.; Tang, Z.; He, J.; Tan, P.; Deng, H.; Huang, W.; Liu, Y. Sarco-Osteoporosis: Prevalence and Association with Frailty in Chinese Community-Dwelling Older Adults. *Int. J. Endocrinol.* **2015**, *2015*, 1-8.

[19] Woo, J.; Leung, J.; Morley, J.E. Defining Sarcopenia in Terms of Incident Adverse Outcomes. *J. Am. Med. Dir. Assoc.* **2015**, *16*, 247-252.

[20] Wang, H.; Hai, S.; Cao, L.; Zhou, J.; Liu, P.; Dong, B. Estimation of prevalence of sarcopenia by using a new bioelectrical impedance analysis in Chinese community-dwelling elderly people. *BMC Geriatr.* **2016**, *16*.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

[21] Han, P.; Kang, L.; Guo, Q.; Wang, J.; Zhang, W.; Shen, S.; Wang, X.; Dong, R.; Ma, Y.; Shi, Y.; Shi, Z.; Li, H.; Li, C.; Ma, Y.; Wang, L.; Niu, K. Prevalence and Factors Associated With Sarcopenia in Suburb-dwelling Older Chinese Using the Asian Working Group for Sarcopenia Definition. *The Journals of Gerontology Series A: Biological Sciences and Medical Sciences*. **2016**, *71*, 529-535.

[22] Han, D.; Chang, K.; Li, C.; Lin, Y.; Kao, T.; Tsai, K.; Wang, T.; Yang, W. Skeletal muscle mass adjusted by height correlated better with muscular functions than that adjusted by body weight in defining sarcopenia. *Sci. Rep.-UK*. **2016**, *6*.

[23] Xia, Z.W.; Meng, L.P.; Man, Q.Q.; Li, L.X.; Song, P.K.; Li, Y.Q.; Gao, Y.X.; Jia, S.S.; Zhang, J. Analysis of the dietary factors on sarcopenia in elderly in Beijing. *J Hygiene Res.* **2016**, *45*, 388-393 (Chinese article).

[24] Chen, Z.; Dai, R.; Gu, X.H. Prevalence of sarcopenia and quality of life in retired cadres of a military unit. *Chin. J. Osteopo. Rosis. & Bone Miner. Res.* **2016**, *9*, 264-268 (Chinese article).

[25] Bai, H.; Sun, J.; Chen, M.; Xu, D.; Xie, H.; Yu, Z.; Bao, Z.; Chen, J.; Pan, Y.; Lu, D.; Cheng, S. Age-related decline in skeletal muscle mass and function among elderly men and women in Shanghai, China: A cross sectional study. *Asia Pac. J. Clin. Nutr.* **2016**, *25*, 326-332..

[26] Hai, S.; Wang, H.; Cao, L.; Liu, P.; Zhou, J.; Yang, Y.; Dong, B. Association between sarcopenia with lifestyle and family function among community-dwelling Chinese aged 60 years and older. *BMC Geriatr.* **2017**, *17*.

[27] Ma, Y.; Bian, A.L.; Wang, X.F.; Wang, J.X.; Zhou, X.Z. Difference of the prevalence rate of sarcopenia in elderly people with different gender and age in Tianjin. *Chin. General Practice* **2017**, *20*, 3267-3271 (Chinese article).

[28] Wu, Z.N.; Wang, Y.B.; Shu, F.Q.; Song, B.; Li, G.; Zhu, H.M. Epidemiological investigation of sarcopenia in the nursing homes in Fengxian District and analysis of their risk factors. *Shanghai Med.* **2017**, *38*, 37-40 (Chinese article).

[29] Huang, L.J.; Liu, Y.C.; Wen, Y.F.; Liu, K. State and associated factors of sarcopenia in Jinzhou community elderly people. *Modern Preve. Med.* **2017**, *44*, 4458-4460 (Chinese article).

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

[30] Yang, M.; Hu, X.; Xie, L.; Zhang, L.; Zhou, J.; Lin, J.; Wang, Y.; Li, Y.; Han, Z.; Zhang, D.; Zuo, Y.; Li, Y.; Wu, L. Screening Sarcopenia in Community-Dwelling Older Adults: SARC-F vs SARC-F Combined With Calf Circumference (SARC-CalF). *J. Am. Med. Dir. Assoc.* **2018**, *19*, 271-277.

[31] Zeng, Y.; Hu, X.; Xie, L.; Han, Z.; Zuo, Y.; Yang, M. The Prevalence of Sarcopenia in Chinese Elderly Nursing Home Residents: A Comparison of 4 Diagnostic Criteria. *J. Am. Med. Dir. Assoc.* **2018**, *19*, 690-695.

[32] Wang, Y.N.; Xu, J.L.; Song, H.L. Influencing factors of sarcopenia among elderly patients. *Modern Clinical Nursing* **2018**, *17*, 45-50 (Chinese article).

[33] Yang, L.; Wu, G.; Yang, Y.; Wu, Y.; Zhang, L.; Wang, M.; Mo, L.; Xue, G.; Wang, C.; Weng, X. Nutrition, Physical Exercise, and the Prevalence of Sarcopenia in Elderly Residents in Nursing Homes in China. *Med. Sci. Monitor*. **2019**, *25*, 4390-4399.

[34] Du, Y.; Wang, X.; Xie, H.; Zheng, S.; Wu, X.; Zhu, X.; Zhang, X.; Xue, S.; Li, H.; Hong, W.; Tang, W.; Chen, M.; Cheng, Q.; Sun, J. Sex differences in the prevalence and adverse outcomes of sarcopenia and sarcopenic obesity in community dwelling elderly in East China using the AWGS criteria. *BMC Endocr. Disord.* **2019**, *19*.

[35] Wang, H.; Hai, S.; Liu, Y.; Liu, Y.X.; Zhou, J.H.; Yang, Y.; Dong, B.R.; Yue, J.R. Prevalence of sarcopenia and associated factors in community-dwelling elderly populations in Chengdu China. *J. Sichuan Univ.* **2019**, *50*, 224-228 (Chinese article).

[36] Zhang, Y.; Chen, X.; Hou, L.; Lin, X.; Qin, D.; Wang, H.; Hai, S.; Cao, L.; Dong, B. Prevalence and Risk Factors Governing the Loss of Muscle Function in Elderly Sarcopenia Patients: A longitudinal Study in China with 4 Years of Follow-Up. *The journal of nutrition, health & aging*. **2020**, *24*, 518-524.

[37] Liu, X.; Hao, Q.; Yue, J.; Hou, L.; Xia, X.; Zhao, W.; Zhang, Y.; Ge, M.; Ge, N.; Dong, B. Sarcopenia, Obesity and Sarcopenia Obesity in Comparison: Prevalence, Metabolic Profile, and Key Differences: Results from WCHAT Study. *The journal of nutrition, health & aging*. **2020**, *24*, 429-437.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

[38] Shafiee, G.; Keshtkar, A.; Soltani, A.; Ahadi, Z.; Larijani, B.; Heshmat, R. Prevalence of sarcopenia in the world: a systematic review and meta-analysis of general population studies. *Journal of Diabetes & Metabolic Disorders*. **2017**, *16*.

[39] Makizako, H.; Nakai, Y.; Tomioka, K.; Taniguchi, Y. Prevalence of sarcopenia defined using the Asia Working Group for Sarcopenia criteria in Japanese community-dwelling older adults: A systematic review and meta-analysis. *Physical therapy research*. **2019**, *22*, 53-57.

[40] Diz, J.B.M.; Leopoldino, A.A.O.; Moreira, B.D.S.; Henschke, N.; Dias, R.C.; Pereira, L.S.M.; Oliveira, V.C. Prevalence of sarcopenia in older Brazilians: A systematic review and meta-analysis. *Geriatr. Gerontol. Int.* **2017**, *17*, 5-16.

[41] Papadopoulou, S.K.; Tsintavis, P.; Potsaki, G.; Papandreou, D. Differences in the Prevalence of Sarcopenia in Community-Dwelling, Nursing Home and Hospitalized Individuals. A Systematic Review and Meta-Analysis. *The journal of nutrition, health & aging*. **2020**, *24*, 83-90.

[42] Han, A.; Sun, T.; Ming, J.; Chai, L.; Liao, X. Are the Chinese moving toward a healthy diet? Evidence from macro data from 1961 to 2017. *Int J Environ Res Public Health,* **2020**, *17*, 5294

Legend

Figure 1. Flowchart of the detailed procedure for the inclusion or exclusion of selected studies.

Figure 2. Forest plots for the total prevalence of sarcopenia among the elderly Chinese population according to the Asian Working Group for Sarcopenia (AWGS) standard. Diamond represents the pooled odds ratio at 95% CI.

Figure 3. Forest plots for the total prevalence of sarcopenia among the elderly Chinese population according to the International Working Group on Sarcopenia (IWGS) standard. Diamond represents the pooled odds ratio at 95% CI.

Figure 4. Forest plots for the total prevalence of sarcopenia among the elderly Chinese population according to the European Working Group on Sarcopenia in Older People (EWGSOP) standard. Diamond represents the pooled odds ratio at 95% CI.

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Figure 5. Forest plots for the prevalence of sarcopenia among male and female elderly people according to the Asian Working Group for Sarcopenia (AWGS) standard. Diamond represents the pooled odds ratio at 95% CI.

Figure 6. Forest plots for the prevalence of sarcopenia among male and female elderly people according to the International Working Group on Sarcopenia (IWGS) standard. Diamond represents the pooled odds ratio at 95% CI.

Figure 7. Forest plots for the prevalence of sarcopenia among male and female elderly people according to the European Working Group on Sarcopenia in Older People (EWGSOP) standard. Diamond represents the pooled odds ratio at 95% CI.

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Table 1 Study characteristics of the studies included in the meta-analysis

Author	Publication Year	Participants(n)			Region	Setting	Measurement	Diagnostic Criteria	Prevalence (%)			NOS
		Total	Male	Female					Total	Male	Female	
Lee WJ	2013	386	223	163	Taiwan	Urban Community	DEXA	IWGS	11.1	10.8	11.7	6
								EWGSOP	16.6	14.9	19.0	
Wu IC	2014	2155	1055	1100	Taiwan	Urban Community	BIA	EWGSOP	3.9	5.4	2.5	6
Wu CH	2014	549	285	264	Taiwan	Rural Community	BIA	EWGSOP	12.7	8.1	17.8	6
Li M	2014	169	169	0	Beijing	Urban Community	DEXA	EWGSOP	33.3	33.3	-	5
Meng NH	2015	771	359	712	Taiwan	Urban Community	DEXA	EWGSOP	9.7	0.7	12.5	6
Gao LL	2015	612	254	358	Sichuan	Rural and urban Community	Calf circumference	AWGS	9.8	6.7	12.0	6
Wen X	2015	286	136	150	Hangzhou	Urban Community	DEXA	IWGS	6.0	7.4	4.7	5
								EWGSOP	0.4	0.8	0.01	
								AWGS	3.1	5.9	0.7	

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Woo J	2015	4000	2000	2000	Hongkong	Urban Community	DEXA	IWGS	20.2	22.1	18.3	6
								EWGSOP	9.0	9.5	8.6	
								AWGS	7.3	5.3	9.3	
Zeng P	2015	395	199	196	Beijing	Rural and urban Community	BIA	AWGS	24.3	18.1	30.6	5
Wang YJ	2015	316	164	152	Changsha	Urban Community	DEXA	AWGS	29.7	26.2	33.6	6
Han PP	2016	1069	467	602	Tianjin	Urban Community	BIA	AWGS	9.3	6.4	17.5	5
Wang H	2016	636	316	320	Chengdu	Urban Community	BIA	AWGS	10.4	8.2	12.5	5
Bai HJ	2016	415	212	203	Shanghai	Urban Community	BIA	AWGS	28.7	33.5	23.6	6
Xia ZW	2016	683	239	444	Beijing	Urban Community	BIA	AWGS	20.1	17.2	21.6	5
Chen Z	2016	281	203	78	Chongqing	Urban Community	BIA	IWGS	27.8	-	-	6
Han DS	2016	878	402	476	Taiwan	Urban Community	BIA	EWGSOP	9.8	4.5	14.3	6
Hai S	2017	834	415	419	Chengdu	Urban Community	BIA	AWGS	10.6	11.3	9.8	6
Huang	2017	193	65	128	Jinzhou	Urban Community	BIA	AWGS	11.4	7.7	13.3	6

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Yang M	2018	384	160	224	Chengdu	Urban Community	BIA	AWGS	15.9	11.9	18.8	6
								IWGS	25.0	24.4	25.4	
								EWGSOP	11.7	10.6	12.5	
Du YP	2019	631	213	418	Shanghai	Urban Community	BIA	AWGS	12.2	19.2	8.6	5
Wang H	2019	947	465	482	Chengdu	Urban Community	BIA	AWGS	7.7	6.5	8.9	5
Zhang Y	2020	474	238	236	Chengdu	Urban Community	BIA	AWGS	15.4	-	-	5
Liu XL	2020	4500	1627	2873	Yunnan, Guizhou, Sichuan, and Xinjiang	Urban Community	BIA	AWGS	19.3	22.1	17.8	6

DEXA, dual-energy x-ray absorptiometry; BIA, bioelectrical impedance analysis; IWGS, International Working Group on Sarcopenia; AWGS, Asia Working Group for Sarcopenia; EWGSOP, European Working Group on Sarcopenia in Older People; NOS, Newcastle–Ottawa scale.

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Table 2 The prevalence of sarcopenia diagnosed by different criteria using DEXA and BIA measurement (%).

	DEXA	BIA
AWGS	13(5−21)	15(12−19)
IWGS	12(3−22)	26(23−29)
EWGSOP	12(6−18)	9(5−14)

DEXA, dual-energy x-ray absorptiometry; BIA, bioelectrical impedance analysis; IWGS, International Working Group on Sarcopenia; AWGS, Asia Working Group for Sarcopenia; EWGSOP, European Working Group on Sarcopenia in Older People.

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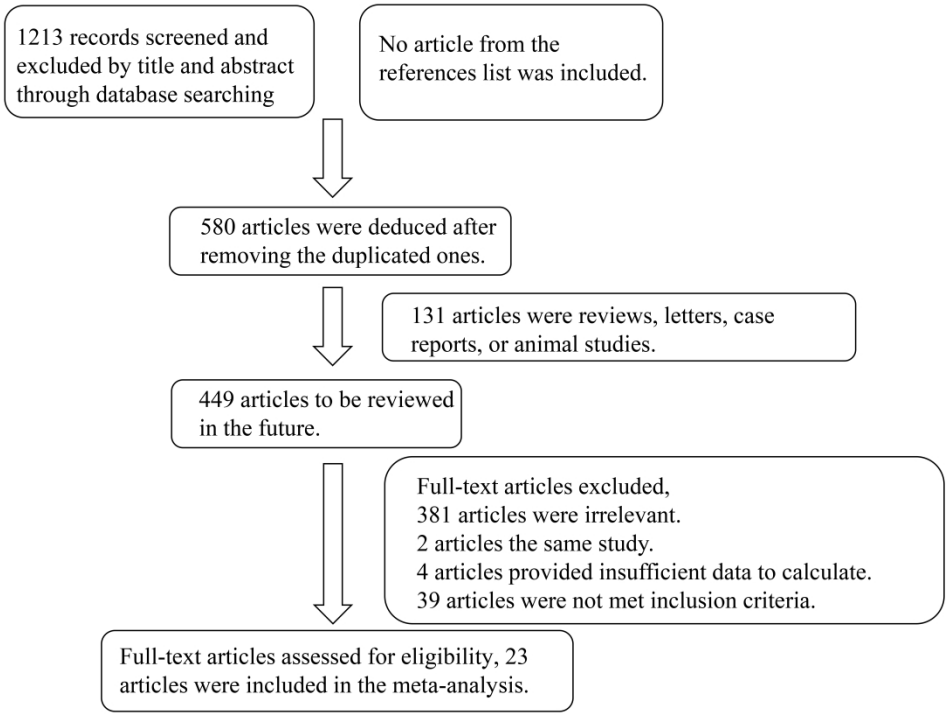


Figure 1. Flowchart of the detailed procedure for the inclusion or exclusion of selected studies.

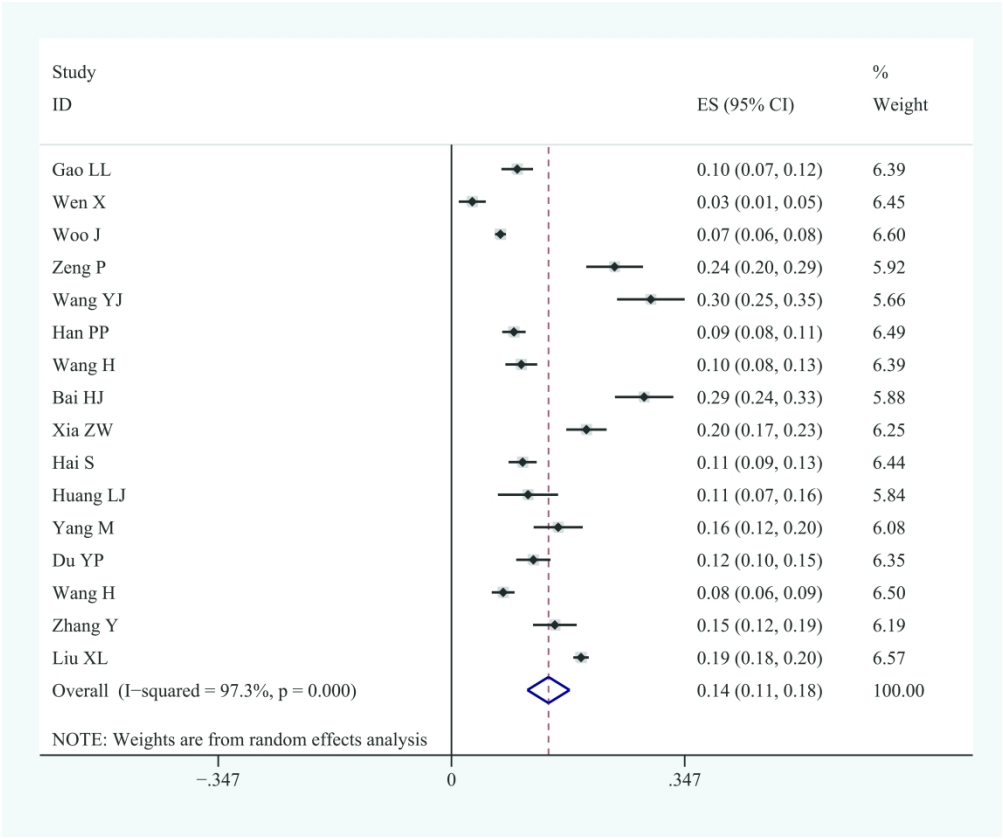


Figure 2. Forest plots for the total prevalence of sarcopenia among the elderly Chinese population according to the Asian Working Group for Sarcopenia (AWGS) standard. Diamond represents the pooled odds ratio at 95% CI.

280x233mm (600 x 600 DPI)

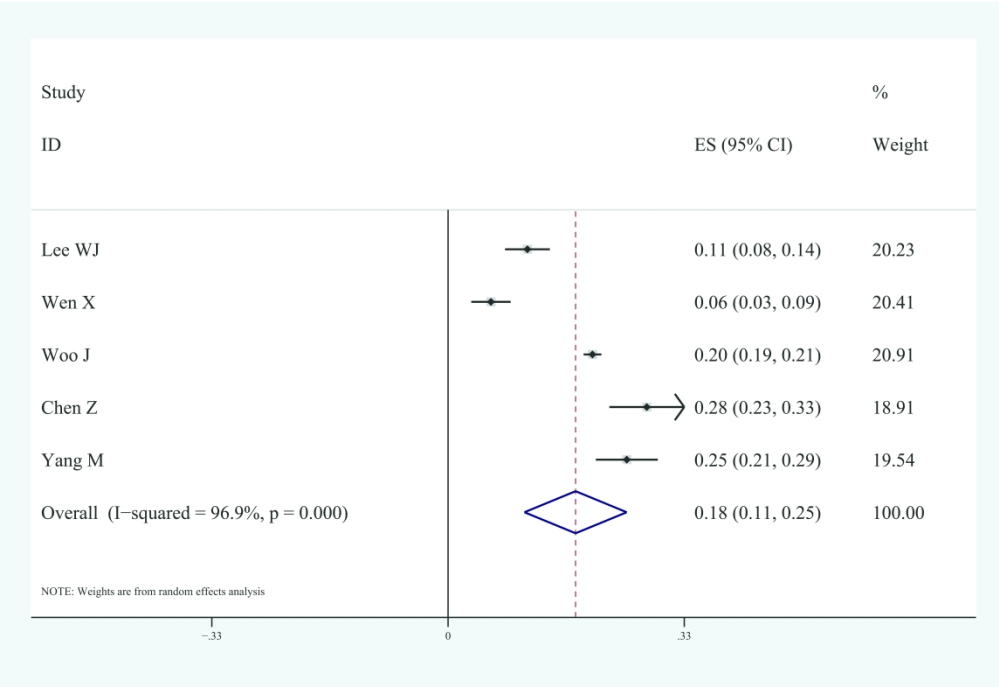


Figure 3. Forest plots for the total prevalence of sarcopenia among the elderly Chinese population according to the International Working Group on Sarcopenia (IWGS) standard. Diamond represents the pooled odds ratio at 95% CI.

272x186mm (600 x 600 DPI)

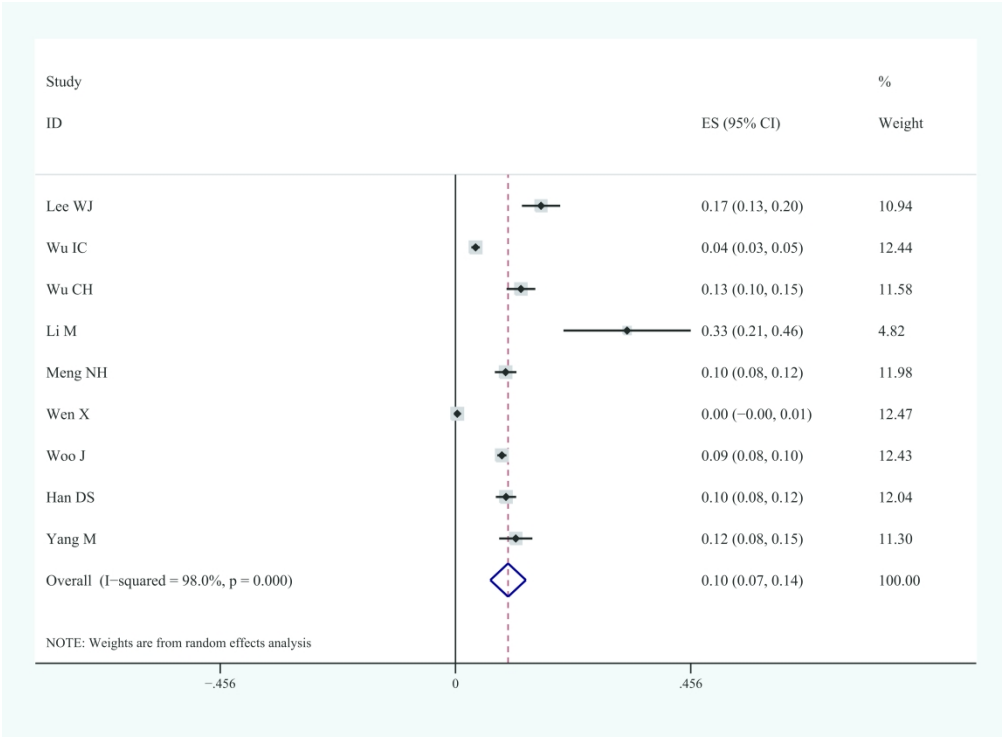


Figure 4. Forest plots for the total prevalence of sarcopenia among the elderly Chinese population according to the European Working Group on Sarcopenia in Older People (EWGSOP) standard. Diamond represents the pooled odds ratio at 95% CI.

274x201mm (600 x 600 DPI)

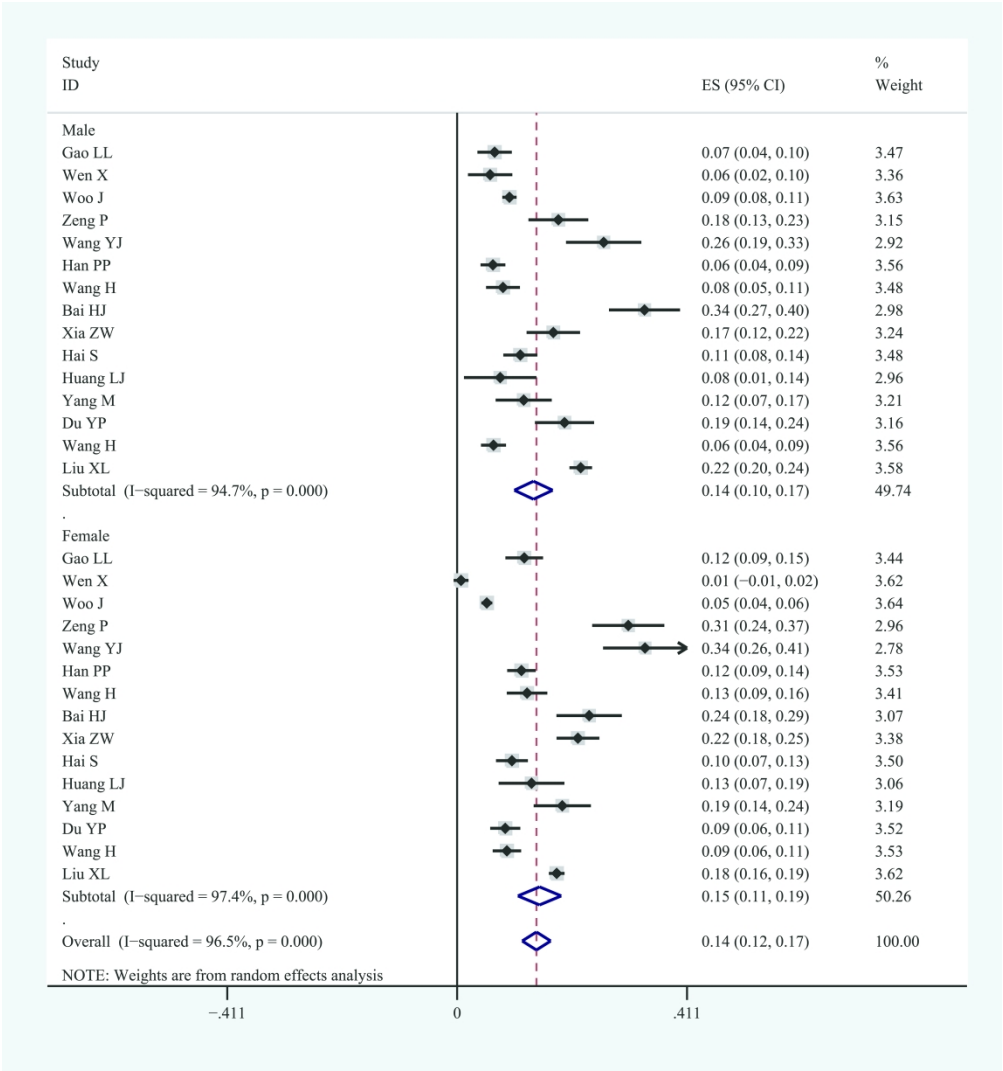


Figure 5. Forest plots for the prevalence of sarcopenia among male and female elderly people according to the Asian Working Group for Sarcopenia (AWGS) standard. Diamond represents the pooled odds ratio at 95% CI.

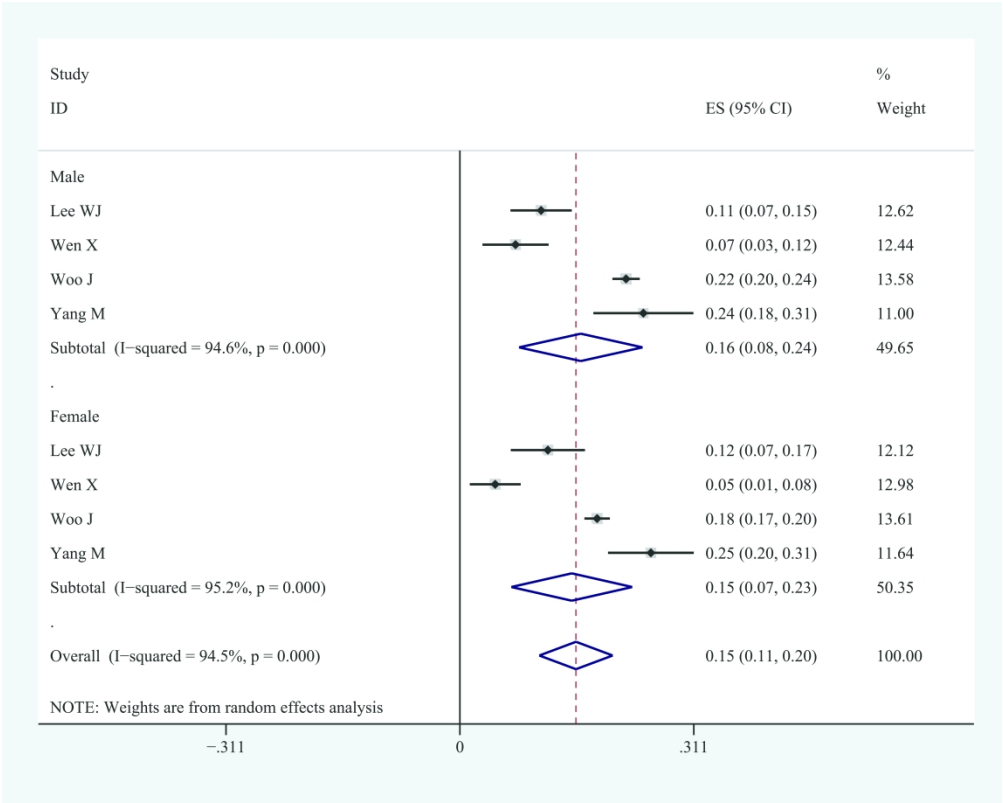


Figure 6. Forest plots for the prevalence of sarcopenia among male and female elderly people according to the International Working Group on Sarcopenia (IWGS) standard. Diamond represents the pooled odds ratio at 95% CI.

278x223mm (600 x 600 DPI)

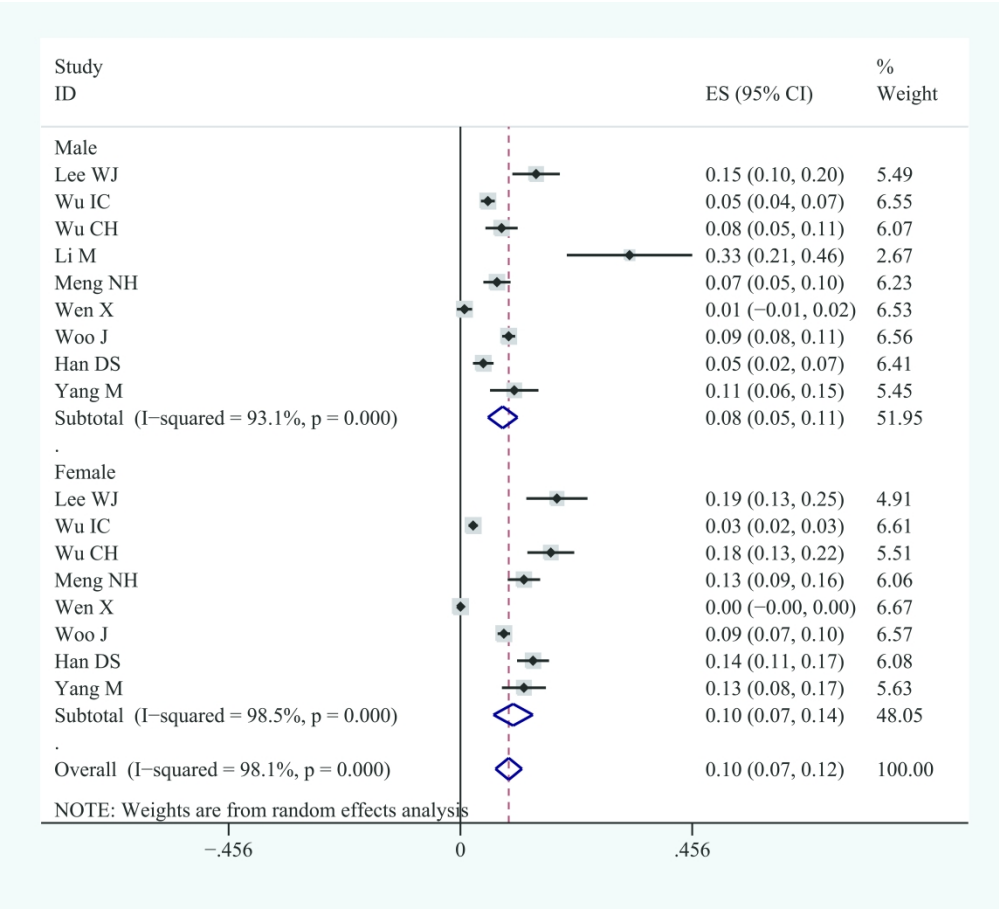


Figure 7. Forest plots for the prevalence of sarcopenia among male and female elderly people according to the European Working Group on Sarcopenia in Older People (EWGSOP) standard. Diamond represents the pooled odds ratio at 95% CI.

284x257mm (600 x 600 DPI)

Figure S1. The sensitivity analysis results according to the Asian Working Group for Sarcopenia (AWGS) standard.

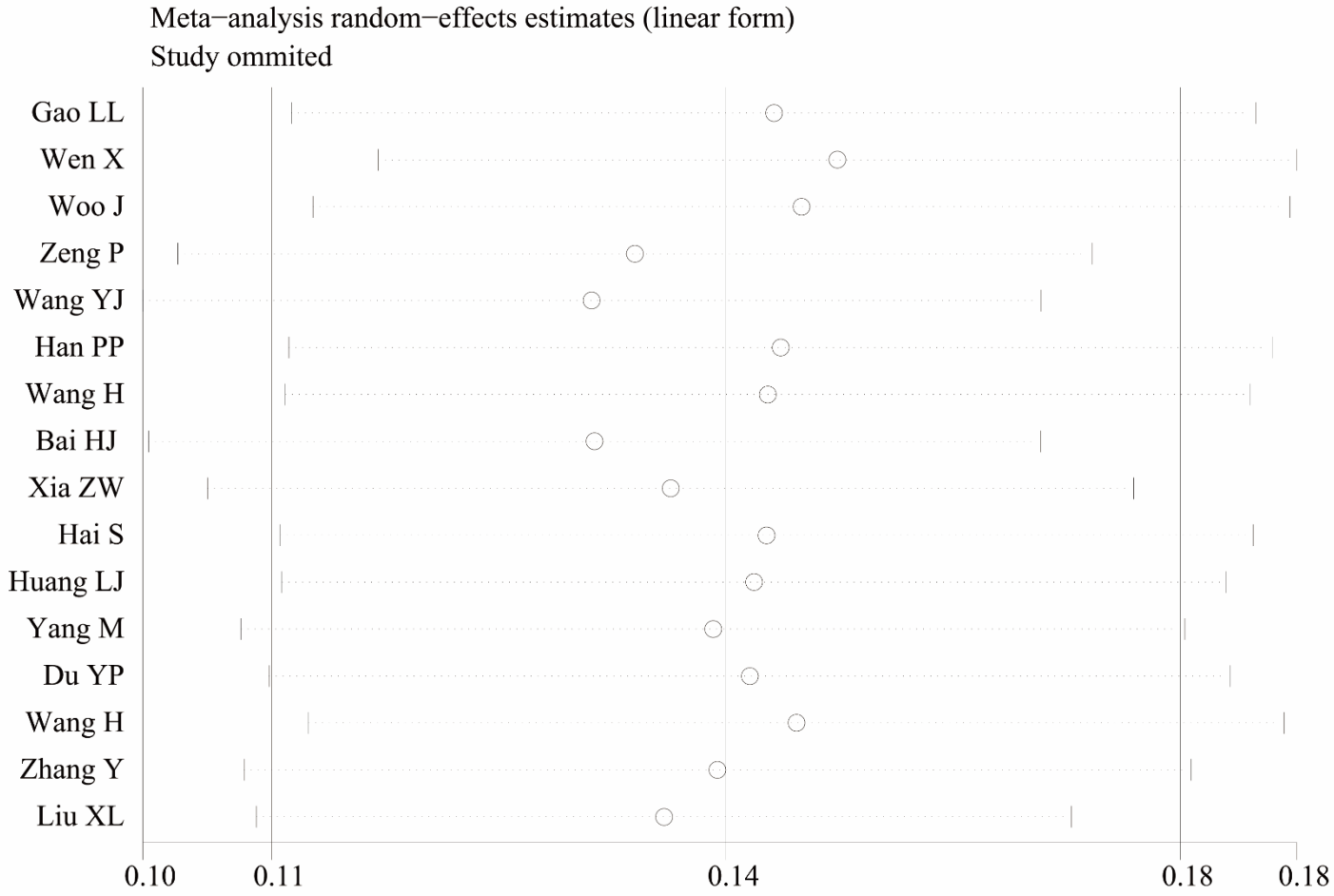


Figure S2. The sensitivity analysis results according to the International Working Group on Sarcopenia (IWGS) IWGS standard.

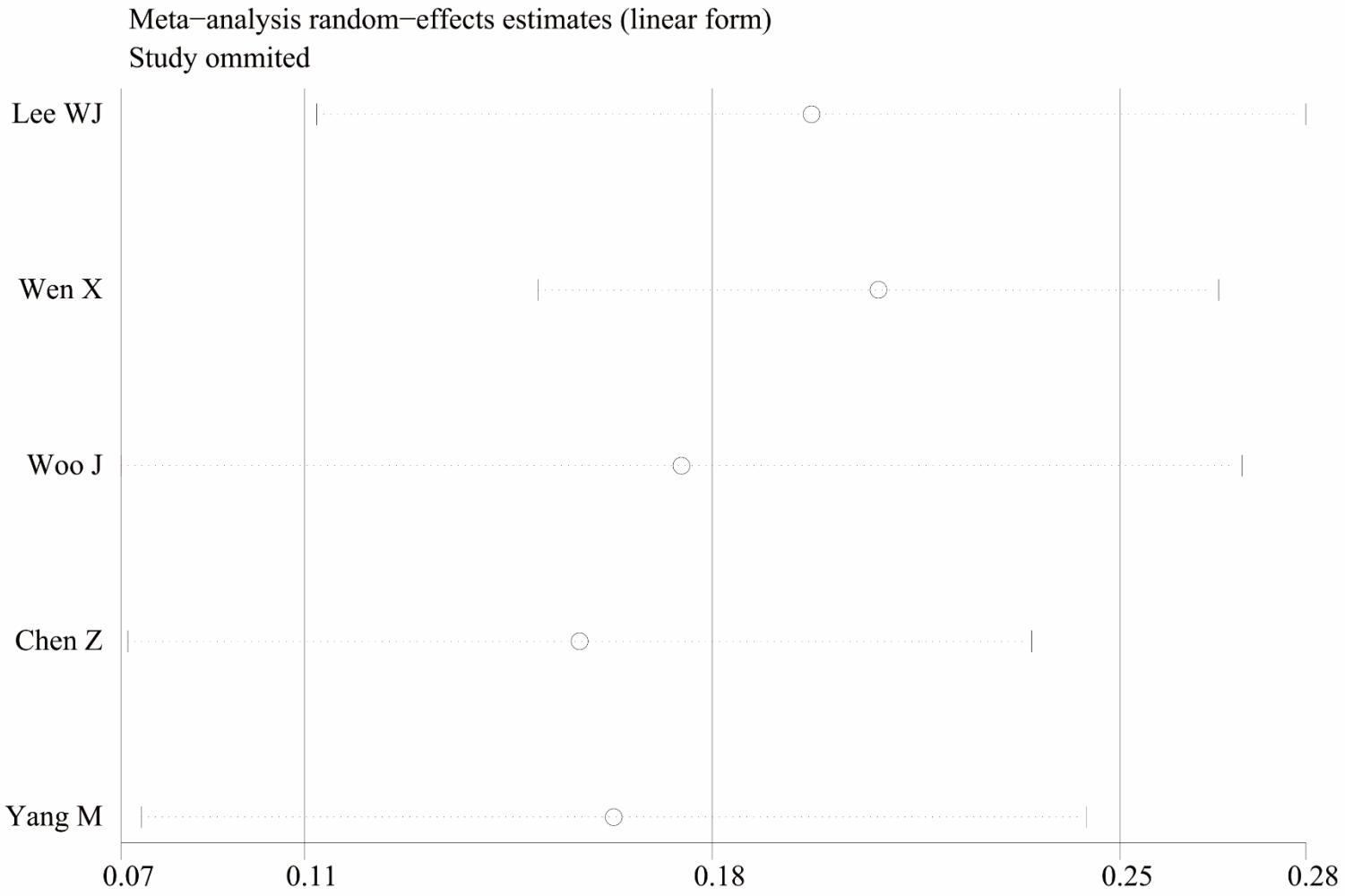


Figure S3. The sensitivity analysis results according to the European Working Group on Sarcopenia in Older People (EWGSOP) standard.

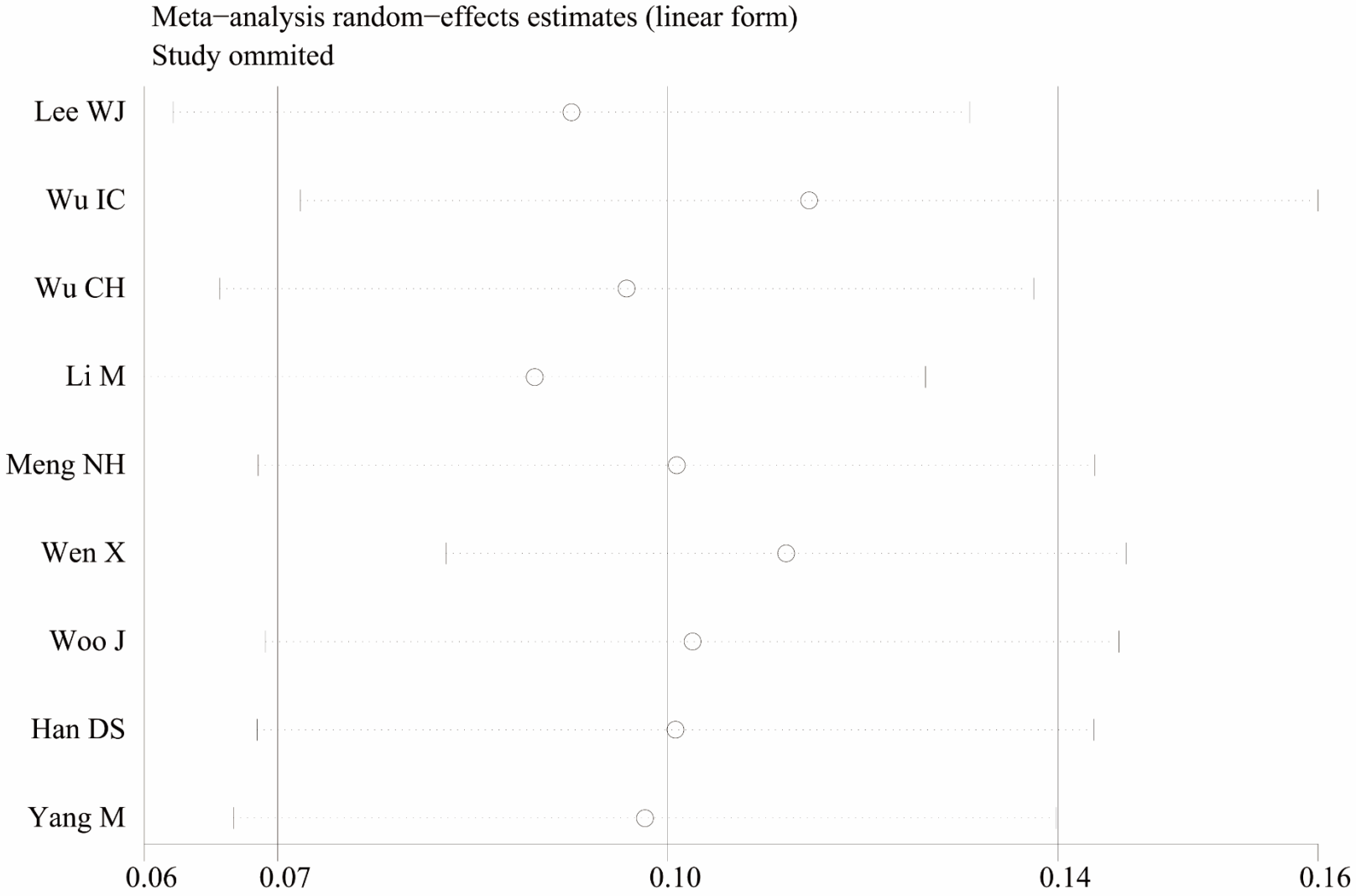


Figure S4. The publication bias results according to the Asian Working Group for Sarcopenia (AWGS) standard.

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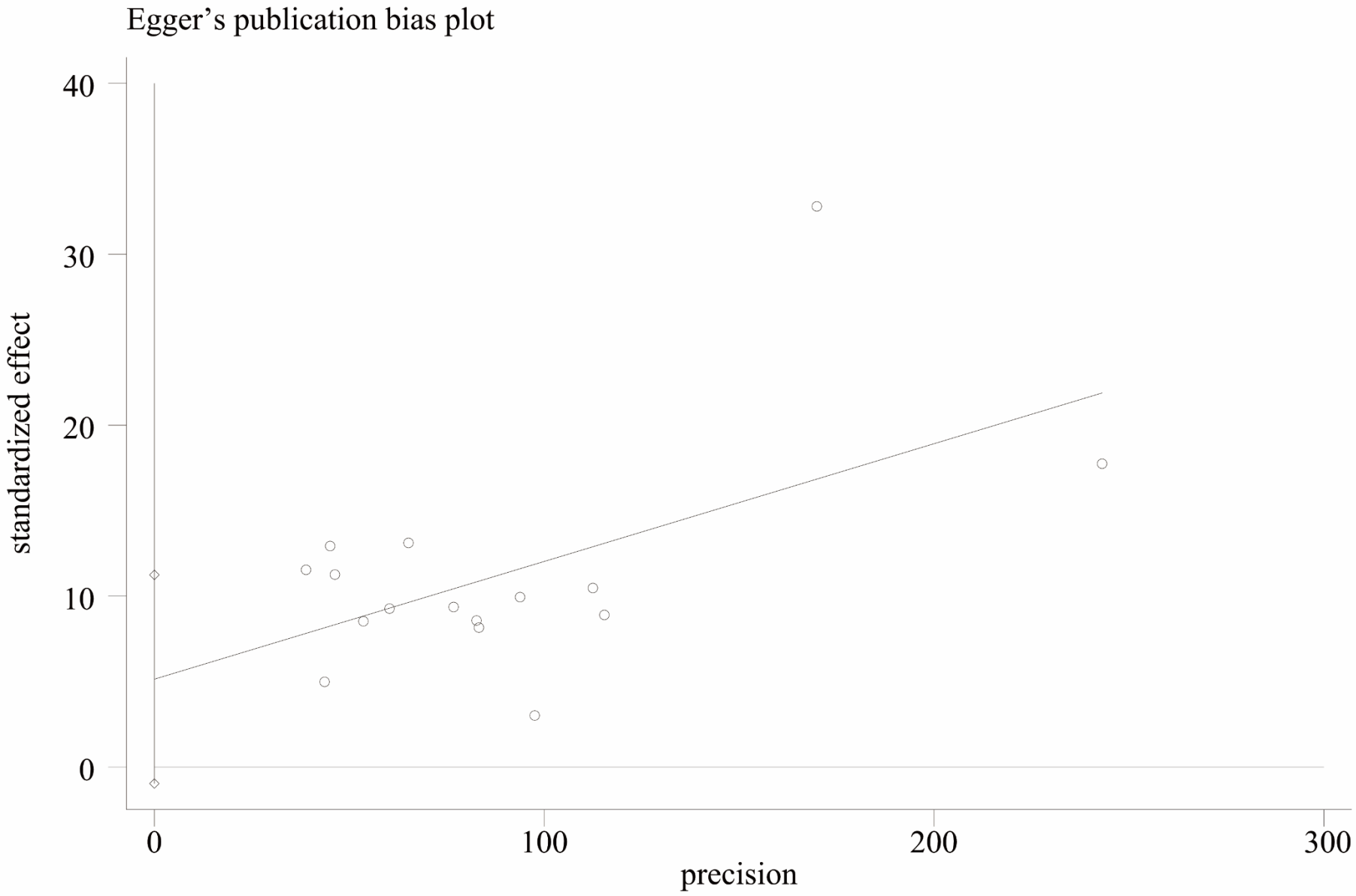


Figure S5. The publication bias results according to the the International Working Group on Sarcopenia (IWGS) standard.

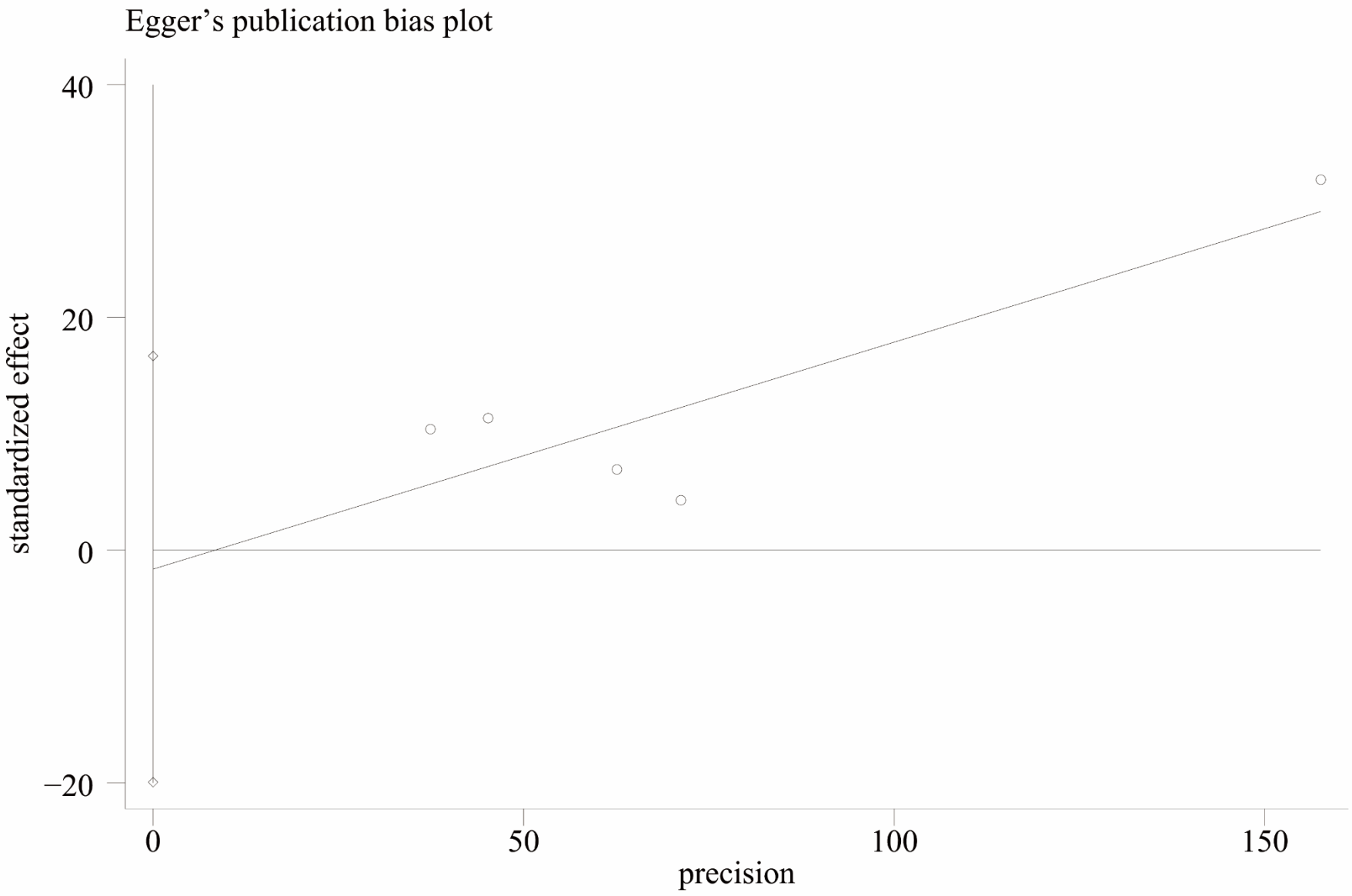
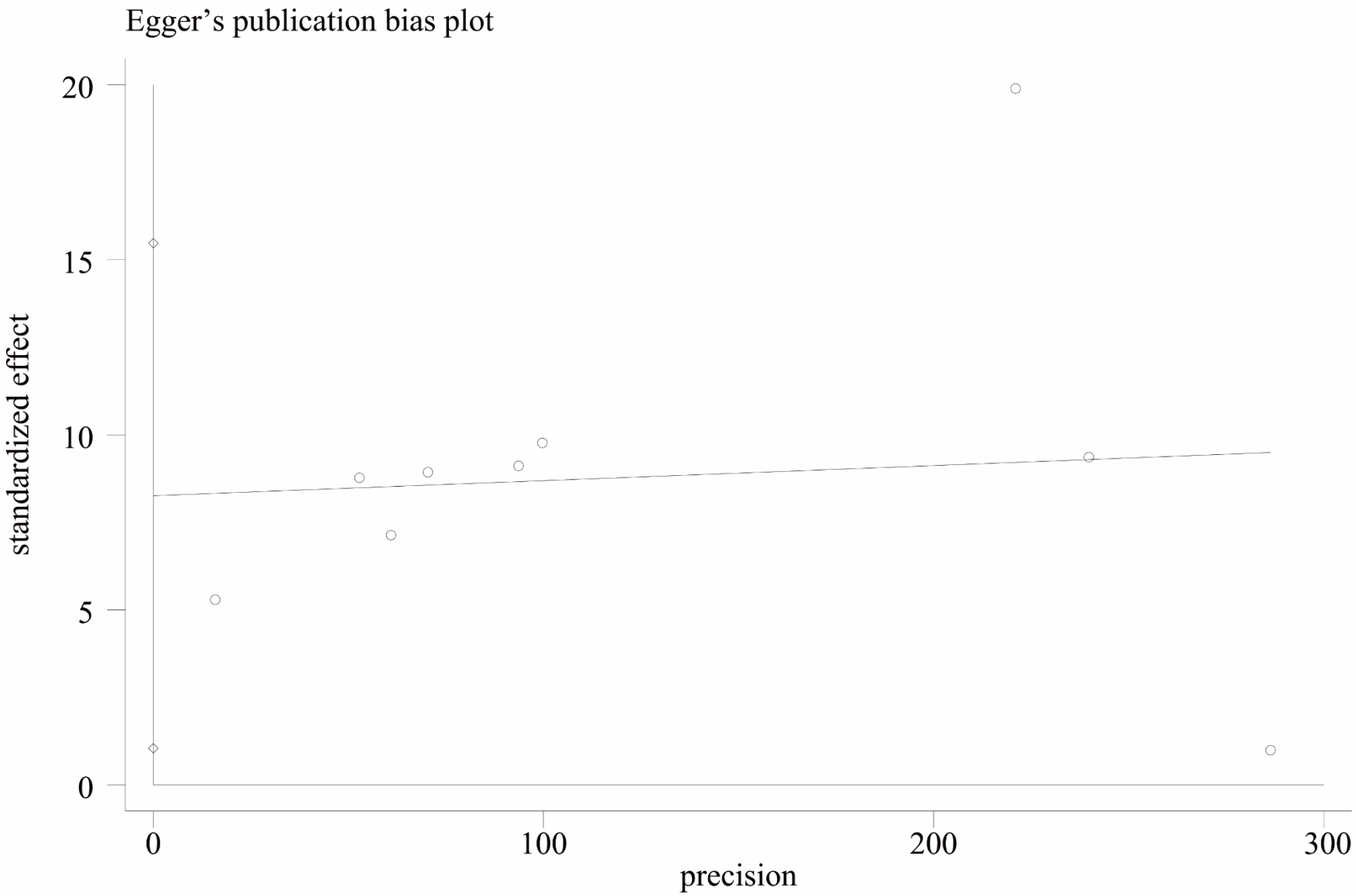


Figure S6. The publication bias results according to the European Working Group on Sarcopenia in Older People (EWGSOP) standard.


Egger's publication bias plot
standardized effect
precision
20
15
10
5
0
0
100
200
300

PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	3
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	3,4
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	4
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	4
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	4
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	4
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	4
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	5
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	5
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	5
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	5
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	5

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	5
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	5
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	6
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	6
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	6
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	6
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	6,7
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	7
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	7
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	7,8
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	8
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	8
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	9

From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(7): e1000097. doi:10.1371/journal.pmed1000097

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Prevalence of Sarcopenia in Older Chinese Adults: A Systematic Review and Meta-analysis

Journal:	*BMJ Open*
Manuscript ID	bmjopen-2020-041879.R5
Article Type:	Original research
Date Submitted by the Author:	15-Jun-2021
Complete List of Authors:	Xin, Caihong; Fourth People's Hospital in Shenyang, Department of Endocrinology and Metabolism Sun, Xin; First Affiliated Hospital of Soochow University, Department of Endocrinology and Metabolism; People's Hospital of Liaoning Province, Department of Endocrinology and Metabolism Lu, Li; People's Hospital of Liaoning Province, Department of Endocrinology and Metabolism Shan, Ling; People's Hospital of Liaoning Province, Department of Endocrinology and Metabolism
Primary Subject Heading:	Geriatric medicine
Secondary Subject Heading:	Epidemiology, Global health, Public health
Keywords:	GERIATRIC MEDICINE, EPIDEMIOLOGY, General endocrinology < DIABETES & ENDOCRINOLOGY

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Prevalence of Sarcopenia in Older Chinese Adults: A Systematic Review and Meta-analysis

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Running title: Prevalence of sarcopenia in

Key words: Prevalence; sarcopenia; China

Word count of the manuscript: 1864

Figures number: 7

Tables number: 2

Disclosure Statement: The authors have nothing to disclose.

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Abstract

Objective: This meta-analysis was conducted to estimate the overall prevalence of sarcopenia in older Chinese adults.

Design: Systematic review and meta-analysis

Participants: A literature research was conducted using the PubMed, Web of Science, CNKI, CQVIP, and Wanfang databases. The following search terms in the abstract were used: "sarcopenia" in combination with the terms "prevalence," "epidemiology," and "China." All studies published from January 2010 to November 2020 were included. The random effect model was used to estimate the prevalence of sarcopenia. The sex-specific prevalence of sarcopenia at a 95% confidence interval (CI) was also calculated using different criteria for defining sarcopenia.

Primary outcome measures: The overall prevalence of sarcopenia in older Chinese adults.

Results: In total, 23 articles were included in this meta-analysis involving 21,564 participants. On the basis of the Asian Working Group for Sarcopenia (AWGS) criterion, the overall prevalence of sarcopenia among the elderly in China was 14% (95% CI, 11%–18%), the prevalence was higher in Chinese females than in males (15% vs. 14%).

Conclusions: This systematic review is the first estimation of the pooled prevalence of sarcopenia in older Chinese adults. Our results suggest that China has a large number of sarcopenia patients. These findings would be useful for sarcopenia prevention in China. There is a high degree of heterogeneity and although there are a large number of cases and could be an emerging public health issue, more research is required to make these claims.

Trial registration: PROSPERO CRD42020223405

Strengths and limitations of this study:

- To our knowledge, this systematic review is the first estimation of the pooled prevalence of sarcopenia in Chinese older people.
- The search languages were limited to English and Chinese.
- We included limited number of high-quality studies.
- A high level of heterogeneity among included studies was observed.

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Introduction

Sarcopenia is characterized by the deterioration of muscle mass and function that occurs with aging. Several studies have shown that as age increases, skeletal muscle mass and strength are lost at a rate of 1%–2% annually, possibly leading to the decline of physical function and increasing the disability and mortality rates of elderly people as well as bringing serious economic and social burdens [1–4].

A practical clinical definition was reported by the European Working Group on Sarcopenia in Older People (EWGSOP) in 2010 [5]. Based on this EWGSOP criteria, slow walking speed (0.8 m/sec) or low grip strength (<30 kg for men and <20 kg for women) are indications of muscle mass decline. Later in 2011, another consensus was published by the International Working Group on Sarcopenia (IWGS) [6]. However, Asia is a vast region comprising different races, countries, and cultures, and developing screening and diagnostic strategies for the Asian population is imperative. The Asian Working Group for Sarcopenia (AWGS) was established in 2013, and the consensus of AWGS was released based on 16 epidemiologic reports on sarcopenia in six countries or regions, leading to the development of a diagnosis strategy for sarcopenia in Asia [7]. The diagnostic criteria recommended by AWGS and EWGSOP are similar, and both include muscle mass, muscle strength, and activity measurements. The only difference lies in the specific cut-off value of muscle strength and muscle mass.

According to statistics, the elderly population (≥60 years old) includes 16.1% of China's total population, which is close to 250 million. In the face of a severe aging situation, people are paying more attention to sarcopenia, but its prevalence is inconsistent across different studies [8–12]. At present, there is no large-scale epidemiologic survey of sarcopenia in China. The purpose of this study is to systematically determine the prevalence of sarcopenia in older Chinese adults.

Methods

Search strategy

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We searched the PubMed, Web of Science, China National Knowledge Infrastructure (CNKI), CQVIP, and Wanfang electronic databases. The search terms in the title or abstract included the following: "sarcopenia," "muscle mass," and "muscle strength" in combination with the terms "prevalence," "epidemiology," and "China." The complete search strategy are below: PubMed: (((sarcopenia [Title/Abstract]) OR muscle mass[Title/Abstract]) OR muscle strength[Title/Abstract]) AND ((prevalence[Title/Abstract]) OR epidemi*[Title/Abstract]) AND ((Chinese[Title/Abstract]) OR China[Title/Abstract]); Web of Science: (((TS=sarcopenia) OR TS=muscle mass) OR TS=muscle strength) AND ((TS=prevalence) OR TS=epidemi*) AND ((TS=Chinese) OR TS=China); CNKI: (AB =sarcopenia OR AB =muscle mass OR AB =muscle strength) AND (AB =prevalence OR AB =epidemiology); CQVIP: (R=sarcopenia +R=muscle mass+ R=muscle strength) *(R=prevalence +R=epidemiology); Wangfang: Abstract: (sarcopenia + muscle mass + muscle strength) * (prevalence + epidemiology). All studies published from January 2010 to November 2020 were included. The references of the retrieved articles were examined to identify additional eligible studies, and unpublished studies were excluded. The search languages were limited to English and Chinese. This systematic review and meta-analysis was registered in PROSPERO (registration number: CRD42020223405). This meta-analysis was conducted to estimate the overall prevalence of sarcopenia in older Chinese adults.

Patient and public involvement

No patients were involved in this study.

Inclusion and exclusion criteria

Inclusion of the selected studies was based on the following criteria: (1) a cross-sectional study or retrospective study, (2) a sample size of ≥100 elderly Chinese participants (age ≥60 years); (3) sufficient information on sarcopenia at the corresponding 95% confidence interval (CI) and *P* value, and (4) the definition of sarcopenia in line with the AWGS, EWGSOP, or IWGS standards.

Exclusion of the selected studies was based on the following criteria: (1) same data from different studies, (2) studies defining sarcopenia only as reduction of muscle mass or strength, and (3) the subjects include a specific population, such as patients living in a nursing home or those with disability. Reviews, case reports, letters, and animal researches were also excluded.

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Data extraction and methodological quality assessment

Information from the selected studies was extracted by two independent researchers, including the first author's name, publication year, sample, diagnostic criteria, participants, setting, and prevalence. Two independent reviewers who were not blinded to the authors or journals assessed the risk of bias in the included studies using the Newcastle–Ottawa scale (NOS) [13]. The NOS is recommended by the Cochrane Handbook for Systematic Reviews of Interventions [14]. Each included study was analyzed in terms of three domains, namely, representativeness of the study group selection, group comparability, and exposure or outcome ascertainment, using the "star system." NOS scores ranged from zero to nine stars. The two researchers independently assessed the studies by discussion, comparison of findings, and resolution of differences through consensus formation. If no consensus was reached, a third researcher was commissioned to resolve the difference.

Statistical analysis

We used a systematic analysis approach to calculate the pooled prevalence of sarcopenia for all eligible studies, and heterogeneity among these studies was assessed using the Cochran's Q test and I^2 statistical index. The random effect model was used to combine the prevalence of sarcopenia. We additionally performed sensitivity analysis to evaluate the influence of any given study on the pooled estimate. Publication bias was evaluated using Egger's test. Subgroup analyses were also performed. The significance level was set at $P < 0.05$. All statistical analyses were performed using Stata software version 12.0 (College Station, Texas, USA).

Results

Study characteristics

In total, 1213 studies were retrieved from the databases and 580 articles were remained after the exclusion of duplicates. After screening for eligibility based on the full-text, a total of 23 articles were included as per the criteria shown in Figure 1 [8–12, 15–37]. Altogether, the 23 articles included 21,564 participants in China between 2010 and 2020. The selected studies' characteristics are summarized in Table 1. On the basis of the NOS scale, the score of selected studies were all above five stars; therefore, all articles were included.

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Results of the meta-analysis

According to the AWGS standard, the total prevalence of sarcopenia among Chinese elderly people was 14% (95% CI, 11%–18%, Figure 2), and total prevalence rates of sarcopenia among the elderly in China based on the IWGS and EWGSOP standards were 18% (95% CI, 11%–25%, Figure 3) and 10% (95% CI, 7%–14%, Figure 4), respectively.

Subgroup analyses were performed to determine the influence of sex on the prevalence of sarcopenia. Sarcopenia prevalence among elderly females was higher than that in males as per the AWGS and EWGSOP standards (Figures 5–7). Table 2 shows the subgroup analyses results based on the different diagnostic criteria. The prevalence of sarcopenia estimated using the bio-electrical impedance analysis (BIA) method was higher than that estimated using the dual energy X-ray absorptiometry (DEXA) method as per the AWGS and IWGS standards.

Sensitivity analysis and publication bias

Sensitivity analysis was performed to examine the influence of each study. We found no significant difference between the results of the sensitivity analysis and our previous estimates, indicating that our statistical results were relatively credible (Figures S1–S3). The articles obtained from the databases were carefully and comprehensively searched. Additionally, Egger's test was conducted to determine whether potential publication bias existed in the reviewed literature. The results ($P > 0.05$) suggested that there were no publication bias (Figures S4–S6).

Discussion

To the best of our knowledge, this study is the first meta-analysis of the prevalence of sarcopenia in older Chinese adults. Sarcopenia is an age-related disease characterized by progressive deterioration in skeletal muscle fiber quality, strength, and function. Currently, sarcopenia has been internationally recognized as a new type of geriatric syndrome and has attracted worldwide attention. In 2017, Shafeee et al. conducted a meta-analysis on the prevalence of sarcopenia. Their results showed that sarcopenia prevalence among the elderly in communities worldwide was approximately 10%; this was higher among non-Asian individuals than Asians in both sexes [38]. A similar study was performed in Japan, where sarcopenia prevalence among the elderly Japanese was 9.9% (95% CI, 6.2%–15.4%) according to the AWGS

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standard [39]. The results of our study showed that the prevalence of sarcopenia was 14% (95% CI, 11%–18%) in older Chinese adults, which was much higher than the results of the aforementioned studies [40, 41]. This high prevalence may be related to the Chinese diet structure. Although protein is the raw material for muscle synthesis, Chinese people, particularly the older population, have higher carbohydrate intake and less protein intake [42]. Therefore, more attention should be given to the prevention and control of sarcopenia in older Chinese adults.

Here, we found that the prevalence of sarcopenia among females was higher than that in males, which was consistent with other meta-analyses [38–40]. This is mainly because men eat and exercise more than women; this supports muscle synthesis. Most of the studies included in our meta-analysis used the BIA method because of its affordability and portable equipment. BIA is a widely used non-invasive method for measuring human body composition. The principle behind this method is to record the different electrical resistances of different tissues using electrodes placed on the body surface and then to measure muscle mass using an image reconstruction method. DEXA is also commonly used to evaluate muscle mass. DEXA can accurately distinguish whole-body and local muscles as well as fats and bones. However, the limited number of studies using DEXA included in this meta-analysis may influence our results. As is known, aging increases the prevalence of sarcopenia; however, due to the different age classifications in the included studies, we cannot obtain any conclusion.

The meta-analysis results show that the heterogeneity is high, which may be related to factors such as age, race, and living environment. In addition, there are differences in the prevalence of sarcopenia in rural and urban areas. China is a country with a vast territory and multi-ethnic people. The findings of this meta-analysis should be interpreted with caution.

The limitation of the present study is that the main data are from urban populations and that studies on the prevalence of sarcopenia in Chinese rural areas are rare; this impacts the calculation of the total sarcopenia prevalence. Second, the prevalence of sarcopenia in people aged 60–70 years and 70–80 years might be different. Not all studies presented their data according to age group, so we cannot accurately estimate the prevalence in terms of age subgroup. Third, most of the studies were performed in large cities or economically developed areas. All of these factors affected our results, and as such, they should be interpreted cautiously as further research is needed.

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Conclusions

To the best of our knowledge, the present systematic review is the first to estimate the pooled prevalence of sarcopenia in older Chinese adults. The results of the present systematic review reveal that China has a large number of sarcopenia cases and sarcopenia is an emerging public health issue in China. There is a high degree of heterogeneity and although there are a large number of cases and could be an emerging public health issue, more research is required to make these claims.

Author contributions statement

SL designed the study. SX and LL searched databases and collected the data. XCH and SL assessed the quality of the study. SL performed the analysis. SX and XCH wrote the manuscript. SX modified the manuscript in the revision. All authors contributed to this systematic review and meta-analysis.

Supporting Information

Figure S1. The sensitivity analysis results according to the Asian Working Group for Sarcopenia (AWGS) standard.

Figure S2. The sensitivity analysis results according to the the International Working Group on Sarcopenia (IWGS) standard.

Figure S3. The sensitivity analysis results according to the European Working Group on Sarcopenia in Older People (EWGSOP) standard.

Figure S4. The publication bias results according to the Asian Working Group for Sarcopenia (AWGS) standard.

Figure S5. The publication bias results according to the the International Working Group on Sarcopenia (IWGS) standard.

Figure S6. The publication bias results according to the European Working Group on Sarcopenia in Older People (EWGSOP) standard.

Patient consent

Not required.

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Ethics approval

This study did not require ethical approval as the data used have been published previously, and hence are already in the public domain.

Funds

This research was financially supported by the National Natural Science Foundation of China (grant No. 81800697).

Acknowledgments

We thank Enago for English language editing.

Conflict of interests

None declared.

Data availability statement

All data relevant to the study are included in the article or uploaded as supplementary information

Reference

[1] Cederholm, T.; Morley, J.E. Sarcopenia: the new definitions. *Current Opinion in Clinical Nutrition and Metabolic Care.* **2015**, *18*, 1–4.

[2] Frontera, W.R.; Hughes, V.A.; Fielding, R.A.; Fiatarone, M.A.; Evans, W.J.; Roubenoff, R. Aging of skeletal muscle: a 12-yr longitudinal study. *J Appl Physiol (1985)*. **2000**, *88*, 1321-1326.

[3] Delmonico, M.J.; Harris, T.B.; Lee, J.S.; Visser, M.; Nevitt, M.; Kritchevsky, S.B.; Tylavsky, F.A.; Newman, A.B. Alternative definitions of sarcopenia, lower extremity performance, and functional impairment with aging in older men and women. *J. Am. Geriatr. Soc.* **2007**, *55*, 769-774..

[4] Woo, J. Sarcopenia. *Clin. Geriatr. Med*. **2017**, *33*, 305–314.

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[5] Cruz-Jentoft, A.J.; Baeyens, J.P.; Bauer, J.M.; Boirie, Y.; Cederholm, T.; Landi, F.; Martin, F.C.; Michel, J.P.; Rolland, Y.; Schneider, S.M.; Topinkova, E.; Vandewoude, M.; Zamboni, M. Sarcopenia: European consensus on definition and diagnosis: Report of the European Working Group on Sarcopenia in Older People. *Age Ageing*. **2010**, *39*, 412-423.

[6] Fielding, R.A.; Vellas, B.; Evans, W.J.; Bhasin, S.; Morley, J.E.; Newman, A.B.; Abellan, V.K.G.; Andrieu, S.; Bauer, J.; Breuille, D.; Cederholm, T.; Chandler, J.; De Meynard, C.; Donini, L.; Harris, T.; Kannt, A.; Keime, G.F.; Onder, G.; Papanicolaou, D.; Rolland, Y.; Rooks, D.; Sieber, C.; Souhami, E.; Verlaan, S.; Zamboni, M. Sarcopenia: an undiagnosed condition in older adults. Current consensus definition: prevalence, etiology, and consequences. International working group on sarcopenia. *J. Am. Med. Dir. Assoc.* **2011**, *12*, 249-256.

[7] Chen, L.K.; Liu, L.K.; Woo, J.; Assantachai, P.; Auyeung, T.W.; Bahyah, K.S.; Chou, M.Y.; Chen, L.Y.; Hsu, P.S.; Krairit, O.; Lee, J.S.; Lee, W.J.; Lee, Y.; Liang, C.K.; Limpawattana, P.; Lin, C.S.; Peng, L.N.; Satake, S.; Suzuki, T.; Won, C.W.; Wu, C.H.; Wu, S.N.; Zhang, T.; Zeng, P.; Akishita, M.; Arai, H. Sarcopenia in Asia: consensus report of the Asian Working Group for Sarcopenia. *J. Am. Med. Dir. Assoc.* **2014**, *15*, 95-101.

[8] Lee, W.; Liu, L.; Peng, L.; Lin, M.; Chen, L. Comparisons of Sarcopenia Defined by IWGS and EWGSOP Criteria Among Older People: Results From the I-Lan Longitudinal Aging Study. *J. Am. Med. Dir. Assoc.* **2013**, *14*, 521-528.

[9] Wu, I.; Lin, C.; Hsiung, C.A.; Wang, C.; Wu, C.; Chan, D.D.; Li, T.; Lin, W.; Huang, K.; Chen, C.; Hsu, C. Epidemiology of sarcopenia among community-dwelling older adults in Taiwan: A pooled analysis for a broader adoption of sarcopenia assessments. *Geriatr. Gerontol. Int.* **2014**, *14*, 52-60.

[10] Wu, C.; Chen, K.; Hou, M.; Chang, Y.; Chang, C.; Liu, P.; Wu, S.; Chiu, C.; Jou, I.; Chen, C. Prevalence and associated factors of sarcopenia and severe sarcopenia in older Taiwanese living in rural community: The Tianliao Old People study 04. *Geriatr. Gerontol. Int.* **2014**, *14*, 69-75.

[11] Li, M.; Hu, Y.; Dong, H.Y.; Zhang, Y.; Fan, L.; Zhang, M.X.; Sun, J.; Han, X.Q.; Liu, Y.X.; Ma, X.N. Compare different measurement for prevalence of sarcopenia in a cohort of healthy community dwellingely in Beijing area. *Chin. J. Health Care Med.* **2014**, *16*, 426-429(Chinese article).

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

[12] Meng, N.; Li, C.; Liu, C.; Lin, C.; Lin, W.; Chang, C.; Li, T.; Lin, C. Comparison of height- and weight-adjusted sarcopenia in a Taiwanese metropolitan older population. *Geriatr. Gerontol. Int.* **2015**, *15*, 45-53.

[13] Wells, G.A.; Shea, B.; O'Connell, D.; Peterson, J.; Welch, V.; Losos, M.; Tugwell, P. The Newcastle-Ottawa Scale (NOS) for assessing the quality of nonrandomised studies in meta-analyses. Available: http://www.ohri.ca/programs/clinical_epidemiology/oxford.asp. Accessed 2014 Aug 5

[14] Higgins, J.P.T.; Green, S. Cochrane Handbook for Systematic Reviews of Interventions Version 5.1.0. Available: http://www.cochrane-handbook.org. Accessed 2014 Aug

[15] Wen, X.; An, P.; Chen, W.C.; Lv, Y.; Fu, Q. Comparisons of sarcopenia prevalence based on different diagnostic criteria in Chinese older adults. *The journal of nutrition, health & aging*. **2015**, *19*, 342-347.

[16] Zeng, P.; Wu, S.; Han, Y.; Liu, J.; Zhang, Y.; Zhang, E.; Zhang, Y.; Gong, H.; Pang, J.; Tang, Z.; Liu, H.; Zheng, X.; Zhang, T. Differences in body composition and physical functions associated with sarcopenia in Chinese elderly: Reference values and prevalence. *Arch. Gerontol. Geriat.* **2015**, *60*, 118-123.

[17] Gao, L.; Jiang, J.; Yang, M.; Hao, Q.; Luo, L.; Dong, B. Prevalence of Sarcopenia and Associated Factors in Chinese Community-Dwelling Elderly: Comparison Between Rural and Urban Areas. *J. Am. Med. Dir. Assoc.* **2015**, *16*, 1001-1003.

[18] Wang, Y.; Wang, Y.; Zhan, J.; Tang, Z.; He, J.; Tan, P.; Deng, H.; Huang, W.; Liu, Y. Sarco-Osteoporosis: Prevalence and Association with Frailty in Chinese Community-Dwelling Older Adults. *Int. J. Endocrinol.* **2015**, *2015*, 1-8.

[19] Woo, J.; Leung, J.; Morley, J.E. Defining Sarcopenia in Terms of Incident Adverse Outcomes. *J. Am. Med. Dir. Assoc.* **2015**, *16*, 247-252.

[20] Wang, H.; Hai, S.; Cao, L.; Zhou, J.; Liu, P.; Dong, B. Estimation of prevalence of sarcopenia by using a new bioelectrical impedance analysis in Chinese community-dwelling elderly people. *BMC Geriatr.* **2016**, *16*.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

[21] Han, P.; Kang, L.; Guo, Q.; Wang, J.; Zhang, W.; Shen, S.; Wang, X.; Dong, R.; Ma, Y.; Shi, Y.; Shi, Z.; Li, H.; Li, C.; Ma, Y.; Wang, L.; Niu, K. Prevalence and Factors Associated With Sarcopenia in Suburb-dwelling Older Chinese Using the Asian Working Group for Sarcopenia Definition. *The Journals of Gerontology Series A: Biological Sciences and Medical Sciences*. **2016**, *71*, 529-535.

[22] Han, D.; Chang, K.; Li, C.; Lin, Y.; Kao, T.; Tsai, K.; Wang, T.; Yang, W. Skeletal muscle mass adjusted by height correlated better with muscular functions than that adjusted by body weight in defining sarcopenia. *Sci. Rep.-UK*. **2016**, *6*.

[23] Xia, Z.W.; Meng, L.P.; Man, Q.Q.; Li, L.X.; Song, P.K.; Li, Y.Q.; Gao, Y.X.; Jia, S.S.; Zhang, J. Analysis of the dietary factors on sarcopenia in elderly in Beijing. *J Hygiene Res.* **2016**, *45*, 388-393 (Chinese article).

[24] Chen, Z.; Dai, R.; Gu, X.H. Prevalence of sarcopenia and quality of life in retired cadres of a military unit. *Chin. J. Osteopo. Rosis. & Bone Miner. Res.* **2016**, *9*, 264-268 (Chinese article).

[25] Bai, H.; Sun, J.; Chen, M.; Xu, D.; Xie, H.; Yu, Z.; Bao, Z.; Chen, J.; Pan, Y.; Lu, D.; Cheng, S. Age-related decline in skeletal muscle mass and function among elderly men and women in Shanghai, China: A cross sectional study. *Asia Pac. J. Clin. Nutr.* **2016**, *25*, 326-332..

[26] Hai, S.; Wang, H.; Cao, L.; Liu, P.; Zhou, J.; Yang, Y.; Dong, B. Association between sarcopenia with lifestyle and family function among community-dwelling Chinese aged 60 years and older. *BMC Geriatr.* **2017**, *17*.

[27] Ma, Y.; Bian, A.L.; Wang, X.F.; Wang, J.X.; Zhou, X.Z. Difference of the prevalence rate of sarcopenia in elderly people with different gender and age in Tianjin. *Chin. General Practice* **2017**, *20*, 3267-3271 (Chinese article).

[28] Wu, Z.N.; Wang, Y.B.; Shu, F.Q.; Song, B.; Li, G.; Zhu, H.M. Epidemiological investigation of sarcopenia in the nursing homes in Fengxian District and analysis of their risk factors. *Shanghai Med.* **2017**, *38*, 37-40 (Chinese article).

[29] Huang, L.J.; Liu, Y.C.; Wen, Y.F.; Liu, K. State and associated factors of sarcopenia in Jinzhou community elderly people. *Modern Preve. Med.* **2017**, *44*, 4458-4460 (Chinese article).

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

[30] Yang, M.; Hu, X.; Xie, L.; Zhang, L.; Zhou, J.; Lin, J.; Wang, Y.; Li, Y.; Han, Z.; Zhang, D.; Zuo, Y.; Li, Y.; Wu, L. Screening Sarcopenia in Community-Dwelling Older Adults: SARC-F vs SARC-F Combined With Calf Circumference (SARC-CalF). *J. Am. Med. Dir. Assoc.* **2018**, *19*, 271-277.

[31] Zeng, Y.; Hu, X.; Xie, L.; Han, Z.; Zuo, Y.; Yang, M. The Prevalence of Sarcopenia in Chinese Elderly Nursing Home Residents: A Comparison of 4 Diagnostic Criteria. *J. Am. Med. Dir. Assoc.* **2018**, *19*, 690-695.

[32] Wang, Y.N.; Xu, J.L.; Song, H.L. Influencing factors of sarcopenia among elderly patients. *Modern Clinical Nursing* **2018**, *17*, 45-50 (Chinese article).

[33] Yang, L.; Wu, G.; Yang, Y.; Wu, Y.; Zhang, L.; Wang, M.; Mo, L.; Xue, G.; Wang, C.; Weng, X. Nutrition, Physical Exercise, and the Prevalence of Sarcopenia in Elderly Residents in Nursing Homes in China. *Med. Sci. Monitor*. **2019**, *25*, 4390-4399.

[34] Du, Y.; Wang, X.; Xie, H.; Zheng, S.; Wu, X.; Zhu, X.; Zhang, X.; Xue, S.; Li, H.; Hong, W.; Tang, W.; Chen, M.; Cheng, Q.; Sun, J. Sex differences in the prevalence and adverse outcomes of sarcopenia and sarcopenic obesity in community dwelling elderly in East China using the AWGS criteria. *BMC Endocr. Disord.* **2019**, *19*.

[35] Wang, H.; Hai, S.; Liu, Y.; Liu, Y.X.; Zhou, J.H.; Yang, Y.; Dong, B.R.; Yue, J.R. Prevalence of sarcopenia and associated factors in community-dwelling elderly populations in Chengdu China. *J. Sichuan Univ.* **2019**, *50*, 224-228 (Chinese article).

[36] Zhang, Y.; Chen, X.; Hou, L.; Lin, X.; Qin, D.; Wang, H.; Hai, S.; Cao, L.; Dong, B. Prevalence and Risk Factors Governing the Loss of Muscle Function in Elderly Sarcopenia Patients: A longitudinal Study in China with 4 Years of Follow-Up. *The journal of nutrition, health & aging*. **2020**, *24*, 518-524.

[37] Liu, X.; Hao, Q.; Yue, J.; Hou, L.; Xia, X.; Zhao, W.; Zhang, Y.; Ge, M.; Ge, N.; Dong, B. Sarcopenia, Obesity and Sarcopenia Obesity in Comparison: Prevalence, Metabolic Profile, and Key Differences: Results from WCHAT Study. *The journal of nutrition, health & aging*. **2020**, *24*, 429-437.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

[38] Shafiee, G.; Keshtkar, A.; Soltani, A.; Ahadi, Z.; Larijani, B.; Heshmat, R. Prevalence of sarcopenia in the world: a systematic review and meta-analysis of general population studies. *Journal of Diabetes & Metabolic Disorders*. **2017**, *16*.

[39] Makizako, H.; Nakai, Y.; Tomioka, K.; Taniguchi, Y. Prevalence of sarcopenia defined using the Asia Working Group for Sarcopenia criteria in Japanese community-dwelling older adults: A systematic review and meta-analysis. *Physical therapy research*. **2019**, *22*, 53-57.

[40] Diz, J.B.M.; Leopoldino, A.A.O.; Moreira, B.D.S.; Henschke, N.; Dias, R.C.; Pereira, L.S.M.; Oliveira, V.C. Prevalence of sarcopenia in older Brazilians: A systematic review and meta-analysis. *Geriatr. Gerontol. Int.* **2017**, *17*, 5-16.

[41] Papadopoulou, S.K.; Tsintavis, P.; Potsaki, G.; Papandreou, D. Differences in the Prevalence of Sarcopenia in Community-Dwelling, Nursing Home and Hospitalized Individuals. A Systematic Review and Meta-Analysis. *The journal of nutrition, health & aging*. **2020**, *24*, 83-90.

[42] Han, A.; Sun, T.; Ming, J.; Chai, L.; Liao, X. Are the Chinese moving toward a healthy diet? Evidence from macro data from 1961 to 2017. *Int J Environ Res Public Health,* **2020**, *17*, 5294

Legend

Figure 1. Flowchart of the detailed procedure for the inclusion or exclusion of selected studies.

Figure 2. Forest plots for the total prevalence of sarcopenia among the elderly Chinese population according to the Asian Working Group for Sarcopenia (AWGS) standard. Diamond represents the pooled odds ratio at 95% CI.

Figure 3. Forest plots for the total prevalence of sarcopenia among the elderly Chinese population according to the International Working Group on Sarcopenia (IWGS) standard. Diamond represents the pooled odds ratio at 95% CI.

Figure 4. Forest plots for the total prevalence of sarcopenia among the elderly Chinese population according to the European Working Group on Sarcopenia in Older People (EWGSOP) standard. Diamond represents the pooled odds ratio at 95% CI.

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Figure 5. Forest plots for the prevalence of sarcopenia among male and female elderly people according to the Asian Working Group for Sarcopenia (AWGS) standard. Diamond represents the pooled odds ratio at 95% CI.

Figure 6. Forest plots for the prevalence of sarcopenia among male and female elderly people according to the International Working Group on Sarcopenia (IWGS) standard. Diamond represents the pooled odds ratio at 95% CI.

Figure 7. Forest plots for the prevalence of sarcopenia among male and female elderly people according to the European Working Group on Sarcopenia in Older People (EWGSOP) standard. Diamond represents the pooled odds ratio at 95% CI.

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394 **Table 1 Study characteristics of the studies included in the meta-analysis**

Author	Publication Year	Participants(n)			Region	Setting	Measurement	Diagnostic Criteria	Prevalence (%)			NOS
		Total	Male	Female					Total	Male	Female	
Lee WJ	2013	386	223	163	Taiwan	Urban Community	DEXA	IWGS	11.1	10.8	11.7	6
								EWGSOP	16.6	14.9	19.0	
Wu IC	2014	2155	1055	1100	Taiwan	Urban Community	BIA	EWGSOP	3.9	5.4	2.5	6
Wu CH	2014	549	285	264	Taiwan	Rural Community	BIA	EWGSOP	12.7	8.1	17.8	6
Li M	2014	169	169	0	Beijing	Urban Community	DEXA	EWGSOP	33.3	33.3	-	5
Meng NH	2015	771	359	712	Taiwan	Urban Community	DEXA	EWGSOP	9.7	0.7	12.5	6
Gao LL	2015	612	254	358	Sichuan	Rural and urban Community	Calf circumference	AWGS	9.8	6.7	12.0	6

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Wen X	2015	286	136	150	Hangzhou	Urban Community	DEXA	IWGS	6.0	7.4	4.7	5
								EWGSOP	0.4	0.8	0.01	
								AWGS	3.1	5.9	0.7	
Woo J	2015	4000	2000	2000	Hongkong	Urban Community	DEXA	IWGS	20.2	22.1	18.3	6
								EWGSOP	9.0	9.5	8.6	
								AWGS	7.3	5.3	9.3	
Zeng P	2015	395	199	196	Beijing	Rural and urban Community	BIA	AWGS	24.3	18.1	30.6	5
Wang YJ	2015	316	164	152	Changsha	Urban Community	DEXA	AWGS	29.7	26.2	33.6	6
Han PP	2016	1069	467	602	Tianjin	Urban Community	BIA	AWGS	9.3	6.4	17.5	5
Wang H	2016	636	316	320	Chengdu	Urban Community	BIA	AWGS	10.4	8.2	12.5	5
Bai HJ	2016	415	212	203	Shanghai	Urban Community	BIA	AWGS	28.7	33.5	23.6	6

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Xia ZW	2016	683	239	444	Beijing	Urban Community	BIA	AWGS	20.1	17.2	21.6	5
Chen Z	2016	281	203	78	Chongqing	Urban Community	BIA	IWGS	27.8	-	-	6
Han DS	2016	878	402	476	Taiwan	Urban Community	BIA	EWGSOP	9.8	4.5	14.3	6
Hai S	2017	834	415	419	Chengdu	Urban Community	BIA	AWGS	10.6	11.3	9.8	6
Huang LJ	2017	193	65	128	Jinzhou	Urban Community	BIA	AWGS	11.4	7.7	13.3	6
Yang M	2018	384	160	224	Chengdu	Urban Community	BIA	AWGS	15.9	11.9	18.8	6
								IWGS	25.0	24.4	25.4	
								EWGSOP	11.7	10.6	12.5	
Du YP	2019	631	213	418	Shanghai	Urban Community	BIA	AWGS	12.2	19.2	8.6	5

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Wang H	2019	947	465	482	Chengdu	Urban Community	BIA	AWGS	7.7	6.5	8.9	5
Zhang Y	2020	474	238	236	Chengdu	Urban Community	BIA	AWGS	15.4	-	-	5
Liu XL	2020	4500	1627	2873	Yunnan, Guizhou, Sichuan, and Xinjiang	Urban Community	BIA	AWGS	19.3	22.1	17.8	6

DEXA, dual-energy x-ray absorptiometry; BIA, bioelectrical impedance analysis; IWGS, International Working Group on Sarcopenia; AWGS, Asia Working Group for Sarcopenia; EWGSOP, European Working Group on Sarcopenia in Older People; NOS, Newcastle–Ottawa scale.

Table 2 The prevalence of sarcopenia diagnosed by different criteria using DEXA and BIA measurement (%).

	DEXA	**BIA**
AWGS	13(5–21)	15(12–19)

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IWGS	12(3–22)	26(23–29)
EWGSOP	12(6–18)	9(5–14)

DEXA, dual-energy x-ray absorptiometry; BIA, bioelectrical impedance analysis; IWGS, International Working Group on Sarcopenia; AWGS, Asia Working Group for Sarcopenia; EWGSOP, European Working Group on Sarcopenia in Older People.

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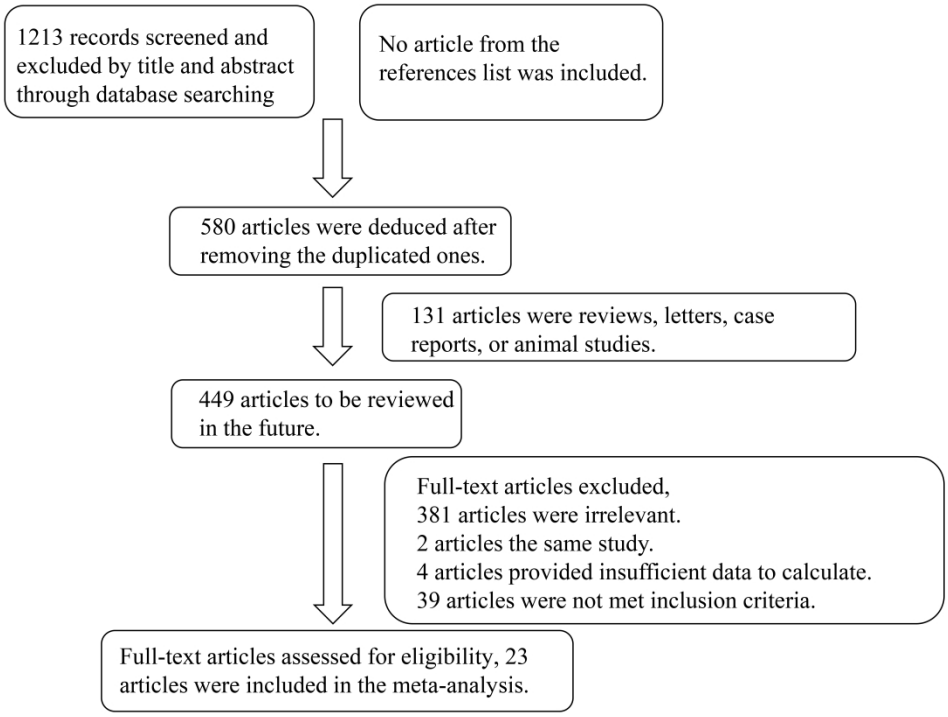


Figure 1. Flowchart of the detailed procedure for the inclusion or exclusion of selected studies.

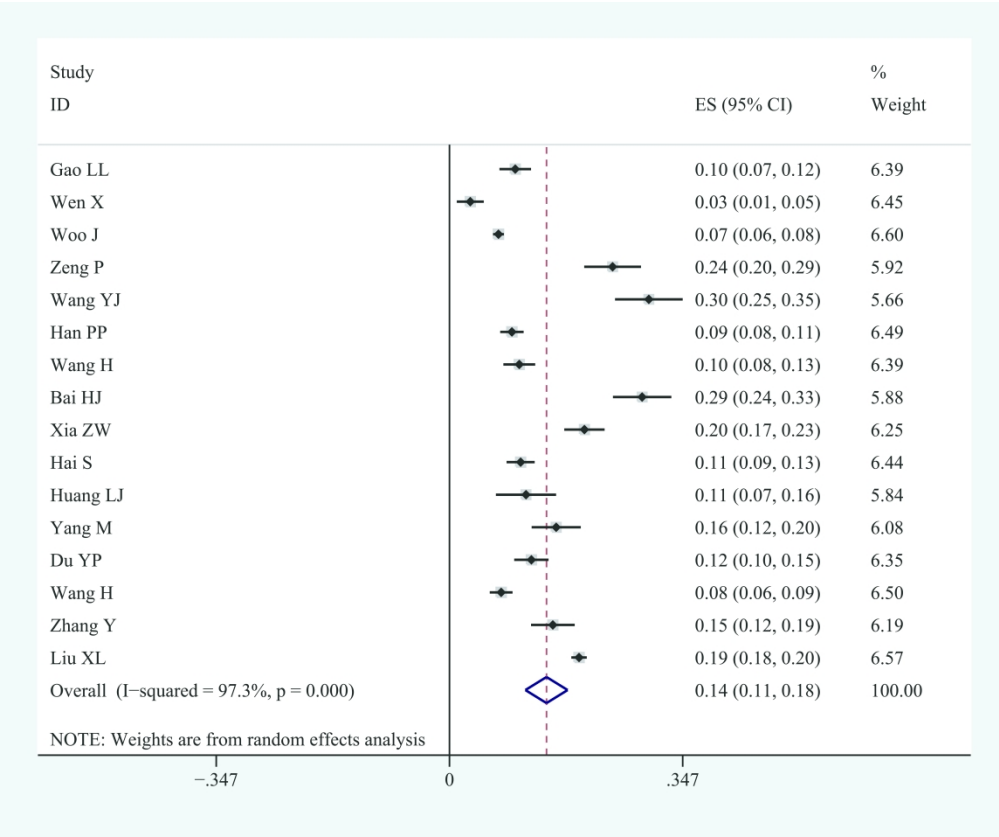


Figure 2. Forest plots for the total prevalence of sarcopenia among the elderly Chinese population according to the Asian Working Group for Sarcopenia (AWGS) standard. Diamond represents the pooled odds ratio at 95% CI.

280x233mm (600 x 600 DPI)

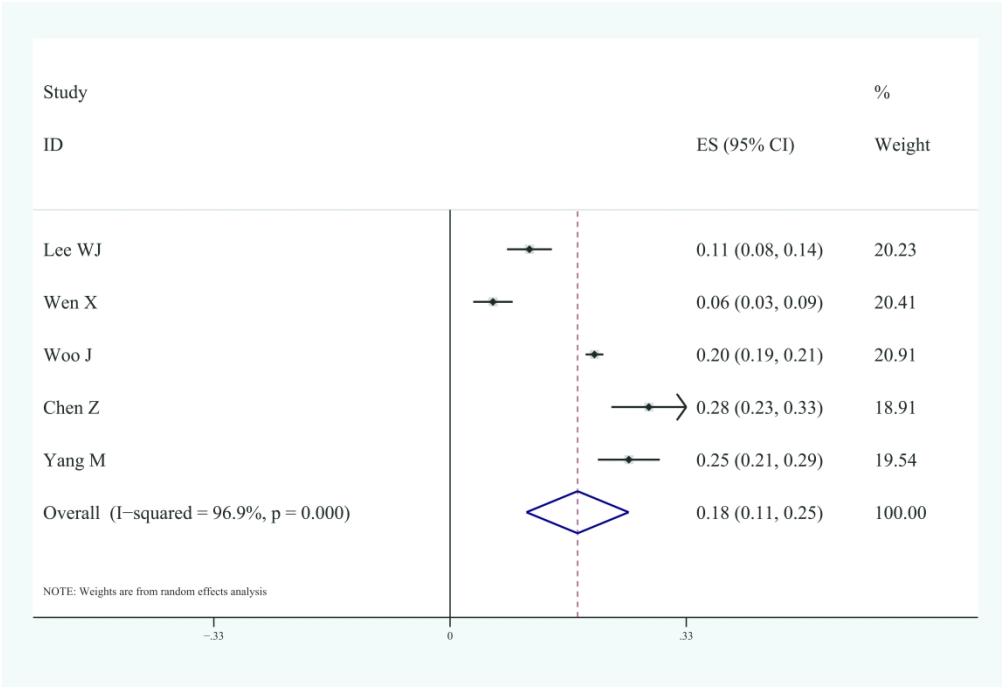


Figure 3. Forest plots for the total prevalence of sarcopenia among the elderly Chinese population according to the International Working Group on Sarcopenia (IWGS) standard. Diamond represents the pooled odds ratio at 95% CI.

272x186mm (600 x 600 DPI)

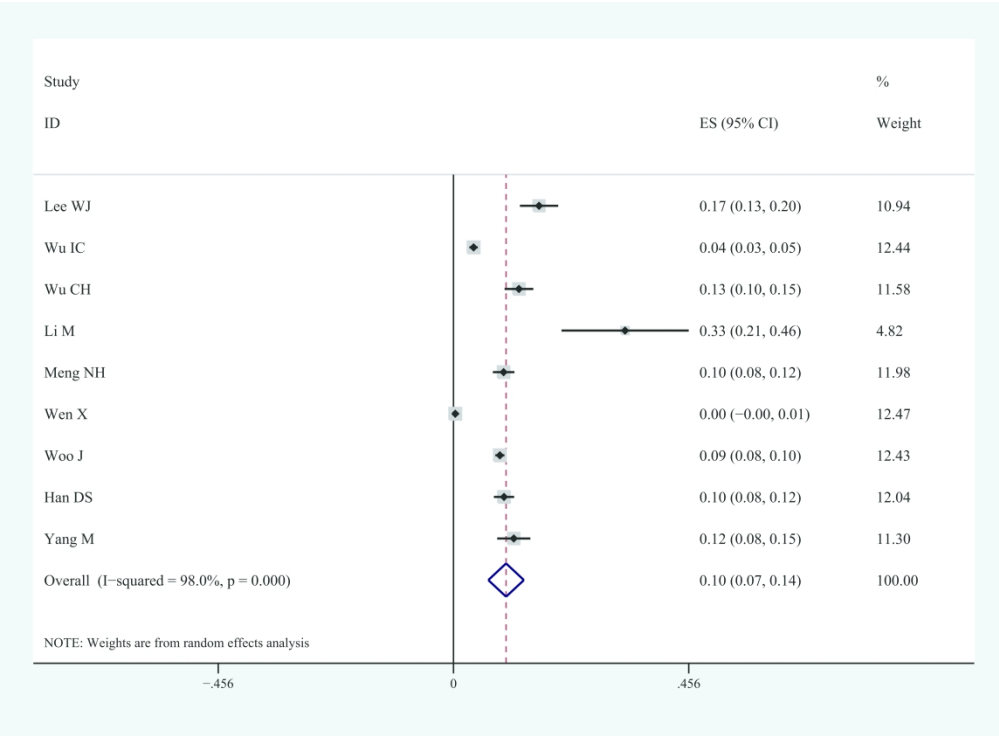


Figure 4. Forest plots for the total prevalence of sarcopenia among the elderly Chinese population according to the European Working Group on Sarcopenia in Older People (EWGSOP) standard. Diamond represents the pooled odds ratio at 95% CI.

274x201mm (600 x 600 DPI)

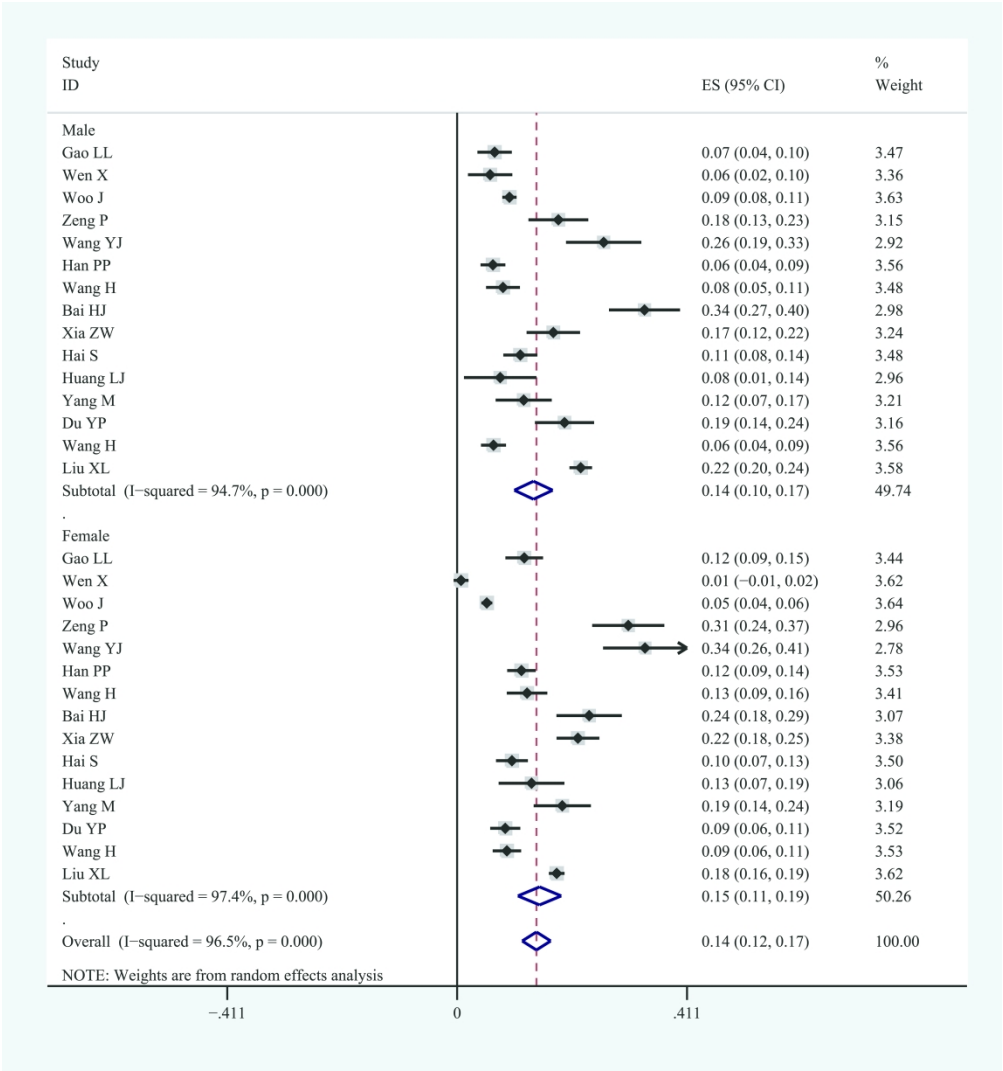


Figure 5. Forest plots for the prevalence of sarcopenia among male and female elderly people according to the Asian Working Group for Sarcopenia (AWGS) standard. Diamond represents the pooled odds ratio at 95% CI.

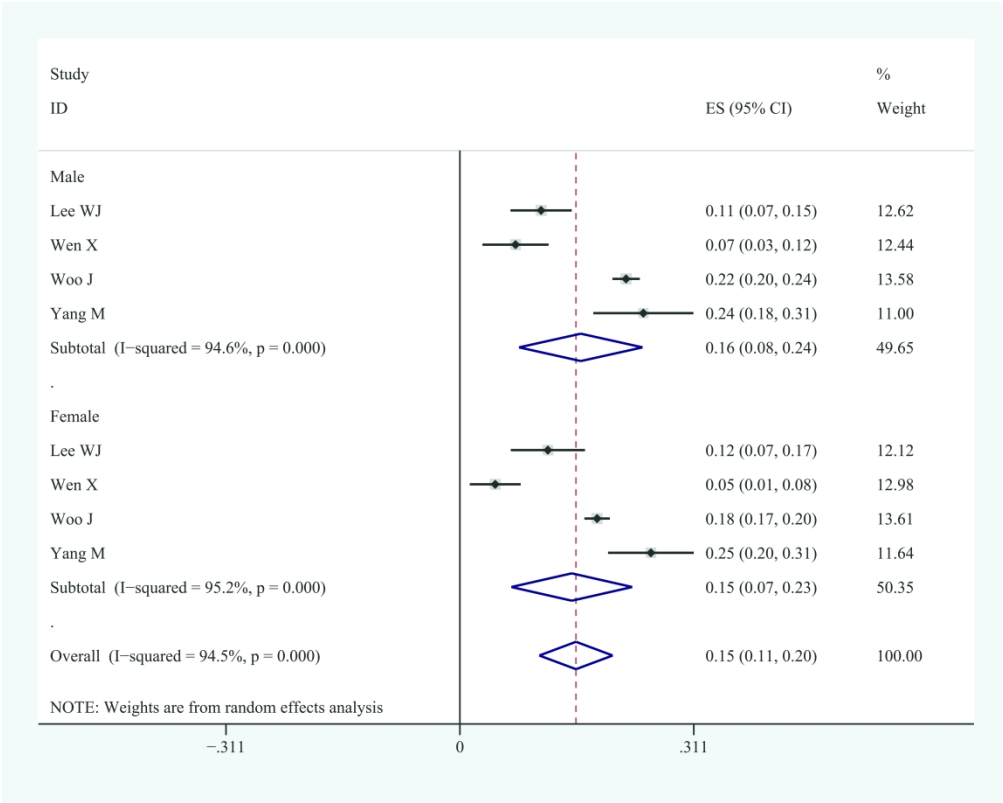


Figure 6. Forest plots for the prevalence of sarcopenia among male and female elderly people according to the International Working Group on Sarcopenia (IWGS) standard. Diamond represents the pooled odds ratio at 95% CI.

278x223mm (600 x 600 DPI)

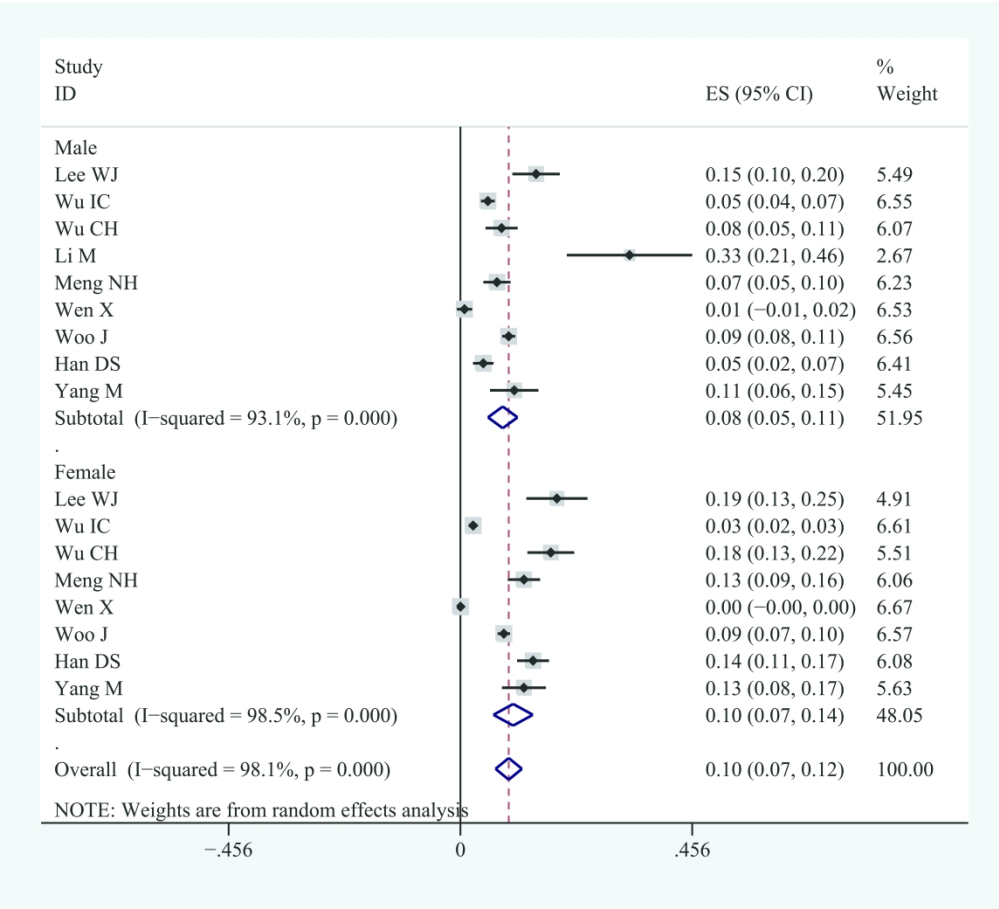


Figure 7. Forest plots for the prevalence of sarcopenia among male and female elderly people according to the European Working Group on Sarcopenia in Older People (EWGSOP) standard. Diamond represents the pooled odds ratio at 95% CI.

284x257mm (600 x 600 DPI)

Figure S1. The sensitivity analysis results according to the Asian Working Group for Sarcopenia (AWGS) standard.

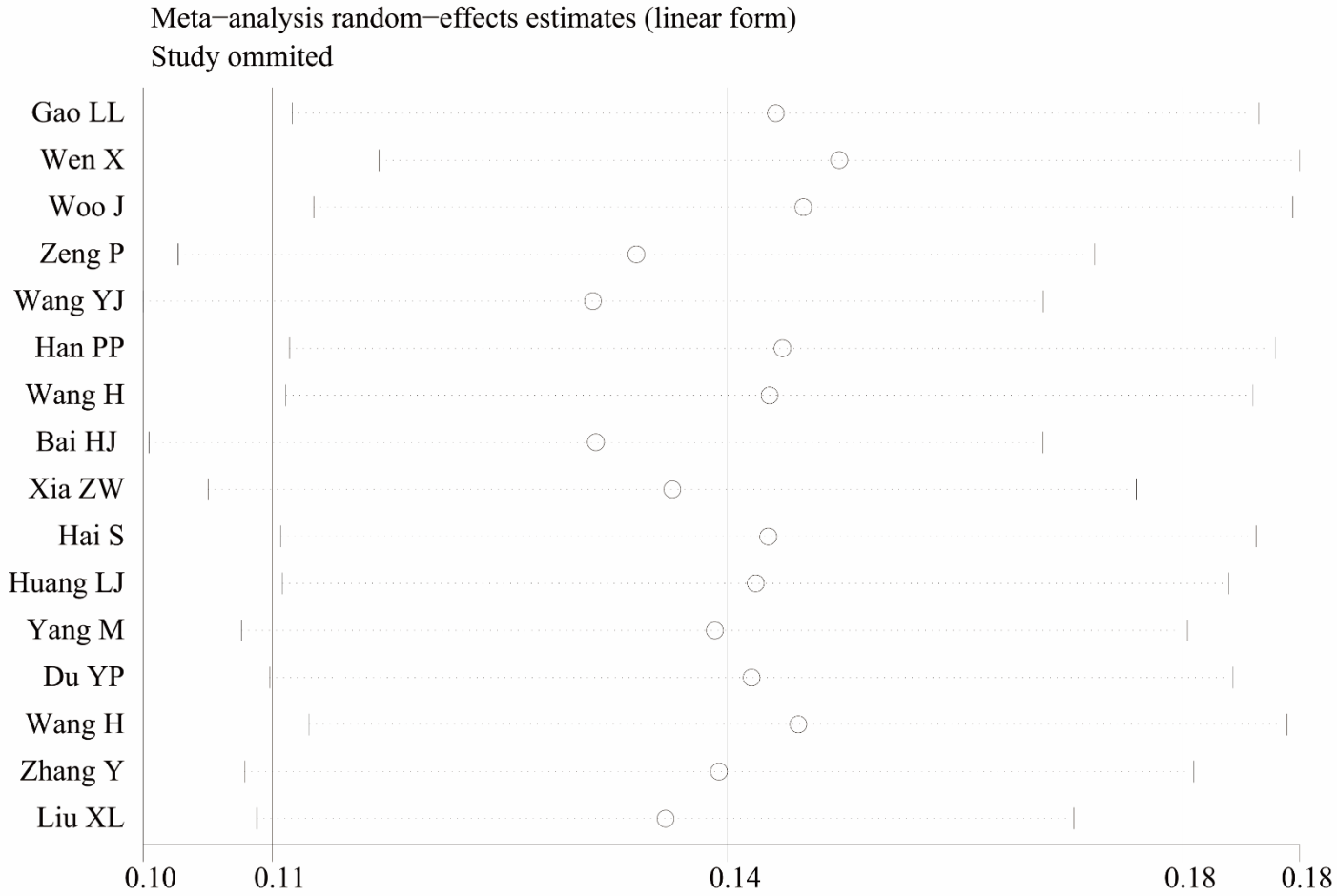


Figure S2. The sensitivity analysis results according to the International Working Group on Sarcopenia (IWGS) IWGS standard.

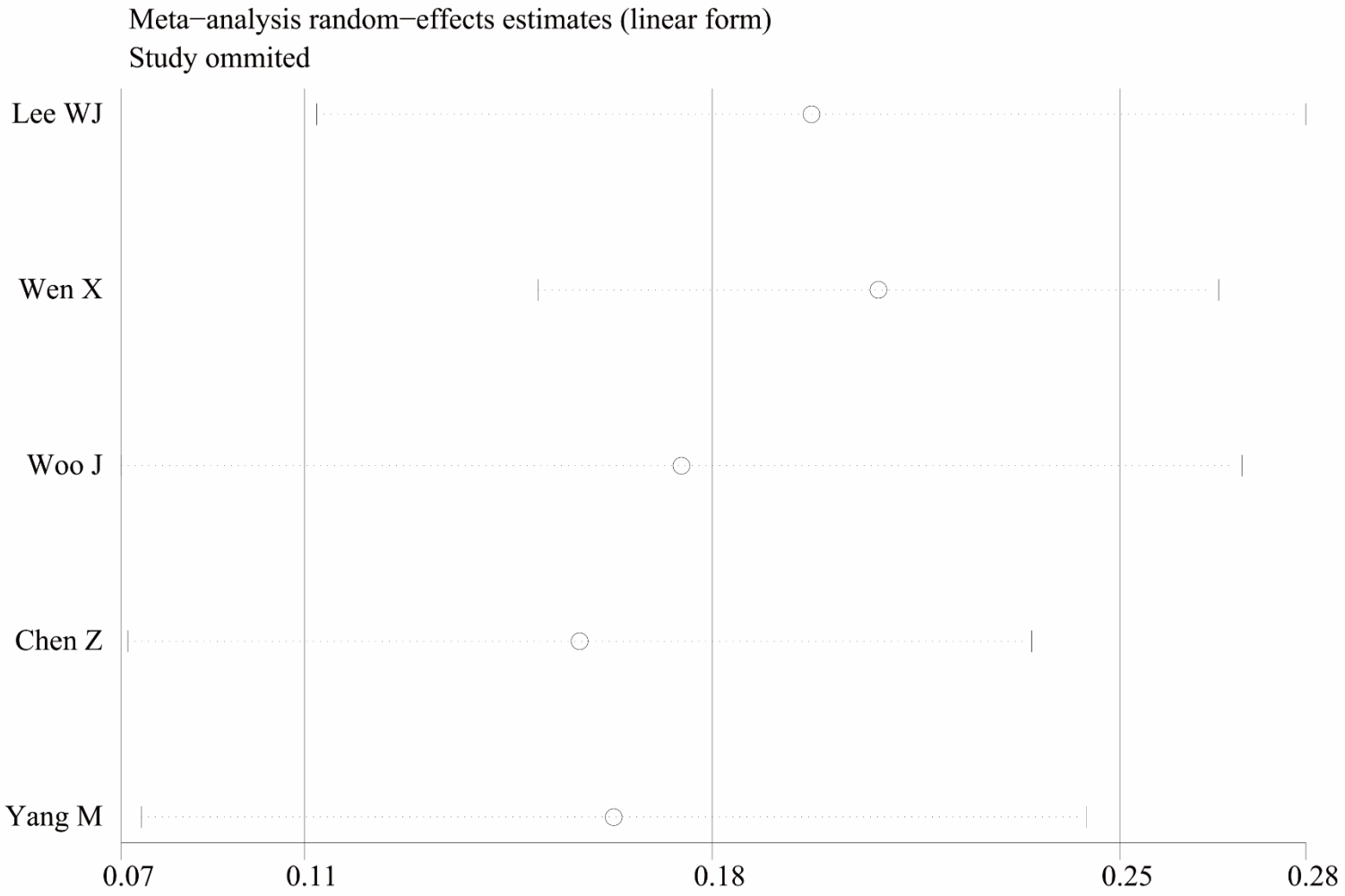


Figure S3. The sensitivity analysis results according to the European Working Group on Sarcopenia in Older People (EWGSOP) standard.

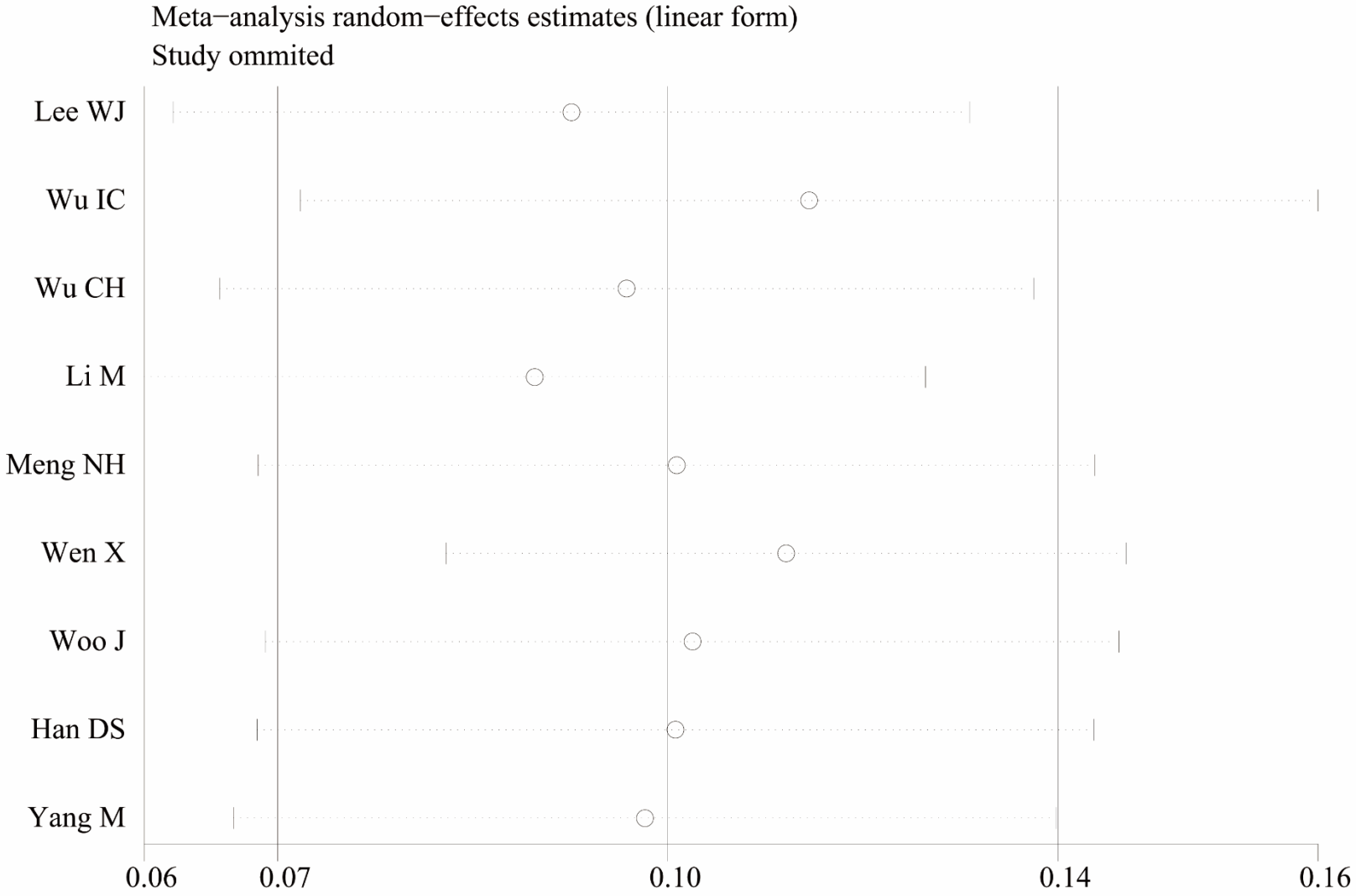


Figure S4. The publication bias results according to the Asian Working Group for Sarcopenia (AWGS) standard.

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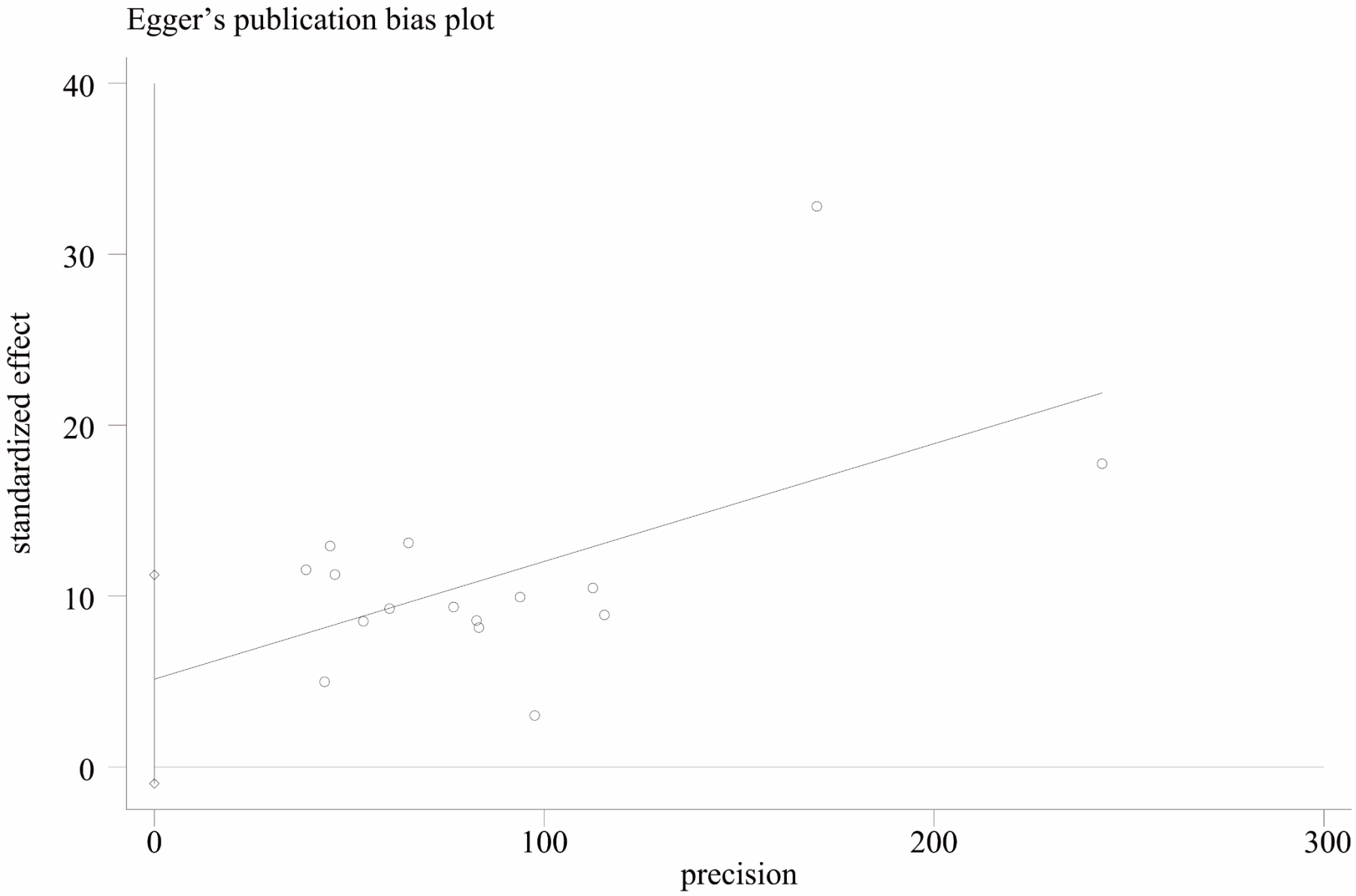


Figure S5. The publication bias results according to the the International Working Group on Sarcopenia (IWGS) standard.

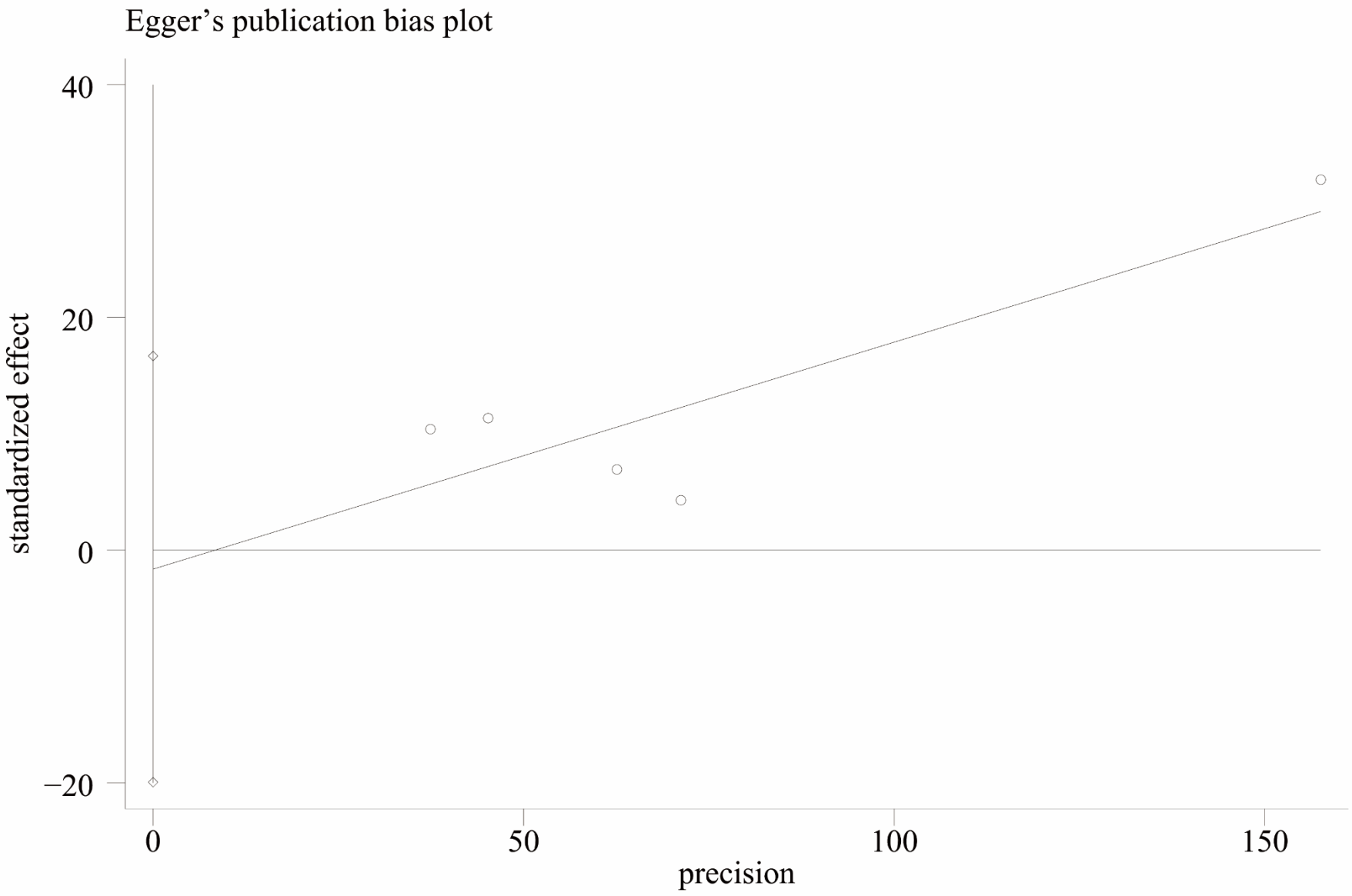
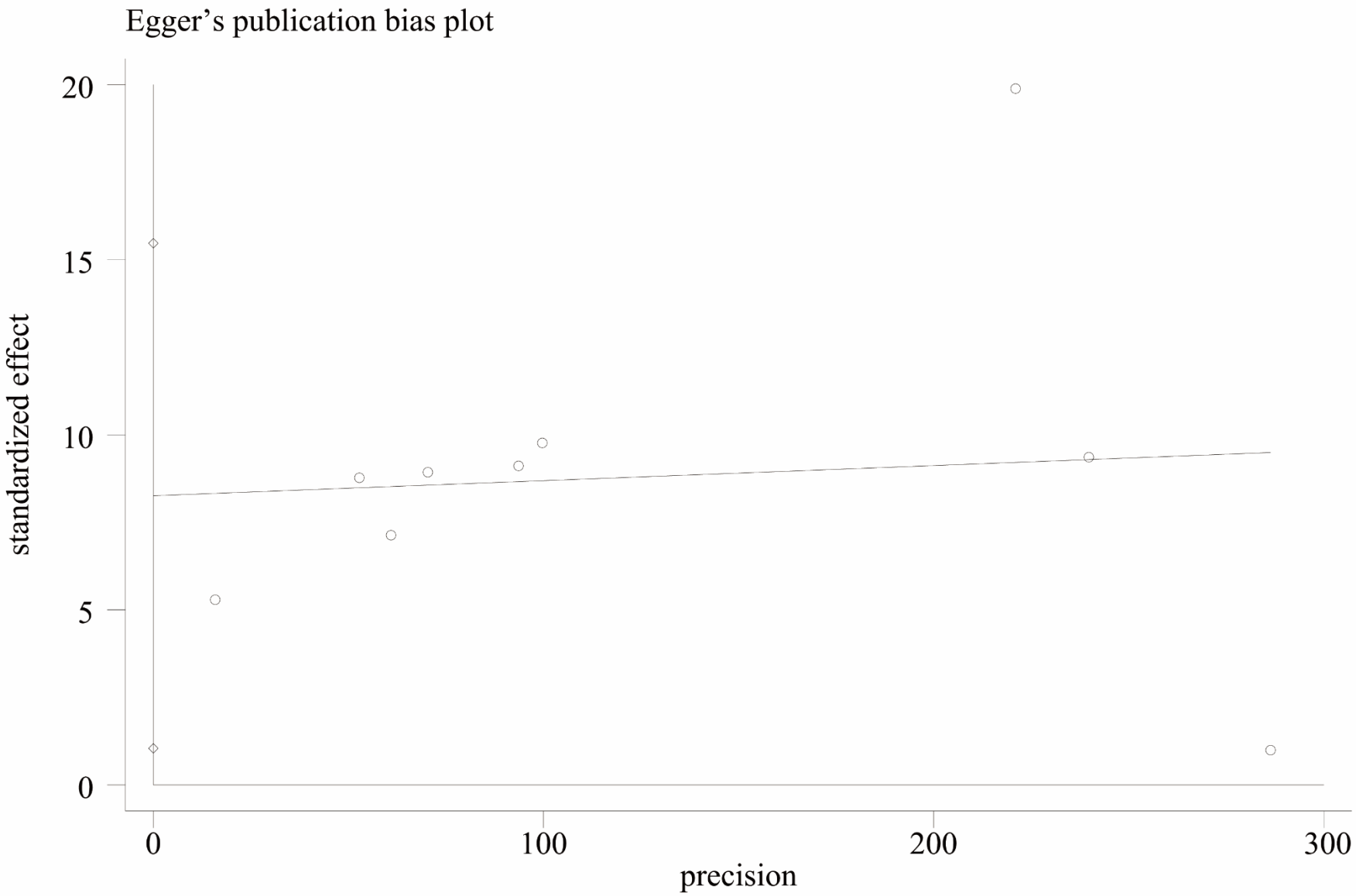


Figure S6. The publication bias results according to the European Working Group on Sarcopenia in Older People (EWGSOP) standard.


Egger's publication bias plot
standardized effect
precision
20
15
10
5
0
0
100
200
300

PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	3
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	3,4
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	4
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	4
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	4
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	4
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	4
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	5
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	5
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	5
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	5
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	5

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	5
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	5
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	6
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	6
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	6
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	6
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	6,7
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	7
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	7
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	7,8
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	8
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	8
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	9

From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(7): e1000097. doi:10.1371/journal.pmed1000097

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